

POINT OPERATIONS AND MATCHING-LINE BUBBLES IN THE ATIYAH–FLOER COMPARISON

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ABSTRACT. We prove compatibility of the integral point operations in the admissible $SO(3)$ Atiyah–Floer comparison of Daemi, Fukaya, and Lipyanskiy, in their embedded-Lagrangian setting. In the positive p_1 normalization, the symplectic point operation is shifted by 8, 4, or 0 according as the marked component has genus one, two, or at least three. One coherent orientation system works for both matching components. Together with the known surface and loop comparisons and integral operation relations, this gives the corrected observable-algebra action over every commutative coefficient ring. When four is invertible, quarter-Pontryagin normalization gives the shifts 2, 1, and 0.

An index-four mixed sequence can develop a matching bubble of normalized energy one-half. Holomorphic doubling and the converse estimates identify compact limiting models: degree-one stable families and single elementary modifications of constant families. A Dirichlet construction reconstructs these models and glues them to regular index-zero mixed solutions. Converse gluing and the incidence conditions identify every bubbling end of the chosen point cut. The relative incidence degrees are eight and four in genera one and two. Comparing the actual gluing derivative with determinant-line excision determines their signs. Ordinary reference orientations agree for the two homologous matching surfaces; a local sphere comparison and a stabilized torus example fix the genus-two normalization even when the ambient rational Floer homology vanishes. The resulting boundary identity uses the original integral geometric count as its homotopy.

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1. INTRODUCTION

1.1. The point operation. Let $(Y_\#, E_\#)$ be an admissible $SO(3)$ pair split along a closed oriented surface Σ . Atiyah’s gauge–symplectic comparison principle predicts an identification of the two resulting Floer theories [Ati88]. In the embedded-Lagrangian setting of Daemi–Fukaya–Lipyanskiy, henceforth DFL, the splitting produces an instanton Floer complex C_G with differential d_G , a Lagrangian Floer complex C_S with differential d_S , and a comparison chain map

$$N : C_G \longrightarrow C_S.$$

Their main theorem identifies the induced homologies [DFL26, Theorem 2]. The comparison is defined by counting solutions of a mixed equation: an anti-self-dual connection on one side of a codimension-one seam and a pseudoholomorphic map on the other, with their boundary values identified in the symplectic moduli space of flat connections [AB83]. The regularity, Fredholm, compactness, and removal theorems for this equation are proved in [DFL24].

The universal bundle over the flat moduli space and the splitting surface gives operations indexed by $H_*(\Sigma)$. DFL prove that N intertwines the operations associated with $H_2(\Sigma)$ and $H_1(\Sigma)$ [DFL26, Section 7.2]. They also explain why the same argument stops at $H_0(\Sigma)$: a point insertion requires four-dimensional mixed moduli spaces, and these can bubble on the matching line. A neighborhood of that boundary was not part of their compactification.

We prove the integral point comparison by first determining the boundary of a suitable point cut. The cylinder calculation shows that uncorrected compatibility is impossible in low genus. The geometric boundary retains local orientation factors. Ordinary determinant transport and a calibration using the signed genus-two quantum relation identify these factors in one coherent system.

An index-four sequence can develop a matching bubble of normalized energy one-half. Holomorphic doubling splits its charge between a stable family and elementary modifications. The compact limiting models are degree-one stable families and single elementary modifications of constant families. We reconstruct these models by a Dirichlet problem and glue them to regular index-zero mixed solutions. Converse gluing and the incidence conditions identify every bubbling end of the chosen point cut. The ordinary broken limits then give Theorem 11.5. Determinant-line comparisons determine the signed local contributions in one coherent normalization. The final homotopy is the original integral count, divided by four for the rational normalization.

Convention 1.1. All moduli spaces use DFL’s orientation, grading, and end conventions. The symbol p_1 denotes the positive integral degeneracy cycle determined by their product orientation. Unless an integral statement is explicitly made, coefficients are rational. A *matching atom* is an atom of the mixed energy measure at a point of the matching line. It is *primitive* when its relative clutching number is one, equivalently when its normalized energy is $\frac{1}{2}$. Normalized energy means DFL’s topological energy, recalled in (3).

1.2. Results. Write

$$\Sigma = \Sigma_0 \sqcup \Sigma_1, \quad g_j = \text{genus}(\Sigma_j).$$

We retain all admissibility, irreducibility, and embeddedness hypotheses of DFL. In particular, the standard secondary perturbations are supported away from the matching line. For $q \in \Sigma_j$, let b_q^G and b_q^S be the gauge and symplectic operations represented by the positive integral degeneracy cycle $p_1/[q]$. Over $\mathbb{Q}$ put

$$m_q^G = \frac{1}{4}b_q^G, \quad m_q^S = \frac{1}{4}b_q^S.$$

This is our quarter-Pontryagin rescaling of DFL's positive integral convention. The sign is fixed by their product orientation on the linear-dependence cycle, rather than by the alternative convention $\mu(q) = -\frac{1}{4}p_1/[q]$ sometimes used in instanton theory [DK90]. For the cylinder splitting of $\Sigma_g \times S^1$, abbreviate the corresponding point operations by b_G and b_S .

Theorem A (integral geometric point comparison; = Theorem 12.13). *There are one coherent orientation normalization and compatible equation and section data for which the original integral point-cut counts $K_q^{p_1}$ satisfy*

$$(1) \quad Nb_q^G - b_q^S N = d_S K_q^{p_1} + K_q^{p_1} d_G + c_{g_j} N, \quad c_g = \begin{cases} 8, & g = 1, \\ 4, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

The data can be chosen for both matching components, or any specified finite set of point insertions, with the same equation and comparison map N . The identity is integral and base-changes to every commutative coefficient ring.

The orientation normalization is specified in Proposition 12.12. It is determined by a local comparison of ordinary closed and mixed determinant lines. When four is invertible, put $m_q = \frac{1}{4}b_q$ and $K_q = \frac{1}{4}K_q^{p_1}$. The corrected symplectic point operation is then

$$\hat{m}_q^S = m_q^S + \kappa_{g_j} \text{id}, \quad (\kappa_1, \kappa_2, \kappa_{g \geq 3}) = (2, 1, 0).$$

Its rational version is Theorem 13.1.

For the observable actions, we use cohomological indexing of the DFL complexes:

$$C_G^r = (C_G)_{-r}, \quad C_S^r = (C_S)_{-r},$$

with indices read modulo four. Thus the differentials have degree one; see Section 13. For a commutative ring R , give $V_R = H_*(\Sigma; \mathbb{Z}) \otimes R$ the degrees $|x| = 4 - k$ for $x \in H_k(\Sigma; \mathbb{Z})$, and set

$$\mathbb{A}_{p_1}(\Sigma; R) = T_R(V_R) / \langle xy - (-1)^{|x||y|}yx \rangle.$$

In particular, an odd generator satisfies $2x^2 = 0$. When two is invertible the algebra is symmetric on the point and surface classes and exterior on the loop classes.

Corollary B (corrected observable action; = Corollary 13.5). *For every commutative ring R , the DFL comparison is an isomorphism of $\mathbb{A}_{p_1}(\Sigma; R)$ -modules after each symplectic point generator is corrected by $c_{g_j} \text{id}$. Its action kernels agree and its image operation algebras are identified by conjugation. For every chosen word in the generators, the intertwining identity has an explicit chain homotopy.*

DFL's surface and loop counts and Theorem A give the integral generator homotopies. The operation relations use two labeled ordinary insertions below the index-eight bubbling threshold. A separate ordinary path-of-points homotopy treats the two-point pair. These relations transfer through N and its integer inverse, and the resulting chain identities base-change to arbitrary R . Wordwise comparison homotopies follow by the graded Leibniz rule.

The geometric boundary result used in the proof is recorded next. Let M_g denote the moduli space of flat connections on the nontrivial $SO(3)$ bundle over a connected genus- g surface. Under the Narasimhan–Seshadri correspondence it is the moduli space of stable rank-two bundles with fixed odd determinant [NS65]. Muñoz's extension space and compact primitive bundle space are defined as follows. Fix the determinant line Λ , and let $\mathcal{E}_g$ be the rank- g vector bundle on $J(\Sigma_g)$ with fiber $H^1(\Sigma_g, L^2 \otimes \Lambda^{-1})$ over L . Then

$$(2) \quad N_g = \mathbb{P}(\mathcal{E}_g^\vee) \longrightarrow J(\Sigma_g), \quad \mathfrak{M}_g = \mathbb{P}((\mathcal{E}_g \oplus \mathcal{E}_g)^\vee) \longrightarrow J(\Sigma_g),$$

and there is a natural map $i_g : N_g \rightarrow M_g$ [Muñ99b]. A point of a fiber of $\mathfrak{M}_g$ is a pair of extension vectors $[v_0 : v_1]$. Independent vectors give a minimal rational curve; dependent vectors give a degree-one quasimap with one basepoint.

For the next statement, a *primitive matching bubble* means a rescaled limit with relative clutching charge one. We call it *full rank* when v_0 and v_1 are independent and *rank drop* when they are dependent. An algebraic limit is *admissible along the seam* when every surface restriction over the compactified matching line is stable and any quasimap basepoint lies on the gauge-theoretic hemisphere.

Theorem C (fixed-point geometric boundary; Theorem 11.5 and Proposition 12.1). *For a fixed point $q \in \Sigma_j$, compatible equation and section choices make the index-four point cut regular. After removal of finitely many primitive gluing neighborhoods, it compactifies to an oriented one-manifold with ordinary Floer boundary and artificial primitive scale boundaries. Its integral counts satisfy*

$$Nb_q^G - b_q^S N = d_S K_q^{p_1} + K_q^{p_1} d_G + B_q.$$

For genus two, $B_q = 4\chi_j N$, where $\chi_j \in \{+1, -1\}$ is the sphere compatibility sign in the chosen coherent system. It equals +1 when this component is selected in DFL’s sphere normalization. For genus at least three, $B_q = 0$. For a torus, each residual x_0 contributes $8n(x_0)\eta_1(x_0)$, where $n(x_0)$ is its oriented count and $\eta_1(x_0) \in \{+1, -1\}$ is the relative primitive sign. The choices can be made for any fixed finite list of points with the same equation and comparison map.

The finite-point assertion is Lemma 12.2. Theorem C concerns the chosen cut. An ambient collar through all primitive and ordinary nodal configurations is a stronger assertion and is not used in its proof.

Theorem 7.2 constructs a smooth gluing map at every sufficiently small positive scale for each regular seam-admissible attaching datum over a regular index-zero residual. Its four real parameters are the center and scale on the original domain and two coordinates on the normalized attaching fiber. The construction includes rank drop in the interior of the gauge hemisphere. Theorem 7.25 proves full-neck convergence. Section 8 proves the converse for degree-one stable families and for families with one elementary modification, together with injectivity and transition limits for regular charts. Proposition 11.4 exhausts the bubbling limits of the chosen point cut by these regular neighborhoods.

Both full-rank and rank-drop primitives occur in the construction. Convergence away from a concentration point on the original domain fixes the rescaled evaluation at infinity; it does not force a rescaled map to be constant at finite points. The generic genus-two bubble is therefore full rank, while rank drop is a compactification stratum. In genus one every pair of extension vectors is dependent and only the rank-drop, purely gauge-theoretic branch remains.

1.3. The cylinder relations. Muñoz’s cylinder computations provide a useful check on the constants and the geometric generators. In their respective published conventions, the genus-two relations read

$$a_G^2 + b_G - 8 = 0, \quad a_S^2 + b_S - 4 = 0,$$

where a is the degree-two surface generator [Muñ99a, Muñ99b]. If both presentations have been identified with the same coherent DFL normalization, surface compatibility would give

$$Nb_G - b_S N = 4N.$$

For $g = 1$, the odd-determinant character variety is a point and the positive torus gauge normalization has $b_G = 8 \text{id}$, giving

$$Nb_G - b_S N = 8N.$$

The common orientation convention is a mathematical input to the first calculation. We do not obtain it by identifying the names of published generators. The proof below retains the sphere sign and uses the unreduced torus operation before taking the gauge quotient. Its resulting cylinder identities show that the uncorrected comparison fails in low genus.

1.4. Method. We call the original, unrescaled domain coordinates *physical* coordinates; normalized coordinates are obtained by translating to the bubble center and dividing by its scale. The point-cut homotopy is zero-dimensional in mixed index three. Its boundary relation is one-dimensional in mixed index four. DFL's energy-index formula and half-integrality imply that any defective index-four sequence loses exactly $\frac{1}{2}$ unit of topological energy and has an index-zero residual solution. This global statement does not by itself prevent several nonquantized matching atoms.

The geometric proof has four stages. First, on a small circle about a matching point, the gauge and symplectic halves are spliced to a connection on $S^1 \times \Sigma_g$. Comparing the original smooth filling with the removable limiting filling identifies their mixed Chern–Simons branch difference with $n/2$, where $n = -p_1/2$ is the relative clutching number of a bundle on $S^2 \times \Sigma_g$. The total defect $1/2$ then forces $n = 1$ at one point. A good-circle Chern–Simons argument controls selected annuli. The subsequent converse and exhaustion arguments supply the full-neck statement for sequences in the chosen cut.

The second step avoids the usual adiabatic projection. After base dilation the vertical-curvature term in the ASD equation is multiplied by the square of the scale, rather than its inverse. The usual adiabatic projection to M_g therefore does not apply; compare [DS94, DS07]. Instead, every smooth finite-scale mixed solution determines a holomorphic bundle on the full disc times Σ . Relative stable reduction associates to these doubled bundles a classifying map and a finite modification cycle. Relative p_1 additivity gives

$$n = d + \ell.$$

The primitive identity $n = 1$ leaves exactly the charge pairs $(d, \ell) = (1, 0)$ and $(0, 1)$. The compact limiting models are a minimal line and one modification of a constant family. The finite doubled family itself may still vary in a small stable chart; the converse estimates control that variation.

The third stage is reconstruction and gluing. Decomposing $\mathbb{P}^1$ into two hemispheres and prescribing the Narasimhan–Seshadri seam metrics puts the gauge hemisphere in the setting of Donaldson's Dirichlet theorem for Hermitian Yang–Mills metrics [Don92]. This gives auxiliary compact models. Scaled Hölder estimates and a formal expansion prove their metric regularity. An interior index excision and smooth infinitesimal doubling prove fixed-scale mixed regularity. Quantified cap metrics and a decreasing fractional exponent permit a geometric right inverse in complete graph spaces. Exact nonlinear seam and end charts give the corrected solutions. Four affine-averaged holonomy functions identify their ordinary moduli parameters. The entire constructed neck has vanishing energy and surface/map diameter, and fixed-domain regularity and holomorphic bundle identifications give the prescribed broken limit. The regular-sector converse and its overlap comparison identify every remaining bubbling end of the chosen point cut.

The fourth stage treats incidence and orientation by a proper relative incidence construction on finite regular primitive neighborhoods. Determinant-line excision is compared with

the actual gluing derivative. For $g \geq 2$, the image of i_g has real codimension $2(g - 2)$, the genus-two map i_2 has degree four, and the actual torus seam-clutching calculation gives degree eight. The relative incidence computation transfers these degrees to the physical scale boundaries, with their coherent orientation factors. An interior sphere comparison determines the selected genus-two sign. Exclusion of the other bubbling alternatives and finite regularization of the remaining compact cut give Theorem C.

The ordinary torus tube fixes the torus reference orientation. For homologous matching surfaces, a reduction to real and complex coefficient operators identifies their closed ordinary references. The ratio of the genus-two sphere reference to its closed reference is local. A verified torus/genus-two splitting of T^3 , together with the signed quantum relation, calibrates that ratio. Determinant locality transfers the calibration to every ambient splitting, including rationally acyclic ones. The resulting positive local coefficients give the original integral geometric homotopy in every genus.

1.5. Relation to prior work. The mixed equation and the admissible comparison theorem are due to DFL [DFL26, DFL24]. The earlier adiabatic comparison of instantons and holomorphic curves is due to Dostoglou–Salamon [DS94, DS07]. Wehrheim established energy quantization and removal for ASD instantons with Lagrangian boundary conditions [Weh05a] and the energy identity on $\mathbb{C} \times \Sigma$ [Weh06]. Owens proved the correspondence between finite-energy instantons on a cylinder and stable bundles on $\mathbb{P}^1 \times \Sigma$ [Owe01]; the stable-family-plus-modification description used here is its relative local form.

Muñoz computed both the instanton Floer ring of $\Sigma \times S^1$ and the quantum cohomology of M_g [Muñ99a, Muñ99b]. His projective-bundle description of primitive product bundles supplies the finite-dimensional compactification and the two low-genus numbers. The appearance of a correction term is analogous in spirit to seam-bubble corrections in pseudoholomorphic quilts [BW18], but the local object here couples an ASD half to a holomorphic half and is not identified with a figure-eight bubble.

2. THE POINT CUT IN THE MIXED MODULI SPACE

2.1. The DFL setting. Let

$$Y_{\#} = Y \cup_{\Sigma} (-Y')$$

be an admissible splitting in the sense of [DFL26, Sections 2–3]. The restriction F of the $SO(3)$ bundle to every component of Σ is nontrivial. We assume, as in the embedded form of the DFL theorem, that the perturbed three-manifold Lagrangians are embedded and intersect transversely and that all flat connections in the construction are irreducible. Put

$$M = \mathcal{M}(\Sigma, F) = \prod_j M_j, \quad M_j = \mathcal{M}(\Sigma_j, F_j).$$

The standard complex structure J_* identifies M_j with the fixed-odd-determinant stable-bundle moduli space. Its real dimension is $6g_j - 6$ for $g_j \geq 2$; M_j is a point for $g_j = 1$.

For generators $\alpha \in C_G$ and $\beta \in C_S$, let

$$\mathcal{M}_{\eta}(\alpha, \beta)_r$$

denote the moduli space of perturbed mixed solutions whose linearized mixed operator has Fredholm index r . The secondary perturbation η is a finite sum of gauge-theoretic holonomy perturbations and domain-dependent almost-complex variations. It is identically zero on a fixed neighborhood of the matching line. DFL construct such perturbations regular through index three [DFL26, Proposition 3.23].

For a mixed pair (A, u) , their normalized topological energy is

$$(3) \quad \mathcal{E}(A, u) = \frac{1}{8\pi^2} \int_X \text{tr}(F_A \wedge F_A) + \frac{1}{4\pi^2} \int_S u^* \Omega,$$

with the perturbation terms incorporated on the cylindrical pieces. The index formula has the form

$$(4) \quad \text{ind } \mathcal{D}_{(A, u)} = 8\mathcal{E}(A, u) + \varepsilon(\beta) - \varepsilon(\alpha).$$

For two mixed pairs with the same ends,

$$(5) \quad 2(\mathcal{E}(A, u) - \mathcal{E}(A', u')) \in \mathbb{Z};$$

see [DFL26, Lemma 3.15]. These two formulas are the numerical source of the index-four threshold.

2.2. The universal degeneracy cycle. Let $\tilde{\mathbb{V}}$ be DFL's interpolating universal adjoint bundle over the mixed configuration space times the glued parameter space $\tilde{X}$. Its complexification has rank three. For two generic sections s_1, s_2 , let

$$\mathcal{D}(s_1, s_2) = \{x : \text{rank}_{\mathbb{C}}(s_1(x), s_2(x)) \leq 1\}.$$

At the top rank-one stratum this has complex codimension two and hence real codimension four. With DFL's product orientation it represents the positive class $p_1(\tilde{\mathbb{V}})$. The rank-zero stratum has larger codimension and is absent from every moduli space considered below.

The interpolation from the gauge insertion to the symplectic insertion is an embedding

$$\iota : \mathbb{R} \times \Sigma \longrightarrow \tilde{X}.$$

The parameter $t \in \mathbb{R}$ tends at the two ends to the representatives used to define b_q^G and b_q^S . If $P \subset \Sigma$ is an oriented smooth k -cycle, define

$$\mathcal{L}_d^P(\alpha, \beta) \subset \mathcal{M}_\eta(\alpha, \beta)_r \times \mathbb{R}_t \times P$$

by the condition

$$s_1([A, u], \iota(t, x)) \wedge s_2([A, u], \iota(t, x)) = 0.$$

Lemma 2.1 (dimension of the comparison cut). *If the mixed moduli space and the degeneracy cycle are transverse, then $\mathcal{L}_d^P(\alpha, \beta)$ has expected dimension d precisely when*

$$(6) \quad r = d + 3 - k.$$

Proof. The mixed moduli, the interpolation parameter, and the cycle P have total dimension $r + 1 + k$. The top degeneracy stratum has real codimension four. Thus the cut has dimension $r + k - 3$, which equals d exactly when (6) holds. $\square$

For a point $q \in \Sigma$, the zero-dimensional comparison homotopy uses index three and the one-dimensional relation uses index four:

$$(7) \quad \begin{aligned} \mathcal{L}_0^q(\alpha, \beta) &\subset \mathcal{M}_\eta(\alpha, \beta)_3 \times \mathbb{R}_t, \\ \mathcal{L}_1^q(\alpha, \beta) &\subset \mathcal{M}_\eta(\alpha, \beta)_4 \times \mathbb{R}_t. \end{aligned}$$

This is the first insertion for which DFL's construction reaches the matching-line bubbling threshold.

2.3. The point-cut homotopy. By Proposition 3.1 below, mixed moduli spaces of index at most three do not bubble. Once transversality is imposed, the oriented zero-manifolds $\mathcal{L}_0^q(\alpha, \beta)$ therefore define

$$(8) \quad K_q^{p_1}(\alpha) = \sum_{\beta} \# \mathcal{L}_0^q(\alpha, \beta) \beta.$$

The rational geometric homotopy is $\frac{1}{4}K_q^{p_1}$.

The ordinary compactification of $\mathcal{L}_1^q$ has four familiar boundary types: gauge and symplectic Floer breaking and escape of t to its gauge and symplectic ends. With DFL's conventions their signed sum is

$$(9) \quad d_S K_q^{p_1} + K_q^{p_1} d_G - N b_q^G + b_q^S N.$$

The only possible additional codimension-one face consists of matching-line bubbles. We denote its signed contribution by B_q until Section 12.7. The desired integral boundary formula is therefore

$$(10) \quad N b_q^G - b_q^S N = d_S K_q^{p_1} + K_q^{p_1} d_G + B_q.$$

The remainder of the paper constructs this face and establishes $B_q = c_{g(q)}N$ with $c_1 = 8$, $c_2 = 4$, and $c_g = 0$ for $g \geq 3$.

3. INDEX-FOUR CONCENTRATION AND LOCAL CHARGE

3.1. Exhaustion of the index budget. Consider a sequence in $\mathcal{M}_\eta(\alpha, \beta)_r$. After DFL chain compactness, write (A_0, u_0) for the residual mixed solution, A_i^G for the nonconstant gauge Floer levels, and v_j^S for the nonconstant symplectic Floer levels. Define the global defect

$$(11) \quad \Delta = \mathcal{E}(A_\nu, u_\nu) - \mathcal{E}(A_0, u_0) - \sum_i \mathcal{E}(A_i^G) - \sum_j \mathcal{E}(v_j^S).$$

It includes all energy not carried by the displayed residual levels: ordinary bubbles, matching atoms, and any neck loss.

Proposition 3.1 (index-four exhaustion). *Assume that the residual moduli spaces are regular and have no negative-index solutions.*

- (1) *A sequence of mixed solutions of index at most three cannot bubble.*
- (2) *If an index-four sequence has $\Delta > 0$, then*

$$(12) \quad \Delta = \frac{1}{2}, \quad \text{ind } \mathcal{D}_{(A_0, u_0)} = 0,$$

and there are no nonconstant gauge or symplectic Floer levels.

- (3) *The defect in part (2) carries all four missing Fredholm-index units.*

Proof. DFL compactness makes Δ nonnegative. If it is positive, the strict form of the topological-energy inequality and (5) give $\Delta \geq \frac{1}{2}$. Additivity of the mixed index under gluing gives

$$(13) \quad \text{ind } \mathcal{D}_{(A_0, u_0)} + \sum_i \text{ind } \mathcal{D}_{A_i^G} + \sum_j \text{ind } \mathcal{D}_{v_j^S} + 8\Delta \leq r.$$

Every term except the last is nonnegative, and each nonconstant Floer level has positive index. If $r \leq 3$, the inequalities $8\Delta \geq 4$ and (13) are incompatible. If $r = 4$, equality is forced

throughout: $8\Delta = 4$ and every residual index other than the index-zero mixed component vanishes. A nonconstant Floer level cannot have index zero, proving the claim. $\square$

An interior four-dimensional instanton has integral charge at least one, hence energy at least one and index at least eight. It cannot occur in Proposition 3.1(2), either as the primary bubble or as a smaller-scale bubble inside a matching configuration. A nonconstant sphere in M_g has normalized energy at least $\frac{1}{2}$ and Chern index four; a nonconstant disc on either three-manifold Lagrangian has Maslov index at least four. Thus the half-unit budget permits at most one ordinary symplectic bubble and no simultaneous breaking.

3.2. Ordinary bubbles and the point incidence. The preceding numerical statement concerns the uncut moduli space. The dimension argument for the point incidence also requires regular compactification strata and compatible limits of the evaluated bundle and sections. For the fixed equation and section choices made below, Proposition 11.3 proves exclusion directly from a compact family of matched minimal spheres and the actual evaluation limits, without requiring regular boundary-disc strata.

The matching line is different. The interpolation path crosses the seam, so the ratio between its normal coordinate and the concentration scale is a parameter in the rescaled problem. Together with the center, scale, and marked-domain directions, it provides the four real directions cut by the incidence. A matching bubble can therefore be a genuine boundary point of the one-dimensional cut.

3.3. The local Chern–Simons sector. DFL’s half-integrality controls the total defect, not each atom. The next theorem is the local quantization needed to rule out several matching centers whose a priori energy measures sum to $\frac{1}{2}$.

Let z be a matching point at which a sequence (A_ν, u_ν) concentrates. Choose a small half-disc on either side of the seam, with no other concentration point in its closure. On the negative semicircle use the restriction of A_ν . On the positive semicircle choose the flat representatives of u_ν supplied by the matching condition. At the two seam endpoints insert the standard flat cylinder. These pieces give a spliced connection

$$B_{\nu,r} \quad \text{on} \quad Y_r \cong S^1 \times \Sigma_g.$$

The removable limit gives a second connection $B_{0,r}$ on the same three-manifold.

DFL’s connection-valued filling construction replaces this boundary data by a smooth connection on $D^2 \times \Sigma_g$. Its Chern–Weil integral contains both the gauge energy and the symplectic area [DFL24, Lemma 4.67]; the equality between the curvature term of the flat-representative family and twice its symplectic area is [DFL26, Lemma 5.10].

Theorem 3.2 (local charge). *Let $\delta_z > 0$ be the atom of the normalized mixed energy measure at z . The original bundle and the removable limiting bundle determine a relative clutching integer n_z such that*

$$(14) \quad \delta_z = \frac{n_z}{2}, \quad n_z \in \mathbb{Z}_{>0}.$$

The sign is normalized so that the DFL positive generator has $n_z = 1$ and $p_1 = -2$.

Proof. Take regular radii $r \downarrow 0$ whose closed discs contain no other concentration point. For fixed r , compactness supplies annular gauges in which the boundary connections converge smoothly. Apply DFL’s filling construction to the original and removable limiting pairs, and denote the resulting bundles with connection over $D^2 \times \Sigma_g$ by $(Q_{\nu,r}, \mathbb{A}_{\nu,r})$ and $(Q_{0,r}, \mathbb{A}_{0,r})$.

Use the limiting framing to identify $Q_{0,r}|_{Y_r}$ with a fixed bundle $Q_r^\partial \rightarrow Y_r$. The annular convergence gauge supplies a projective bundle isomorphism

$$\gamma_{\nu,r} : Q_{\nu,r}|_{Y_r} \longrightarrow Q_r^\partial$$

for which $\gamma_{\nu,r*}B_{\nu,r} \rightarrow B_{0,r}$ smoothly. No extension of $\gamma_{\nu,r}$ over either filling is chosen or required; a nonextendible gauge class will be the clutching class below. With DFL’s normalization, the filling identities just cited are

$$\mathcal{E}_\nu(r) = \frac{1}{8\pi^2} \int_{D^2 \times \Sigma_g} \text{tr}(F_{\mathbb{A}_{\nu,r}} \wedge F_{\mathbb{A}_{\nu,r}}), \quad \mathcal{E}_0(r) = \frac{1}{8\pi^2} \int_{D^2 \times \Sigma_g} \text{tr}(F_{\mathbb{A}_{0,r}} \wedge F_{\mathbb{A}_{0,r}}).$$

Interpolate linearly, in a fixed affine chart of connections on Q_r^∂ , between $\gamma_{\nu,r*}B_{\nu,r}$ and $B_{0,r}$ on $[0,1] \times Y_r$. Smooth convergence makes its normalized Chern–Weil integral $\epsilon_{\nu,r}$ tend to zero. Glue the original filling through $\gamma_{\nu,r}$, this interpolation, and the reverse of the limiting filling. The result is an $SO(3)$ bundle $P_{\nu,r}$ over $S^2 \times \Sigma_g$. Thus every component of the annular gauge, including a possible large-gauge component, is recorded in $P_{\nu,r}$. Replacing $\gamma_{\nu,r}$ by a convergence gauge in the same local slice changes the clutching map by a homotopy and leaves $p_1(P_{\nu,r})$ unchanged. Chern–Weil excision, with the sign convention of [DFL26, Proposition 5.14], gives

$$(15) \quad \mathcal{E}_\nu(r) - \mathcal{E}_0(r) + \epsilon_{\nu,r} = -\frac{1}{4}p_1(P_{\nu,r}).$$

The manifold $S^2 \times \Sigma_g$ is spin. The Wu formula and $p_1(P) \equiv w_2(P)^2 \pmod{2}$ therefore show that $p_1(P_{\nu,r})$ is even. Put

$$n_{\nu,r} := -\frac{1}{2}p_1(P_{\nu,r}) \in \mathbb{Z}.$$

For fixed r , weak convergence of the energy measures and $\epsilon_{\nu,r} \rightarrow 0$ show that the half-integers $n_{\nu,r}/2$ converge. They are consequently eventually constant; write their limit as $n(r)/2$. As regular radii tend to zero,

$$\frac{n(r)}{2} = \lim_{\nu \rightarrow \infty} (\mathcal{E}_\nu(r) - \mathcal{E}_0(r)) \longrightarrow \delta_z,$$

because the removable limit is smooth. The convergent half-integers $n(r)/2$ are again eventually constant. Calling the eventual integer n_z proves (14). Positivity of the atom gives $n_z > 0$. DFL’s disc generator in Section 2.2 has normalized curvature integral $1/2$, so it has $p_1 = -2$ and fixes the stated sign.

All Chern–Simons lifts in this argument are the real-valued lifts selected by the displayed fillings. For two regular radii, restricting the original and limiting fillings to the intervening annulus gives the same clutching bundle on their common boundary. Hence changing the radius cannot transfer an integer to an independently chosen boundary branch; the only change is the Chern–Weil integral of that actual annular restriction, already present in (15). $\square$

The construction does not pass through M_g . It therefore retains the clutching class when $g = 1$, even though M_1 is a point. It is also additive over disconnected components of Σ , since the relative Pontryagin number splits componentwise.

Corollary 3.3 (unique primitive matching center). *Suppose an index-four sequence contributing to the point cut develops a matching atom. Then there is exactly one matching*

concentration point z , and

$$(16) \quad n_z = 1, \quad \delta_z = \Delta = \frac{1}{2}.$$

There is no second matching point and no ordinary bubble or nonconstant Floer level away from z .

Proof. Proposition 3.1 gives total defect $\frac{1}{2}$. Theorem 3.2 assigns every matching atom a positive multiple of $\frac{1}{2}$. Thus the first atom already carries the whole defect and has charge one. Every ordinary sphere or disc also costs at least $\frac{1}{2}$, and every interior instanton costs at least one. Energy exhaustion excludes each such component away from the matching center. $\square$

For the section choices of Proposition 11.3, Corollary 3.3 therefore leaves a single charge-one matching center attached to an index-zero residual as the possible finite- t concentration of the point cut. The next sections construct the primitive charts and prove the converse and neck estimates needed to cover these ends.

4. HOLOMORPHIC COMPACTIFICATION OF A PRIMITIVE ATOM

4.1. The local equations and base dilation. Choose coordinates (s, θ) centered at the matching point, with $D_- = \{s \leq 0\}$ the gauge half-disc and $D_+ = \{s \geq 0\}$ the symplectic half-disc. In a gauge adapted to the s direction, write the connection on $D_- \times \Sigma_g$ as

$$A = \beta(s, \theta) + \psi(s, \theta)d\theta.$$

The ASD equation is

$$(17) \quad *_\Sigma F_\beta + \partial_s \psi = 0,$$

$$(18) \quad -\partial_\theta \beta + *_\Sigma \partial_s \beta + d_\beta \psi = 0.$$

On D_+ the map u satisfies

$$(19) \quad \partial_\theta u - J_* \partial_s u = 0.$$

Along $s = 0$, the connection $\beta(0, \theta)$ is flat and represents $u(0, \theta)$.

For a concentration scale $\rho \rightarrow 0$, set

$$\beta_\rho(S, \Theta) = \beta(\rho S, \rho \Theta), \quad \psi_\rho(S, \Theta) = \rho \psi(\rho S, \rho \Theta).$$

Equations (17)–(18) become

$$(20) \quad \rho^2 *_\Sigma F_{\beta_\rho} + \partial_S \psi_\rho = 0,$$

$$(21) \quad -\partial_\Theta \beta_\rho + *_\Sigma \partial_S \beta_\rho + d_{\beta_\rho} \psi_\rho = 0.$$

The coefficient of vertical curvature is ρ^2 , not ρ^{-2} . This is the opposite of the Dostoglou–Salamon adiabatic regime and does not force the interior surface slices to be flat. The correct intermediate object is a holomorphic bundle on the product, rather than a direct map of the negative half-plane to M_g .

4.2. Finite-scale doubling. Give the base the complex coordinate $w = s + i\theta$.

Lemma 4.1 (finite-scale holomorphic doubling). *Every smooth mixed solution on a disc neighborhood of the matching line determines a holomorphic rank-two bundle on the full disc times Σ_g . It is canonical up to fixed-determinant complex gauge and tensoring by a line bundle pulled back from the disc.*

Proof. On the negative half, the product metric is Kähler. The ASD equation implies $F_A^{0,2} = 0$, so $A^{0,1}$ is an integrable Dolbeault operator.

Every seam fiber is stable. Stability is open, so the negative holomorphic family has stable fibers on a collar of the seam. The fine moduli property [Owe01, Section 4] gives a holomorphic classifying map f_- on that collar and identifies its bundle, up to a line bundle from the base, with the pullback of the universal bundle. The positive half similarly has the holomorphic classifying map $f_+ = u$.

The matching condition identifies the continuous boundary values of f_- and f_+ on the seam. In a complex chart of M_g , their union is continuous on a full collar and holomorphic on its two open halves. Morera’s theorem shows that it is holomorphic across the seam. Pull back the universal bundle by this glued map and use the resulting collar isomorphisms as transition maps between the negative and positive families. This gives the required holomorphic bundle. The fine-moduli ambiguity is a line bundle from the base; changing its trivialization and the collar isomorphisms has exactly the ambiguity stated in the lemma. $\square$

The point of Lemma 4.1 is that it is applied before taking the rescaled limit. No unproved trace convergence for a limiting connection-valued half-plane equation is required.

4.3. Stable reduction over a disc. The following one-parameter statement is the local stable-reduction input. It is the bundle-theoretic mechanism appearing in Owens’s cylindrical correspondence [Owe01].

Lemma 4.2 (disc stable reduction). *Let $E \rightarrow D \times \Sigma_g$ be a holomorphic rank-two bundle whose surface fibers have fixed determinant of degree one, and suppose at least one fiber is stable. Then:*

- (1) *the unstable fibers form a finite analytic subset of every compact subdisc;*
- (2) *the classifying map from the stable locus to M_g extends holomorphically across the unstable fibers without base change;*
- (3) *the original and pullback stable families differ by a finite Langton sequence supported on those fibers; all terms are locally free, retain the fixed surface-fiber determinant, and agree with E off those fibers;*
- (4) *if one step is the kernel of a quotient line G_j of degree $q_j \leq 0$, its positive charge is*

$$\ell_j = \frac{1}{2}(p_1(\text{ad } E^{j+1}) - p_1(\text{ad } E^j)) = 1 - 2q_j > 0.$$

Thus $\ell = \sum_j \ell_j$ is a nonnegative integer, and it vanishes exactly when no modification occurs.

Proof. The family E is flat over D . Toma’s relative Harder–Narasimhan theorem makes semistability Zariski open in such an analytic family [Tom20, Theorem 2.4]. Here semistability is stability, so the existence of one stable fiber makes the unstable locus a proper analytic subset of D , hence finite on compact subdiscs.

Fix an unstable point and a smaller disc Δ containing no other one. Toma’s analytic Langton theorem produces, without base change, a coherent subsheaf $E' \subset E$ that equals

E over Δ^* , is flat over Δ , and has stable central fiber [Tom20, Theorem 3.1]. It is locally free. Indeed, all its fibers are torsion-free sheaves on the smooth curve and hence are vector bundles. At a stalk with base parameter t , lift a basis of E'/tE' to a map $R^2 \rightarrow E'$. Nakayama makes it surjective; flatness gives $\mathrm{Tor}_1^R(E', R/(t)) = 0$, so Nakayama applied to the kernel makes the map an isomorphism.

In rank two Toma's construction is the finite Langton sequence

$$0 \longrightarrow E^{j+1} \longrightarrow E^j \longrightarrow i_* G_j \longrightarrow 0,$$

where, on the central curve, $B_j \subset E_0^j$ is the saturated destabilizing line and $G_j = E_0^j/B_j$. The transformed central fiber fits into

$$0 \longrightarrow G_j \longrightarrow E_0^{j+1} \longrightarrow B_j \longrightarrow 0.$$

Thus every step retains the determinant $B_j \otimes G_j$ and equals its predecessor off the central fiber. The final stable family supplies the holomorphic extension of the classifying map by the fine-moduli property; it differs from the pullback universal family only by a line bundle from Δ . In particular, the comparison preserves the outer attaching identification. This is also the local data in [Owe01, Proposition 5.2].

Put $q_j = \deg G_j$. Since $\deg E_0^j = 1$ and B_j destabilizes, $q_j \leq 0$. For the displayed elementary transform, the fiber divisor has square zero and

$$c_1(E^{j+1}) = c_1(E^j) - [\Sigma_g], \quad \mathrm{ch}_2(E^{j+1}) = \mathrm{ch}_2(E^j) - q_j.$$

It follows that

$$p_1(\mathrm{ad} E^{j+1}) - p_1(\mathrm{ad} E^j) = 2 - 4q_j.$$

This proves the charge formula and positivity. Repeating at the finitely many unstable fibers proves the lemma. $\square$

4.4. The primitive charge splitting. Let $E \rightarrow D \times \Sigma_g$ be one doubled family supplied by Lemma 4.1. The convergence to the removable residual solution fixes an outer projective framing. Apply Lemma 4.2, denote its final stable family by F , its classifying map by $f : D \rightarrow M_g$, and its total modification charge by ℓ .

Choose the outer circle so that $f(\partial D)$ lies in a contractible fine-moduli chart. Cap it in that chart and put

$$(22) \quad d = \left\langle -\frac{1}{2} p_1(\mathrm{ad} \mathcal{U}) / [\Sigma_g], [\widehat{f}(S^2)] \right\rangle.$$

This is the positive generator normalization: the pullback of $\mathrm{ad} \mathcal{U}$ by a minimal line has Pontryagin number -2 [Muñ99b, Proposition 3].

Proposition 4.3 (fixed-family framed splitting). *If $n(E)$ is the relative clutching charge in the outer framing, then*

$$(23) \quad n(E) = d + \ell.$$

The identity is independent of the cap within the chosen chart, the universal-bundle representative, the base-line ambiguity in the fine-moduli problem, and the factorization into elementary transforms. In the primitive sector, $n(E) = 1$. If the signed normalized area of the cap has absolute value less than $\frac{1}{2}$, the individual doubled family has either $(d, \ell) = (1, 0)$ or $(0, 1)$. In particular this applies eventually to the shrinking caps used below.

Proof. For one elementary transform with quotient line G of degree q , Grothendieck–Riemann–Roch and the triviality of the normal bundle of the fiber give

$$\mathrm{ch}(i_*G) = [\Sigma_g] + q[\mathrm{pt}].$$

Thus

$$c_1(E') = c_1(E) - [\Sigma_g], \quad \mathrm{ch}_2(E') = \mathrm{ch}_2(E) - q[\mathrm{pt}].$$

Since

$$p_1(\mathrm{ad} E) = 4 \mathrm{ch}_2(E) - c_1(E)^2, \quad [\Sigma_g]^2 = 0, \quad c_1(E) \cdot [\Sigma_g] = 1,$$

one obtains

$$p_1(\mathrm{ad} E') - p_1(\mathrm{ad} E) = 2 - 4q = 2(1 - 2q).$$

All transforms are supported away from the framed outer circle, and hence

$$(24) \quad p_1(\mathrm{ad} F) = p_1(\mathrm{ad} E) + 2\ell$$

as a relative identity. The fine-moduli property identifies

$$F \cong (\mathrm{id}_{\Sigma_g} \times f)^* \mathcal{U} \otimes p_D^* L_D.$$

The adjoint removes the base-line factor. After capping, (22) therefore says $p_1(\mathrm{ad} \widehat{F}) = -2d$. By definition $p_1(\mathrm{ad} \widehat{E}) = -2n(E)$, so (24) gives $n(E) = d + \ell$.

Two caps in the chosen chart are homotopic relative to the boundary. A base-line twist leaves the adjoint bundle unchanged, and $\ell = (p_1(\mathrm{ad} F) - p_1(\mathrm{ad} E))/2$ depends only on the framed endpoints. These observations prove all independence assertions. The normalized area identity is

$$\frac{d}{2} = \mathrm{Area}(f) + \mathrm{Area}(\mathrm{cap}).$$

The first term is nonnegative. Under the stated cap bound, the integer d is greater than -1 , hence nonnegative. Together with $n(E) = 1$ and nonnegativity of ℓ , this proves the final dichotomy. Proposition 8.17 below identifies this framed charge with the analytical local charge and supplies caps of vanishing area for physical primitive sequences. $\square$

The compactness distinction. The pair (d, ℓ) belongs to one individual doubled family and need not be continuous under degeneration. For the compact extension pencils of Section 5, a sequence with $(d, \ell) = (1, 0)$ can converge in the quasimap compactification to $(0, 1)$; the missing degree then appears as a sphere component in the stable-map compactification. That sphere and the quasimap basepoint are two records of the same primitive charge and cannot be added as independent components.

Near rank drop in this compact pencil space, write a full-rank pencil in an affine base coordinate as $s_\nu(z) = a_\nu z + b_\nu$. If its limiting basepoint is $z_\nu = x_\nu + iy_\nu$, choose a linear functional on the limiting image line, absorb the phase of its transverse coefficient into w_ν , and write exactly

$$(25) \quad s_\nu(z) = a_\nu(z - z_\nu) + \varepsilon_\nu w_\nu, \quad \varepsilon_\nu > 0, \quad \varepsilon_\nu \rightarrow 0, \quad a_\nu \rightarrow v, \quad w_\nu \rightarrow w, \quad v \wedge w \neq 0.$$

After the seam-preserving change $z = \varepsilon_\nu \zeta + iy_\nu$, the pencil is

$$[a_\nu(\zeta - \lambda_\nu) + w_\nu], \quad \lambda_\nu = x_\nu/\varepsilon_\nu.$$

If λ_ν has a finite limit, this converges to a full-rank mixed bubble. If $\lambda_\nu \rightarrow +\infty$ or $-\infty$, it is constant on seam-centered compact sets and recentering at z_ν extracts, respectively, a positive holomorphic sphere or a negative cylindrical instanton. Thus a seam basepoint is not a fourth type: center-scale normalization either recovers a mixed bubble or one of the two pure ends.

For the point theorem these algebraic alternatives are combined with analytic convergence in the first-order norms used below. Negative recentering returns to the normalized gauge rank-drop chart. At positive infinite ratio an ordinary minimal sphere is extracted; its evaluation is excluded from the chosen incidence cut. Sections 6.4–8.16 give the neck estimates, gluing and converse arguments used in this exhaustion.

5. THE ALGEBRAIC PRIMITIVE MODULI

5.1. Muñoz’s extension spaces. Fix a line bundle Λ of degree one on Σ_g . Let $J = J(\Sigma_g)$ be the Jacobian and let $\mathcal{L}$ be a Poincaré line bundle on $\Sigma_g \times J$. If $p : \Sigma_g \times J \rightarrow J$ is the projection, put

$$(26) \quad \mathcal{E}_g = R^1 p_*(\mathcal{L}^2 \otimes \Lambda^{-1}).$$

With the standard principal polarization $\omega \in H^2(J; \mathbb{Z})$, the Grothendieck–Riemann–Roch calculation is

$$(27) \quad \text{ch}(\mathcal{E}_g) = g + 4\omega.$$

The projective bundle

$$N_g = \mathbb{P}(\mathcal{E}_g^\vee) \longrightarrow J$$

parametrizes nonsplit extensions

$$(28) \quad 0 \longrightarrow L \longrightarrow V \longrightarrow \Lambda \otimes L^{-1} \longrightarrow 0.$$

The middle term is stable and determines a holomorphic map

$$(29) \quad i_g : N_g \longrightarrow M_g.$$

Muñoz studies stable rank-two bundles E on $\mathbb{P}^1 \times \Sigma_g$ with

$$(30) \quad c_1(E) = [\mathbb{P}^1] + [\Sigma_g], \quad c_2(E) = 1, \quad p_1(\text{ad } E) = -2,$$

for a polarization close to the Σ_g boundary of the ample cone. For $g \geq 2$, he proves that their moduli space is the smooth projective bundle

$$(31) \quad \mathfrak{M}_g = \mathbb{P}((\mathcal{E}_g \oplus \mathcal{E}_g)^\vee) \longrightarrow J, \quad \dim_{\mathbb{C}} \mathfrak{M}_g = 3g - 1.$$

See [Muñ99b, Sections 3–4]. For $g = 1$, we use the same projective bundle to parametrize the displayed extensions; the deformation calculation in Proposition 6.4 treats their seam-admissible locus directly. The two summands are the coefficients of a section of $\mathcal{O}_{\mathbb{P}^1}(1) \otimes \mathcal{E}_g$.

Fix $L \in J$, write $V = (\mathcal{E}_g)_L$, and represent a point of the fiber by

$$[v_0 : v_1] \in \mathbb{P}(V \oplus V).$$

At $x = [x_0 : x_1] \in \mathbb{P}^1$, the corresponding surface extension has class

$$(32) \quad v(x) = x_0 v_0 + x_1 v_1.$$

Proposition 5.1 (rank stratification). *The moduli space $\mathfrak{M}_g$ has the following two strata.*

- (1) *If v_0, v_1 are independent, (32) never vanishes and defines a nonconstant parameterized minimal rational curve in M_g . Every parameterized minimal rational curve arises uniquely in this way.*
- (2) *If v_0, v_1 are dependent, write $(v_0, v_1) = (a_0 v, a_1 v)$. The induced quasimap is constant away from the single basepoint $[-a_1 : a_0]$. The rank-drop locus is the Segre bundle*

$$(33) \quad \mathcal{R}_g = N_g \times_J \mathbb{P}^1 \longrightarrow \mathfrak{M}_g$$

of complex codimension $g - 1$.

For a compact extension pencil on $\mathbb{P}^1 \times \Sigma_g$, the two strata have respectively $(d, \ell) = (1, 0)$ and $(0, 1)$.

Proof. If the two vectors are independent, no nonzero pair (x_0, x_1) kills their linear combination. Thus (32) defines a line in the projective extension fiber $\mathbb{P}(V)$, followed by the map i_g . Muñoz’s classification identifies this full-rank open locus with the space of parameterized minimal curves.

If the two vectors are dependent, every nonzero extension away from $[-a_1 : a_0]$ represents the same point $[v] \in \mathbb{P}(V)$. At the unique zero the surface bundle splits, and stable reduction records one elementary modification. Fiberwise, the dependent tensors form the Segre image $\mathbb{P}(V) \times \mathbb{P}(\mathbb{C}^2)$ in $\mathbb{P}(V \otimes \mathbb{C}^2)$. Its dimension is $(g - 1) + 1 = g$, whereas the ambient projective space has dimension $2g - 1$, so its codimension is $g - 1$. In the independent case the stable classifying map is a minimal line and there is no modification. In the dependent case it is constant and the simple zero contributes one modification of charge one. This proves the last assertion. $\square$

The statement concerns compact extension pencils. A finite doubled family on $D \times \Sigma_g$ need not be such a pencil: the cap in Proposition 4.3 is smooth, and does not extend its classifying map holomorphically to $\mathbb{P}^1$. In particular $d = 0$ permits a nonconstant stable replacement contained in a small chart. The comparison pencils are constructed from the actual finite families in Propositions 8.2 and 8.3, with quantitative control of the remaining variation.

For $g = 1$, the extension space V is one-dimensional, and every pair is dependent. The full-rank locus is empty and $\mathfrak{M}_1 = \mathcal{R}_1$. This is the algebraic reason the genus-one primitive class is invisible in the point-valued character variety.

5.2. Evaluation at infinity and dimensions of attaching spaces. Choose $\infty = [1 : 0] \in \mathbb{P}^1$. On the locus $v_0 \neq 0$, evaluation of the induced stable map or quasimap is

$$(34) \quad \text{ev}_\infty([v_0 : v_1]) = i_g([v_0]).$$

Hence every primitive bubble, full rank or rank drop away from infinity, attaches through $i_g(N_g)$.

Fix $[v] \in N_g$. After the overall projective scalar is used to set $v_0 = v$, the full-rank fiber of ev_∞ is

$$V \setminus \mathbb{C}v.$$

The real affine group preserving the two half-planes and fixing infinity is

$$(35) \quad G_{\mathbb{R}} = \{w \mapsto aw + ib : a > 0, b \in \mathbb{R}\}.$$

It fixes the translation and scale of a generic mixed bubble. The reduced fiber therefore has real dimension $2g - 2$.

For $g \geq 2$, the dimensions of the attachment spaces are

$$(36) \quad \dim_{\mathbb{C}} N_g = 2g - 1, \quad \dim_{\mathbb{C}} M_g = 3g - 3.$$

Muñoz proves that i_g is generically finite onto its image. Thus

$$(37) \quad \text{codim}_{\mathbb{R}} i_g(N_g) = 2(g - 2).$$

Let $e_0 : \mathbb{R} \rightarrow M_g$ be the matching evaluation path of a residual index-zero mixed solution. Whenever it meets the regular part of $i_g(N_g)$, the reduced full-rank primitive stratum has dimension

$$(38) \quad (2g - 2) + 1 - 2(g - 2) = 3.$$

The interpolation parameter adds one dimension and the integral p_1 incidence removes four, leaving isolated boundary points. For $g = 1$, instead, $N_1 = J(\Sigma_1)$ and M_1 is a point; there is no full-rank stratum and the preceding codimension and full-rank attachment formulas do not apply.

For $g \geq 3$, a one-dimensional path transverse to a stratification of the image misses it by (37). Proposition 9.2 establishes avoidance for actual index-zero residual seams after a finite perturbation enlargement. Deducing the vanishing of the matching correction also requires the attachment-support and relative-incidence assertions. In genus two, N_2 and M_2 have the same dimension and i_2 has a nonzero degree, computed next.

5.3. The genus-two degree. Let $h = c_1(\mathcal{O}_{N_2}(1))$, and normalize the principal polarization by

$$(39) \quad \int_{J(\Sigma_2)} \frac{\omega^2}{2} = 1.$$

If $\alpha \in H^2(M_2; \mathbb{Z})$ is the standard generator, Muñoz's universal-bundle formulas give

$$(40) \quad i_2^* \alpha = 4\omega + h, \quad h^2 + 4\omega h + 8\omega^2 = 0.$$

Proposition 5.2. *The map $i_2 : N_2 \rightarrow M_2$ has degree four.*

Proof. Using (40) and $\omega^3 = 0$ gives

$$h^3 = 8\omega^2 h, \quad 12\omega h^2 = -48\omega^2 h.$$

Therefore

$$(41) \quad \int_{N_2} (4\omega + h)^3 = \int_{N_2} 8\omega^2 h = 16,$$

where the last equality uses (39) and integration over the projective-line fiber. The genus-two moduli space M_2 is a smooth intersection of two quadrics in $\mathbb{P}^5$, and α is its hyperplane class. Hence

$$\int_{M_2} \alpha^3 = 4.$$

The ratio of (41) to this number is the degree: $\deg i_2 = 16/4 = 4$. $\square$

All local algebraic multiplicities are positive in the complex orientation. Thus, after the analytic incidence determinant is identified with its algebraic counterpart, the magnitude of the integral genus-two contribution is four. Proposition 9.7 below instead computes the actual relative seam-frame degree on each covered finite-center branch. Its physical orientation and the global boundary identification remain separate.

5.4. Special genus-two lines and their attaching values. In this subsection Σ has genus two and $\deg \Lambda = 1$. Write $J = \text{Pic}^0(\Sigma)$ and $\pi : N = \mathbb{P}(\mathcal{V}) \rightarrow J$, where $\mathcal{V}_L = H^1(\Sigma, L^2 \Lambda^{-1})$; this is the extension space N_2 in the line convention for projectivization. Let $i : N \rightarrow M_2$ be its middle-bundle map. Put

$$(42) \quad S = \{L \in J : H^0(\Sigma, \Lambda L^{-2}) \neq 0\}, \quad R = \pi^{-1}(S), \quad B = i(R).$$

We call the extension line $i(N_L)$ special when $L \in S$. A stable bundle V is *very stable* if it admits no nonzero nilpotent Higgs field, namely no nonzero holomorphic map $\phi : V \rightarrow V \otimes K_\Sigma$ with $\phi^2 = 0$. The locus B is the genus-two non-very-stable locus [PP21, Section 2 of the preprint]. We give the argument in our subline convention to identify precisely the exceptional set in the converse.

Proposition 5.3. *The set S is a smooth curve and R is exactly the critical locus of i . The differential of i has complex rank two on R and rank three elsewhere. The map i embeds each projective fiber N_L , including the special fibers. Its critical image B is a proper closed algebraic subset of pure complex codimension one, and consists exactly of the bundles that are not very stable.*

Proof. For $A_L = \Lambda L^{-2}$, the map $L \mapsto A_L$ is an étale map from J to $\text{Pic}^1(\Sigma)$. The effective degree-one locus is the embedded Abel curve $q \mapsto \mathcal{O}(q)$. A degree-one line bundle on Σ has at most one section: otherwise its sections would give a degree-one map to $\mathbb{P}^1$. Thus S is smooth and, for $L \in S$, $A_L = \mathcal{O}(q)$ for a unique q , with $h^0(A_L) = 1$.

A nonzero extension

$$0 \longrightarrow L \longrightarrow V_e \longrightarrow Q = \Lambda L^{-1} \longrightarrow 0$$

is stable. Indeed a destabilizing line would have degree at least one, map isomorphically to Q , and split the extension. Hence i is defined everywhere and is proper. Infinitesimal subline variations with fixed middle bundle are graphs in $\text{Hom}(L, Q)$, so

$$(43) \quad \ker di_{(L, [e])} = H^0(\Sigma, A_L).$$

Scalar stable automorphisms add no directions. Since N and M_2 have dimension three, this proves the rank assertion.

The chosen subline of isomorphism class L is unique in V_e . When $L \notin S$, tensoring the extension by L^{-1} gives $h^0(V_e L^{-1}) = 1$ directly. When $L \in S$, the sequence $0 \rightarrow \mathcal{O}(-q) \rightarrow \mathcal{O} \rightarrow \mathbb{C}_q \rightarrow 0$ gives an isomorphism

$$(44) \quad H^1(\mathcal{O}(-q)) \longrightarrow H^1(\mathcal{O}).$$

Indeed $H^0(\mathcal{O}) \rightarrow H^0(\mathbb{C}_q)$ is an isomorphism. The connecting map $H^0(\mathcal{O}(q)) \rightarrow H^1(\mathcal{O})$ for the tensored extension sends its generator to the nonzero image of e under (44). Again $h^0(V_e L^{-1}) = 1$. Thus i is injective on N_L . For $L \in S$, the projection of the kernel direction represented by $s \in H^0(A_L)$ to $T_L J = H^1(\mathcal{O})$ is $\pm s \cup e$. This follows by differentiating the transition functions of the graph subline. It is nonzero by (44); hence the vertical tangent of N_L is not in the kernel. The restriction of i to N_L is therefore an injective immersion, hence an embedding by properness. Its image is a projective line by the primitive classification [Muñ99b, Propositions 2–3].

Properness makes B closed, with dimension at most two. As $[e]$ varies over a special fiber, the lines $\mathbb{C}(s \cup e)$ range over all of $\mathbb{P}(H^1(\mathcal{O}))$. Exactly one of them equals $T_L S$. Except at that point of the fiber, $\ker di$ is transverse to T_R , and $di|_{T_R}$ has rank two. Each component of R therefore has a two-dimensional image. This proves the codimension assertion; it does not assert that B is smooth.

Finally, a nonzero nilpotent Higgs field has a saturated kernel line L of degree $d \leq 0$. It induces a nonzero map $\Lambda L^{-1} \rightarrow LK_\Sigma$, so $1 + 2d \geq 0$ and $d = 0$. Conversely such a map, composed with the quotient and inclusion, gives a nonzero nilpotent Higgs field. Riemann–Roch gives

$$h^0(K_\Sigma L^2 \Lambda^{-1}) = h^1(A_L) = h^0(A_L).$$

This identifies its locus with B . □

Proposition 5.4. *Over $U = M_2 \setminus B$, the map $i : i^{-1}(U) \rightarrow U$ is an unramified cover of degree four. For every compact $K \subset U$, its inverse branches have uniform bounds in every prescribed finite derivative order on a finite cover of K by smaller coordinate neighborhoods. In particular, if $p_\nu \rightarrow p \in U$ and $i(L_\nu, [e_\nu]) = p_\nu$, every subsequential limit of the extension points is nonspecial.*

Proof. The restriction is proper with invertible differential, hence a finite unramified cover. Proposition 5.2 gives its degree. The compact set $i^{-1}(K)$ is disjoint from R , so the smallest singular value of di has a positive lower bound there. Choose finitely many evenly covered neighborhoods and shrink them inside slightly larger ones. Smooth local inverses and compactness give the derivative bounds. Properness puts every limiting extension point over p , where no point is critical. This argument allows monodromy among the branches. $\square$

Very stability of an attaching value is stronger than regularity of a chosen extension line through it: a different line through the same value can be special. No genericity of a residual evaluation path is asserted here. That issue is separated in Section 9.1.

5.5. Cylinder identities and conventions. Let a_G, b_G denote Muñoz's surface and point generators in the instanton Floer ring of $\Sigma_g \times S^1$, and let a_S, b_S be the geometric generators of $QH^*(M_g)$. We distinguish this dictionary of universal classes from a comparison of the orientation systems used for their operations.

We record the normalization dictionary. Muñoz uses $\mu(q) = -\frac{1}{4}p_1/[q]$ and defines his point generator by $b = -4\mu(q) = p_1/[q]$; his surface generator is $a = 2\mu(\Sigma) = -\frac{1}{2}p_1/[\Sigma]$ [Muñ99a, equation (4)]. Thus b_G, b_S are exactly the positive integral point classes used here. DFL's comparison for its own operations associated to $p_1/[\Sigma] = -2a$ implies, over $\mathbb{Q}$,

$$(45) \quad Na_G = a_S N.$$

For genus two, in the complex sphere orientation, Muñoz's corrected quantum generator is $\widehat{b}_S = b_S + 4$, and his relation $a_S^2 + \widehat{b}_S - 8 = 0$ is therefore $a_S^2 + b_S - 4 = 0$ in the unshifted geometric generator [Muñ99b, Example 24]. On the gauge side his corresponding relation is $a_G^2 + b_G - 8 = 0$ [Muñ99a, Example 22]. These formulas belong to their respective published conventions. The DFL quantum relation in an arbitrary coherent system is $a_S^2 + b_S = 4\chi$, as proved in Lemma 12.8. An additional convention comparison is needed before substituting an external Floer presentation into (45). In genus one the published gauge relation is $b_G = 8 \text{ id}$ [Muñ99a, Lemma 11], whereas the geometric class b_S vanishes because M_1 is a point [Muñ99b, Example 23].

Proposition 5.5 (low-genus cylinder identities). *On rational Floer homology, use the positive integral p_1 classes and the coherent normalization specified in Theorem 12.13. Then*

$$(46) \quad Nb_G - b_S N = \begin{cases} 8N, & g = 1, \\ 4N, & g = 2. \end{cases}$$

Proof. Apply Theorem 12.13 to the two-component cylinder splitting, whose Lagrangians are diagonal. It gives respectively $8N$ and $4N$. In genus one the symplectic point class vanishes because M_1 is a point; the ordinary gauge normalization is also computed directly in Proposition 12.4. No identification with an external Floer-ring normalization is needed. $\square$

The cylinder groups are nonzero, so the proposition proves that uncorrected point compatibility is false in low genus. Nonvanishing follows from the cylinder Floer groups in [Muñ99a]. Either lift of the degree-four involution has a nonzero invariant subspace: on the unreduced complexified homology, multiplication in degree r by $e^{\pi i r/4}$ interchanges its two eigenspaces. The proposition is a consequence of the proved comparison, not an independent calibration of the local orientation used to prove that comparison. The positive p_1 convention fixes the incidence coorientation; it does not alone fix the index-four gauge-path orientation. The dependence on that remaining choice and the scope of DFL's sphere normalization are recorded in Section 10.2.

6. DIRICHLET RECONSTRUCTION AND CAP ESTIMATES

6.1. Seam-admissible product bundles. Decompose the base sphere into closed hemispheres

$$(47) \quad \mathbb{P}^1 = H_- \cup_{\Gamma} H_+,$$

where Γ is the compactified matching line and $\infty \in \Gamma$ is the attaching point. The gauge equation is placed on $H_- \times \Sigma_g$ and the holomorphic-map equation on H_+ . A primitive bundle $E \in \mathfrak{M}_g$ induces on the stable fibers of the base a classifying map to M_g .

Definition 6.1. A bundle $E \in \mathfrak{M}_g$ is *seam admissible* if every restriction $E|_{\{z\} \times \Sigma_g}$ with $z \in \Gamma$ is stable and the classifying quasimap has no basepoint in H_+ . We denote the seam-admissible locus by $\mathfrak{M}_g^{\text{sa}}$.

Every full-rank pair is seam admissible. A rank-drop pair is seam admissible precisely when its unique basepoint lies in the interior of H_- . A basepoint in H_+ makes the positive map undefined, while a basepoint on Γ cannot occur in a smooth seam-admissible object. For a sequence approaching Γ , the center–scale normal form (25) either recovers a full-rank mixed bubble at finite seam ratio or converges to one of the two pure limits. The real Möbius group preserving (47) acts transitively on the interior of H_- ; after fixing infinity, its affine subgroup $G_{\mathbb{R}}$ in (35) normalizes a gauge-side basepoint to a compact set.

6.2. Reconstruction on a compact product. For $\rho > 0$, equip the gauge hemisphere with the product Kähler form

$$(48) \quad \omega_{\rho} = \rho^2 \omega_{H_-} + \omega_{\Sigma_g}.$$

Here ω_{H_-} is smooth on the compact hemisphere. This defines an auxiliary boundary problem. Physical base dilation gives $\rho^2 |dw|^2 + g_{\Sigma_g}$ on an expanding Euclidean half-disc. In the coordinate $\zeta = 1/w$ its base metric is $\rho^2 |\zeta|^{-4} |d\zeta|^2$, which is singular at the attaching point. Replacing it by a smooth compact metric changes the four-dimensional ASD equation. The two problems have the same base-flat zero-scale equation on finite charts, but their positive-scale solutions cannot be identified without an additional comparison.

Theorem 6.2 (compact-domain Dirichlet reconstruction). *Let $E \in \mathfrak{M}_g^{\text{sa}}$. For every $\rho > 0$ there is a unique fixed-determinant projectively ASD connection $A_{E,\rho}$ on $E|_{H_- \times \Sigma_g}$ whose boundary metric is the family of Narasimhan–Seshadri metrics on the stable seam fibers. Together with the holomorphic classifying map*

$$u_E : H_+ \longrightarrow M_g,$$

it is an exact solution of the auxiliary compact mixed equation for (48). It has relative clutching charge one and normalized mixed energy $\frac{1}{2}$. For $\rho > 0$ the solution depends smoothly on (E, ρ) .

Proof. Each stable odd-degree bundle on a seam fiber has a unique fixed-determinant Hermitian–Einstein metric. Simplicity removes every noncentral ambiguity, and elliptic parameter dependence makes these metrics a smooth family H_{Γ} on $E|_{\Gamma \times \Sigma_g}$.

Donaldson’s boundary-value theorem [Don92], in the fixed-determinant form recalled in [Zha24, Theorem 3.8 and Proposition 3.9], gives a unique Hermitian metric $H_{E,\rho}$ with fixed determinant satisfying

$$(49) \quad H_{E,\rho}|_{\Gamma \times \Sigma_g} = H_{\Gamma}, \quad i\Lambda_{\omega_{\rho}} F_{H_{E,\rho}}^{\perp} = 0.$$

On a Kähler surface, integrability and the trace-free Hermitian Yang–Mills equation are equivalent to projective anti-self-duality. The Chern connection of $H_{E,\rho}$ is therefore the required $A_{E,\rho}$. No stability of E with respect to ω_ρ is needed: the Dirichlet theorem applies to every holomorphic bundle on a compact Kähler manifold with boundary. This point is essential because the polarization (48) approaches the chamber opposite to the one used to define $\mathfrak{M}_g$.

On H_+ , seam admissibility gives the holomorphic map u_E . Its pullback universal bundle is $E|_{H_+ \times \Sigma_g}$, and its canonical flat representative along Γ is the Chern connection of H_Γ . Thus the two sides satisfy the exact matching condition. Holomorphicity gives the positive equation.

The Chern–Weil filling identity and $p_1(\text{ad } E) = -2$ give relative charge one and energy $\frac{1}{2}$. The linearization of (49) in the Hermitian metric direction is the Dirichlet Chern Laplacian on trace-free self-adjoint endomorphisms. Its kernel is zero by integration by parts. Elliptic parameter dependence and the implicit-function theorem give smooth dependence on (E, ρ) . $\square$

For a full-rank E , the positive map is part of a basepoint-free minimal line and the negative half is its unique Dirichlet ASD completion. For a rank-drop E with basepoint in H_- , the positive map is constant and the entire modification lies on the gauge hemisphere. Thus both charge possibilities in Proposition 4.3 have compact algebraic models realized by exact local mixed solutions. In genus one the second type is the pure gauge bubble.

6.3. The compact mixed index and kernel. All operators in this subsection use a smooth compact product metric and one fixed scale $\rho > 0$. We use the real $W^{1,2}$ domain and L^2 target. The boundary conditions are

$$(51) \quad \iota_{\mathbf{n}} a = 0, \quad d_\alpha a_\Sigma = 0, \quad \Pi_\alpha a_\Sigma = \kappa_\alpha \nu,$$

where Π_α is harmonic projection and κ_α identifies the tangent space of M_g with harmonic surface one-forms. These are DFL’s canonical linearized matching condition and normal gauge condition. Their local estimate and weak-adjoint regularity [DFL24, Theorem 5.9], patched with interior estimates, give Fredholmness on the compact mixed domain. Their bounded identifications of nearby domains [DFL24, Proposition 5.27] give continuous Fredholm families when the smooth coefficients and correspondence vary at fixed scale. These statements also apply to smooth pairs that do not solve the mixed equation.

Lemma 6.3 (compact mixed index). *Let (A, u) be a smooth pair with the canonical seam correspondence, and let $\hat{P}$ be the closed adjoint bundle obtained by adjoining the pullback universal bundle on $H_+ \times \Sigma_g$, with its prescribed boundary identification. Then*

$$(51) \quad \text{ind}_{\mathbb{R}} \mathcal{D}_{A,u}^{\text{mix}} = 6g - 6 - 2\langle p_1(\hat{P}), [\mathbb{P}^1 \times \Sigma_g] \rangle.$$

Proof. First take the pullback of an irreducible flat surface connection α on the negative side and the constant map $[\alpha]$ on the positive side. Denote this operator by D_0 . Surface Hodge decomposition splits it into harmonic and nonharmonic sectors. The harmonic negative variation and the positive variation have equal traces, so they form a single $W^{1,2}$ section on $\mathbb{P}^1$ with values in $V = \mathcal{H}_\alpha^1$. Their equation is $\bar{\partial}$ on this trivial complex bundle, up to invertible target normalizations. It is surjective, with the constant sections as kernel, of real dimension $6g - 6$. For $g = 1$, $V = 0$.

For a nonharmonic mode $d_\alpha^* d_\alpha e = \mu e$, $\mu > 0$, write

$$a = a_{Be} + d_\alpha(\xi e) + *_\Sigma d_\alpha(\eta e).$$

The boundary conditions are $\iota_{\mathbf{n}}a_B = 0$ and $\eta|_{\Gamma} = 0$. The mixed curvature rows of the homogeneous equation give $a_B = d_B\xi + *_{B}d_B\eta$, and the remaining rows give

$$(52) \quad (\Delta_B + \rho^2\mu)\xi = 0, \quad (\Delta_B + \rho^2\mu)\eta = 0, \quad \partial_{\mathbf{n}}\xi = 0, \quad \eta|_{\Gamma} = 0.$$

Here Δ_B is the nonnegative scalar Laplacian. The Neumann condition follows from the zero normal component of a_B and the zero tangential derivative of the Dirichlet trace of η . Integration by parts gives $\xi = \eta = 0$, hence this mode has no kernel.

Its index is zero. For this one fixed surface eigenmode, multiply the zero-order surface coupling by $t \in [0, 1]$, retaining the boundary condition. This is a continuous family of elliptic boundary problems on the base disc. At $t = 0$ it splits into two operators. The divergence-curl operator on base one-forms with zero normal trace has zero kernel and a one-dimensional cokernel: the divergence must have integral zero. The Neumann–Dirichlet Hodge decomposition $a_B = df + *_{B}dg$ proves that this is its only range constraint. Its index is -1 . The other operator is

$$(\xi, \eta) \longmapsto d_B\xi + *_{B}d_B\eta, \quad \eta|_{\Gamma} = 0.$$

Its kernel consists of constant ξ , and it is onto. Indeed, solve a Dirichlet Poisson equation for η to prescribe the exterior derivative of the target; the remaining closed one-form is exact on the disc. Its index is $+1$. Thus the original mode has index zero, and its zero kernel implies surjectivity. The full constant operator is Fredholm, so its range is closed. Surjectivity on every finite sum of vertical modes and on the harmonic sector proves surjectivity by density. Consequently

$$(53) \quad \text{ind } D_0 = 6g - 6, \quad \ker D_0 = V, \quad \text{coker } D_0 = 0.$$

The modewise homotopy is not a homotopy of the complete four-dimensional symbol with its surface derivatives removed.

For the general pair, contract u smoothly to $u(\infty)$ by a contraction of the positive disc fixing ∞ . Pull back the universal projective bundle along this homotopy and transport the original seam identification through that bundle family. Extend the resulting flat surface boundary connections into the negative region. At the endpoint, arrange that the full negative connection is the pullback connection $\alpha = u(\infty)$ on a collar of the seam, using the transported identification. The base components can be changed freely in this homotopy. These data give a continuous Fredholm family. Its endpoint D_1 agrees with D_0 near the seam and on the positive component.

The closed bundles obtained by adjoining the moving positive caps form a bundle family, so the Pontryagin number is unchanged. In particular, the endpoint boundary identification need not extend over the negative interior. It is retained as the clutching map.

Choose a hypersurface $N = S^1 \times \Sigma_g$ strictly inside the common negative collar. Complete the gauge regions of D_1 and D_0 by the same four-dimensional product cap to closed ASD deformation operators Q_1 and Q_0 . Their bundles are $\hat{P}$ and the surface pullback P_0 . The operators agree in an annulus about N . Exchanging the two interiors in $D_1 \oplus Q_0$ produces $D_0 \oplus Q_1$.

For completeness, this excision uses the actual mixed domain. In the common gauge annulus use the source and target identifications

$$R(\vartheta) = \begin{pmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{pmatrix},$$

where ϑ is zero at the outer edge and $\pi/2$ at the inner edge. Extend by the identity on the exteriors and by the signed interchange on the interiors. These maps are bounded and

invertible on the direct-sum Sobolev spaces. They preserve the mixed boundary domain because they are the identity near the seam and on the positive map component. After identification, the direct-sum operators differ by a zero-order commutator supported in the annulus. Rellich compactness therefore gives

$$(54) \quad \text{ind } D_1 + \text{ind } Q_0 = \text{ind } D_0 + \text{ind } Q_1.$$

This is the interior relative-index argument; compare [BB16, Section 6.2]. No replacement of the seam by an ordinary finite-rank transmission condition is used.

The closed ASD index formula [DK90] is

$$\text{ind } Q_P = -2\langle p_1(P), [X] \rangle - 3(1 - b_1(X) + b^+(X)), \quad X = \mathbb{P}^1 \times \Sigma_g.$$

Since $b_1(X) = 2g$, $b^+(X) = 1$, and $p_1(P_0) = 0$, subtracting the two closed indices in (54) and using (53) proves the lemma. $\square$

Proposition 6.4 (fixed-scale mixed regularity). *For $E \in \mathfrak{M}_g^{\text{sa}}$ and $\rho > 0$, infinitesimal holomorphic doubling gives a natural real-linear isomorphism*

$$(55) \quad \ker \mathcal{D}_{E,\rho}^{\text{mix}} \cong H^1(\mathbb{P}^1 \times \Sigma_g, \text{End}_0 E).$$

The real kernel dimension is $6g - 2$ and the mixed operator is surjective. Its determinant line can therefore be oriented by the complex orientation on the right. This includes gauge-side rank drop and genus one. Evaluation at infinity and quotienting by $G_{\mathbb{R}}$ are separate finite-dimensional operations.

Proof. We first compute the holomorphic deformation groups. The primitive extension has the form

$$(56) \quad 0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(1) \boxtimes L \longrightarrow E \longrightarrow \mathcal{O}_{\mathbb{P}^1} \boxtimes (\Lambda L^{-1}) \longrightarrow 0;$$

see [Muñ99b, Proposition 2] for $g \geq 2$. The same extension construction applies when $g = 1$, and the following calculation treats that case directly. On each horizontal line $C_x = \mathbb{P}^1 \times \{x\}$ the extension splits, since $H^1(\mathbb{P}^1, \mathcal{O}(1)) = 0$. Hence

$$(\text{End}_0 E \otimes K_X)|_{C_x} \cong \mathcal{O}(-1) \oplus \mathcal{O}(-2) \oplus \mathcal{O}(-3).$$

It has no section, so Serre duality gives $H^2(X, \text{End}_0 E) = 0$. A trace-free holomorphic endomorphism vanishes on the open set of stable surface fibers containing the seam, hence $H^0(X, \text{End}_0 E) = 0$. Riemann–Roch gives

$$\chi(\text{End}_0 E) = 3(1 - g) - 2 = 1 - 3g, \quad h^1(X, \text{End}_0 E) = 3g - 1.$$

Vanishing of H^2 makes the fixed-determinant deformation problem unobstructed. Equivalently, the integrability equation in a Dolbeault slice has surjective linearization onto $(0, 2)$ -forms, and the implicit-function theorem integrates every H^1 class.

Now take a mixed kernel element (a, ν) . DFL regularity makes it smooth. The integrability row of the ASD linearization gives a Dolbeault cocycle $v_- = a^{0,1}$ on the negative half. The positive equation makes ν a holomorphic infinitesimal map; pulling back the universal bundle over the dual numbers gives its Dolbeault cocycle v_+ . Identify the original holomorphic bundles on both halves with the restrictions of E . Matching says that their vertical cohomology classes agree on the seam. Since the seam fibers are simple, there is a unique smooth trace-free ξ there such that

$$(v_- - v_+)_{\Sigma} = \bar{\partial}_{\alpha} \xi.$$

Extend ξ smoothly to the negative side and subtract $\bar{\partial}_E \tilde{\xi}$ from v_- . The resulting piecewise $(0, 1)$ -form has equal vertical traces and is distributionally $\bar{\partial}_E$ -closed. Indeed, the possible

seam delta term is $d\bar{z}$ wedged with the jump in the vertical component. A jump in the coefficient of $d\bar{z}$ contributes zero because $d\bar{z} \wedge d\bar{z} = 0$. The L^2 Dolbeault Hodge decomposition on the closed surface therefore gives a class in $H^1(X, \text{End}_0 E)$. Different extensions give global $W^{1,2}$ zero-forms whose Dolbeault derivatives are the differences. Changing cocycle representatives has the same effect. This defines (55) without an extension on the complete mixed Sobolev domain.

Suppose the class is zero. Write the corrected piecewise cocycle as $\bar{\partial}_E \epsilon$ with a global $W^{1,2}$ zero-form ϵ . Its vertical class on the positive side is zero, so $\nu = 0$. Choose $v_+ = 0$. Then ϵ is holomorphic on the positive side and vanishes there by simplicity of its stable surface fibers. Its seam trace is zero. On the negative side,

$$a^{0,1} = \bar{\partial}_E \eta, \quad \eta = \epsilon + \tilde{\xi}, \quad \eta|_\Gamma = \xi.$$

The real matching condition now says $a_\Sigma = d_\alpha \xi$ with ξ skew-Hermitian. Uniqueness of its vertical Dolbeault primitive identifies it with the same ξ . Dirichlet elliptic regularity makes η smooth. Decompose $\eta = \chi + s$ into skew-Hermitian and self-adjoint parts; then $s|_\Gamma = 0$. Removing the infinitesimal unitary gauge direction χ leaves a Hermitian-metric variation of the fixed holomorphic bundle. The linearized HYM equation is its Dirichlet Chern Laplacian, so integration by parts gives $s = 0$. Thus $a = d_A \chi$. The Coulomb row and normal boundary condition give

$$d_A^* d_A \chi = 0, \quad \nabla_{\mathbf{n}} \chi = 0.$$

A second integration by parts gives $d_A \chi = 0$, and hence $a = 0$. This proves injectivity, including the unitary gauge direction.

Conversely, integrate an H^1 class to a family of holomorphic bundles. Seam admissibility persists for sufficiently small parameters by openness of stability over the compact positive hemisphere. Reconstruct by Theorem 6.2 at the fixed scale and differentiate. Impose Coulomb gauge and zero normal component by solving the covariant Neumann equation for an infinitesimal unitary gauge transformation. Its kernel consists of parallel trace-free endomorphisms and vanishes by irreducibility of a seam fiber. This gauge change preserves the linearized equations and matching, and doubling the resulting variation returns the prescribed H^1 class. The kernel identification is onto.

Finally, Lemma 6.3 and $p_1(\text{ad } E) = -2$ give real index $6g - 2$, equal to the kernel dimension. The cokernel is zero, and the complex orientation of the kernel orients the determinant line. $\square$

On compact families with scales bounded away from zero, the kernel bundle has locally bounded complementary right inverses. Uniform estimates as $\rho \rightarrow 0$ require the original-coordinate construction in Section 7. The comparison with DFL's coherent orientations is established in Sections 10 and 12.

The rank-drop normal can be read explicitly. At a tensor $[v \otimes a] \in \mathbb{P}(V \otimes \mathbb{C}^2)$, let $\ell_v = \mathbb{C}v$ and $\ell_a = \mathbb{C}a$. The normal to the Segre locus is

$$(57) \quad \text{Hom}(\ell_v, V/\ell_v) \otimes \text{Hom}(\ell_a, \mathbb{C}^2/\ell_a),$$

of complex dimension $g - 1$. In genus two it is a complex line; after its basepoint is fixed, both real components remain as smoothing directions from rank drop to a full-rank bubble. The seam-preserving affine group has no rotational factor and does not remove the phase. Thus the genus-two Segre locus has real codimension two inside the primitive face. In genus one the normal has rank zero, leaving the elliptic excess family.

6.4. Primitive no-neck at good circles. Let (A_ν, u_ν) be a sequence with one primitive matching center $z_\nu \rightarrow z_0$ and concentration scales $\rho_\nu \rightarrow 0$, and suppose its doubled bundles have a limit E in the three-colored master compactification described after Proposition 4.3. Write $\mathcal{E}_\nu(a, b)$ for the normalized mixed energy in the paired half-annulus $a < |z - z_\nu| < b$.

Proposition 6.5 (primitive good-circle no-neck). *After passage to a subsequence, there are radii*

$$\rho_\nu \ll a_\nu \ll b_\nu \longrightarrow 0$$

such that

$$(58) \quad \mathcal{E}_\nu(a_\nu, b_\nu) \longrightarrow 0.$$

The capped region inside a_ν has clutching sector one and energy tending to $\frac{1}{2}$. Hence no positive-energy atom or further analytic scale lies between the primitive core and the residual solution. On the two cut circles, the spliced representatives converge in a common seam gauge to the same flat attaching representative.

Proof. The stable limit is regular at infinity. For a sufficiently large rescaled radius R , cap its tail in a contractible fine-moduli chart. The finite charge identity in Proposition 4.3 shows that, for R and then ν large, the capped region $|z - z_\nu| \leq R\rho_\nu$ has clutching sector one. This includes rank drop because the lost map degree is exactly the length-one elementary modification. The local charge theorem gives the same sector for the outer atom. Every annulus between the two caps therefore has sector zero.

Put $E_\nu(r) = \mathcal{E}_\nu(0, r)$. First choose the outer cuts from smooth convergence on compact subsets of the punctured disc to the residual solution. DFL removability extends that solution over z_0 , so a diagonal choice gives $b_\nu \rightarrow 0$, with $b_\nu/\rho_\nu \rightarrow \infty$, such that the outer-tail energy relative to the residual solution and $b_\nu E'_\nu(b_\nu)$ both tend to zero.

Since

$$\int_{r_1}^{r_2} r E'_\nu(r) d \log r = E_\nu(r_2) - E_\nu(r_1),$$

choose $R_\nu \rightarrow \infty$ slowly enough that $R_\nu^2 \rho_\nu < b_\nu$, and subdivide the logarithmic region from $R_\nu \rho_\nu$ to $R_\nu^2 \rho_\nu$ into dyadic shells. Bounded total energy and coarea give a shell of energy tending to zero and a radius a_ν in it for which

$$(59) \quad a_\nu E'_\nu(a_\nu) + b_\nu E'_\nu(b_\nu) \longrightarrow 0.$$

At either good circle form DFL's spliced connection $B_{\nu,r}$ on $S^1 \times \Sigma_g$: it joins the gauge semicircle to a connection representing the holomorphic semicircle through the flat interpolation strip. The small-energy mean-value estimate on the selected shell and the proof of [DFL24, Lemma 4.62] give the localized slice bound

$$|\text{CS}_{\beta_{\nu,r}}(B_{\nu,r})| \leq C r E'_\nu(r).$$

The flat references at the two radii are joined by the seam path, so the Chern–Simons functional is independent of that choice. The Stokes calculation of [DFL24, Lemma 4.67], now on the annulus, gives

$$\mathcal{E}_\nu(a_\nu, b_\nu) = \text{CS}(B_{\nu,b_\nu}) - \text{CS}(B_{\nu,a_\nu})$$

with the normalization of (3). The possible integral change of lift is zero because the annulus has clutching sector zero. Equation (59) proves (58).

The total local energy tends to $\frac{1}{2}$, the annular energy vanishes, and the outer-tail energy tends to that of the removable residual solution. Thus the sector-one inner core has energy $\frac{1}{2}$ in the limit. A further atom would have positive half-integral charge, whereas a diffuse

remainder is excluded by the same Chern–Simons identity. Finally, DFL’s small-energy Coulomb gauges on the two selected shells converge to the flat seam representatives; stable reduction at infinity identifies both with the attaching value. $\square$

6.5. Inverse-adiabatic Dirichlet estimates. Put $\tau = \rho^2$. Since the Kähler form is a product, the trace-free HYM equation is equivalent to

$$(60) \quad i\Lambda_{H_-} F_H^\perp + \tau i\Lambda_{\Sigma_g} F_H^\perp = 0.$$

Thus the limit $\tau = 0$ is elliptic in the base variable and regards the surface variable as a parameter. This is the useful direction of the inverse-adiabatic limit.

For the following statements, $W_{B,\Sigma}^{r,s;p}$ controls mixed derivatives with at most r base derivatives and at most s surface derivatives in fixed product charts. On the HYM-equation component we use the equivalent ρ -weighted target norm obtained by multiplying the equation by ρ^2 , as in (60). The next theorem concerns the fixed compact base; estimates at the physical attaching end are established in Section 7.

Theorem 6.6 (Dirichlet regularity on a fixed compact product). *Let $C \subset \mathfrak{M}_g^{\text{sa}}/G_{\mathbb{R}}$ be compact, represented in smooth local families of bundles and boundary metrics on the fixed compact hemisphere. The auxiliary Dirichlet metrics $H_{E,\rho}$ extend smoothly in (E, τ) to $\tau = 0$ in every fixed-coordinate $W_{B,\Sigma}^{r,s;p}$ space. The limiting metric is the unique fixed-determinant solution of*

$$(61) \quad i\Lambda_{H_-} F_{H_{E,0}}^\perp = 0, \quad H_{E,0}|_{\Gamma \times \Sigma_g} = H_\Gamma.$$

Moreover, $H_{E,0}^{-1} H_{E,\rho} - 1 = O(\tau)$ in every fixed-coordinate C^k norm, uniformly over C . Interior rank drop is allowed. This statement concerns the smooth compact metric (48).

Proof. Write $B = H_-$. For each $x \in \Sigma_g$, Donaldson’s Dirichlet theorem on the complex disc gives the solution $H_{E,0}$ of (61). Its linearization is the invertible base Dirichlet Chern Laplacian. Ordinary parameter-dependent elliptic theory therefore gives smooth dependence on x and E . No surface derivative is inverted at this step.

We use norms which include pointwise control. Fix $0 < \alpha < 1$ and use the metric

$$\widehat{g}_\rho = g_B + \rho^{-2} g_{\Sigma_g}.$$

Define $\|\cdot\|_{k,\alpha;\rho}$ as the supremum of local $C^{k,\alpha}$ norms in product charts of a fixed small base radius and surface radius proportional to ρ , pulling back the surface coordinate by $x = x_0 + \rho y$. Include the fixed base boundary charts. Thus base derivatives have weight one and surface derivatives have weight ρ . In particular,

$$\|s\|_\infty \leq \|s\|_{k,\alpha;\rho}, \quad \|st\|_{k,\alpha;\rho} \leq C_k \|s\|_{k,\alpha;\rho} \|t\|_{k,\alpha;\rho}.$$

These estimates follow in each fixed rescaled chart and then by taking the supremum. Their constants do not depend on the number of charts.

Put

$$\Phi_\tau(H) = i\Lambda_B F_H^\perp + \tau i\Lambda_{\Sigma_g} F_H^\perp.$$

We first establish a linear estimate at any smooth approximate family K_τ whose contracted curvature $\Phi_\tau(K_\tau)$ tends uniformly to zero. Assume that K_τ and its inverse have uniform bounds in all fixed-coordinate derivatives. Transport the moment map to the fixed metric by

$$\Psi_{K,\tau}(s) = e^{s/2} \Phi_\tau(K_\tau e^s) e^{-s/2}.$$

Its real linearization with zero Dirichlet boundary value is

$$\begin{aligned} L_{K,\tau}s &= i\Lambda_{\hat{g}_\rho}\bar{\partial}\partial_K s + \frac{1}{2}[s, \Phi_\tau(K_\tau)] \\ &= \frac{1}{2}\nabla_K^* \nabla_K s. \end{aligned}$$

Indeed, the Kähler identity for the first term adds $\frac{1}{2}[\Phi_\tau(K_\tau), s]$ to half the covariant Laplacian; transport cancels that commutator. This holds also at the approximate metric. Kato's inequality therefore gives

$$(\Delta_B + \tau\Delta_{\Sigma_g})|s| \leq C|L_{K,\tau}s|,$$

where the scalar Laplacians have nonnegative spectrum. If $\Delta_B\varphi = 1$ and $\varphi|_{\partial B} = 0$, comparison with $C\varphi\|L_{K,\tau}s\|_\infty$ gives a uniform C^0 estimate. Interior and Dirichlet boundary Schauder estimates in the rescaled charts then give

$$(62) \quad \|s\|_{k+2,\alpha;\rho} \leq C_k \|L_{K,\tau}s\|_{k,\alpha;\rho}.$$

The coefficients, scaled coefficient derivatives, and ellipticity constants are uniform. The global zeroth-order bound uses the fixed base barrier. For each positive τ the Dirichlet operator has index zero, by deformation to its connection Laplacian. The estimate gives injectivity and hence surjectivity.

Construct formal coefficients before applying contraction. The derivative of Φ_0 at $H_{E,0}$ is the base Dirichlet Chern Laplacian L_0^B . For each integer $N \geq 0$, successive equations for this operator give smooth sections s_j , zero on the boundary, such that

$$H_\tau^{[N]} = H_{E,0} \exp \left(\sum_{j=1}^N \tau^j s_j \right), \quad \Phi_\tau(H_\tau^{[N]}) = O(\tau^{N+1})$$

in every fixed-coordinate C^k norm. At order j , the unknown term is $L_0^B s_j$; all other terms involve already constructed smooth coefficients. This avoids an implicit-function theorem at $\tau = 0$ with loss of surface derivatives.

Apply (62) at $H_\tau^{[N]}$. In the scaled Hölder spaces, the exponential-coordinate nonlinear remainder obeys

$$\|\mathcal{N}_\tau(s) - \mathcal{N}_\tau(t)\|_{0,\alpha;\rho} \leq C(\|s\|_{2,\alpha;\rho} + \|t\|_{2,\alpha;\rho})\|s - t\|_{2,\alpha;\rho}.$$

Indeed, in the fixed rescaled charts the principal coefficients are smooth matrix functions of the metric variable, and the remaining nonlinear terms are products of at most two first derivatives. The preceding product estimate makes the constant uniform. Contraction produces

$$H_{E,\rho} = H_\tau^{[N]} e^{r_{N,\tau}}, \quad \|r_{N,\tau}\|_{2,\alpha;\rho} \leq C_N \tau^{N+1}.$$

Dirichlet uniqueness identifies the solutions constructed for different N . Higher Schauder estimates for the difference equation give the same remainder order in every scaled $C^{k,\alpha}$ norm: keep the highest derivatives in the elliptic linearization, and use the previously bounded lower derivatives of the remainder on the right. The initial C^0 bound comes from contraction, rather than from this derivative induction.

For positive τ ordinary Dirichlet theory gives smooth parameter dependence. Differentiating the difference equation in τ contributes one unweighted surface Laplacian, which costs at most $\rho^{-2} = \tau^{-1}$ in the scaled norms. The highest parameter derivative remains in the same invertible linearization. Induction, using the higher Schauder estimates, gives for fixed k, j and sufficiently large N

$$\|\partial_\tau^j r_{N,\tau}\|_{k,\alpha;\rho} \leq C_{N,k,j} \tau^{N+1-j}.$$

Bundle-parameter derivatives are treated in the same way. Converting k derivatives to unscaled coordinates loses at most ρ^{-k} . Since N is arbitrary, the desired unscaled remainder and its parameter derivatives tend to zero at every prescribed finite order. The formal coefficients are consequently the Taylor coefficients of the exact family at zero. This proves the smooth extension and the asserted $O(\tau)$ estimates. All arguments are uniform on the compact family. An unstable surface fiber in the interior does not change any Dirichlet hypothesis used here. $\square$

6.6. The normalized limiting surface operator. There is a Fredholm problem on the compact base at zero scale. It keeps the surface transmission equation and fixes the residual surface gauge at the attaching point. Let B_- and B_+ be the compact negative and positive discs, with common boundary Γ and attaching point $z_0 = \infty \in \Gamma$. For a seam-admissible primitive bundle E , use its base-flat metric H_0 . In the unitary parallel identification from z_0 ,

$$(63) \quad A_B^{(0)} = 0, \quad A_\Sigma^{(0)} = \beta(z), \quad \bar{\partial}_B \beta^{0,1} = 0.$$

Only the boundary surface connections are required to be flat. Put $\alpha = \beta(z_0)$ and $u = u_E$. Write $H_\Sigma^k = H^k(\Sigma; T^*\Sigma \otimes \text{ad})$ using the attaching connection and the complex structure $*_\Sigma$. For $q > 2$ and an integer $k \geq 0$, define the closed real domain

$$(64) \quad \begin{aligned} \mathcal{X}_E^{q,k} = \{ & (b, \nu) \in W^{1,q}(B_-; H_\Sigma^k) \oplus W^{1,q}(B_+; u^*TM_g) : \\ & C_{\beta(z)} b(z) = 0, \quad H_{\beta(z)} b(z) = \ell_z \nu(z) \quad (z \in \Gamma), \\ & E_\alpha b(z_0) = 0 \}. \end{aligned}$$

Here ℓ_z gives harmonic representatives. The first Sobolev norm takes one base derivative and k surface derivatives; evaluation at z_0 is bounded because $q > 2$. Set

$$(65) \quad \begin{aligned} \mathcal{Y}_E^{q,k} &= L^q(B_-; \Lambda_B^{0,1} \otimes H_\Sigma^k) \oplus L^q(B_+; \Lambda_B^{0,1} \otimes u^*TM_g), \\ \mathcal{C}_E(b, \nu) &= (\bar{\partial}_B b, D_u \nu). \end{aligned}$$

The real form b is identified with its $(0, 1)$ part in the first equation, and D_u is the Dolbeault operator on u^*TM_g . The last row of (64) fixes only the exact attaching value. Without it, $(d_{\beta(z)} \zeta, 0)$ for base-constant skew-Hermitian ζ would give an infinite-dimensional kernel.

Proposition 6.7 (normalized limiting surface problem). *The operator $\mathcal{C}_E : \mathcal{X}_E^{q,k} \rightarrow \mathcal{Y}_E^{q,k}$ is surjective and Fredholm. Its kernel is smooth, and doubling gives a real-linear isomorphism*

$$(66) \quad \ker \mathcal{C}_E \cong H^1(\mathbb{P}^1 \times \Sigma, \text{End}_0 E), \quad \dim_{\mathbb{R}} \ker \mathcal{C}_E = 6g - 2.$$

It has uniformly bounded right inverses on compact smooth families of these data. For a bounded projection Π_E onto its kernel,

$$(67) \quad \|v - \Pi_E v\|_{\mathcal{X}_E^{q,k}} \leq C \|\mathcal{C}_E v\|_{\mathcal{Y}_E^{q,k}}.$$

All norms here use the fixed compact base metrics.

Proof. We first describe the constant infinite-dimensional boundary problem. If a complex Hilbert space has a real Lagrangian L , then

$$(68) \quad \bar{\partial} : \{v \in W^{1,q}(D; L \oplus iL) : v|_{\partial D} \in L, v(z_0) = 0\} \longrightarrow L^q(D; \Lambda^{0,1} \otimes (L \oplus iL))$$

is an isomorphism. For scalar data, take a Cauchy–Green solution, cancel its imaginary boundary trace by a holomorphic correction obtained from the harmonic Dirichlet solution and its conjugate, and subtract its remaining real value at z_0 . The scalar singular integral and harmonic extension estimates give the $W^{1,q}$ bound. A holomorphic function with real

boundary trace is constant, which proves uniqueness. To extend this bound to Hilbert-valued data, apply the scalar real-linear inverse to $\sum_j \epsilon_j f_j$ and average over independent random signs. The Khintchine inequalities give the bound for pointwise ℓ^2 norms, also after a base derivative. Density proves (68). The corresponding local linear boundary regularity is also covered by [Weh04, Theorem 1.3 and Corollary 1.4], in the author's corrected version. A separable Hilbert space satisfies its Banach-space hypothesis through the Rademacher embedding into L^q .

Fold the positive disc onto the negative one and choose a unitary trivialization of u^*TM_g , with fiber V . The Hilbert space and complex structure become

$$\mathbb{H} = H_\Sigma^0 \oplus V, \quad \mathbb{J} = (*_\Sigma, -J_V), \quad \dim_{\mathbb{C}} V = 3g - 3.$$

The sign accounts for the orientation reversal. The positive zeroth-order coefficients have finite rank; target identifications absorb the nonzero conformal factors. Use the harmonic metric on V , so that ℓ_z is an isometry. The orthogonal projection onto the boundary subspace is

$$(69) \quad P(z) = \begin{pmatrix} E_{\beta(z)} + \frac{1}{2}H_{\beta(z)} & \frac{1}{2}\ell_z \\ \frac{1}{2}\ell_z^* & \frac{1}{2}I_V \end{pmatrix}, \quad \mathbb{J}P(z)\mathbb{J}^{-1} = I - P(z).$$

For $G = (d_\beta^* d_\beta)^{-1}$ on surface zero-forms, $E_\beta = d_\beta G d_\beta^*$ and

$$(70) \quad \dot{E}_\beta = \dot{d} G d^* + d G \dot{d}^* - d G (\dot{d}^* d + d^* \dot{d}) G d^*.$$

Here dotted differentials have surface order zero and $\dot{G} = -G(\dot{d}^* d + d^* \dot{d})G$. Each displayed term maps H_Σ^k to H_Σ^{k+1} : for example $G d^*$ gains one derivative, while $d G$ gains one and its remaining middle factors lose at most one. The same conclusion holds after every fixed number of base derivatives, by repeated resolvent differentiation on the compact irreducible family. The formulas $C_\beta = *E_\beta*^{-1}$ and $H_\beta = I - E_\beta - C_\beta$ give the other projections. Thus $P(z) - P(z_0)$ and all its base derivatives gain one surface derivative. Its harmonic and off-diagonal terms are smooth finite-rank operators.

Put $P_\infty = P(z_0)$. Choose finite spectral projections F_N containing V , the harmonic surface space, and whole attaching nonharmonic eigenspaces. They commute with P_∞ and $\mathbb{J}$. Compactness of the surface operator differences gives uniform operator-norm convergence of their finite compressions, including any specified finite collection of base derivatives and surface Sobolev orders. Hence

$$\tilde{P}_N = P_\infty + F_N(P - P_\infty)F_N$$

is close to P , self-adjoint, and has the symmetry in (69). Spectral projection onto the cluster near 1 gives an orthogonal projection P_N with the same symmetry. Its difference from P_∞ is supported in F_N , and $P_N(z_0) = P_\infty$.

The polar map

$$(71) \quad \begin{aligned} S &= P P_N + (I - P)(I - P_N), & U &= S(S^* S)^{-1/2}, \\ S - I &= (P - P_N)(2P_N - I), & S^* S &= I - (P - P_N)^2 \end{aligned}$$

commutes with $\mathbb{J}$ and satisfies $U P_N U^{-1} = P$. It equals I at z_0 . Its difference from I and all base derivatives gain one surface derivative: increase N so the inverse-square-root series converges in each specified Sobolev operator norm, then use the last row of (71). Extend its small logarithm over a collar and cut it off toward the center of the disc. This extends the near-identity map U , while retaining the possibly nontrivial finite-dimensional Lagrangian loop P_N .

Conjugation by U changes the constant principal operator by a smooth zeroth-order operator family $K(z)$ gaining one surface derivative. Multiplication by K is compact from $W^{1,q}(D; H_\Sigma^k \oplus V)$ to $L^q(D; H_\Sigma^k \oplus V)$. Indeed, approximate K uniformly by finite spectral compressions and apply finite-dimensional base Rellich to each compression. No infinite-dimensional base Rellich embedding is used. The positive zeroth-order term has the same compactness property.

The boundary problem for P_N splits into a finite-dimensional Riemann–Hilbert problem and a constant infinite-dimensional tail. On that tail the allowed boundary value is exact, so $E_\alpha b(z_0) = 0$ is precisely the normalization in (68). On the finite part it imposes only finitely many point conditions. Local complex-linear straightening, constant real boundary estimates, weak-adjoint regularity, and finite-dimensional Rellich prove Fredholmness of the finite part. Compact perturbation now proves Fredholmness of $\mathcal{C}_E$.

It also gives smoothness of the kernel. A generalized inverse for the finite part, together with (68), preserves each fixed surface Sobolev order; its kernel is finite-dimensional and smooth. In the homogeneous equation the term K gains one surface derivative. Iterating this generalized-inverse identity gives all surface orders. Differentiating the local constant-boundary estimates in the base, with smooth coefficients, then gives all base orders.

We prove surjectivity by dense image and the established closed range. Take smooth targets f_-, f_+ supported away from the seam in their respective discs. Interpret f_- as an $\text{End}_0 E$ -valued $(0, 2)$ form and lift f_+ by the positive harmonic representatives. Their union is a smooth global $(0, 2)$ form F on $X = \mathbb{P}^1 \times \Sigma$. The vanishing $H^2(X, \text{End}_0 E) = 0$ proved in Proposition 6.4 gives a smooth $(0, 1)$ form v with $\bar{\partial}_E v = F$. On the positive side set $\nu(z) = [v_\Sigma(z)]$. The natural Dolbeault operator on $R^1\pi_* \text{End}_0 E = u^*TM_g$ gives $D_u \nu = f_+$; the term $\bar{\partial}_\Sigma v_B$ has zero vertical class.

On the boundary there is a unique smooth complex trace-free potential ξ such that

$$v_\Sigma - (\ell\nu)^{0,1} = \bar{\partial}_\beta \xi.$$

This is the surface Dolbeault decomposition and simplicity of the stable boundary fibers. Let $s = (\xi + \xi^*)/2$. In the base-parallel frame solve the scalar real boundary problem

$$(72) \quad \bar{\partial}_B \eta = v_B, \quad \frac{1}{2}(\eta + \eta^*)|_\Gamma = s, \quad b^{0,1} = v_\Sigma - \bar{\partial}_\beta \eta.$$

The same harmonic Dirichlet and conjugate construction used above is smooth in the surface parameter. Integrability in (63) gives $\bar{\partial}_B b = f_-$. On the boundary $\xi - \eta$ is skew-Hermitian, so $b = \ell\nu + d_\beta(\xi - \eta)$ and both seam rows hold. Finally subtract $(d_{\beta(z)}\zeta, 0)$ with $\zeta = G_\alpha d_\alpha^* b(z_0)$. This is a smooth homogeneous solution and sets $E_\alpha b(z_0) = 0$. Thus every chosen smooth target is in the image. Such targets are dense in each $\mathcal{Y}_E^{q,k}$, proving surjectivity. No bounded extension of arbitrary transmission traces into a Dolbeault domain is required.

For a kernel element, double the negative cocycle $b^{0,1}$, whose base component is zero, with the cocycle of the positive infinitesimal map. As in Proposition 6.4, subtract the Dolbeault derivative of an extension of their boundary difference potential. Equal vertical traces make the piecewise cocycle distributionally closed; the possible delta term is $d\bar{z}$ wedged with the vertical difference, and a jump in the base component contributes zero. This defines the class in (66) independently of choices.

If its class is zero, then $\nu = 0$ and the positive cocycle can be chosen zero. The global exact potential vanishes on the positive half by stability of its surface fibers. On the negative half one obtains

$$b^{0,1} = \bar{\partial}_\beta \eta, \quad \bar{\partial}_B \eta = 0, \quad \eta|_\Gamma \text{ skew-Hermitian.}$$

The Hermitian part is base-harmonic with zero Dirichlet data, hence zero. A base-holomorphic skew-Hermitian function is base-constant. Thus $b = d_\beta \zeta$, and the point normalization gives $d_\alpha \zeta = 0$. Irreducibility implies $\zeta = 0$, proving injectivity. Conversely, integrate an $H^1(X, \text{End}_0 E)$ class to a bundle family, using $H^2 = 0$. Differentiate its smoothly varying base-flat Dirichlet metric in the parallel identification, together with its positive map, and normalize by a base-constant surface gauge. This gives a kernel element doubling to that class. The dimension is $2(3g - 1)$ as computed in Proposition 6.4.

The smooth kernel bundle has bounded L^2 projections on the fixed compact base. Nearby boundary domains are identified by small polar maps followed by the homogeneous gauge correction that restores the point normalization. Bounded evaluation and the smooth surface Green inverse make these bounded local identifications. Surjectivity then gives locally bounded right inverses, uniformly over a compact parameter family. This proves (67). $\square$

The harmonic attaching value in this proposition remains free. The physical operator also contains base one-form components and, after rescaling, terms ρd_Σ . Those terms are unbounded from $W_B^{1,q} H_\Sigma^k$ to $L_B^q H_\Sigma^k$ at every fixed $\rho > 0$: surface frequency λ contributes $\rho \lambda$. Consequently the proposition does not give a positive-scale inverse by small bounded perturbation, or the required comparison of compact and expanding physical norms. We next restore the base equations on the fixed compact product in a different, fractional norm.

Let B be a fixed smooth compact oriented disc, with $z_0 \in \partial B$. For an irreducible flat surface connection α , put $\Lambda = (d_\alpha^* d_\alpha)^{1/2}$ on trace-free zero-forms. Write a nonharmonic surface form as

$$b = d_\alpha \Lambda^{-1} x + *_\Sigma d_\alpha \Lambda^{-1} y.$$

If the actual base variation is ra , set $m = r\Lambda$. For $\delta = d_B^*$ and $\text{curl} = *_B d_B$, a fixed normalization of the flat ASD–Coulomb rows is

$$(73) \quad \mathcal{D}_m(a, x, y) = (\delta a + mx, -\text{curl} a + my, dx + *_B dy - ma).$$

Fix $0 < \varepsilon < 1/2$ and an integer $k \geq 0$. Define

$$(74) \quad \begin{aligned} \|U\|_{\mathcal{X}_r^{\varepsilon,k}} &= \|(a, x, y)\|_{H_B^{1+\varepsilon} H_\Sigma^k} + \|ma\|_{H_B^\varepsilon H_\Sigma^k} + \|my\|_{H_B^\varepsilon H_\Sigma^k}, \\ \|F\|_{\mathcal{Y}^{\varepsilon,k}} &= \|(h, v)\|_{H_B^\varepsilon H_\Sigma^k}. \end{aligned}$$

These are Hilbert tensor Sobolev norms. The graph domain satisfies

$$(75) \quad \delta a = 0, \quad a(n) = 0, \quad y|_{\partial B} = 0, \quad x(z_0) = 0.$$

Point evaluation is bounded at base order $1 + \varepsilon > 1$. There is no mx term in the norm: the weighted surface derivative of b is its exterior derivative, which controls its coexact part.

6.7. Fractional cap inverses.

Lemma 6.8 (flat fractional estimate). *The last two rows of (73) give an isomorphism $\mathcal{A}_r : \mathcal{X}_r^{\varepsilon,k} \rightarrow \mathcal{Y}^{\varepsilon,k}$ with*

$$(76) \quad \|\mathcal{A}_r^{-1} F\|_{\mathcal{X}_r^{\varepsilon,k}} \leq C \|F\|_{\mathcal{Y}^{\varepsilon,k}} \quad (0 \leq r \leq 1).$$

The constant is uniform over compact smooth irreducible attaching families and over all surface frequencies.

Proof. Use the scalar Dirichlet and Neumann Laplacians Δ_D, Δ_N on B , and the absolute one-form Laplacian Δ_{abs} , whose boundary conditions are $a(n) = \text{curl} a = 0$. The Dirichlet and absolute spectra have positive gaps because B is a disc. Integration by parts gives

$$\mathcal{D}_m^*(g, h, v) = (dg + *_B dh - mv, mg + \delta v, mh - \text{curl} v), \quad h|_{\partial B} = 0, \quad v(n) = 0.$$

The two squares, with their induced boundary domains, are

$$(77) \quad \begin{aligned} \mathcal{D}_m^* \mathcal{D}_m &= \text{diag}(\Delta_{\text{abs}} + m^2, \Delta_N + m^2, \Delta_D + m^2), \\ \mathcal{D}_m \mathcal{D}_m^* &= \text{diag}(\Delta_N + m^2, \Delta_D + m^2, \Delta_{\text{abs}} + m^2). \end{aligned}$$

For example, the extra boundary condition on x is Neumann, since the normal trace of $dx + *_B dy - ma$ reduces to $\partial_n x$. For a target $(0, h, v)$ put

$$(78) \quad \begin{aligned} H &= (\Delta_D + m^2)^{-1} h, & V &= (\Delta_{\text{abs}} + m^2)^{-1} v, \\ a &= *_B dH - mV, & x &= \delta V, & y &= mH - \text{curl } V. \end{aligned}$$

Then $\mathcal{D}_m(a, x, y) = (0, h, v)$, with $a(n) = 0$, $y = 0$, and x of base mean zero. Only the two positive-gap resolvents are used, including at $m = 0$.

At order $\varepsilon < 1/2$ their spectral Sobolev spaces agree with ordinary H^ε . At order $1 + \varepsilon$, the Dirichlet and absolute spaces impose $y = 0$ and $a(n) = 0$, respectively; the Neumann derivative condition is not imposed below order $3/2$. These are the noncritical interpolation ranges of [Gru16, Definition 2.1 and Theorem 2.2, pp. 833–834], including the recalled vector-bundle version. The maps $d, \delta, *$ intertwine the appropriate base Laplacians. On a positive base eigenspace of eigenvalue t , the first-order factors have norm at most $\sqrt{t}$, and

$$\frac{t}{t + m^2} \leq 1, \quad \frac{m\sqrt{t}}{t + m^2} \leq \frac{1}{2}, \quad \frac{m^2}{t + m^2} \leq 1.$$

Orthogonal summation over base and surface spectra gives

$$(79) \quad \|(a, x, y)\|_{H_B^{1+\varepsilon} H_\Sigma^k} + \|m(a, x, y)\|_{H_B^\varepsilon H_\Sigma^k} \leq C\|(h, v)\|_{H_B^\varepsilon H_\Sigma^k}.$$

Let Ta be the mean-zero weak Neumann potential defined by $\int_B \langle dTa, d\zeta \rangle = \int_B \langle a, d\zeta \rangle$. The base Hodge projection gives $T : H^s \rightarrow H^{s+1}$ for the orders used here. For $\varphi = Ta$ set

$$a' = a - d\varphi, \quad x' = x - m\varphi, \quad y' = y.$$

Both ASD rows are unchanged, while $\delta a' = 0$ and $a'(n) = 0$. The estimate for x' uses

$$\|m\varphi\|_{H_B^{1+\varepsilon} H_\Sigma^k} \leq C\|ma\|_{H_B^\varepsilon H_\Sigma^k};$$

the weighted bound for a' uses the bounded base Hodge projection on H^ε . Thus (79) controls (74). Subtract the base constant $x'(z_0)$ to impose the point condition. This is bounded by point evaluation and changes neither ASD row nor either weighted term. It would not give a bound for an additional mx' term, which is why that term is omitted.

For uniqueness at a fixed positive surface mass, the Neumann gauge equation $\Delta_N + m^2$ puts a homogeneous solution into the full Coulomb domain. It then vanishes by (77). The original solution is pure gauge, $a = d\psi$, $x = m\psi$, $y = 0$. Base Coulomb gauge makes ψ constant, and $x(z_0) = 0$ removes it. This is a modewise argument and uses no uniform bound for that gauge equation. At $r = 0$, the base divergence-curl problem has zero kernel and the real Cauchy–Riemann kernel consists of constants removed at z_0 . The graph domain is complete by closedness of the surface operators and continuity of the traces. Finite surface-mode approximation extends the construction to that domain. $\square$

Return to the exact compact reconstruction $(A_{E,\rho}, u_E)$. Use the fixed smooth metric $\rho^2 g_B + g_\Sigma$ throughout the next proposition. For $0 < \varepsilon < 1/2$ and integer $k \geq 2$, write the actual negative variation as $\rho a + b$ and put

$$(80) \quad \begin{aligned} \|U\|_{X_\rho} &= \|(a, b, \nu)\|_{H_B^{1+\varepsilon} H_\Sigma^k} + \rho \|d_\alpha a\|_{H_B^\varepsilon H_\Sigma^k} + \rho \|d_\alpha b\|_{H_B^\varepsilon H_\Sigma^k}, \\ \delta_B a &= 0, \quad a(n) = 0, \quad C_{\beta_\rho} b = 0, \quad H_{\beta_\rho} b = \ell_\rho \nu \quad \text{on } \Gamma, \quad E_\alpha b(z_0) = 0. \end{aligned}$$

The positive summand has only its finite-dimensional base norm. The weighted derivative of b is exterior differentiation. Let Y be the $H_B^\varepsilon H_\Sigma^k$ space of normalized negative scalar and mixed rows, together with the positive H^ε target.

Proposition 6.9 (uniform compact estimate). *On a compact smooth seam-admissible primitive bundle family there are unitary representatives with*

$$(81) \quad \delta_B A_{E,\rho,B} = 0, \quad A_{E,\rho,B}(n) = 0, \quad A_{E,\rho,B} = O(\rho^2), \quad \beta_\rho = \beta_0 + O(\rho^2), \quad \beta_\rho(z_0) = \alpha.$$

The bounds hold in every fixed smooth norm, with bundle parameters. In these representatives, the normalized linearized mixed operator $A_{E,\rho} : X_\rho \rightarrow Y$ is surjective Fredholm of real index $6g - 2$, for $0 \leq \rho \leq \rho_0$. Its smooth kernel is naturally $H^1(X, \text{End}_0 E)$. There are uniformly bounded kernel projections $\Pi_{E,\rho}$ and right inverses satisfying

$$(82) \quad \|U - \Pi_{E,\rho} U\|_{X_\rho} \leq C \|A_{E,\rho} U\|_Y.$$

At zero scale the base variable is the independent base divergence-curl block. The harmonic attaching value remains free.

Proof. The smooth compact metric family gives $A_{E,\rho,B} = O(\rho^2)$ in the base-flat parallel trivialization. Apply the ordinary base Neumann Hodge potential T pointwise on Σ . In logarithmic gauge coordinates the equation is $\chi = -T(A_B + N(A_B, \chi))$ on mean-zero potentials, with N quadratic. Fixed-base Hölder estimates give a contraction in a small $C^{j+1,\gamma}$ ball and $\chi = O(\rho^2)$. Regularity and parameter differentiation give this at every order. This is a two-dimensional problem with smooth surface parameters. Multiply the gauge by its value at z_0 inverse; base-constant conjugation preserves the base conditions and makes the gauge the identity at z_0 . This proves (81).

First extend Proposition 6.7 to $H_B^{1+\varepsilon} H_\Sigma^k \rightarrow H_B^\varepsilon H_\Sigma^k$. The scalar Cauchy–Green inverse has this bound. Its real boundary correction uses harmonic extension of an $H^{1/2+\varepsilon}$ trace, and its point subtraction is bounded. Hilbert summation gives the constant infinite-dimensional inverse. Smooth local complex frames reduce the finite Lagrangian boundary problem to that scalar boundary estimate with zeroth-order terms. A finite partition, the interior estimate, and the corresponding adjoint estimate give Fredholmness and smooth homogeneous kernels. Its finite point conditions are imposed at $H^{1+\varepsilon}$, where evaluation is bounded. The smoothing remainders in the proof of that proposition are compact by finite surface compression followed by $H^{1+\varepsilon}(B) \subseteq H^\varepsilon(B)$. Since $\varepsilon < 1/2$, smooth targets supported away from the seam are dense in H^ε . The same smooth Dolbeault construction gives dense image, hence surjectivity by closed range, and identifies the kernel with $H^1(X, \text{End}_0 E)$. The base block $-\text{curl}$ on $\delta_B a = 0$, $a(n) = 0$ is an isomorphism with inverse $*_B d \Delta_D^{-1}$, so adds index zero. This proves the zero-scale assertion separately.

Use the finite Lagrangian projection P_N and the complex-linear polar map U in the preceding limiting proof. Include in its finite spectral space the corresponding surface zero-form modes for a . The positive-scale boundary projection differs by $O(\rho^2)$ from its limit, with one surface derivative gain. A second polar map U_ρ , equal to I at z_0 , identifies this domain with the limit domain. The composite $W_\rho = U_\rho U$ maps the P_N domain to the actual surface/positive domain and leaves a unchanged. Both maps and their inverses preserve the graph norms uniformly: $W_\rho - I$ gains one surface derivative, so $\rho d_\alpha(W_\rho - I)$ and $(W_\rho^{-1} - I)\rho d_\alpha$ are $O(\rho)$ maps of surface order zero. Off-diagonal positive components are finite rank with smooth surface range. The point condition is unchanged.

Let R_ρ have constant- α flat differential expression, constant harmonic/positive Cauchy–Riemann expression, and boundary projection P_N . Its infinite tail has the uniform inverse of

Lemma 6.8. Its finite part is $R_{0,N} + \rho B_N$ on a fixed base domain, with B_N bounded. If a_N, c_N are its zero-scale kernel and cokernel dimensions, choose smooth columns $F : \mathbb{R}^{c_N} \rightarrow Y_N$ spanning the cokernel and bounded coordinates $L : X_N \rightarrow \mathbb{R}^{a_N}$ on the smooth kernel. Smooth columns exist by density; no regularity of cokernel functionals is needed. Then

$$(83) \quad T_\rho(U, \eta) = (R_\rho U + F\eta, LU)$$

is invertible at zero and remains uniformly invertible on the finite part for small ρ . Together with the tail estimate,

$$(84) \quad \|T_\rho^{-1}\|_{Y \oplus \mathbb{R}^{a_N} \rightarrow X_\rho \oplus \mathbb{R}^{c_N}} \leq C.$$

Here and below, until returning by W_ρ , X_ρ denotes the reference graph domain with projection P_N .

For clarity, compare the varying graph spaces in a fixed product:

$$(85) \quad J_\rho U = (U, \rho d_\alpha a, \rho d_\alpha b) \in X_0 \oplus Y \oplus Y.$$

For each fixed target f , the vectors $(J_\rho, I)T_\rho^{-1}f$ converge to $(U_0, 0, 0, \eta_0)$, where $(U_0, \eta_0) = T_0^{-1}f$. Indeed, for finitely many surface modes this is bounded perturbation on a fixed base problem, and the weighted derivatives tend to zero. Such targets are dense; (84) extends the convergence to every target. This is strong convergence on fixed sources.

The normalized negative rows of the exact operator, before the boundary identification, have the form

$$(86) \quad \begin{aligned} S_\rho(a, b) &= - *_B (d_B a + [A_{E,\rho,B} \wedge a]) - \rho *_\Sigma d_{\beta_\rho} b, \\ M_\rho(a, b) &= (d_B b + [A_{E,\rho,B}, b] - \rho d_{\beta_\rho} a)^{0,2}. \end{aligned}$$

The scalar row is the negative contracted variation of $F_{BB} + \rho^2 F_{\Sigma\Sigma}$ divided by ρ , with base variation ρa . Indeed $*_\Sigma d_\alpha b = -\Lambda y$ in the flat coexact component, giving $-\text{curl } a + my$. The mixed row has no further singular division. Thus $\beta_\rho - \alpha$ enters only as ρ times smooth multiplication, and the base connection as $O(\rho^2)$ multiplication.

Conjugate the mixed target by W_ρ^{-1} . Complex linearity preserves the base principal symbol. The derivatives of U and the finite-dimensional positive connection give a fixed term K gaining one surface derivative. All other terms just described, including those with $\rho d_\alpha(U - I)$, give

$$(87) \quad \widehat{A}_{E,\rho} = R_\rho + K + S_\rho, \quad K : X_0 \rightarrow Y \text{ compact}, \quad \|S_\rho\|_{X_\rho \rightarrow Y} \leq C\rho.$$

Compactness again uses finite surface approximation before base Rellich. Augment with F, L and compose with T_ρ^{-1} . The result is the identity plus a compact operator plus an operator of norm $O(\rho)$. Absorbing the last term proves Fredholmness of index zero for the augmented exact operator. Its unaugmented index is $a_N - c_N = 6g - 2$, by the zero-scale result and compact perturbation invariance. The unbounded surface perturbation has never been treated as a bounded map on X_0 .

We next identify the kernel without using the uniform estimate. For fixed $\rho > 0$, $k \geq 2$ implies $X_\rho \subset W^{1,2}$ on the four-dimensional product. A fixed-scale covariant Neumann solve puts a homogeneous field into full Coulomb gauge with zero normal component. It preserves ASD and matching, so Proposition 6.4 gives a smooth representative of a class in $H^1(X, \text{End}_0 E)$. To return a smooth representative v to the base gauge, solve

$$(88) \quad \Delta_B \chi + \delta_B [A_{E,\rho,B}, \chi] = -\delta_B v_B, \quad \partial_n \chi = -v_B(n).$$

The compatibility condition is the divergence theorem. Pointwise on Σ , the mean-zero Neumann operator is an invertible small perturbation of Δ_B . Its remaining kernel consists exactly

of base-constant surface potentials, since $A_{E,\rho,B}$ has zero base divergence and normal trace. Such a potential changes $b(z_0)$ by $d_\alpha \zeta$; the unique choice $\zeta = -G_\alpha d_\alpha^* b(z_0)$ imposes the point condition. The same reasoning proves regularity of an initially rough field: after subtracting the smooth solution of the base equation, its potential is base-constant as a distribution, and the point normalization identifies it with a smooth surface Green solution. It does not evaluate an arbitrary $W^{2,2}$ potential at z_0 . Thus the kernel has real dimension $6g - 2$ and the cokernel is zero.

For the kernel projections, integrate a basis of $H^1(X, \text{End}_0 E)$ to a local bundle deformation family using $H^2 = 0$. Differentiate its smooth compact Dirichlet reconstruction in the base gauge. Normalize the attaching connection in a local surface Coulomb slice, or use the base-constant correction after differentiating. The resulting kernel basis satisfies

$$(89) \quad e_{\rho,j} = (a_{\rho,j}, b_{\rho,j}, \nu_j), \quad a_{\rho,j} = O(\rho), \quad b_{\rho,j} \longrightarrow b_{0,j} \quad \text{in } C^\infty.$$

The point correction changes the normalized base component by $\rho^{-1}[A_{E,\rho,B}, \zeta] = O(\rho)$. The basis therefore converges in the fixed graph embedding, with weighted components tending to zero. Its unweighted L^2 Gram matrix stays invertible, giving uniformly bounded projections that converge in the fixed base norms. This construction also holds at rank drop and in genus one; it uses the unobstructed bundle family, not constant rank of the derivative of the positive map.

Suppose now that (82) fails for $\rho_j \rightarrow 0$. In the reference domain choose U_j of graph norm one with $\Pi_{E,\rho_j} U_j = 0$ and $\hat{\mathcal{A}}_{E,\rho_j} U_j \rightarrow 0$. The identity

$$T_{\rho_j}(U_j, 0) = (\hat{\mathcal{A}}_{E,\rho_j} U_j - K U_j - S_{\rho_j} U_j, L U_j)$$

has a convergent subsequence on the right: K is compact, $S_{\rho_j} U_j \rightarrow 0$, and L is finite dimensional. Strong convergence of the stabilized inverses and their uniform bound imply $J_{\rho_j} U_j \rightarrow (U_0, 0, 0)$, with $\|U_0\|_{X_0} = 1$. Passing to the operator and to the independently convergent kernel projections gives $\mathcal{A}_{E,0} U_0 = 0$ and $\Pi_{E,0} U_0 = 0$, a contradiction. A convergent sequence of bundle parameters proves uniformity on compact families. Surjectivity and the estimate give the right inverse on the projected complement. Returning by the uniform domain identifications completes the proof. $\square$

This proposition concerns a fixed smooth compact base metric. The Euclidean dilation metric is singular at its compactifying point. Replacing it by a scale-dependent smooth cap introduces a variable surface coefficient, which is not covered by the constant-mass diagonalization above. The following estimates treat that coefficient explicitly. They still leave the outer attaching equations and the global nonlinear problem to be solved.

Let $f \geq 1$ be smooth on the fixed disc B , and set

$$M = \|f\|_\infty, \quad L = 1 + \|d \log f\|_\infty.$$

Use the metric $\rho^2 f^2 g_B + g_\Sigma$ and the base variation ρa , so $m = \rho \Lambda$ remains constant in the base. For the nonharmonic surface coordinates x, y , define

$$(90) \quad \mathcal{A}_{\rho,f}(a, x, y) = (f^{-1}(-\text{curl } a + m f^2 y), dx + *_{B} dy - ma) = (H, v).$$

The unweighted scalar numerator is $h = fH$. Its invariant form is $-*_{B} d_B a - \rho f^2 *_{\Sigma} d_\alpha b$. Impose (75), and use the complete graph norm

$$(91) \quad \begin{aligned} \|U\|_{X_{\rho,f}^{\varepsilon,k}} &= \|(x, y)\|_{H_B^{1+\varepsilon} H_\Sigma^k} + M^{-1} \|a\|_{H_B^{1+\varepsilon} H_\Sigma^k} \\ &\quad + \|ma\|_{H_B^\varepsilon H_\Sigma^k} + \|m f y\|_{H_B^\varepsilon H_\Sigma^k}. \end{aligned}$$

6.8. Euclidean cap metrics and their inverse.

Lemma 6.10 (variable conformal factor). *For $0 < \varepsilon < 1/2$ and integer $k \geq 0$, (90) is an isomorphism from (91) to the $H_B^\varepsilon H_\Sigma^k$ target, with*

$$(92) \quad \|\mathcal{A}_{\rho,f}^{-1} F\|_{X_{\rho,f}^{\varepsilon,k}} \leq CL^\varepsilon \|F\|_{H_B^\varepsilon H_\Sigma^k}.$$

The constant is independent of $\rho \geq 0$, all surface frequencies, M , and higher derivatives of f . It is uniform on compact smooth irreducible attaching families. The factor M^{-1} in the base norm is part of this assertion.

Proof. Work first in one surface mode. The base Hodge decomposition is

$$v = dx_0 + *_B dz, \quad \int_B x_0 = 0, \quad z|_{\partial B} = 0, \quad x_0 = Tv, \quad z = -\Delta_D^{-1} \operatorname{curl} v.$$

The maps have their usual $H^s \rightarrow H^{s+1}$ bounds. No boundary condition on v is imposed; Tv uses its natural Neumann data. Write $a = *_B d\psi$, $\psi|_{\partial B} = 0$, by base Coulomb gauge. The equations become

$$(93) \quad \begin{aligned} x &= x_0 + \text{constant}, & y - m\psi &= z, \\ (\Delta_D + m^2 f^2)\psi &= fH - mf^2 z, & (\Delta_D + m^2 f^2)y &= \Delta z + mfH. \end{aligned}$$

The zero Dirichlet traces remove the constant in $y - m\psi = z$. The positive Dirichlet operator supplies existence, including at $m = 0$. At the fractional order, set $x = x_0 - x_0(z_0)$.

Pair the last equation with y . Then

$$(94) \quad \|dy\|_2^2 + \|mfy\|_2^2 = \langle dz, dy \rangle + \langle H, mfy \rangle.$$

Dirichlet Poincaré, the base Hodge bound, and $ma = *_B d(y - z)$ give

$$\|y\|_{H^1} + \|mfy\|_2 + \|ma\|_2 \leq C\|(H, v)\|_2.$$

Since $\Delta\psi = f(H - mfy)$, Dirichlet regularity also gives $M^{-1}\|a\|_{H^1} \leq C\|(H, v)\|_2$. These estimates are independent of f, m , with mean normalization for x_0 .

For the next endpoint take $H \in H_0^1(B)$ and $v \in H^1(B; T^*B)$. Pair the last equation of (93) with Δy . Integration by parts gives

$$(95) \quad \|\Delta y\|_2^2 + \|mf dy\|_2^2 = \langle \Delta z, \Delta y \rangle + m\langle d(fH), dy \rangle - 2m^2 \langle fy df, dy \rangle.$$

The scalar source boundary term vanishes because H has zero trace. The last two terms are bounded by

$$(\|dH\|_2 + L\|H\|_2 + 2L\|mfy\|_2)\|mf dy\|_2.$$

Absorption and the preceding energy bound imply

$$\|y\|_{H^2} + \|mf dy\|_2 \leq CL\|(H, v)\|_{H^1}.$$

Differentiating mfy adds $(d \log f)(mfy)$, so its H^1 norm has the same bound, as does $ma = *_B d(y - z)$. Moreover

$$\|f(H - mfy)\|_{H^1} \leq M(\|H - mfy\|_{H^1} + L\|H - mfy\|_2) \leq CML\|(H, v)\|_{H^1}.$$

Thus $M^{-1}\|a\|_{H^2}$ has the same CL bound.

Interpolate the same linear inverse, still with mean-normalized x_0 , between the source spaces $L^2 \oplus L^2(T^*B)$ and $H_0^1 \oplus H^1(T^*B)$. For $\varepsilon < 1/2$ their interpolation space is ordinary $H^\varepsilon \oplus H^\varepsilon(T^*B)$, with no boundary condition; this is the noncritical Dirichlet range in [Gru16, Definition 2.1 and Theorem 2.2]. Interpolate the fixed output product $(x_0, y, M^{-1}a, ma, mfy)$ at orders $(1, 1, 1, 0, 0)$ and $(2, 2, 2, 1, 1)$. The constants C and CL give CL^ε . The final point

subtraction is bounded because $1 + \varepsilon > 1$; it is not made at the H^1 endpoint. Summing both endpoint estimates over surface modes, with the commuting H^k weights, before interpolation proves the Hilbert-valued assertion.

For uniqueness the homogeneous Hodge data have $x = z = 0$, and the first Dirichlet equation in (93) gives $\psi = 0$, hence $a = y = 0$. Closedness of Λ and $f\Lambda$, together with continuity of the traces, gives completeness. Finite surface-mode density extends the inverse to the stated domain. Its forward operator norm may depend on M, L ; only the displayed inverse bound is asserted uniformly. $\square$

We now construct the backgrounds to which this estimate will be applied. Near $z_0 = \infty$, use $\zeta = 1/w$ and write the Euclidean conformal factor as $f_\infty = c(\zeta)|\zeta|^{-2}$, with c smooth and positive. Choose the fixed g_B sufficiently small that $f_\infty \geq 2$. For a smooth cutoff χ equal to zero on $[0, 1]$ and one on $[2, \infty)$, set near z_0

$$(96) \quad f_R = c(\zeta)(\chi(R|\zeta|)|\zeta|^{-2} + (1 - \chi(R|\zeta|))R^2),$$

and use f_∞ away from this chart. For large R it is smooth, is at least one, and equals f_∞ on $|w| \leq R/2$. Rescaling the transition annulus gives

$$(97) \quad \|f_R\|_\infty \leq CR^2, \quad |D^j f_R| \leq C_j R^j f_R, \quad 1 + \|d \log f_R\|_\infty \leq CR.$$

Lemma 6.11 (Dirichlet metrics for the caps). *For $R \leq \rho^{-1/6}$ and sufficiently small ρ , the Dirichlet reconstruction for $\rho^2 f_R^2 g_B + g_\Sigma$ satisfies, at every specified finite derivative order,*

$$(98) \quad \|\log(H_0^{-1} H_{\rho,R})\|_{C^j} \leq C_j \rho^2 R^{4+j}.$$

For fixed J , when $\rho^2 R^{J+8}$ is sufficiently small it admits base-gauged unitary representatives with

$$(99) \quad \begin{aligned} \delta_B A_B &= 0, \quad A_B(n) = 0, \quad \beta(z_0) = \alpha, \\ \|A_B\|_{C^J} + \|\beta - \beta_0\|_{C^J} &\leq C_J \rho^2 R^{J+8}. \end{aligned}$$

They satisfy the canonical matching with u_E . Constants are uniform on compact smooth seam-admissible bundle families; a finite number of bundle-parameter derivatives is obtained by increasing the derivative order used in the estimate.

Proof. Put $\tau = \rho^2$ and $\eta = R^{-1}$. The normalized equation is

$$\Phi_{\tau,R}(H) = i\Lambda_B F_H^\perp + \tau f_R^2 i\Lambda_\Sigma F_H^\perp = 0, \quad H|_{\partial B} = H_0|_{\partial B}.$$

Use base charts of radius $c\eta$ and surface charts of radius $c\eta\rho f_R(z)$ at the base center z , including boundary charts. Their pulled-back Hölder norms contain the supremum norm and have uniform product constants. Equation (97) controls the ratio $f_R/f_R(z)$ and all its rescaled derivatives on each chart. The equation multiplied by η^2 is uniformly elliptic there.

The transported metric linearization is

$$(100) \quad L_{K,\tau,R} = \frac{1}{2}(\nabla_{K,B}^* \nabla_{K,B} + \tau f_R^2 \nabla_{K,\Sigma}^* \nabla_{K,\Sigma}).$$

Indeed, apply the Kähler identity to the actual metric $\tau f_R^2 g_B + g_\Sigma$, cancel the transported commutators, and multiply its covariant Laplacian by τf_R^2 . The base volume factor cancels the inverse base metric, and the surface derivative does not differentiate f_R . Thus no derivative of f_R occurs in (100). Kato's inequality and the fixed base barrier give $\|s\|_\infty \leq C\|L_{K,\tau,R}s\|_\infty$ for zero Dirichlet data. If K has bounded adapted derivatives, local Schauder estimates therefore give

$$(101) \quad \|s\|_{j+2,\gamma;\text{ad}} \leq C_j(\|s\|_\infty + \eta^2 \|L_{K,\tau,R}s\|_{j,\gamma;\text{ad}}) \leq C_j \|L_{K,\tau,R}s\|_{j,\gamma;\text{ad}}.$$

The fixed-positive-parameter index-zero Dirichlet argument supplies surjectivity. Undoing the factor η^2 in the nonlinear equation gives a quadratic constant at most CR^2 .

In the base-flat identification the zero-scale inverse is the scalar fixed-base Dirichlet inverse. Construct formal coefficients s_i , zero on the boundary, for $H^{[N]} = H_0 \exp(\sum_{i=1}^N \tau^i s_i)$. They obey

$$(102) \quad \|s_i\|_{C^j} \leq C_{i,j} R^{6i-2+j}.$$

For $i = 1$ the source is $f_R^2 i \Lambda_\Sigma F_{H_0}^\perp$, of size R^{4+j} . At higher order, a base nonlinear monomial has at least $q = 2$ positive-order factors and at most two derivatives, so its exponent is at most $6i-2q+j+2 \leq 6i-2+j$. A surface monomial has the extra τf_R^2 , giving $4+6(i-1)-2q+j+2 \leq 6i-2+j$ for $q \geq 1$. The base inverse preserves the fixed C^j estimate, with no surface derivative loss. This proves the induction.

In the adapted charts every derivative of these coefficients is bounded by the same power R^{6i-2} : base derivatives have factor R^{-1} , and surface derivatives cost at most $C\rho R^2$ after (102). One more derivative controls the Hölder seminorm. Consequently $H^{[N]}$ and its inverse have bounded adapted derivatives when $\rho^2 R^6$ is small. The uncanceled residual is bounded by

$$e_N = C_N \rho^{2N+2} R^{6N+4}$$

at every fixed adapted order. The same monomial count bounds the analytic tail by a convergent power series in $\rho^2 R^6$. Using (101), contraction has smallness $R^2 e_N \leq C_N (\rho^2 R^6)^{N+1}$ and gives an exact correction r_N of adapted size $C e_N$. Higher adapted Schauder estimates give this bound at every fixed order; the supremum bound is already supplied by contraction. Dirichlet uniqueness identifies the solutions obtained with different N .

Unscaling j derivatives costs at most $(R/\rho)^j$, hence

$$\|r_N\|_{C^j} \leq C_{N,j} \rho^{2N+2-j} R^{6N+4+j}.$$

For $R \leq \rho^{-1/6}$ its ratio to $\rho^2 R^{4+j}$ is at most ρ^{N-j} . Choose $N \geq j+1$ and use the finite sum (102) to prove (98). This is a finite-order statement; it does not assert smallness of all derivatives at one fixed growth exponent.

The connection formula uses one derivative, giving $\|A_B\|_{C^j} + \|\beta - \beta_0\|_{C^j} \leq C_j \rho^2 R^{5+j}$ before base gauge. Apply the ordinary base Neumann gauge construction, treating surface variables as parameters. At parameter order $J+1$ its contraction controls $\chi, d_B \chi$ in $C^{J+1,\gamma}$; no surface derivative occurs in that gauge equation. The extra parameter derivative controls $d_\Sigma \chi$ in the transformed surface connection. An integer C^{J+2} norm of A_B suffices, supplied by metric order $J+3$ with size $\rho^2 R^{J+7}$. Allowing one further power gives (99). The final base-constant normalization makes the gauge equal to the identity at z_0 and preserves the two base conditions. The Dirichlet surface metric supplies the exact canonical matching. $\square$

For the full mixed cap use the closed domain in (80), with its nonflat matching, but replace its norm by

$$(103) \quad \begin{aligned} \|U\|_{X_{\rho,f}} &= \|(b, \nu)\|_{H_B^{1+\varepsilon} H_\Sigma^k} + M^{-1} \|a\|_{H_B^{1+\varepsilon} H_\Sigma^k} \\ &\quad + \rho \|d_\alpha a\|_{H_{\bar{B}}^\varepsilon H_\Sigma^k} + \rho \|f d_\alpha b\|_{H_{\bar{B}}^\varepsilon H_\Sigma^k}. \end{aligned}$$

Normalize the scalar target by f^{-1} as in (90); the mixed and positive targets retain their complex base form. Denote their $H_B^\varepsilon H_\Sigma^k$ space by Y .

Proposition 6.12 (mixed inverse on the Euclidean caps). *Fix $0 < \varepsilon < 1/2$ and integer $k \geq 2$. Choose a fixed sufficiently large derivative order J for the surface Hodge and boundary identifications. For*

$$(104) \quad 0 < a < \min\{1/6, 2/(J + 10 + 2\varepsilon)\}, \quad R = \rho^{-a},$$

the exact capped reconstruction has a surjective normalized mixed operator $\mathcal{A}_{\rho, f_R} : X_{\rho, f_R} \rightarrow Y$, of index $6g - 2$ and smooth kernel $H^1(X, \text{End}_0 E)$. It has a kernel projection Π and right inverse Q such that

$$(105) \quad \|U - \Pi U\|_{X_{\rho, f_R}} \leq CR^\varepsilon \|\mathcal{A}_{\rho, f_R} U\|_Y, \quad \|Q\| + \|\Pi\| \leq CR^\varepsilon.$$

Constants are uniform on compact smooth seam-admissible bundle families. The harmonic attaching value remains free.

Proof. We prove a criterion for a general $f \geq 1$. Suppose its exact base-gauged background has $\|A_B\|_{C^J} + \|\beta - \beta_0\|_{C^J} \leq \delta$ and the same attaching connection. It suffices that

$$(106) \quad \rho ML^{4\varepsilon} + \delta ML^{2\varepsilon} < c.$$

Multiplication by f/M and $1/f$ has H^ε norm at most CL^ε : interpolate their L^2 bound one and their H^1 bound CL .

Make the fixed limiting boundary reduction once, with projection P_{N_0} and polar map U . A second polar map differing from I by $O(\delta)$ identifies the exact boundary; both are the identity at z_0 . Their composite acts on b, ν and leaves a unchanged. The smoothing commutators with the mass terms are bounded by $C\rho ML^\varepsilon$ in (103), so these are uniform domain identifications under (106). The exact normalized rows before identification are

$$-f^{-1} *_B (d_B a + [A_B \wedge a]) - \rho f *_\Sigma d_\beta b, \quad (d_B b + [A_B, b] - \rho d_\beta a)^{0,2}.$$

Conjugating the mixed target gives

$$(107) \quad \hat{\mathcal{A}}_{\rho, f} = R_{\rho, f} + K + \mathcal{E}_{\rho, f}, \quad \|\mathcal{E}_{\rho, f}\|_{X_{\rho, f} \rightarrow Y} \leq CML^\varepsilon(\rho + \delta).$$

Here $R_{\rho, f}$ has the flat variable-factor expression and boundary P_{N_0} . The fixed term K gains one surface derivative and acts only on b, ν . The term $f^{-1}[A_B, a]$ costs $C\delta ML^\varepsilon \|a/M\|$, and the two surface connection differences cost $C\rho ML^\varepsilon$. The remaining terms are the derivatives of the small polar map and its smoothing mass commutators, giving the stated bound.

For a surface cutoff $N \geq N_0$, let F_N contain the positive, harmonic, and complete exact/coexact modes up to frequency N , with corresponding zero-form modes for a . It commutes with the reference boundary and expression. The fixed smoothing term obeys

$$K_N = F_N K F_N, \quad \|K - K_N\|_{H_B^{1+\varepsilon} H_\Sigma^k \rightarrow Y} \leq C/N.$$

The right truncation uses $K : H_\Sigma^{k-1} \rightarrow H_\Sigma^k$, the left one $K : H_\Sigma^k \rightarrow H_\Sigma^{k+1}$. There is no factor M because K does not act on a . Let $d = 6g - 2$ and choose fixed smooth coordinates ℓ on the limiting surface kernel, so its operator augmented by ℓ is an isomorphism. The coordinates $\ell_N = \ell F_N$ converge in operator norm. Thus $(R_0^{\text{surface}} + K_N, \ell_N)$ is invertible uniformly for all sufficiently large N , by the already proved limiting theorem. Its high tail is the constant real boundary isomorphism, so the finite part has that uniform bound as well.

The zero-scale base inverse is $a = *_B d\Delta_D^{-1}(fH)$, bounded by CL^ε in the M^{-1} -weighted norm. On the finite space, the positive mass perturbation has norm $C\rho MNL^\varepsilon$, and the extra graph terms have this bound relative to the zero-scale norm. Hence, if $\rho MNL^{2\varepsilon}$ is small, the

finite augmented inverse has bound CL^ε in the full graph norm. On the infinite tail apply Lemma 6.10. Their direct sum

$$T_{\rho,f,N}U = (R_{\rho,f}U + K_N U, \ell_N U)$$

is therefore invertible with norm CL^ε .

Choose N comparable to $AL^{2\varepsilon}$, with A a sufficiently large fixed constant and $N \geq N_0$. After this reference inverse the error in the exact augmented operator is bounded by

$$C(L^\varepsilon/N + ML^{2\varepsilon}(\rho + \delta)).$$

The choice of A controls its first term. Condition (106) controls the others and ensures the finite perturbation is small. A Neumann series gives an exact augmented inverse $B = (\hat{\mathcal{A}}_{\rho,f}, \ell_N)^{-1}$ with norm CL^ε . Set $QF = B(F, 0)$ and $\Pi U = B(0, \ell_N U)$. These give surjectivity, kernel dimension d , and the stated estimate. At fixed ρ, f , the regularity and base-gauge argument in Proposition 6.9 identifies the smooth kernel with $H^1(X, \text{End}_0 E)$.

Finally (97) and (99) bound the two smallness terms by $C\rho R^{2+4\varepsilon}$ and $C\rho^2 R^{J+10+2\varepsilon}$. Both tend to zero in (104). Also $L^\varepsilon \leq CR^\varepsilon$, which proves (105). $\square$

The cap agrees with the physical Euclidean metric on $|w| \leq R/2$. Its outer metric is auxiliary. Proposition 7.13 combines this cap inverse with compatible cutoffs, nonzero neck data and the residual inverse on the full preglued configuration. The following proposition first gives the nonlinear chart on the cap. The physical comparison used in the converse is proved in Section 8.

6.9. Nonlinear coordinates on an exact cap.

Proposition 6.13 (nonlinear coordinates on the cap). *Use the background, domain, and norm of Proposition 6.12. There are uniform local coordinates on the nonlinear canonical matching domain, with derivative $(a, b, \nu) \mapsto (\rho a, b, \nu)$ at zero, preserving ordinary base Coulomb gauge, the full zero-normal condition, and exact attaching normalization. In these coordinates the exact normalized equation is*

$$\mathcal{F}(\Psi(U)) = \mathcal{A}_{\rho,f}U + \mathcal{N}_{\rho,f}(U), \quad \mathcal{N}_{\rho,f}(0) = D\mathcal{N}_{\rho,f}(0) = 0.$$

On a uniform ball in $X_{\rho,f}$ its remainder satisfies

$$(108) \quad \|\mathcal{N}_{\rho,f}(U) - \mathcal{N}_{\rho,f}(V)\|_Y \leq C(1 + \rho M^2 L^\varepsilon)(\|U\|_X + \|V\|_X)\|U - V\|_X,$$

where $M = \|f\|_\infty$ and $L = 1 + \|d \log f\|_\infty$. The constant is uniform for the caps in (104), since $\rho M^2 L^\varepsilon \leq C\rho R^{4+\varepsilon} \rightarrow 0$. The coordinate assertion concerns the configuration space with these gauge conditions, not the existence of a uniform slice for physical gauge orbits.

Proof. We first give a surface construction at an irreducible flat connection α . Let $E_\alpha, H_\alpha, C_\alpha$ be its Hodge projections. Choose a smooth flat Coulomb slice $t \mapsto \alpha_t$ satisfying

$$[\alpha_t] = \text{Exp}_{[\alpha]} t, \quad E_\alpha(\alpha_t - \alpha) = 0, \quad D\alpha_t|_0 = \ell t.$$

For example, solve flatness of $\alpha + h + c$ for the coexact component c , with small harmonic h . The derivative $d_\alpha : C_\alpha H^k \rightarrow H^{k-1}(\Omega^2)$ is invertible because $H_\alpha^2 = H_\alpha^0 = 0$. The implicit function theorem gives the slice, and a finite-dimensional change of coordinates gives the stated exponential parameter. For small $e \in E_\alpha H^k$ solve

$$(109) \quad E_\alpha(\exp(\zeta)^* \alpha_t - \alpha) = e, \quad \zeta \in H^{k+1}(\Omega^0).$$

Use the gauge-action convention with derivative $d_\alpha \zeta$. Its inverse is $G_\alpha^0 d_\alpha^*$, so this is another Banach implicit-function problem. Put

$$F_\alpha(e, t) = \exp(\zeta(e, t))^* \alpha_t, \quad S_\alpha(e, t) = F_\alpha(e, t) - \alpha - e - \ell t.$$

Then S_α is smooth and quadratic in H^k , and $E_\alpha S_\alpha = 0$. Flatness and Hodge decomposition give

$$(110) \quad \begin{aligned} d_\alpha S_\alpha &= -\frac{1}{2}[q \wedge q], & q &= e + \ell t + S_\alpha, \\ S_\alpha &= H_\alpha S_\alpha - \frac{1}{2}d_\alpha^* G_\alpha^2 [q \wedge q]. \end{aligned}$$

Since $k > 1$ on the surface, the wedge product belongs to H^k . The last inverse has order minus one, and the harmonic term has smooth finite-dimensional range. Thus S_α is a smooth quadratic map into H^{k+1} , including its difference and derivative estimates. Fixing the exact component in (109) is essential here.

Write $s = 1 + \varepsilon$, $\mathcal{H}_j = H_B^s H_\Sigma^j$, and $\mathcal{W}_j = H_B^\varepsilon H_\Sigma^j$. For $j \geq k$, the product estimates are

$$(111) \quad \mathcal{H}_j \mathcal{H}_j \subset \mathcal{H}_j, \quad \mathcal{H}_j \mathcal{W}_j \subset \mathcal{W}_j.$$

They follow from the surface Hilbert algebra and base dyadic decomposition. For the second estimate, the low-frequency strong factor is bounded in L^∞ . At frequency 2^n , the weak low-frequency sum is bounded in L^∞ by $C2^{(1-\varepsilon)n} \|v\|_{\mathcal{W}_j}$. The weighted contribution of a strong block u_n is therefore $C2^{-\varepsilon n} (2^{sn} \|u_n\|_2) \|v\|_{\mathcal{W}_j}$. The comparable-frequency term is bounded in H^ε by $C2^{-\varepsilon n} (2^{sn} \|u_n\|_2) (2^{\varepsilon n} \|v_n\|_2)$ and is summable. These arguments apply to Hilbert algebra valued blocks, with constants independent of surface spectral truncation. For the underlying scalar paraproduct and finite-dimensional composition estimates, see [Tay91, §§3.1, 3.5, especially (3.1.20) and (3.5.1)–(3.5.13)].

Pull the actual flat boundary family back to a fixed collar, and denote this flat family by α_z . Its needed fixed-order derivatives are uniformly bounded by Lemma 6.11. Apply (109) directly in the mixed spaces: its derivative d_{α_z} has bounded inverse $G_{\alpha_z}^0 d_{\alpha_z}^*$ from $E_{\alpha_z} \mathcal{H}_k$ to $\mathcal{H}_{k+1}$. The gauge products are smooth by (111). The slice depends on finitely many parameters, so its composition with H^s base sections is smooth. One may first use its analytic harmonic coordinates and then the smooth finite-dimensional exponential change. The implicit function theorem initially gives S_z in $\mathcal{H}_k$; (110), with the varying Hodge operators, improves its range to $\mathcal{H}_{k+1}$. For a fixed collar cutoff $\chi = 1$ on the seam, set

$$S(b)(z) = \chi(z) S_{\alpha_z}(E_{\alpha_z} b, \ell_z^{-1} H_{\alpha_z} b),$$

extended by zero outside the collar. We have

$$(112) \quad \|S(b) - S(c)\|_{\mathcal{H}_{k+1}} \leq C(\|b\|_{\mathcal{H}_k} + \|c\|_{\mathcal{H}_k}) \|b - c\|_{\mathcal{H}_k}.$$

This is proved in the mixed spaces themselves; it does not require a composition theorem for arbitrary nonlinear maps between infinite-dimensional Sobolev spaces.

Define the configuration chart by

$$(113) \quad \Psi(a, b, \nu) = (A_B + \rho a, \beta + b + S(b), \text{Exp}_{u_E} \nu).$$

At the seam its surface connection is $F_\beta(e, t) + c$, where $c = C_\beta b$ and $t = \ell^{-1} H_\beta b$. If $c = 0$ and $t = \nu$, it is flat with the correct class. Conversely, for small e, t , the derivative $d_{F_\beta(e, t)} : C_\beta H^k \rightarrow H^{k-1}(\Omega^2)$ remains invertible. Its unique small flat correction is $c = 0$. Equality of classes then implies $t = \nu$ in the exponential coordinate ball. This proves both implications of nonlinear matching. The boundary traces have order $1/2 + \varepsilon$ with values in H_Σ^k and are continuous; curvature there is interpreted in H_Σ^{k-1} . No boundary trace of the

weak graph derivative is used. At z_0 one has $\alpha_z = \alpha$ and $E_\alpha S(b)(z_0) = 0$, so the exact attaching normalization is unchanged. The base conditions remain affine because the first component of (113) is $A_B + \rho a$. This chart is not a four-dimensional gauge transformation.

The multiplier bound for f/M and (112) give

$$(114) \quad \rho \|f d_\alpha(S(b) - S(c))\|_{\mathcal{W}_k} \leq C \rho M L^\varepsilon (\|b\|_{\mathcal{H}_k} + \|c\|_{\mathcal{H}_k}) \|b - c\|_{\mathcal{H}_k}.$$

Here $\rho M L^\varepsilon \leq 1$ for sufficiently small cap scale. Thus $b \mapsto b + S(b)$ has a uniformly small Lipschitz remainder on a uniform graph ball. Contraction gives its inverse on a uniform smaller ball, and the Banach implicit function theorem gives smoothness. Restricting this ambient chart by the equivalences above gives precisely the stated linear domain.

It remains to estimate the equation. Put $B = b + S(b)$. In the scalar sign convention already fixed, the negative remainders are

$$(115) \quad \begin{aligned} \mathcal{N}_H &= -\frac{\rho}{2f} *_B [a \wedge a] - \rho f *_\Sigma d_\beta S(b) - \frac{\rho f}{2} *_\Sigma [B \wedge B], \\ \mathcal{N}_V &= (d_B S(b) + [A_B, S(b)] + \rho[a, B])^{0,2}. \end{aligned}$$

Indeed, the base curvature increment is $\rho d_{A_B} a + \rho^2[a \wedge a]/2$ and its scalar normalization divides by ρf . The three scalar terms have quadratic difference constants $C \rho M^2 L^\varepsilon$, $C \rho M L^\varepsilon$, and $C \rho M L^\varepsilon$. For the middle one use the surface derivative gained in (112). The mixed terms have constants $C, C, C \rho M$. The norm controls $a/M, b, \nu$ in the strong space; base differentiation of S takes it to the weak space. All background coefficients in these estimates are uniformly bounded. On the positive side, smooth exponential coordinates and a smooth target trivialization give a remainder consisting of a coefficient vanishing at zero times $d_B \nu$, and a zeroth-order quadratic term. Equation (111) and finite-dimensional composition give its uniform difference bound. Combining the estimates, and using $M, L \geq 1$, proves (108). $\square$

This proposition concerns an exact cap. The complete preglued chart and its estimates are proved in Proposition 7.16.

6.10. Localization and annular source errors.

Lemma 6.14 (localization of the flat cap inverse). *Choose the fixed base metric flat in a half-disc coordinate $\zeta = s + it$ at $z_0 = 0$. Use the cap $f = f_{R_1}$ of (96), with $r_1 = R_1^{-1}$. Let $Q_{\rho,f}$ be the flat nonharmonic inverse of Lemma 6.10, with its domain (91). Suppose the target $F = (H, v)$ vanishes for $|\zeta| < r_0$, where $r_1 < c r_0$ for a fixed sufficiently small $c > 0$ and r_0 is smaller than the fixed coordinate radius. For $U = (a, x, y) = Q_{\rho,f} F$, write $a = *_B d\psi$ with $\psi|_{\partial B} = 0$. Choose a radial cutoff χ , zero for $|\zeta| \leq 4r_1$ and one for $|\zeta| \geq 8r_1$, with fixed rescaled profile. Then*

$$(116) \quad \mathcal{C}_1 U = (*_B d(\chi\psi), \chi x, \chi y)$$

belongs to the same domain, vanishes near the changed cap metric, and satisfies

$$(117) \quad \|\mathcal{C}_1 U - U\|_{X_{\rho,f}} + \|\mathcal{A}_{\rho,f} \mathcal{C}_1 U - F\|_Y \leq C r_0^{-6} r_1^{1-2\varepsilon} \|F\|_Y.$$

The constant is independent of scale and every surface frequency. The norms in this statement are the compact-base norms; a global physical norm is not asserted.

Proof. Set $m = \rho \Lambda$, interpreted initially in a single surface mode. The cap obeys $M \asymp r_1^{-2}$, $L \leq C r_1^{-1}$, and $f(\zeta) \asymp (|\zeta|^2 + r_1^2)^{-1}$ near zero. The function $\hat{f}(Z) = r_1^2 f(r_1 Z)$ has uniform

positive lower bound and fixed-order derivative bounds on every fixed rescaled half-disc. The base Hodge decomposition and (93) give

$$(118) \quad \begin{aligned} v &= dx + *_B dz, & y &= z + m\psi, & z|_{\partial B} &= 0, \\ (\Delta + m^2 f^2)\psi &= fH - mf^2 z, & (\Delta + m^2 f^2)y &= \Delta z + mfH. \end{aligned}$$

Here $x(z_0) = 0$ is retained. The Hodge estimate bounds x, z in $H^{1+\varepsilon}$ by $C\|F\|_Y$; the flat inverse bounds y there by $CL^\varepsilon\|F\|_Y$. Where $F = 0$, x, z are conjugate harmonic functions, with Neumann and Dirichlet boundary conditions respectively, and both vanish at zero. Their fixed-order rescaled norms on D_{64}^+ are bounded by $C_j(r_1/r_0)\|F\|_Y$. This follows from reflection and harmonic derivative estimates, decreasing c if necessary.

The norm of the homogeneous solution y is subharmonic by Kato's inequality. Comparison with the harmonic function zero on the diameter and one on the semicircle gives

$$(119) \quad \|y(\zeta)\|_{H_\Sigma^k} \leq C \frac{\operatorname{Im} \zeta}{r_0} L^\varepsilon \|F\|_Y, \quad |\zeta| < r_0/2.$$

To use derivatives uniformly in the mass, consider $(-\Delta_{\text{Eucl}} + n^2 g^2)w = 0$ on nested fixed half-discs, with zero trace on the straight edge and g uniformly positive and smooth. Caccioppoli controls dw and nw on a smaller half-disc by the outer L^2 norm. Apply it repeatedly to $n^j w$, which satisfies the same equation. This controls every fixed power of n . Ordinary local Dirichlet regularity for the Laplacian then gives

$$\|w\|_{H^{j+2}(\text{inner})} \leq C_j (\|n^2 w\|_{H^j(\text{middle})} + \|w\|_{L^2(\text{middle})}).$$

Induction, applied also to $n^2 w$, controls every fixed number of derivatives and powers of n by the outer norm. No reflection of the potential is used. The proof applies to a nonnegative diagonal operator n on a Hilbert space, by finite spectral truncation and passage to the limit.

Split at $m = r_1$. On the low-frequency subspace, Kato and Dirichlet comparison bound $\|\psi(\zeta)\|_{H_\Sigma^k}$ by the scalar Green integral with source $f\|H\|_{H^k} + r_1 f^2 \|z\|_{H^k}$. For $|\zeta| \leq 64r_1$ and $|\eta| \geq r_0/2$, the fixed-disc Green function is bounded by $Cr_1 r_0^{-2}$. Since $f \leq Cr_0^{-2}$ there, this part contributes at most

$$(120) \quad C(r_1 r_0^{-4} + r_1^2 r_0^{-6}) \|F\|_Y.$$

Use Cauchy–Schwarz for H and the supremum bound for z . In the inner region the harmonic Dirichlet estimate gives $\|z(\eta)\| \leq C \operatorname{Im}(\eta) r_0^{-1} \|F\|_Y$. Rescaling bounds the remaining contribution by $Cr_0^{-1} \|F\|_Y$ times

$$\int_{\mathbb{H}} G_{\mathbb{H}}(X, Y) \frac{\operatorname{Im} Y}{(1 + |Y|^2)^2} dY, \quad G_{\mathbb{H}}(X, Y) = \frac{1}{4\pi} \log \left(1 + \frac{4 \operatorname{Im} X \operatorname{Im} Y}{|X - Y|^2} \right).$$

This integral is bounded for $|X| \leq 64$: the singularity is logarithmic and the integrand is $O(|Y|^{-4})$ at infinity. The smooth local Green correction is bounded by $C \operatorname{Im}(\zeta) \operatorname{Im}(\eta)$; its contribution is at most $Cr_1^2 \log(1 + r_0/r_1) r_0^{-1} \|F\|_Y$. Together with (120), this proves

$$\sup_{D_{64r_1}^+} \|\psi\|_{H_\Sigma^k} \leq Cr_0^{-6} \|F\|_Y \quad \text{on } m \leq r_1.$$

The rescaled equation is

$$(-\Delta_Z + (m/r_1)^2 \hat{f}^2) \psi(r_1 Z) = -\frac{m}{r_1^2} \hat{f}^2 z(r_1 Z).$$

Its potential is bounded on this subspace, and the right side has fixed-order Sobolev norms at most $C_j r_0^{-1} \|F\|_Y$. Local Dirichlet regularity gives the same coarse $C_j r_0^{-6} \|F\|_Y$ bound for every required derivative of $\psi(r_1 \cdot)$ on D_{16}^+ .

On $m > r_1$, apply the homogeneous parameter estimate with $n = m/r_1$ and $g = \widehat{f}$. Equation (119) bounds all required rescaled derivatives of y and ny by $C_j(r_1/r_0)L^\varepsilon \|F\|_Y$. Now $\psi = m^{-1}(y - z)$ may be used, since $\|m^{-1}\| \leq r_1^{-1}$ on this subspace. Combining the two ranges, and setting $A = r_0^{-6} L^\varepsilon \|F\|_Y$, gives

$$(121) \quad \begin{aligned} & \|(x, y, a/M)(r_1 \cdot)\|_{H^j(D_{16}^+; H_\Sigma^k)} \leq C_j r_1 A, \\ & \|(ma, mfy)(r_1 \cdot)\|_{H^j(D_{16}^+; H_\Sigma^k)} + \|\psi(r_1 \cdot)\|_{H^j(D_{16}^+; H_\Sigma^k)} \leq C_j A. \end{aligned}$$

The first line uses $a = r_1^{-1} *_B d_Z \psi$ and $M \asymp r_1^{-2}$. For the second, use $ma = r_1^{-1} *_B d_Z(y - z)$ and $mfy = r_1^{-1} n \widehat{f} y$ in the high range, and $m/r_1 \leq 1$ in the low range. All estimates are in the surface spectral Hilbert norm equivalent to H^k , so the constants are independent of the number of modes.

The stream cutoff preserves the domain since $\delta_B *_B d = 0$ and $\chi\psi$ has zero boundary trace. Its exact commutators, with positive Δ , are

$$(122) \quad \begin{aligned} & (\mathcal{A}_1 U - \chi \mathcal{A} U)_H = f^{-1}(\psi \Delta \chi - 2\langle d\chi, d\psi \rangle), \\ & (\mathcal{A}_1 U - \chi \mathcal{A} U)_V = x d\chi + z *_B d\chi. \end{aligned}$$

Indeed $a_\chi = \chi a + \psi *_B d\chi$, and the mixed commutator contains $y - m\psi = z$. The difference $\mathcal{C}_1 U - U$ is supported on $D_{8r_1}^+$. A strong $H^{1+\varepsilon}$ norm after rescaling costs $r_1^{-\varepsilon}$, and a weak H^ε norm costs $r_1^{1-\varepsilon}$. Bounded extension from a fixed half-disc and (121) give $\|\mathcal{C}_1 U - U\|_X \leq C r_1^{1-\varepsilon} A$, including the derivative of ψ in the cutoff. On the commutator annulus, f^{-1} is r_1^2 times a uniformly smooth function. Equation (122) therefore gives the same bound for the source error. Here $\chi F = F$. Finally $L^\varepsilon \leq C r_1^{-\varepsilon}$ proves (117). Approximation with a fixed slightly smaller source-free neighborhood and the closed graph norm gives the assertion for the stated Sobolev targets. $\square$

Writing $r_i = R_i^{-1}$, the bound is $C R_0^6 R_1^{-1+2\varepsilon}$. It tends to zero for $R_1 = R_0^p$ if $p(1-2\varepsilon) > 6$. The new source is on the cutoff annulus, outside the original separated source class; the lemma alone cannot be iterated on that class. A global construction must handle this annular source, the nonflat seam, and the free harmonic attaching value.

Lemma 6.15 (Dirichlet estimate with a positive mass potential). *Let D be a fixed smooth bounded planar domain, $\mathcal{H}$ a Hilbert space, and N a nonnegative self-adjoint operator on $\mathcal{H}$. Suppose g is real, uniformly positive, and has bounded C^2 norm. For $0 < \varepsilon < 1/2$, the full Dirichlet inverse $T_N = (\Delta + g^2 N^2)^{-1}$ satisfies*

$$(123) \quad \|T_N G\|_{H^{2+\varepsilon}} + \|N T_N G\|_{H^{1+\varepsilon}} + \|N^2 T_N G\|_{H^\varepsilon} \leq C \|G\|_{H^\varepsilon}.$$

The constant is independent of N and $\mathcal{H}$; no source boundary condition is imposed. On nested fixed half-discs with zero trace only on their straight edge, the corresponding local estimate is

$$(124) \quad \|y\|_{H^{2+\varepsilon}(D_a^+)} + \|Ny\|_{H^{1+\varepsilon}(D_a^+)} + \|N^2 y\|_{H^\varepsilon(D_a^+)} \leq C(\|y\|_{L^2(D_b^+)} + \|G\|_{H^\varepsilon(D_b^+)}),$$

where $(\Delta + g^2 N^2)y = G$, $a < b$, and g has uniformly bounded smooth derivatives through the required fixed orders.

Proof. Use the closed form $\|du\|_2^2 + \|gNu\|_2^2$ on $H_0^1(D; \mathcal{H}) \cap L^2(D; \text{Dom } N)$. Dirichlet Poincaré gives coercivity also at mass zero. First restrict the spectrum of N to a bounded interval. Pairing $(\Delta + g^2 N^2)u = G$ with u and $N^2 u$ gives

$$\|u\|_{H^1} + \|Nu\|_2 \leq C\|G\|_2, \quad \|dNu\|_2^2 + \|gN^2 u\|_2^2 = \text{Re}(G, N^2 u).$$

Ordinary Dirichlet regularity applied to $\Delta u = G - g^2 N^2 u$ therefore yields

$$(125) \quad \begin{aligned} \|u\|_{H^2} + \|Nu\|_{H^1} + \|N^2 u\|_2 &\leq C\|G\|_2, \\ \|u\|_{H^3} + \|Nu\|_{H^2} + \|N^2 u\|_{H^1} &\leq C\|G\|_{H^1}, \quad G \in H_0^1. \end{aligned}$$

For the second inequality, write $q = g^2$ and pair the equation with $N^2 \Delta u$. Both G and $qN^2 u$ have zero trace, so

$$(126) \quad \begin{aligned} \|N\Delta u\|_2^2 + \int_D q |dN^2 u|^2 &= \text{Re} \int_D \langle dG, dN^2 u \rangle \\ &\quad - \text{Re} \int_D \langle N^2 u, (dq) \lrcorner dN^2 u \rangle. \end{aligned}$$

Absorb the derivative term by Young's inequality and use the first endpoint to bound $N^2 u$ in L^2 . This controls $N\Delta u$ and $dN^2 u$. Dirichlet regularity for Nu , and then the equation for u , prove the second endpoint. The estimates are Hilbert estimates before interpolation and are independent of the spectral truncation. Closedness of N and the form solution give their limit.

Interpolate the same map $G \mapsto (u, Nu, N^2 u)$ between these two endpoints. For $\varepsilon < 1/2$ the source space $[L^2, H_0^1]_\varepsilon$ is the unrestricted H^ε space, by [Gru16, Definition 2.1 and Theorem 2.2]. The target spaces are the three spaces in (123). This proves that estimate; the zero trace in the second endpoint has not been asserted for a general fractional source.

For (124), choose a smooth domain inside the outer half-disc with a straight boundary segment containing the inner half-disc. Multiply G by a fixed cutoff equal to one near the inner half-disc and zero near the remaining boundary. Subtract its full Dirichlet solution. The difference is homogeneous near the inner half-disc and has zero straight-edge trace. The homogeneous positive-potential argument in the proof of Lemma 6.14 controls every required fixed derivative and power of N by its outer L^2 norm. Combining with (123) proves the local estimate. No reflection of the potential or forcing is used. $\square$

Proposition 6.16 (localization on a nonflat cap). *Use the exact cap, domain, and operator of Proposition 6.12, with its smallness condition and $r_1 = R_1^{-1}$. Let $U = (a, b, \nu)$ and suppose $F = \mathcal{A}_{\rho, f} U$ vanishes on both half-discs of radius r_0 at z_0 , where $r_1 < cr_0$ for a sufficiently small fixed $c > 0$. In addition to the domain's exact normalization, suppose*

$$(127) \quad H_\alpha b(z_0) = 0.$$

*Write $a = *_B d\psi$, $\psi|_{\partial B} = 0$, and use the radial cutoff in (116), on both sides of the seam. Then $\mathcal{C}_1 U = (*_B d(\chi\psi), \chi b, \chi\nu)$ belongs to the same domain, vanishes near the changed cap, and satisfies*

$$(128) \quad \begin{aligned} \|\mathcal{C}_1 U - U\|_X + \|\mathcal{A}_{\rho, f} \mathcal{C}_1 U - F\|_Y \\ \leq Cr_0^{-8} r_1^{1-\varepsilon} (\|U\|_X + \|F\|_Y) + C(\rho + \delta) r_1^{-6} \|U\|_X. \end{aligned}$$

The constants are uniform over the stated smooth compact families and all surface frequencies.

Proof. The ordinary base gauge gives a Dirichlet stream potential. Its two estimates, with $s = 1 + \varepsilon$, are

$$(129) \quad M^{-1} \|\psi\|_{H_B^{s+1} H_\Sigma^k} \leq C \|U\|_X, \quad \rho \|\psi\|_{H_B^s H_\Sigma^{k+1}} \leq C \|U\|_X.$$

For the second, commute d_α through $a = *_B d\psi$. The graph term $\rho d_\alpha a$ controls the base gradient of $\rho d_\alpha \psi$ in H^ε . Dirichlet Poincaré and the irreducible surface zero-form estimate give the assertion without a factor M . Define

$$(130) \quad c = b - \rho *_\Sigma d_\beta \psi, \quad c^{0,1} = b^{0,1} - i \rho \bar{\partial}_\beta \psi.$$

Use the convention $*d\bar{\zeta} = i d\bar{\zeta}$. Mixed integrability of the background gives $[D_B, D_\Sigma] = 0$ for its covariant antiholomorphic derivatives. Since $a^{0,1} = i \bar{\partial}_B \psi$, the mixed linearization V satisfies the exact identity

$$(131) \quad D_B c^{0,1} = V - i \rho D_\Sigma [A_B^{0,1}, \psi].$$

The final term is bounded by $C\delta \|U\|_X$ in $H_B^s H_\Sigma^k$, by (129). Also $\|c\|_{H_B^s H_\Sigma^k} \leq C \|U\|_X$. The stronger surface order in that potential estimate justifies the trace $d_\beta \psi = 0$ on the seam. Thus c and b have the same boundary trace and attaching value.

We need the local gain

$$(132) \quad \|(c, \nu)\|_{H^{2+\varepsilon}(D_{r_0/2}^+; H_\Sigma^k \oplus V)} \leq C r_0^{-3} \|U\|_X.$$

Fold the positive side as in Proposition 6.7. Near zero, the polar map from the canonical real boundary projection $P(0)$ to $P(s)$ is complex linear and has bounded smooth derivatives on each required surface Sobolev space. Extend it independently of the normal coordinate. This is a local boundary trivialization. The transformed field is $p + Jq$, with p, q in a constant real Hilbert space and $q = 0$ on the diameter. Its equation is

$$(\partial_s + J\partial_t)(p + Jq) = F_R + JF_I, \quad \Delta q = \partial_t F_R - \partial_s F_I.$$

On the source-free half-disc the new forcing is a sum of smooth bounded coefficients times c, ν and the controlled potential in (129). It is the restriction of an H^s field on a fixed neighborhood with norm $C \|U\|_X$. Localize q with a cutoff of radius r_0 . The Dirichlet forcing has H^ε norm at most $C r_0^{-2-\varepsilon} \|U\|_X$: at most two derivatives fall on the cutoff, and q is already H^s . Lemma 6.15 at $N = 0$ bounds q in $H^{2+\varepsilon}$. The two first derivatives of p follow from its first-order equation. Together with its L^2 norm this proves (132). Local rescaling and Sobolev embedding give a C^1 bound $C r_0^{-5} \|U\|_X$ on a smaller half-disc.

Pull the flat seam connection β_s over a fixed collar and put

$$(133) \quad S = \chi_{\text{col}} C_\alpha (C_\alpha - C_{\beta_s}) c, \quad c^* = c - S, \quad h = H_\alpha c^* = H_\alpha c.$$

The projection difference gains one surface derivative, so S has global $H_B^s H_\Sigma^{k+1}$ norm $C \|U\|_X$ and the local gain (132). On the seam $C_\beta c = 0$, hence $C_\alpha c^* = 0$; at zero $S = 0$. Compare with the flat nonharmonic field

$$b^* = P_{\alpha, \text{nh}} c^* + \rho *_\Sigma d_\alpha \psi, \quad b - b^* = S + h + \rho *_\Sigma [\beta - \alpha, \psi].$$

It has zero coexact boundary value and normalized exact value at zero. Its graph norm is at most $C \|U\|_X$: $d_\alpha h = 0$, the derivative of S uses its surface gain, and the final term uses the second estimate in (129). The flat scalar source is $H + E_H$, where direct subtraction gives

$$(134) \quad \begin{aligned} E_H &= f^{-1} *_B [A_B \wedge a] + \rho f *_\Sigma [\beta - \alpha, b] + \rho f *_\Sigma d_\alpha S \\ &\quad + \rho^2 f *_\Sigma d_\alpha *_\Sigma [\beta - \alpha, \psi], \quad \|E_H\|_Y \leq C M L^\varepsilon (\delta + \rho) \|U\|_X. \end{aligned}$$

After expanding the connection difference, the last term has only one surface derivative of ψ .

Decompose $P_{\alpha, \text{nh}} c^* = d_\alpha \Lambda^{-1} x + *_\Sigma d_\alpha \Lambda^{-1} z$. Here $z|_{\partial B} = 0$, $x(0) = z(0) = 0$, and x, z have the local gain (132). The flat mixed source is $v^* = dx + *dz$. Subtract $W = Q_{\rho, f}(E_H, 0)$ from (a, b^*) . Its Hodge variables x_W, z_W vanish exactly, and

$$(135) \quad \|W\|_X \leq CML^{2\varepsilon}(\delta + \rho)\|U\|_X.$$

The remaining field $\widehat{U}$ has the same x, z , scalar source H , and norm at most $C\|U\|_X$ by the cap smallness condition.

Set $B = \|U\|_X + \|F\|_Y$. The stream equations are

$$(\Delta + m^2 f^2) \widehat{\psi} = fH - mf^2 z, \quad (\Delta + m^2 f^2) \widehat{y} = \Delta z + mfH, \quad \widehat{y} = z + m\widehat{\psi}.$$

Near zero $H = 0$. The local C^1 estimate gives $|x| + |z| \leq C|\zeta|r_0^{-5}B$ and $|z| \leq Ctr_0^{-5}B$. Kato's inequality gives $\Delta|\widehat{y}| \leq |\Delta z|$ there. On a smaller smooth domain with a straight segment, compare with the sum of a harmonic function having boundary values $|\widehat{y}|$ and the full zero-boundary Poisson solution with source $|\Delta z|$. Since $H^\varepsilon \subset L^{2/(1-\varepsilon)}$ and this exponent exceeds two, the Poisson solution is C^1 . Rescaling gives $|\widehat{y}| \leq Ctr_0^{-5}B$ on a smaller half-disc: the H^ε bound for Δz costs r_0^{-3} , the local L^p embedding at most r_0^{-1} , and the Poisson boundary-gradient scaling costs $r_0^{1-2/p} \leq 1$. The harmonic contribution costs $r_0^{-1}\|\widehat{y}\|_\infty \leq Cr_0^{-1}B$.

Split the surface spectrum at $m = r_1$. For $m \leq r_1$, the Green comparison in Lemma 6.14 requires only linear vanishing of z . Its remote contribution is bounded by $C(r_1 r_0^{-4} + r_1^2 r_0^{-6})B$ and its local contribution by $Cr_0^{-5}B$. Thus $\|\widehat{\psi}\|_{C^0(D_{64r_1}^+; H_\Sigma^k)} \leq Cr_0^{-6}B$. The rescaled low-mass potential is bounded, and its forcing has H^ε norm at most $Cr_0^{-5}B$. Local Dirichlet regularity therefore bounds $\widehat{\psi}(r_1 \cdot)$ in $H^{2+\varepsilon}$ by $Cr_0^{-6}B$. For $m > r_1$, take $N = m/r_1$ and $g(Z) = r_1^2 f(r_1 Z)$. The forcing $r_1^2 \Delta z(r_1 \cdot)$ has H^ε norm at most $Cr_1 r_0^{-3}B$, since its L^2 and fractional scaling factors are r_1 and $r_1^{1+\varepsilon}$. The linear boundary bound controls the outer rescaled L^2 norm of $\widehat{y}$ by $Cr_1 r_0^{-5}B$. Lemma 6.15 gives

$$\|\widehat{y}(r_1 \cdot)\|_{H^{2+\varepsilon}} + \|N\widehat{y}(r_1 \cdot)\|_{H^{1+\varepsilon}} + \|N^2\widehat{y}(r_1 \cdot)\|_{H^\varepsilon} \leq Cr_1 r_0^{-5}B.$$

The rescaled $H^{2+\varepsilon}$ norm of z has the same bound: its value and first derivative use its C^1 bound, and its second-derivative norms scale with r_1 and $r_1^{1+\varepsilon}$. Division by m is now bounded by r_1^{-1} , giving the required $H^{2+\varepsilon}$ bound for $\widehat{\psi}$.

In both spectral ranges the rescaled strong norms of $x, \widehat{y}, \widehat{a}/M$ are bounded by $Cr_1 r_0^{-6}B$. The rescaled weak norms of $m\widehat{a}, mf\widehat{y}$, and the $H^{2+\varepsilon}$ norm of $\widehat{\psi}$, are bounded by $Cr_0^{-6}B$. The weighted identities are $m\widehat{a} = r_1^{-1} *_Z dz(\widehat{y} - z)$ and $mf\widehat{y} = r_1^{-1} gN\widehat{y}$. The stream cutoff therefore gives graph and flat source error at most $Cr_0^{-6}r_1^{1-\varepsilon}B$. The mixed source v^* need not vanish near zero: the additional term $(\chi - 1)v^*$ has the same bound by (132). The scalar source H does vanish there.

For the removed particular field, a crude bound suffices:

$$\|\mathcal{C}_1 W - W\|_X + \|\mathcal{A}_{\text{flat}}(\mathcal{C}_1 W - W)\|_Y \leq Cr_1^{-3}\|W\|_X.$$

Indeed ψ/M is strong with one extra base derivative; the weak weighted potential is controlled by (129). Multiplication by χ and $d\chi$ costs at most r_1^{-2} and r_1^{-3} in the strong norm. In the scalar commutator $f^{-1}\Delta\chi$ has size $O(1)$ and $f^{-1}d\chi$ has size $O(r_1)$, with fractional multiplier loss $r_1^{-\varepsilon}$. Its mixed commutator is zero since $x_W = z_W = 0$. Equation (135) and $5 + 2\varepsilon < 6$ give the second term of (128).

Finally S, h, ν vanish at zero by (127) and have the local $H^{2+\varepsilon}$ estimate. Their localized strong norms are at most $Cr_0^{-5}r_1^{1-\varepsilon}\|U\|_X$. The weighted derivative of h is zero, and that of S

is bounded by $C\rho r_1^{-3}\|U\|_X$ using its global surface gain. The term $\rho *_{\Sigma} [\beta - \alpha, \psi]$ contributes at most $C\rho r_1^{-4}\|U\|_X$, by the two potential estimates. These prove the graph bound for the actual field. Its exact mixed commutator is

$$(136) \quad \begin{aligned} V(\mathcal{C}_1 U) - \chi V(U) &= (\bar{\partial}_B \chi) c^{0,1}, \\ H(\mathcal{C}_1 U) - \chi H(U) &= f^{-1}(\psi \Delta \chi - 2\langle d\chi, d\psi \rangle) - f^{-1} *_B [A_B \wedge \psi *_B d\chi]. \end{aligned}$$

The positive commutator is the derivative of χ times ν . Their bounds follow from the local gain and zero attaching values. The final scalar term is bounded by $C\delta r_1^{-1-\varepsilon}\|U\|_X$, smaller than the second term in (128). Here $\chi F = F$ for the actual source. Combining the estimates proves the assertion; r_0^{-8} allows the fixed further restrictions of half-discs. $\square$

The additional harmonic condition (127) is not automatic for a cap right inverse. It requires a controlled kernel correction or a retained matching profile. The new annular source also remains outside the original separated source class. The harmonic condition is addressed next; the complete-source inverse is constructed in Proposition 7.13.

6.11. Attaching evaluation and kernel normalization.

Proposition 6.17 (attaching evaluation and kernel normalization). *For the primitive extension (56), put $D_L = \Lambda L^{-2}$, of degree one. The differential*

$$e_E : H^1(X, \text{End}_0 E) \longrightarrow H^1(\Sigma_g, \text{End}_0 E_{\infty}), \quad X = \mathbb{P}^1 \times \Sigma_g,$$

of evaluation at infinity has

$$(137) \quad \dim_{\mathbb{C}} \ker e_E = g + h^0(D_L), \quad \dim_{\mathbb{C}} \text{coker } e_E = g - 2 + h^0(D_L).$$

For a fixed regular index-zero residual solution, every regular nonempty primitive attaching fiber product has e_E onto. On any compact smooth family of such points, the cap operator has right inverses $Q_{\rho,R}^0$ satisfying

$$(138) \quad \mathcal{A}_{\rho,f_R} Q_{\rho,R}^0 = I, \quad H_{\alpha}(Q_{\rho,R}^0 F)_{\Sigma}(z_0) = 0, \quad \|Q_{\rho,R}^0\|_{Y \rightarrow X} \leq CR^{\varepsilon}$$

when R grows polylogarithmically in ρ^{-1} . The additional zero-evaluation kernel has real dimension four and admits uniformly bounded projections in X .

Proof. Write the line subbundle and quotient of the primitive extension as $A = \mathcal{O}(1) \boxtimes L$ and $B = \mathcal{O} \boxtimes (\Lambda L^{-1})$. Let $P \subset \text{End}_0 E$ preserve A , and let $F_{\infty} = \{\infty\} \times \Sigma_g$. In a local splitting, P consists of the upper triangular trace-free matrices. The off-diagonal and diagonal entries give exact sequences

$$0 \longrightarrow \mathcal{O}(1) \boxtimes D_L^{-1} \longrightarrow P \longrightarrow \mathcal{O}_X \longrightarrow 0, \quad 0 \longrightarrow P \longrightarrow \text{End}_0 E \longrightarrow \mathcal{O}(-1) \boxtimes D_L \longrightarrow 0.$$

These are locally free sequences; a global splitting of E is unnecessary. Twist by $\mathcal{O}(-F_{\infty})$ and project to Σ_g . Since $\mathcal{O}(-1)$ has no cohomology on $\mathbb{P}^1$, whereas $H^0(\mathcal{O}) = \mathbb{C}$ and $H^1(\mathcal{O}(-2)) = \mathbb{C}$, the direct-image sequences give

$$(139) \quad \pi_*(\text{End}_0 E(-F_{\infty})) = D_L^{-1}, \quad R^1 \pi_*(\text{End}_0 E(-F_{\infty})) = D_L.$$

The second equality includes the fixed one-dimensional factor $H^1(\mathbb{P}^1, \mathcal{O}(-2))$. Leray over the curve and Riemann–Roch give

$$h^1(X, \text{End}_0 E(-F_{\infty})) = g + h^0(D_L), \quad h^2(X, \text{End}_0 E(-F_{\infty})) = g - 2 + h^0(D_L).$$

The restriction sequence identifies these spaces with the kernel and cokernel of e_E : the attaching fiber is simple, and $H^2(X, \text{End}_0 E) = 0$ by Proposition 6.4. Restriction of a first-order bundle deformation is evaluation of that deformation at infinity, so this is the stated

differential. The doubling construction identifies its harmonic representative with the mixed attaching value.

The image of e_E is complex. A proper image has real cokernel dimension at least two. A fixed regular index-zero residual solution has no kernel, and its center moves in one real direction along the matching evaluation path. Thus surjectivity of the fiber-product matching map $(\dot{s}, \xi) \mapsto de_0(\dot{s}) - e_E(\xi)$ forces e_E itself to be onto. In genus at least three this cannot occur by (137). In genus two it requires $h^0(D_L) = 0$. In genus one $h^1(D_L) = 0$ and the target harmonic space is zero. This argument concerns the fixed residual solution, not higher-parameter residual families.

We next choose uniform representatives of genuine cap kernel directions. Integrate $H^1(X, \text{End}_0 E)$ in finitely many smooth local bundle charts over the compact family. In the base-flat parallel trivializations the limiting base connection is zero throughout each chart. At infinity choose the flat Coulomb representative of the varying attaching class, as in the proof of Proposition 6.13. The cap metric construction of Lemma 6.11 permits an additional bundle-parameter derivative at each fixed base and surface order. Increasing that finite derivative count and its polynomial exponent if needed gives, for a fixed B_0 ,

$$(140) \quad \|\partial_E A_B\|_{C^j} \leq C\rho^2 R^{B_0}, \quad \|\partial_E \beta\|_{C^j} \leq C(1 + \rho^2 R^{B_0}), \quad \|\partial_E u\|_{C^j} \leq C.$$

The extra surface derivative in the base gauge is included in this count. A final surface gauge constant in the base restores the chosen attaching representative while preserving ordinary base Coulomb gauge and zero normal trace.

Differentiating the reconstructed family gives $a = \rho^{-1} \partial_E A_B$, $b = \partial_E \beta$, and $\nu = \partial_E u$. These components satisfy the normalized cap domain and the exact linearized equation. The four graph terms are bounded respectively by

$$(141) \quad C\rho R^{B_0}/M, \quad C(1 + \rho^2 R^{B_0}), \quad C\rho^2 R^{B_0}, \quad C\rho R^{2+\varepsilon}(1 + \rho^2 R^{B_0}).$$

They are uniform at polylogarithmic radii, with B_0 fixed first. Doubling returns the original class, so this is the actual kernel identification $I_{\rho, R, E} : H^1(X, \text{End}_0 E) \rightarrow \ker \mathcal{A}_{\rho, f_R}$. Its evaluation is exactly $\ell_\infty e_E$, because the boundary Dirichlet metric and the chosen flat surface slice fix the attaching representative. It is not only a limiting identity. Uniqueness of the normalized kernel representative makes these local constructions agree on overlaps.

Choose smooth Hermitian metrics on the finite tangent bundles. Since e_E is onto on the compact family, $r_E = e_E^* (e_E e_E^*)^{-1}$ is uniformly bounded. Then $B_{\rho, R, E} = I_{\rho, R, E} r_E \ell_\infty^{-1}$ is a uniformly bounded kernel-valued right inverse of harmonic evaluation. For the cap right inverse Q set

$$Q^0 F = QF - B_{\rho, R, E} H_\alpha(QF)_\Sigma(z_0).$$

Point evaluation is bounded at base order $1 + \varepsilon$. This proves (138), without altering the source. The zero-evaluation kernel has complex dimension two by (137). For a bounded projection, use finite bases supplied by $I_{\rho, R, E}$ and their surface/positive L^2 Gram matrices. Those components converge to the normalized limiting kernel of Proposition 6.7, by (140). Its identification is injective, so the limiting Gram matrices are positive and their inverses stay bounded. The basis graph bounds (141) then give a uniform projection. Local projections patch over the compact family. $\square$

Combining (138) with Proposition 6.16 removes its additional harmonic hypothesis on these regular families. For $R_1 = R_0^p$ the resulting separated-source error tends to zero if $p(1 - 2\varepsilon) > 8$ and the radii grow polylogarithmically. The error still lies on a new annulus.

The comparison with the residual source and the solution of the complete global problem are given in Section 7.

6.12. Decreasing fractional exponents.

Proposition 6.18 (small fractional exponents). *Use fixed Bessel-potential norms on the base, and let $0 < \varepsilon \leq 1/4$. Constants below are independent of ε and uniform on the stated compact smooth families. The conclusion of Proposition 6.12 holds under the quantified condition*

$$(142) \quad \varepsilon^{-1} \rho M L^{4\varepsilon} + \varepsilon^{-1/2} \delta M L^{2\varepsilon} < c, \quad \|Q\| + \|\Pi\| \leq C \varepsilon^{-1/2} L^\varepsilon.$$

On the regular attaching families of Proposition 6.17, at polylogarithmic radii, its zero-evaluation inverse satisfies

$$(143) \quad \|Q_{\rho,R}^0\| \leq C \varepsilon^{-1} R^\varepsilon.$$

The localization estimate of Proposition 6.16 can be taken as

$$(144) \quad \|\mathcal{C}_1 U - U\|_X + \|\mathcal{A} \mathcal{C}_1 U - F\|_Y \leq C \varepsilon^{-2} r_0^{-8} r_1^{1-\varepsilon} (\|U\|_X + \|F\|_Y) \\ + C(\rho + \delta) r_1^{-6} \|U\|_X.$$

If $\rho M L^\varepsilon \leq 1$, the nonlinear chart of Proposition 6.13 has a local inverse on a graph ball of radius $c\sqrt{\varepsilon}$, and its quadratic difference constant is at most $C \varepsilon^{-1/2} (1 + \rho M^2 L^\varepsilon)$.

Proof. Choose a fixed extension operator at the integer base orders zero through three. Interpolation gives uniform bounds at the intermediate orders. The identity $[L^2, H_0^1]_\varepsilon = H^\varepsilon$ has uniform equivalence constants for $\varepsilon \leq 1/4$: first use the fixed identity at $1/4$ from [Gru16, Definition 2.1 and Theorem 2.2], then reiterate between that space and L^2 . Thus no boundary threshold is approached. On $\mathbb{R}^2$,

$$\int (1 + |\xi|^2)^{-1-\varepsilon} d\xi = \pi/\varepsilon.$$

Fourier Cauchy–Schwarz gives $\|v\|_{C^0} \leq C \varepsilon^{-1/2} \|v\|_{H^{1+\varepsilon}}$, and the same estimate for C^1 from $H^{2+\varepsilon}$. Weighted convolution gives the strong algebra bound with constant $C \varepsilon^{-1/2}$. The strong-times-weak bound has the same constant. Indeed, for dyadic coefficients $A_j = 2^{(1+\varepsilon)j} \|u_j\|_2$ and $B_j = 2^{\varepsilon j} \|v_j\|_2$, the low–high term uses

$$\sum_{i < j} 2^{-\varepsilon i} A_i \leq C \varepsilon^{-1/2} \|A\|_{\ell^2}.$$

The high–low term uses $\|S_j v\|_\infty \leq C 2^{(1-\varepsilon)j} \|B\|_{\ell^2}$. Comparable high frequencies have H^ε product norm at most $C 2^{-\varepsilon j} A_j B_j$, summable by Cauchy–Schwarz. All constants apart from the first geometric sum are uniform. These arguments apply to the fixed surface Sobolev algebra H_Σ^k , $k \geq 2$.

The flat energy endpoints of Lemma 6.10 therefore give $C \varepsilon^{-1/2} L^\varepsilon$ after exact point normalization. For scalar sources $(H, 0)$, its variables x, z are identically zero, so the bound remains $C L^\varepsilon$. The endpoints and local homogeneous estimate of Lemma 6.15 give a uniform forced Dirichlet bound in this range of exponents. For Kato comparison use

$$p_\varepsilon = \frac{2}{1 - \varepsilon/2}, \quad \|G\|_{L^{p_\varepsilon}} \leq C \|G\|_{H^\varepsilon}.$$

Hausdorff–Young and Fourier Hölder prove the uniform constant: the weight $\langle \xi \rangle^{-\varepsilon}$ is used in $L^{4/\varepsilon}$, whose power integral is the fixed integral of $\langle \xi \rangle^{-4}$. The fixed-domain Dirichlet inverse has uniform $L^p \rightarrow W^{2,p}$ bounds between its $p = 2$ and $p = 4$ estimates. The local Bessel

kernel $|x|^{-1}$ gives $W^{1,p} \rightarrow C^0$ cost at most $C(p-2)^{-1}$. Its zero-boundary Poisson solution thus has C^1 norm at most $C\varepsilon^{-1}\|G\|_{H^\varepsilon}$. In Proposition 6.16, the local $H^{2+\varepsilon}$ estimates have uniform constants; their C^1 bounds and the Kato comparison have the costs just computed. The scalar-only particular correction has no point cost. The Green integral, homogeneous mass estimates, and fixed-profile rescalings are uniform. These give (144); its factor ε^{-2} also bounds the preceding ε^{-1} loss.

We next quantify the limiting inverse. Fix its boundary reduction U and finite projection P_{N_0} from Proposition 6.7, independently of ε . Write $v = (b, \nu) = Uw$ after folding. Let E_∞ be the attaching exact projection, zero on the positive factor, and choose a smooth density χ of integral one supported in a sufficiently small fixed neighborhood of z_0 . Replace point normalization temporarily by

$$(145) \quad \mathcal{L}v := E_\infty \int \chi(z) U(z)^{-1} v(z) dz = 0.$$

This is bounded on base L^2 . In the w coordinates it splits exactly into the fixed finite block and constant tail. For $e \in E_\alpha H_\Sigma^k$, set $T_E e = (d_\beta G_\alpha^0 d_\alpha^* e, 0)$. This is a homogeneous unnormalized seam solution. Since $U(z_0) = I$ and $\beta(z_0) = \alpha$, the operator $S_E = \mathcal{L}T_E$ is close to I , and hence invertible, when the support of χ is small. The required operator continuity follows from the one-surface-derivative gain in $G_\alpha^0 d_\alpha^*$ and the smooth coefficients.

The constant real-boundary Cauchy–Riemann problem, normalized by the average of its real part, is an isomorphism $H^1 \rightarrow L^2$. Solve its imaginary part by the Dirichlet inverse with H^{-1} forcing, recover the real part from its closed L^2 gradient, and fix its constant by the average. Dirichlet coercivity and Poincaré give dimension-independent Hilbert estimates. The finite block with its finitely many average conditions is Fredholm. The surface-smoothing term $K : H_B^1 H_\Sigma^k \rightarrow L_B^2 H_\Sigma^k$ is compact by finite surface compression and finite-dimensional base Rellich. Thus the mean-normalized limiting operator is Fredholm at this endpoint. Smooth targets supported away from the seam have smooth solutions by the doubled-bundle construction already proved. Subtracting $T_E S_E^{-1} \mathcal{L}v$ imposes (145) without changing the source. Density and closed range give surjectivity.

The finite-block generalized inverse and constant-tail inverse preserve surface orders; iteration with the smoothing K gives smooth kernel elements. Doubling identifies this kernel with $H^1(X, \text{End}_0 E)$. Its injectivity now uses $S_E e = 0$ in place of the exact point condition, and its surjectivity uses the same mean correction of differentiated bundle charts. Choose the right inverse Q^{av} with image orthogonal to that finite kernel in L^2 . It has a uniform $H^1 \leftarrow L^2$ bound on the compact family. It also has an $H^2 \leftarrow H^1$ bound. Locally straighten the real boundary and write the field $p + Jq$, with $q = 0$ on the edge. For $(\partial_s + J\partial_t)(p + Jq) = F$, one has $\Delta q = \partial_t F_R - \partial_s F_I$. Dirichlet regularity puts q in H^2 , and the first-order equations put p there. Smooth cutoffs and coefficients add terms controlled by the known H^1 norm. No source boundary condition is imposed.

Interpolate this same inverse between its two endpoints in the unrestricted target Sobolev spaces. It has a uniform $H^{1+\varepsilon} \leftarrow H^\varepsilon$ bound; approximation retains its seam and average. Now set

$$Q_E^{\text{pt}} F = Q^{\text{av}} F - T_E (E_\alpha (Q^{\text{av}} F)_\Sigma(z_0)).$$

This restores exact point normalization, costs $C\varepsilon^{-1/2}$, and leaves the source unchanged. Smooth L^2 coordinates on the finite point-normalized kernel give the same bound for the augmented limiting inverse.

In Proposition 6.12, the two-sided surface truncation of K and truncation of the smooth kernel coordinates have error C/N , uniformly in ε . The finite zero-scale inverse therefore

costs $C\varepsilon^{-1/2}L^\varepsilon$. The finite mass perturbation costs $C\rho MNL^\varepsilon$ before inversion, so it requires $C\varepsilon^{-1/2}\rho MNL^{2\varepsilon}$ small. The infinite tail has the same inverse bound. Take $N \asymp A\varepsilon^{-1/2}L^{2\varepsilon}$, with A fixed large. The final reference-inverse error is

$$C\varepsilon^{-1/2} \left(\frac{L^\varepsilon}{N} + ML^{2\varepsilon}(\rho + \delta) \right).$$

This and the finite mass condition are controlled by (142). The augmented Neumann inverse proves its assertion. Actual cap kernel representatives from Proposition 6.17 have fixed smooth bounds, uniform over these fractional orders. A further harmonic point evaluation costs $\varepsilon^{-1/2}$, proving (143).

Finally multiply the strong Sobolev norm by $A_\varepsilon = C\varepsilon^{-1/2}$. Its algebra constant is uniform. The quadratic flatness equation, gauge exponential, and exact-coordinate implicit equation in Proposition 6.13 work on a fixed small ball in this rescaled norm. Smooth finite-dimensional coordinate changes have uniform composition bounds at orders $1 \leq s \leq 5/4$. To check the latter directly, telescope $F(u) = F(S_0u) + \sum_j m_j \Delta_j u$, where $m_j = \int_0^1 F'(S_j u + t \Delta_j u) dt$. On a fixed L^∞ ball, Bernstein gives $\|D^r m_j\|_\infty \leq C_r 2^{rj}$. Output frequencies $l \leq j$ give the weighted factor $2^{-s(j-l)}$, and two derivatives for $l > j$ give $2^{-(2-s)(l-j)}$. Both sums are uniformly bounded in this range. The low block is bounded by $C\|u\|_2$. Thus $\|F(u) - F(0)\|_{H^s} \leq C\|u\|_{H^s}$. The integral difference formula and the strong product bound then give a uniform Lipschitz bound and quadratic difference constant CA_ε on the ball of radius c/A_ε . This also applies to smooth finite parameter maps into the fixed surface Sobolev spaces.

The identity (110) gives the same bound with one additional surface derivative for S . Its weighted graph term costs at most an additional $\rho ML^\varepsilon \leq 1$. The inverse chart therefore has radius $c\sqrt{\varepsilon}$. In the normalized curvature, the base wedge term has constant $C\rho M^2 L^\varepsilon A_\varepsilon$; the other terms have constant $C(1 + \rho ML^\varepsilon + \rho M)A_\varepsilon$. The strong-times-weak bound treats their differentiated factors. This proves the final assertion. $\square$

For $T = \log(1/\rho)$, take $\varepsilon = T^{-1}$ and any fixed polylogarithmic cap radius. Condition (142) follows from Lemma 6.11; the normalized cap inverse is $O(T)$, its nonlinear chart has radius at least $cT^{-1/2}$, and its quadratic constant is $O(T^{1/2})$. For separated sources, combining (143) and (144) gives error coefficient

$$(146) \quad C(\varepsilon^{-3} R_0^8 R_1^{-1+2\varepsilon} + \varepsilon^{-1}(\rho + \delta) R_1^{6+\varepsilon}).$$

With $R_0 = T^K$ and $R_1 = R_0^p$, this tends to zero if $K(p-8) > 3$. These statements concern the cap norms only. On an annulus $|\zeta| \asymp r_1$, converting a normalized source to a physical H^ε source contributes $\rho^{-\varepsilon} r_1^{2\varepsilon}$: its amplitude changes by r_1^2/ρ and its base length scale is ρ/r_1 . This factor is bounded for the decreasing exponent. Section 7 supplies the complete global domain and annular construction. Its positive-scale parameter argument uses affine-averaged holonomy functions rather than a derivative estimate for a rough Sobolev dilation.

6.13. The physical flat model and auxiliary annular estimates.

Lemma 6.19 (nonharmonic flat-seam estimate). *At a fixed irreducible flat surface connection α , a nonharmonic eigenmode of the physical mixed operator has the form*

$$D_\mu = \partial_\theta - T_\mu, \quad T_\mu = A\partial_s + \sqrt{\mu}B, \quad s \leq 0, \quad \mu > 0,$$

where, in coordinates $U = (\varphi, \tau_0, x, y)^T$,

$$A = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

Its domain has $\tau_0 = y = 0$ at $s = 0$, and

$$(147) \quad \|D_\mu U\|_2^2 = \|\partial_\theta U\|_2^2 + \|\partial_s U\|_2^2 + \mu \|U\|_2^2.$$

The same identity holds for its adjoint. Consequently the nonharmonic half-plane operator is invertible in L^2 , with inverse bound $\mu_0^{-1/2}$, where μ_0 is the first surface zero-form eigenvalue.

Proof. Let $d_\alpha^* d_\alpha e = \mu e$ and use the orthonormal exact and coexact forms $\mu^{-1/2} d_\alpha e$ and $\mu^{-1/2} *_\Sigma d_\alpha e$. Substitution into [DFL24, (5.5)–(5.7)] gives the matrices and the boundary conditions. They satisfy $A^T = -A$, $B^T = B$, $A^2 = -I$, $B^2 = I$, and $AB + BA = 0$. The boundary space $L = \{\tau_0 = y = 0\}$ has $AL = L^\perp$, so the Green form gives the same domain for T_μ and its adjoint. Maximality follows directly from the distributional normal derivative equation and integration by parts. The square domain imposes additionally $\partial_s \varphi = \partial_s x = 0$, and

$$T_\mu^2 = -\partial_s^2 + \mu, \quad D_\mu^* D_\mu = -\partial_\theta^2 - \partial_s^2 + \mu.$$

The θ cross term integrates to zero by self-adjointness, and the s boundary term vanishes since $ABL \subset L^\perp$. This proves (147) and its adjoint version. Closed range and the trivial adjoint kernel give surjectivity. Summing the orthogonal surface modes proves the stated bound. $\square$

In rescaled base variables the uniformly normalized operator is $\hat{D}_{\mu,\rho} = \partial_\Theta - A\partial_S - \rho\sqrt{\mu}B$. Its square is $-\partial_\Theta^2 - \partial_S^2 + \rho^2\mu$ on all four components, with the above Dirichlet/Neumann division; its L^2 inverse bound is $(\rho\sqrt{\mu})^{-1}$. The harmonic surface sector instead joins the positive map variation to give a full-plane Cauchy–Riemann operator. Equal traces identify its weak domain across the seam. When the harmonic space is nonzero, this operator has no positive L^2 gap, as shown by dilating a compactly supported test function. The following extension treats the nonharmonic sector in physical L^p spaces. The nonflat core remains a separate problem.

Corollary 6.20 (physical L^p inverse of the flat model). *For $1 < p < \infty$, the nonharmonic flat operator D_α on $\mathbb{R}_-^2 \times \Sigma$ has an inverse satisfying*

$$(148) \quad \|Q_\alpha f\|_{W_{\text{phys}}^{1,p}} \leq C_p \|f\|_{L_{\text{phys}}^p}.$$

The output $Q_\alpha f$ has zero normal base component and zero coexact surface trace at the seam. Both f and the output have zero harmonic surface projection. No boundary trace is required of f . The physical $W^{1,p}$ norm includes the zeroth-order term and all first base and surface covariant derivatives. Constants are uniform on compact smooth families of irreducible flat attaching connections.

Proof. In the mode coordinates of Lemma 6.19, put $E = \text{diag}(1, -1, 1, -1)$. The identities $EA = -AE$ and $EB = BE$ show that D_α commutes with reflection $U(s, \theta) \mapsto EU(-s, \theta)$. Its invariant sections have even exact surface and tangential base components, and odd coexact surface and normal base components. Their traces are exactly the asserted boundary conditions. The surface Hodge projections are order-zero operators, bounded on L^p and $W^{1,p}$ on the fixed compact surface. Thus reflection gives a bounded extension of an arbitrary L^p

source. The inverse below produces a reflection-invariant $W^{1,p}$ output, whose odd components have precisely the asserted zero traces.

On the full physical product, in differential-form coordinates,

$$D_\alpha^* D_\alpha = D_\alpha D_\alpha^* = L = -\Delta_{\mathbb{R}^2} + \Delta_{\Sigma, \alpha}.$$

Here $-\Delta_{\mathbb{R}^2} = -\partial_s^2 - \partial_\theta^2$; $\Delta_{\Sigma, \alpha}$ is the zero-form Laplacian on the base components and the one-form Hodge Laplacian on the surface component. In particular its curvature terms are included. Its restriction to the nonharmonic sector has a positive gap.

The heat kernel factors into the Euclidean and surface heat kernels. For $0 < t \leq 1$, the surface kernel and its first derivative have bounds $Ct^{-1} \exp(-d^2/(ct))$ and $Ct^{-3/2} \exp(-d^2/(ct))$. These follow from [Lud19, Theorems 1.1 and 3.5]: use the differentiated expansion near the diagonal and absorb the extra polynomial factor into a weaker Gaussian off a fixed diagonal neighborhood. Removing the harmonic projection adds a smooth bounded kernel. Integration in each kernel variable gives the corresponding L^p bounds. For $t \geq 1$, fixed-time surface smoothing, the L^2 spectral gap, and another fixed-time smoothing factor give exponential L^p decay, also after one derivative. The base heat semigroup is a contraction and its gradient has norm $Ct^{-1/2}$. Consequently

$$\|D_\alpha^* e^{-tL}\|_{p \rightarrow p} \leq C_p \begin{cases} t^{-1/2}, & 0 < t \leq 1, \\ e^{-ct}, & t \geq 1. \end{cases}$$

It follows that $Q = \int_0^\infty D_\alpha^* e^{-tL} dt$ converges as an L^p operator. The two square identities give $DQ = QD = I$ on smooth data, hence by density on their respective domains. Ordinary first-order elliptic estimates for the differential-form operator on fixed product charts give

$$\|u\|_{W^{1,p}} \leq C_p (\|D_\alpha u\|_p + \|u\|_p).$$

A cover with bounded overlap gives the global estimate and hence (148). The inverse commutes with reflection, so extension and restriction prove the half-plane assertion. Compactness of the smooth attaching family makes the constants uniform. $\square$

Under $z = \rho w$, retaining physical orthonormal components and multiplying every equation row by ρ , this estimate reads

$$(149) \quad \rho \|U\|_p + \|\nabla_w U\|_p + \rho \|\nabla_\Sigma U\|_p \leq C_p \|\widehat{D}_{\alpha, \rho} U\|_p,$$

where all norms use $dw d\Sigma$. Thus the normalized zeroth-order inverse bound still has order ρ^{-1} .

Remark 6.21 (unaveraged traces of homogeneous solutions). Even zero normal data do not bound the stronger unaveraged trace by the physical $W^{1,p}$ norm. To see this, fix $p > 2$, $r > 0$, and use the flat strip $[-r, 0] \times (\mathbb{R}/2\pi r\mathbb{Z})$, with seam $s = 0$. Let $N_\alpha e = \lambda^2 e$, $\|e\|_{L^p(\Sigma)} = 1$, and put $u = \lambda^{-1} d_\alpha e$, $v = *_\Sigma u$. For $\xi = m/r$ and $k = (\lambda^2 + \xi^2)^{1/2}$, define

$$(150) \quad \begin{aligned} h &= \varphi e dt + \tau e ds + xu + yv, \\ \varphi &= e^{-kr} (\lambda/k) \cosh(ks) \cos(\xi t), & \tau &= 0, \\ x &= -e^{-kr} (\xi/k) \cosh(ks) \sin(\xi t), & y &= e^{-kr} \sinh(ks) \cos(\xi t). \end{aligned}$$

Substitution in the four rows of Lemma 6.19 gives $D_\alpha h = 0$: three rows cancel termwise, and the other uses $k^2 = \lambda^2 + \xi^2$. Both normal traces vanish, and $y = 0$ at the seam, so matching holds with zero positive variation.

Take eigenpairs with $\lambda_j \rightarrow \infty$ and integers m_j with $\xi_j/\lambda_j \rightarrow 1$. Fixed-surface elliptic estimates give $\|e_j\|_{W^{2,p}} \leq C\lambda_j^2$ and $\|d_\alpha e_j\|_p \leq C\lambda_j$. The profiles are bounded by

$Ce^{-k_j(s+r)}$; their L_s^p norm is $O(k_j^{-1/p})$, and first derivatives cost Ck_j . Since $d_\alpha^* u_j = \lambda_j e_j$ and $e^{-k_j r} \cosh(k_j r) \geq 1/2$, this proves

$$(151) \quad \|h_j\|_{W^{1,p}} \leq C_r \lambda_j^{1-1/p}, \quad \|E_\alpha \gamma_o h_{j,\Sigma}\|_{L_t^p W_\Sigma^{1,p}} \geq c_r \lambda_j.$$

Their ratio is unbounded. The outer components satisfy

$$x_o = -k^{-1} \coth(kr) \partial_t y_o, \quad \varphi_o = -(\lambda/k) \coth(kr) y_o.$$

For comparable large base and surface frequencies, the exact and coexact components therefore have the same order of regularity. The outer data here are nonzero. This is a flat boundary test, not an obstruction to an inverse with suitably controlled outer data on the primitive core. The example does not concern the source norms used in the cap construction.

Lemma 6.22 (harmonic annulus estimate). *Let $V = \mathcal{H}_\alpha^1$ with its complex structure, and write $z = r_- e^{t+i\vartheta}$ on $r_- < |z| < r_+$, where $T = \log(r_+/r_-) \geq 1$. The constant matching condition joins the harmonic variations into a V -valued field on the full annulus. Put*

$$(152) \quad C = \partial_t + i\partial_\vartheta = 2|z|e^{-i\vartheta}\bar{\partial}, \quad \|Ch\|_{L_{\text{cyl}}^p}^p = 2^p \int |z|^{p-2} |\bar{\partial}h|^p dA.$$

For $1 < p < \infty$ and fields with zero radial end traces,

$$(153) \quad \|h\|_{W_{\text{cyl}}^{1,p}} \leq C_p(1+T)\|Ch\|_{L_{\text{cyl}}^p}.$$

If the angular average has prescribed end values $a, b \in V$, its compatibility equation for $Ch = f$ is

$$(154) \quad b - a = \int_0^T P_0 f(t) dt,$$

where P_0 is angular averaging. When $V \neq 0$, the order T loss in (153) cannot be removed. For genus one, $V = 0$ and this sector is absent.

Proof. The coordinate and measure identities give (152). Let $P_\pm$ project onto modes $\pm k > 0$. For $k \neq 0$, the equation is $h'_k - kh_k = f_k$. With $P_-h(0) = 0$ and $P_+h(T) = 0$, its inverse is

$$h_k(t) = \begin{cases} \int_0^t e^{k(t-s)} f_k(s) ds, & k < 0, \\ -\int_t^T e^{k(t-s)} f_k(s) ds, & k > 0. \end{cases}$$

The Hardy projections are bounded on $L^p(S^1; V)$. On their nonzero modes the Poisson semigroup has bound $C_p e^{-cs}$: use the Poisson contraction for $s \leq 1$ and its absolutely convergent Fourier series for $s \geq 1$. Young's inequality in t gives an L^p inverse bound independent of T . Extending f by zero gives the decaying solution on the full cylinder; local elliptic estimates on unit charts then give the same uniform bound into $W^{1,p}$.

For $m = P_0 h$, one has

$$m(t) = m(0) + \int_0^t P_0 f(s) ds, \quad \|m - m(0)\|_{L^p(0,T)} \leq T \|P_0 f\|_{L^p(0,T)}.$$

This proves the estimate and the compatibility equation. Taking $m(t) = \sin(\pi t/T)v$, $v \neq 0$, gives $\|m\|_p/\|m'\|_p = T/\pi$. This field has zero end values and is approximable in $W^{1,p}$ by fields vanishing near the ends. $\square$

Equation (154) records the constant attaching mode, of real dimension $6g - 6$. It does not establish surjectivity of the evaluation map between the physical core and residual kernels.

The coefficient normal form in (25) describes the algebraic alternatives when a basepoint approaches the seam. Uniform analytic comparison at that degeneration does not follow from a compact family contained in the seam-admissible locus.

6.14. An expanding half-disc estimate.

Lemma 6.23 (comparison on an expanding half-disc). *Let $B_R^+ = \{|w| < R, \operatorname{Im} w > 0\}$ and let $\delta, M, \epsilon \geq 0$. Suppose $v \geq 0$ is continuous on its closure times Σ_g and smooth inside, with*

$$(\Delta_w + \rho^2 \Delta_{\Sigma_g})v \leq \epsilon.$$

Use scalar Laplacians with nonnegative spectrum. If $v \leq \delta$ on the diameter and $v \leq M$ on the semicircle, then, for $r < R/2$,

$$\sup_{B_r^+ \times \Sigma_g} v \leq \delta + CMr/R + \epsilon Rr/2.$$

Proof. The harmonic function

$$P_R(w) = \frac{2}{\pi} \operatorname{Im} \log \frac{R+w}{R-w}$$

is zero on the open diameter and one on the open semicircle. Its power series gives $0 \leq P_R(w) \leq Cr/R$ for $|w| \leq r < R/2$. Writing $y = \operatorname{Im} w$, the function $V_R = y(R-y)/2$ is nonnegative on the half-disc, vanishes on the diameter, and satisfies $\Delta_w V_R = 1$. Comparison with $\delta + MP_R + \epsilon V_R$ proves the result. The two corners are handled by exhaustion, or by the boundary upper limits of this barrier. $\square$

7. LOCAL GLUING ON REGULAR PRIMITIVE STRATA

7.1. Interpolation and regular attaching data.

Lemma 7.1 (local interpolation preserving the seam). *Near an irreducible flat surface connection α , the nonlinear matching correspondence admits smooth coordinates in which its equations are linear. Interpolating two sufficiently close smooth configurations in these coordinates, with the same cutoff on the two seam traces, preserves the matching condition exactly. This is a fixed-collar statement.*

Proof. Fix $k \geq 2$ and $p > 2$. Let $\mathcal{H}_\alpha^1$ be the harmonic one-forms, and let $V_\alpha = \operatorname{im} d_\alpha^*$ on two-forms, completed in $W^{k,p}$. Irreducibility gives $H_\alpha^0 = H_\alpha^2 = 0$. Thus $d_\alpha : V_\alpha \rightarrow W^{k-1,p}(\Lambda^2 \otimes \operatorname{ad} F)$ is invertible. The implicit-function theorem for $(h, v) \mapsto F_{\alpha+h+v}$ gives a smooth $\eta(h) \in V_\alpha$, with $\eta(0) = D\eta(0) = 0$, for which

$$\alpha(h) = \alpha + h + \eta(h), \quad h \in \mathcal{H}_\alpha^1,$$

is flat. Write $q(h) = [\alpha(h)]$. The local unitary slice theorem gives coordinates for all nearby connections by

$$\mathcal{C}(\chi, h, v) = \exp(\chi)^*(\alpha(h) + v), \quad v \in V_\alpha.$$

Its derivative at zero is the Hodge isomorphism $(\chi, h, v) \mapsto d_\alpha \chi + h + v$, up to the gauge-action sign. Uniqueness in the curvature implicit-function theorem says that this connection is flat exactly when $v = 0$. The matching pairs are therefore exactly $(\mathcal{C}(\chi, h, 0), q(h))$.

For a cutoff b , interpolate χ, h, v linearly on the negative collar and the q -coordinate on the positive collar. Their traces still satisfy $v = 0$ and equality of the two h coordinates. Interpolate the remaining connection components in the fixed trivialization. This constructs

the required smooth configuration. In any fixed strong norms in which these charts have bounded derivatives, its estimates follow from

$$d((1-b)z_0 + bz_1) = (1-b)dz_0 + b dz_1 + (db)(z_1 - z_0)$$

and Taylor's formula for the chart. Compact families of irreducible attaching connections are covered by finitely many such charts. No bound on the weak anisotropic trace spaces is asserted by this argument. $\square$

Let $x_0 = (A_0, u_0)$ be a regular index-zero residual mixed solution and let

$$e_0 : \mathbb{R} \longrightarrow M_g$$

be its evaluation along the matching line. The primitive bubble stratum over x_0 is

$$(155) \quad \mathcal{B}_g(x_0) = (\mathbb{R}_{e_0} \times_{\text{ev}\infty} \mathfrak{M}_g^{\text{sa}}) / G_{\mathbb{R}}.$$

When the fiber product is regular, the dimension calculation (38) gives

$$(156) \quad \dim_{\mathbb{R}} \mathcal{B}_g(x_0) = 3.$$

For $g \geq 3$ the generic fiber product is empty by (37); the construction below applies on every regular nonempty stratum before that final generic choice.

7.2. The local gluing statement. Fix a compact coordinate patch of the regular fiber product above. All primary perturbations are chosen in the summability class specified in Lemma 7.15; this choice is made before the regular residual data are selected. It retains all finite cylinder perturbations. The secondary perturbations and their dependency regions are compact and disjoint from a fixed matching collar, and the geometry there is the unperturbed Euclidean product. These are the support and metric conventions of [DFL26, Proposition 3.23(i)] and [DFL24, Section 1].

Theorem 7.2 (local gluing on regular primitive strata). *Let x_0 be a smooth regular index-zero residual mixed solution. Every regular seam-admissible primitive attaching datum admits a sufficiently small, relatively compact coordinate neighborhood U in the three-dimensional affine quotient of the attaching fiber product, a number $\rho_0 > 0$, and a smooth map*

$$\text{Glue}_{\text{loc}} : U \times (0, \rho_0) \longrightarrow \mathcal{M}_{\eta}(\alpha, \beta)_4$$

which is a local diffeomorphism at every parameter point. It is defined by exact solutions in the prescribed mixed topological component. In the complete graph norm (169) the correction to its seam-preserving pregluing is bounded by

$$C(\log(1/\rho))^{-37}.$$

The assertion is local on the chosen coordinate patch. It includes an interior rank-drop basepoint, and does not assert injectivity on a whole collar, exhaustion of nearby solutions, or convergence at a basepoint on the seam.

We prove this theorem in the following subsections. The construction first gives a geometric inverse, without a global Coulomb condition. Its kernel contains infinitesimal gauge transformations. The ordinary Fredholm tangent space is identified only after correcting the equation and passing to a smooth representative. Four local holonomy functions then give coordinates on that four-dimensional tangent space.

Put

$$(157) \quad T = \log(1/\rho), \quad \varepsilon = T^{-1}, \quad (R_-, R_0, R_2, R_1) = (T^4, T^8, T^{40}, T^{80}).$$

The roles of the four radii are different. The residual homogeneous region has physical radius comparable to ρR_0 . Its attaching profiles are replaced at radius ρR_- . The backgrounds are interpolated at radius ρR_2 . The cap inverse is localized inside radius $\rho R_1/4$. The definition of the graph norm retains this whole localized output in its cap region.

The cap metric of Lemma 6.11 is Euclidean for affine $|w| \leq R_1/2$. Write $f = f_{R_1}$, $M_1 \asymp R_1^2$, and choose the fixed derivative order J large enough for the calculations below. Its actual unitary background has

$$(158) \quad \delta \leq C_J \rho^2 R_1^{J+8}.$$

This includes one derivative in each of the finite bundle directions used later. Every fixed required order is available before choosing ρ_0 . In the compact coordinate $\zeta = 1/w$ the Euclidean factor is $f_\infty = |\zeta|^{-2}$, up to the fixed smooth coordinate factor. All normalized variations below retain the convention $\delta A_B = \rho a$.

Lemma 7.3 (based residual base gauge). *A smooth residual background can be put, on a fixed smaller matching half-disc, in a unitary gauge equal to the identity on the attaching surface and satisfying*

$$\delta_B A_B = 0, \quad A_B(n)|_{\text{seam}} = 0, \quad A_B(0) = 0.$$

In particular $A_B(z) = O(|z|)$, with all fixed surface derivatives. The gauge and its fixed-order bounds are uniform on the compact residual data and independent of ρ . Its artificial outer value is free.

Proof. Use tangential and normal coordinates (s, t) on this half-disc, with seam $t = 0$. The unperturbed scalar ASD row and surface flatness imply $F_B(s, 0) = 0$ on the whole seam segment. Radial parallel transport in the base, with value I at the attaching surface, gives

$$B_i(z) = \int_0^1 \tau z^j F_{ji}(\tau z) d\tau.$$

It follows that $B_s = B_t = 0$ and $\partial_t B_s = 0$ at $t = 0$. On a fixed small half-disc of radius d , the rescaled one-form $B'(Z) = dB(dZ)$ has $C^{1,\gamma}$ norm $O(d^2)$, also after fixed surface derivatives. Reflect B'_s evenly and B'_t oddly. The reflected coefficients are $C^{1,\gamma}$, which suffices for the following full-disc solve; a smooth reflection of every jet is not asserted.

For a small surface zero-form c , write $\eta = sc + h$, with h zero on the unit circle. The equation

$$\delta_B((B')^{\exp \eta}) = 0$$

has derivative Δ_D in h at zero. The Dirichlet implicit function theorem gives $h = h(B', c)$ in $C^{2,\gamma}$, smoothly with respect to the surface point and finite parameters. Reflection and uniqueness make η even, so its normal derivative vanishes at the seam. The tangential value of the transformed connection at zero has derivative I in c : at $B' = c = 0$, differentiation in c gives $\Delta(D_c h) = -\Delta(sc) = 0$ with zero boundary value, so $D_c h = 0$. A second implicit equation therefore sets this tangential value to zero. This uses the free harmonic outer value, not an extra condition on a fully fixed Dirichlet problem.

If $k = \exp \eta$, replace it by $k k(0, \cdot)^{-1}$. The second factor is base-constant and only conjugates the base coefficients. It preserves all three base conditions and makes the total gauge based at zero. Equivariance under unitary changes of surface frame makes the construction global on the surface bundle. On the original half-disc the coefficients are smooth; one-sided Neumann regularity for the elliptic gauge equation supplies every fixed derivative on a smaller half-disc. Undoing the fixed dilation by d and applying Taylor's theorem proves the stated

bounds. The local gauge may be extended through a larger fixed collar; all later constructions use the smaller region on which the displayed conditions hold. $\square$

Use this residual gauge and the based cap representatives. They have the same attaching surface connection and the same positive value. On $\rho R_2 \leq |z - c| \leq 2\rho R_2$, interpolate the negative surface and positive fields in the smooth matching coordinates of Lemma 7.1; interpolate the base connection affinely. Use the exact cap inside and the exact residual outside. This gives a smooth pregluing x_{pre} with exact nonlinear matching and zero normal variation condition. The entire interpolation lies in the Euclidean part of the cap and is disjoint from all perturbation outputs and dependencies. Its geometric equation error is supported on this single transition annulus.

7.3. The fixed residual inverse and surface derivatives. Use DFL's mixed-end weight $\omega > 0$, chosen in its allowed interval. The notation H_ω^s means positive exponential weight on the mixed ends only; the ordinary gauge and symplectic ends have weight zero [DFL26, Definition 3.8]. Let E_ω^1 be the actual augmented first-order domain, with zero normal trace, linear matching and outer Lagrangian conditions, and the finite critical kernel variables at the two mixed ends. Higher and fractional orders retain those same finite variables. Subtract their fixed smooth end profiles before taking a weighted norm and add the norm of their coefficients. We indicate this convention by the subscript as when a localized norm could otherwise be ambiguous.

Let $\mathcal{D}_0 = (d_{A_0}^*, D_0^{\text{geo}})$ be the augmented operator at the residual. Regularity and index zero make it an isomorphism $E_\omega^1 \rightarrow L_\omega^2$. The same inverse has $H_\omega^1 \rightarrow E_\omega^2$ bounds by the fixed mixed regularity theorem [DFL26, Theorem 4.12], [DFL24, Theorem 5.9]. The compact secondary perturbation preserves these statements. At a smooth background its linearization in local frames has the form

$$(159) \quad (Kv)(z) = \omega_0(z) \int_{S^1} C_1(t, z) v_{\gamma(t, z)} (\partial_t \gamma) C_2(t, z) dt.$$

Here $\gamma : S^1 \times D^4 \rightarrow X$ is a submersion, its output chart is an embedding, and ω_0 has compact support [DFL26, Section 6.2, equation (6.13)]. Cauchy–Schwarz in t and coarea give L^2 boundedness: the pushforward of a smooth compactly supported density is smooth and bounded, as seen by integration in submersion charts. Differentiating gives $H^j \rightarrow H^j$ bounds at each fixed order. The adjoint is a fiber integral in the same charts and has those bounds as well. Rellich on the full compact dependency set makes $K : E_\omega^1 \rightarrow L_\omega^2$ compact. The primary Hessians are included in the fixed DFL operator. Thus interpolating this single inverse gives

$$(160) \quad \|\mathcal{D}_0^{-1} F\|_{E_\omega^{1+\varepsilon}} \leq C \|F\|_{H_\omega^\varepsilon}, \quad 0 \leq \varepsilon \leq 1/4.$$

Extra surface regularity must preserve the finite end variables. On the fixed product collar put $\alpha_\theta = \beta(0, \theta)$. Define Γ_X^0 on negative base zero-form coefficients by $(d_{\alpha_\theta}^* d_{\alpha_\theta})^{1/2}$ and on surface one-forms by the square root of their covariant Hodge Laplacian. It kills harmonic surface forms, and is zero on the positive factor. Multiply it by a nonnegative normal cutoff $\chi(s)$ equal to one on a smaller collar, with support disjoint from all perturbation outputs and dependencies. The paired operator Γ_Y^0 uses the same surface Laplacians in the augmented target. Their surface principal symbols are $\chi(s)|\xi_\Sigma|$ times the identity on the negative factors. At the seam they preserve exact and coexact spaces and kill the harmonic and positive values together, so they preserve the matching and normal domain.

On a mixed end identify domain and target with DFL's cross-sectional Hilbert space. Let $P_\pm$ be the projection onto the smooth kernel of the limiting cross-sectional operator $D_\pm$. Correct the limiting surface operator to

$$(161) \quad \Gamma_\pm = (I - P_\pm)\Gamma_\pm^0(I - P_\pm) + P_\pm\Gamma_\pm^0P_\pm.$$

The difference is smooth finite rank. At finite θ first transport the limiting first-order domain to the actual one. The complex polar map for the nearby matching projectors does this in a seam collar; a finite-dimensional unitary does the same in a disjoint positive outer-boundary collar. Their composite U_θ tends exponentially to I and leaves the negative normal component unchanged. Use $P_\theta = U_\theta P_\pm U_\theta^{-1}$, whose range is in the actual domain, and subtract the off-diagonal blocks of Γ_θ^0 relative to this projection, with a cutoff equal to one on the far end. Make the same correction in the target. Adding a common scalar constant if necessary gives nonnegative self-adjoint operators Γ_X, Γ_Y . They preserve the actual first-order boundary domain and the finite asymptotic type. The projection P_θ is a transport of the limiting kernel, not a claim about the kernel of the varying operator.

Lemma 7.4 (fixed residual inverse in surface norms). *Fix $k \geq 2$. On the augmented target define*

$$(162) \quad \|F\|_{Y_{\varepsilon,k}^{\text{res}}} = \|F\|_{H_\omega^\varepsilon} + \sum_{j=1}^k \|\Gamma_Y^j F\|_{L_\omega^2} + \|\varphi F\|_{H_B^\varepsilon H_\Sigma^k; \omega},$$

where φ is a fixed collar cutoff supported strictly where $\chi = 1$ and equal to one near the whole matching line. Define

$$(163) \quad \begin{aligned} \|V\|_{X_{\varepsilon,k}^{\text{res}}} &= \|V\|_{E_\omega^{1+\varepsilon}} + \sum_{j=1}^k \|\Gamma_X^j V\|_{E_\omega^1} \\ &\quad + \|\varphi V\|_{H_B^{1+\varepsilon} H_\Sigma^k; \omega, \text{as}} + \|\varphi V\|_{H_B^\varepsilon H_\Sigma^{k+1}; \omega, \text{as}}. \end{aligned}$$

These are complete spaces on the actual domains. The augmented operator is an isomorphism between them, and

$$(164) \quad \|\mathcal{D}_0^{-1} F\|_{X_{\varepsilon,k}^{\text{res}}} \leq C_k \|F\|_{Y_{\varepsilon,k}^{\text{res}}}, \quad 0 < \varepsilon \leq 1/4.$$

For geometric sources insert a zero Coulomb row in (162). The resulting right inverse has its image in the residual Coulomb slice. The highest surface source order in this norm is in L^2 , not in an additional isotropic H^ε norm.

Proof. The paired commutator $B_1 = \mathcal{D}_0 \Gamma_X - \Gamma_Y \mathcal{D}_0$ is bounded $E_\omega^1 \rightarrow L_\omega^2$. The normal cutoff contributes $\chi' \Lambda_\Sigma$ of surface order one. A tangential derivative of Λ_{α_θ} has order zero because its principal symbol is independent of the connection. Covariant surface differentiation intertwines the flat zero- and one-form Laplacians; replacing the flat connection by the actual collar connection adds lower-order terms. The base principal complex structures commute with the surface Laplacians. These facts exclude a second surface derivative. Repeated paired commutators

$$B_{j+1} = B_j \Gamma_X - \Gamma_Y B_j$$

remain first-order bounded maps: the principal surface symbol is scalar and the normal-cutoff leading term commutes with the same fiber Laplacian. The finite-rank corrections have smooth vectors. On the ends their limiting commutators kill the critical kernel by

(161); the remaining coefficients decay. Thus the estimates include the finite end variables. Expanding $\mathcal{D}_0 \Gamma_X^j V$ and inducting with the fixed inverse proves

$$(165) \quad \sum_{j=0}^k \|\Gamma_X^j \mathcal{D}_0^{-1} F\|_{E_\omega^1} \leq C_k \sum_{j=0}^k \|\Gamma_Y^j F\|_{L_\omega^2}.$$

The weak differentiations in this induction can be justified by $R_\tau^X = (I + \tau \Gamma_X)^{-1}$ and its target counterpart. They preserve the actual domain and asymptotic type. They are uniformly bounded in E^1 : differentiating the uncorrected fiber resolvent in s gives $\tau R_\tau^0 \chi' \Lambda R_\tau^0$, bounded by $C\tau$ times the input's full first surface norm. For a bounded smooth finite-rank correction C , the identity $R_\tau = R_\tau^0 - \tau R_\tau^0 C R_\tau$ retains this bound because ΛC is bounded. This does not divide by χ where it vanishes. Tangential errors have order zero. The paired resolvent identity is

$$\mathcal{D}_0 R_\tau^X - R_\tau^Y \mathcal{D}_0 = -\tau R_\tau^Y B_1 R_\tau^X.$$

Consequently applying the fixed inverse to the equation for $\Gamma_X R_\tau^X V$ and passing by weak compactness proves the first differentiation; repetition proves (165). On a finite end kernel R_τ tends to $(I + \tau \Gamma_\pm)^{-1}$, and the difference on its smooth vectors decays by the same resolvent identity. Hence the finite end terms are included in the limiting argument.

On the wider collar where $\chi = 1$, (165) controls $H_B^1 H_\Sigma^k$ and $L_B^2 H_\Sigma^{k+1}$. Cover it by fixed small base patches of finite overlap, with uniform geometry on the ends. The polar matching transport reduces the local operator to its constant-seam full first-order model plus surface order-zero terms. The latter and the fixed cutoff commutators applied to V have $H_B^1 H_\Sigma^k$ bounds from the integer estimate. For the flat first-order model D_f , use the Bessel identity

$$(166) \quad v = D_f^* (D_f D_f^* + 1)^{-1} D_f v + (D_f^* D_f + 1)^{-1} v.$$

Its squares have Dirichlet, Neumann, and absolute blocks $\Delta + m^2$, including $m = 0$ in the harmonic sector. Even and odd reflection is bounded on unrestricted H^ε for $\varepsilon \leq 1/4$. Fourier transformation gives $H^{1+\varepsilon}$ and mH^ε bounds for the first term and the same bounds from L^2 for the second, uniformly in all surface modes. No source boundary trace is imposed. Localizing φV and summing gives

$$\begin{aligned} & \|\varphi V\|_{H_B^{1+\varepsilon} H_\Sigma^k; \omega, \text{as}} + \|\varphi V\|_{H_B^\varepsilon H_\Sigma^{k+1}; \omega, \text{as}} \\ & \leq C \left(\|\varphi F\|_{H_B^\varepsilon H_\Sigma^k; \omega} + \sum_{j=0}^k \|\Gamma_Y^j F\|_{L_\omega^2} \right). \end{aligned}$$

The derivatives of φ use the wider collar, not stronger fractional regularity of F on an unstated region. Finite end cutoff terms are smooth and controlled by their coefficients. Together with (160) this proves (164).

For completeness take a Cauchy sequence in either displayed norm. Its baseline Sobolev component and finite parameters converge. Testing against compact smooth fields identifies the limits of every Γ power and localized component with the corresponding distributional operator on that baseline limit. Subtract the finite profiles on the ends before this argument. First-order trace continuity preserves the closed boundary conditions. This proves completeness. The commutator expansion and localized derivative bounds also show that $\mathcal{D}_0$ is bounded between these spaces. Its original inverse therefore remains their actual inverse. $\square$

7.4. The complete preglued domain and source. Choose smooth target frames I_ρ , equal to the residual frames outside the gluing region. These are identifications of target bundles, not transports preserving a seam condition. Put

$$(167) \quad \|F\|_{Y_\rho} = \|I_\rho F\|_{Y_{\varepsilon,k}^{\text{res}}},$$

with zero Coulomb row inserted on the right. The source space is complete. On a shrinking attaching neighborhood the cutoffs χ and φ equal one and the finite end corrections vanish. Thus the local norm in (167) is equivalent to the physical $H_B^\varepsilon H_\Sigma^k$ norm. The isotropic H^ε term is controlled by its separate base and surface norms, and $k \geq 2$ supplies the required surface order.

Let θ_1 equal one for $|z - c| \leq \rho R_1/3$ and zero for $|z - c| \geq \rho R_1/2$. The pregluing is already the residual on the support of $1 - \theta_1$. The cap inverse localization is supported where $\theta_1 = 1$. For a physical variation define its cap conversion by

$$C_\rho(a_{\text{phys}}, b_{\text{phys}}, \nu_{\text{phys}}) = (f_\infty O_a a_{\text{phys}}, b_{\text{phys}}, O_\nu \nu_{\text{phys}}) \circ (c + \rho \Phi).$$

Here Φ maps the compact cap to the affine half-plane, and the fixed orthogonal coefficient rotations include the base conformal phase. The positive frames are smooth at infinity. The conversion is used only on the retained support, away from the compactifying point. Define the raw cap norm

$$(168) \quad \|V\|_{X_{\text{cap}}^{\text{raw}}} = \|(a/M_1, b, \nu)\|_{H_B^{1+\varepsilon} H_\Sigma^k} + \|(\rho d_\alpha a, \rho f_\infty d_\alpha b)\|_{H_B^\varepsilon H_\Sigma^k}.$$

The last derivative of b is exterior differentiation. This norm requires neither ordinary base Coulomb gauge nor point normalization. It imposes no full additional surface derivative on the exact part of b .

Let X_ρ be the space of actual distributional physical variations with the linear matching, normal and outer Lagrangian conditions at x_{pre} and norm

$$(169) \quad \|V\|_{X_\rho} = \|C_\rho(\theta_1 V)\|_{X_{\text{cap}}^{\text{raw}}} + \|(1 - \theta_1)V\|_{X_{\varepsilon,k}^{\text{res}}}.$$

Its finite end parameters are exactly those of its residual factor. In particular, no stronger residual norm is imposed on a cap inverse output whose support lies where $\theta_1 = 1$.

Lemma 7.5 (complete geometric domain). *The space X_ρ is complete for every fixed $\rho > 0$, and the geometric linearization $D_\rho : X_\rho \rightarrow Y_\rho$ is bounded. No global Coulomb equation is imposed on this domain.*

Proof. Realize a field by $v = C_\rho(\theta_1 V)$ and $w = (1 - \theta_1)V$. In the product of the two complete localized ambient graph spaces impose their closed support conditions and

$$(1 - \theta_1)C_\rho^{-1}v = \theta_1 w$$

as an equality of distributions. The inverse on this closed subspace is $V = C_\rho^{-1}v + w$. At fixed ρ all coefficient maps are smooth and finite on the retained supports, so convergence in the two graph norms passes through this distributional equality. The strong base order gives the seam and normal traces on the inner factor; the exterior has the ordinary residual traces. Their actual linear boundary conditions are closed. The finite end variables converge in the residual factor. This proves completeness.

Boundedness of D_ρ follows from its actual rows. The cap base derivatives are controlled by the strong norm; its surface terms are $\rho d_\alpha a$ and $\rho f_\infty d_\alpha b$ plus bounded coefficient terms. The exact part of b is not differentiated by an unstated surface divergence. On the residual factor use Lemma 7.4. The cutoff annulus is at a fixed positive size for this assertion and

both localizations fit the stated source. The geometric source inserts zero in the Coulomb row; no bound for a missing raw Coulomb equation is used. $\square$

7.5. Attaching profiles and intermediate cap annuli. The cap inverse is normalized at infinity in both the exact and harmonic surface directions. To join it to a residual output we must restore those directions by actual kernel fields. The exact directions are gauge directions.

Lemma 7.6 (cap attaching profiles). *Let $e \in E_\alpha H_\Sigma^k$ and $h \in \mathcal{H}_\alpha^1$. On the regular attaching patch there is a linear cap field $\mathcal{B}_\rho(e, h)$ in the geometric kernel, with ordinary base divergence and normal trace zero, whose surface and positive values at infinity are $e + h$ and $\ell_\alpha^{-1}h$. At the radii (157),*

$$\|\mathcal{B}_\rho(e, h)\|_X \leq C(\|e\|_{H_\Sigma^k} + |h|).$$

The exact summand is a pure gauge field; it is not a further holomorphic deformation parameter.

Proof. For a base-constant skew-Hermitian surface zero-form $\zeta \in H_\Sigma^{k+1}$ define

$$T_\rho \zeta = (\rho^{-1}[A_B, \zeta], d_\beta \zeta, 0).$$

Gauge covariance at the exact cap gives $\mathcal{A}_{\rho, f} T_\rho \zeta = 0$. Ordinary base divergence and the normal trace vanish because $\delta_B A_B = A_B(n) = 0$. At the seam $d_\beta \zeta$ is exact, so its coexact and harmonic parts vanish. Its attaching value is $d_\alpha \zeta$. The four terms of the cap graph norm are bounded, respectively, by

$$C\left(\frac{\delta}{\rho M_1}, 1, \delta, \rho M_1 L^\varepsilon\right) \|\zeta\|_{H_\Sigma^{k+1}}.$$

In the last term use flatness of α to write $d_\alpha d_\beta \zeta = d_\alpha[\beta - \alpha, \zeta]$. It differentiates ζ only once. Multiplication by f costs $CM_1 L^\varepsilon$. Thus the total coefficient $1 + \delta/(\rho M_1) + \delta + \rho M_1 L^\varepsilon$ is bounded at the stated radii. Take $\zeta = G_\alpha^0 d_\alpha^* e$. Add the genuine harmonic kernel lift from Proposition 6.17, already normalized in its exact attaching value. This proves the assertion.

The exact lift is also compatible with the nonlinear gauge action. For $g = \exp \zeta$, the actual normalized difference is

$$(\rho^{-1}(g^{-1}A_B g - A_B), g^{-1}\beta g + g^{-1}d_\Sigma g - \beta, 0).$$

Writing $\gamma = \beta - \alpha$ and $q = g^{-1}d_\alpha g$, its surface component is $q + g^{-1}\gamma g - \gamma$, with $d_\alpha q = -[q \wedge q]/2$. The surface algebra therefore gives quadratic difference bounds with the same bounded coefficient and no second derivative of ζ . The map $\zeta \mapsto E_\alpha(\alpha^{\exp \zeta} - \alpha)$ has invertible derivative d_α , so small exact attaching coordinates can be prescribed. The attaching flat class stays fixed; the harmonic projection of a finite nonlinear displacement need not vanish. $\square$

The localization estimate at the cap radius does not alone control the intermediate annuli on which the backgrounds differ. We give the additional estimate, keeping the surface mass arbitrary.

Lemma 7.7 (intermediate flat annuli). *Write $r_1 = R_1^{-1}$ and use a boundary coordinate $\zeta = s + it$ at infinity, with $t \geq 0$. Let $f = f_{R_1}$, so*

$$f(\zeta) \leq C(r_1^2 + |\zeta|^2)^{-1}, \quad f(\zeta) = c(\zeta)|\zeta|^{-2} \quad (|\zeta| \geq 2r_1),$$

where c is fixed smooth and positive. For a surface mass $m \geq 0$ suppose
(170)

$$\begin{aligned} a &= *d\psi, \quad y - z = m\psi, \quad (\Delta + m^2 f^2)\psi = fH - mf^2 z, \\ (\Delta + m^2 f^2)y &= \Delta z + mfH, \quad \psi|_{\partial B} = y|_{\partial B} = z|_{\partial B} = 0, \quad x(0) = 0, \quad H|_{D_{r_0}} = 0. \end{aligned}$$

Assume $x, z \in H^{2+\varepsilon}(D_{r_0/2})$ and set

$$B_m = \|H\|_{H^\varepsilon} + \|(x, z, y)\|_{H^{1+\varepsilon}} + r_0^3 \|(x, z)\|_{H^{2+\varepsilon}(D_{r_0/2})}, \quad A_m = C\varepsilon^{-2}r_0^{-8}B_m.$$

On a fixed half-annulus in the variable $Z = \zeta/r$, for $64r_1 \leq r \leq r_0/64$, uniformly in m ,

$$\begin{aligned} (171) \quad & \|\psi(r\cdot)\|_{H^{2+\varepsilon}} \leq A_m, \\ & \|(a/f, x, y)(r\cdot)\|_{H^{1+\varepsilon}} \leq CrA_m, \\ & \|(ma, mfy)(r\cdot)\|_{H^\varepsilon} \leq CA_m. \end{aligned}$$

The same estimates hold after square summation with the fixed surface Sobolev weights.

Proof. Put $A_m^0 = C\varepsilon^{-1}r_0^{-5}B_m$. The local $H^{2+\varepsilon}$ estimate, point normalization and zero trace give $|x(\zeta)| \leq A_m^0|\zeta|$ and $|z(s, t)| \leq A_m^0 t$. Kato comparison in the second equation of (170), on a fixed smooth subdomain of $D_{r_0/2}$, gives $|y(s, t)| \leq CA_m^0 t$ on a smaller half-disc. The comparison uses the global C^0 bound of y for its artificial boundary and the local $H^{2+\varepsilon}$ norm of z for the forcing. The Sobolev and Poisson constants are those of Proposition 6.18; the displayed $\varepsilon^{-1}r_0^{-5}$ dominates them. On an enlarged r annulus,

$$\|(x, z)(r\cdot)\|_{H^{2+\varepsilon}} \leq CrA_m^0.$$

Indeed the value and first derivative use the C^1 estimate; the second derivative has L^2 scaling r , and its fractional part has scaling $r^{1+\varepsilon}$.

First let $m \geq r/16$ and put $N = m/r$. On these annuli $g_r(Z) = r^2 f(rZ)$ and its inverse have uniformly bounded fixed derivatives. The equation for $y(rZ)$ is

$$(\Delta_Z + N^2 g_r^2)y(rZ) = r^2 \Delta z(rZ).$$

Its outer L^2 norm and right-hand H^ε norm are at most CrA_m^0 . Lemma 6.15 on interior or Dirichlet-seam patches yields

$$\|y(r\cdot)\|_{H^{2+\varepsilon}} + \|Ny(r\cdot)\|_{H^{1+\varepsilon}} + \|N^2 y(r\cdot)\|_{H^\varepsilon} \leq CrA_m^0.$$

For $1/16 \leq N \leq 1$ the same assertion is ordinary local regularity with bounded potential. Since $m^{-1} \leq 16/r$, $\psi = (y - z)/m$ has the required strong bound. The identities

$$ma(rZ) = r^{-1} * d_Z(y(rZ) - z(rZ)), \quad mf(rZ)y(rZ) = r^{-1}g_r(Z)Ny(rZ)$$

give the two weak bounds.

Now let $0 < m < r/16$. If $m \leq r_1$, Kato comparison in the first stream equation bounds $|\psi|$ by the Dirichlet Green potential of $f|H| + mf^2|z|$ on B . If $m > r_1$, perform this comparison on $B \setminus D_{4m}$. On the new semicircle $|\psi| = |y - z|/m \leq CA_m^0$, so its harmonic contribution is bounded by CA_m^0 . Domain monotonicity bounds the punctured Green function by G_B . No regularity at the artificial corners is needed for this weak scalar comparison.

Locally G_B is bounded by the half-plane kernel plus $Ct_\zeta t_\eta$, where

$$G_{\mathbb{H}}(\zeta, \eta) = \frac{1}{4\pi} \log \left(1 + \frac{4t_\zeta t_\eta}{|\zeta - \eta|^2} \right).$$

For $|\zeta| \asymp r$ and $|\eta| < r/4$ it is at most Ct_η/r . Consequently the inner contribution when $m \leq r_1$ is bounded by

$$C \frac{m}{r} A_m^0 \int_0^{r/4} \frac{s^3}{(r_1^2 + s^2)^2} ds \leq C \frac{m}{r} (1 + \log(r/r_1)) A_m^0 \leq C A_m^0.$$

When $m > r_1$, the lower limit is $4m$ and the bound is $C(m/r) \log(r/(4m))_+ A_m^0$. The comparable-radius part has an integrable logarithmic singularity and bound $C(m/r) A_m^0$. For $|\eta| > 4r$ in the local half-disc, $G_{\mathbb{H}} \leq Crt_\eta/|\eta|^2$, giving $Cmr \int_{4r}^\infty s^{-3} ds A_m^0 \leq C(m/r) A_m^0$. The smooth Green correction contributes at most $Cmr(1 + \log(r_0/r_1)) A_m^0$ in the first case and $Cmr(1 + \log(r_0/m)) A_m^0$ in the second. These are bounded because $x \log(1/x)$ is bounded. Finally, outside a fixed multiple of D_{r_0} , the bounds $G_B \leq Crr_0^{-2}$ and $f \leq Cr_0^{-2}$ give

$$C(rr_0^{-4} + mrr_0^{-6}) B_m.$$

All of H lies in this remote region. The same remote calculation handles $m = 0$. We have proved $\|\psi(r \cdot)\|_{C^0} \leq C\varepsilon^{-1} r_0^{-6} B_m$ on enlarged annuli.

There the equation is

$$(\Delta_Z + (m/r)^2 g_r^2) \psi(rZ) = -(m/r^2) g_r^2 z(rZ).$$

The potential is uniformly bounded and the right-hand side has H^ε norm at most $C(m/r) A_m^0$. Local interior or Dirichlet regularity gives the asserted $H^{2+\varepsilon}$ bound. The identity $y = z + m\psi$ gives the strong bound for y ; $ma = (m/r) * d_Z \psi$ and $mfy = (m/r^2) g_r y$ give the weak bounds. In both mass ranges $a/f = rg_r^{-1} * d_Z \psi$, proving its strong estimate. The loose factor in A_m includes all preceding restrictions.

For a finite set of surface modes, square the inequalities with each mode's own B_m . Parseval bounds the square sum of these data by the Hilbert-valued input norms. The mode-dependent puncture was only used for a uniform scalar bound; it is not a common operator domain. Completeness then permits the full spectral limit. $\square$

Lemma 7.8 (nonflat annular estimate). *Let $U = (a, b, \nu)$ solve $\mathcal{A}_{\rho, f} U = F$ on the cap, with both attaching values zero and $F = 0$ on D_{r_0} on both sides. Write $a = *d\psi$, with zero Dirichlet trace. For $64r_1 \leq r \leq r_0/64$ put*

$$(172) \quad \begin{aligned} Z_r(U) = & \|\psi(r \cdot)\|_{H_B^{2+\varepsilon} H_\Sigma^k} + r^{-1} \|(a/f, b, \nu)(r \cdot)\|_{H_B^{1+\varepsilon} H_\Sigma^k} \\ & + \|(\rho d_\alpha a, \rho f d_\alpha b)(r \cdot)\|_{H_B^\varepsilon H_\Sigma^k}. \end{aligned}$$

Then

$$(173) \quad Z_r(U) \leq C\varepsilon^{-2} r_0^{-8} (\|U\|_X + \|F\|_Y) + C\varepsilon^{-2} (\rho + \delta) R_1^6 \|U\|_X.$$

The norms are on fixed rescaled half-annuli. Fixed enlargements are understood in applications of local estimates.

Proof. Use the flat comparison and scalar-only correction W in the proof of Proposition 6.16. The data B_m of Lemma 7.7 have square sum at most $C(\|U\|_X + \|F\|_Y)$: the factor r_0^3 cancels that comparison's local r_0^{-3} bound. Its flat part therefore satisfies the first term of (173). The scalar correction has $x_W = z_W = 0$ and

$$\|W\|_X \leq CM_1 L^{2\varepsilon} (\rho + \delta) \|U\|_X, \quad \|\psi_W\|_{H^{2+\varepsilon}} \leq CM_1 \|W\|_X.$$

Its potential has zero trace. The C^1 estimate and rescaling give $\|\psi_W(r \cdot)\|_{H^{2+\varepsilon}} \leq C\varepsilon^{-1/2} r M_1 \|W\|_X$. Its strong surface term in Z_r is at most $Cr^{-2} \|W\|_X$, and each weak term

at most $Cr^{-1}\|W\|_X$. Since $r \geq r_1$, $M_1 \leq CR_1^2$ and $L \leq CR_1$, these fit the second term of the asserted bound.

The restored coexact correction S , its harmonic part h , and ν have zero attaching values and local $H^{2+\varepsilon}$ bounds $Cr_0^{-3}\|U\|_X$. Their strong annular norms are at most $C\varepsilon^{-1/2}rr_0^{-5}\|U\|_X$. Here $d_\alpha h = 0$, while the surface derivative gain of S bounds its weighted derivative by $C\rho r^{-3}\|U\|_X$. The remaining surface term is $T_\Sigma = \rho *_\Sigma [\beta - \alpha, \psi]$. Its global strong and weak norms are bounded by $C\rho M_1\|U\|_X$ and $C\rho M_1 L^\varepsilon\|U\|_X$. Restriction gives $C\rho r^{-2}M_1\|U\|_X$ and $C\rho r^{-1}M_1 L^\varepsilon\|U\|_X$ in Z_r . These also fit the printed R_1^6 loss. The restriction estimate used here is $\|v(r \cdot)\|_{H^{j+\varepsilon}} \leq Cr^{-1}\|v\|_{H^{j+\varepsilon}}$, $j = 0, 1$, on fixed enlargements. If $v(0) = 0$ and $v \in H^{2+\varepsilon}$, its strong rescaled norm instead has the extra factor $C\varepsilon^{-1/2}r$. These follow from the value, derivative and fractional scalings used in the preceding proof. All restored terms have thus been included. $\square$

The annular norm has a direct physical interpretation. At $|\zeta| \asymp r$ the physical base length is ρ/r , and the physical base coefficient is a/f . Both its strong $H_B^{1+\varepsilon}H_\Sigma^k$ norm and the weak exterior surface derivative norms of all negative coefficients are at most

$$(174) \quad C\rho^{-\varepsilon}r^{1+\varepsilon}Z_r(U).$$

For the weak rows the coefficient multiplier is r^2/ρ and the physical H^ε scale is $(\rho/r)^{1-\varepsilon}$. This controls $d_\alpha b$, not an additional full derivative of its exact component. The stream commutator has the same physical source bound. In fact the zero trace of ψ and the fixed-annulus Poincaré estimate give

$$\|\rho d_\alpha \psi(r \cdot)\|_{H_B^{1+\varepsilon}H_\Sigma^k} \leq CrZ_r(U).$$

Consequently $c = b - \rho *_\Sigma d_\beta \psi$ has rescaled strong norm at most $CrZ_r(U)$. The mixed commutator is $(\bar{\partial}\chi)c$, the positive one is $(\bar{\partial}\chi)\nu$, and the scalar one contains $f^{-1}\Delta\chi\psi$ and $f^{-1}d\chi d\psi$. Their rescaled source norms are bounded by $CZ_r(U)$; the extra base-connection term is $O(\delta r)\psi$. On dyadic annuli the physical bounds sum because $\sum r_j^{1+\varepsilon} \leq Cr_{\max}^{1+\varepsilon}$. Smooth annular cutoffs before extension control the fractional cross terms. The finitely many innermost annuli use (144).

7.6. Residual homogeneous fields and the inner cutoff.

Lemma 7.9 (residual homogeneous estimate). *On the fixed Euclidean product collar, let $V = (a, b, \nu)$ solve both the residual homogeneous geometric equation and full covariant Coulomb equation on $D_{2\ell}$, with actual linear matching and $a(n) = 0$ on the seam. Here a is a physical coefficient. There is no condition on the semicircular outer arc. Put $N_\ell = \ell\Lambda_\alpha$, with Λ_α the square root of the flat surface Laplacian, zero on the harmonic and positive factors. For each fixed j, k ,*

$$(175) \quad \sum_{a+b \leq j} \|N_\ell^a V(\ell \cdot)\|_{H_B^b H_\Sigma^k(D_1)} \leq C_{j,k} \|V(\ell \cdot)\|_{L_B^2 H_\Sigma^k(D_2)}.$$

If $B_\varepsilon = \|V\|_{H_B^{1+\varepsilon}H_\Sigma^k(D_d)}$ on a fixed collar neighborhood, the right-hand side is at most $C\varepsilon^{-1/2}B_\varepsilon$.

Proof. Use Lemma 7.3. After dilation by ℓ , the base connection is $O(\ell^2)$ and the surface connection differs from α by $O(\ell)$. Fold the positive side in a smooth complex frame based at the attaching value. The polar unitary for its matching projector and the constant projector at zero maps the constant seam condition to the actual one. Its difference from I , with fixed rescaled base derivatives, gains one surface derivative and is $O(\ell)$; this follows from (70).

Extend it normally constantly and transform the mixed target as well as the domain. Its commutation with the folded complex structure preserves the base principal part. Leave the base coefficients and the Coulomb and scalar targets unchanged. The transformed operator is $D_{0,\ell} + K_\ell$. On each nonharmonic surface mode, $D_{0,\ell}$ is

$$(a, x, y) \mapsto (\delta_Z a + mx, -\text{curl}_Z a + my, d_Z x + *d_Z y - ma), \quad m = \ell\lambda,$$

with $a(n) = 0$ and $y = 0$. On the finite harmonic and positive factors it is the constant Cauchy–Riemann transmission operator.

The remainder K_ℓ is surface order zero. A base derivative of the polar map is $O(\ell)$. Both $\ell d_\alpha(W - I)$ and $\ell(W^{-1} - I)d_\alpha$ are order zero and $O(\ell^2)$ because $W - I$ gains one derivative. The changes $d_\beta - d_\alpha$ and the rescaled base connection are also $O(\ell^2)$; positive frame terms are $O(\ell)$. The principal symbol of Λ_α is scalar, and finite blocks have smooth surface vectors. Thus, for fixed orders,

$$\|\partial_Z^b \text{ad}_{N_\ell}^a K_\ell\|_{H_\Sigma^k \rightarrow H_\Sigma^k} \leq C_{a,b,k}\ell.$$

These bounds also show that the polar maps preserve the norms in (175).

For a field v supported away from the outer arc and satisfying the constant seam condition, expansion of the flat first-order rows gives

$$\|d_Z v\|_2^2 + \|N_\ell v\|_2^2 = \|D_{0,\ell} v\|_2^2.$$

The mass boundary terms vanish by $a(n) = y = 0$; the base Hodge boundary term vanishes on the straight seam. On the harmonic part the symplectic pairing of the trace and its tangential derivative vanishes because both lie in the fixed Lagrangian subspace. This identity also holds with the surface Sobolev weights. A cutoff on nested half-discs now gives

$$\|dv\|_{L^2(D')} + \|N_\ell v\|_{L^2(D')} \leq C\|v\|_{L^2(D'')}.$$

Tangential differentiation and N_ℓ preserve the constant boundary domain and commute with $D_{0,\ell}$. Their differentiated equations have $-K_\ell$ times the differentiated field plus the lower derivatives multiplied by the bounded commutators above. Induction on nested discs adds one tangential, normal, or weighted surface derivative. The coefficient of the normal derivative in $D_{0,\ell}$ is invertible; solving the equation for that derivative then recovers every higher normal order. This proves the asserted integer estimates.

For weak fields, tangential difference quotients and $(I + \tau N_\ell)^{-1}$ justify the differentiations. This regularizer preserves the constant seam, and $[R_\tau, K_\ell] = -\tau R_\tau[N_\ell, K_\ell]R_\tau$ bounds its error. Cutoff energy estimates, weak compactness and the normal equation give the same induction. No higher-order compatibility of an arbitrary forcing is used. Finally the fixed-base embedding $H^{1+\varepsilon} \rightarrow C^0$ gives $\|V(\ell \cdot)\|_{L^2(D_2)} \leq C\varepsilon^{-1/2}B_\varepsilon$. $\square$

Lemma 7.10 (based residual variation and stream cutoff). *Let V be a residual inverse output homogeneous on $D_{2\ell}$, where $\ell \asymp \rho R_0$, and put $\mathfrak{b} = C\varepsilon^{-1/2}B_\varepsilon$. There is a compactly supported infinitesimal gauge change giving V^G with the same geometric source and attaching surface value. On $D_{\ell/4}$ it satisfies ordinary base divergence zero, normal trace zero, and $a^G(0) = 0$. Write*

$$e = E_\alpha b(0), \quad h = H_\alpha b(0), \quad P = P_e^E + P_h^H.$$

Here P_e^E is an exact geometric gauge field and P_h^H is a fixed smooth finite-dimensional matching profile with bounded source. The field $V^0 = V^G - P$ has all three attaching values zero

and can be written $a^0 = *d\psi$, $\psi|_{\text{seam}} = 0$. For $r = \rho R_-$, $q_0 = r/\ell$, and $\psi(\ell Z) = \ell\Psi(Z)$,

$$(176) \quad \begin{aligned} \|V^0(r\cdot)\|_{H_B^{1+\varepsilon}H_\Sigma^k} + \ell\|V^0(r\cdot)\|_{H_B^{1+\varepsilon}H_\Sigma^{k+1}} &\leq Cq_0\mathfrak{b}, \\ \|\Psi(q_0\cdot)\|_{H_B^{2+\varepsilon}H_\Sigma^k} + \ell\|\Psi(q_0\cdot)\|_{H_B^{2+\varepsilon}H_\Sigma^{k+1}} &\leq Cq_0^2\mathfrak{b}. \end{aligned}$$

Let χ_- vanish on D_r and equal one outside D_{2r} , and set $\mathcal{C}_-V^0 = (*d(\chi_- \psi), \chi_- b^0, \chi_- \nu^0)$. The commutator $E_- = D_0^{\text{geo}}\mathcal{C}_-V^0 - \chi_- D_0^{\text{geo}}V^0$ has cap source norm at most

$$(177) \quad \|E_-\|_{Y_{\text{cap}}} \leq CR_-^\varepsilon (R_-/R_0)\mathfrak{b}.$$

The separate inner profile term obeys

$$(178) \quad \|(1 - \chi_-)D_0^{\text{geo}}P_h^H\|_{Y_{\text{cap}}} \leq C\rho R_-^{1+\varepsilon}\mathfrak{b}.$$

These source norms include the physical-to-cap source multiplier.

Proof. The preceding lemma and the flat spectral gap give, for all fixed orders needed here,

$$\ell^q\|V(\ell\cdot)\|_{H_B^jH_\Sigma^{k+q}(D_1)} \leq C_{j,q}\mathfrak{b}.$$

Put $B(Z) = \ell A_B(\ell Z)$ and seek a physical gauge potential $\sigma(z) = \ell\xi(z/\ell)$. The transformed base coefficient is $a(\ell Z) + d_Z\xi + [B, \xi]$. Linearized flatness on the seam and the homogeneous scalar ASD row imply $\partial_T a_S = 0$ there; B_S satisfies the same condition. Reflect a_S, B_S evenly and a_T, B_T oddly. This has the $C^{1,\gamma}$ regularity needed for a full-disc Dirichlet solve. Write $\xi = Sc + h$ with $h = 0$ on the unit circle, and solve

$$\delta_Z(d_Z h + [B, h]) = -\delta_Z a(\ell Z) - \delta_Z[B, Sc].$$

The operator is a small first-order perturbation of Δ_D . Reflection and uniqueness give $\partial_T \xi = 0$. The map from c to the tangential value of $d_Z(Sc + h_c) + [B, Sc + h_c]$ at zero is $I + O(\ell^2)$. Choose c to cancel the attaching base value, then replace ξ by $\xi - \xi(0)$. Since $\delta B = 0$, $B(n) = 0$ and $B(0) = 0$, this preserves all the conditions. Subtraction as a whole surface section also preserves $b(0)$.

One-sided Neumann regularity on the original smooth half-disc, followed by surface differentiation, gives $\ell^q\|\xi\|_{H_B^{j+1}H_\Sigma^{k+q}} \leq C_{j,q}\mathfrak{b}$ on a smaller disc. Cut the potential off radially inside $D_{\ell/2}$, with cutoff one on $D_{\ell/4}$, and set $V^G = V + (d_A\sigma, 0)$. Its geometric source is unchanged by gauge covariance, including the perturbations. Normal trace is preserved. Rescaling the strong coefficients and using $q = 1, 2$ for their extra surface derivatives gives

$$(179) \quad \|V^G - V\|_{X_{\varepsilon,k}^{\text{res}}} \leq C\ell^{-\varepsilon}\mathfrak{b}.$$

The support lies where the collar cutoff is one. There the complete residual norm is controlled by the strong base and full extra surface norms used in this calculation. In particular the second surface derivative of the potential is estimated from the homogeneous output; it is not required of an arbitrary cap field.

With $\zeta = G_\alpha^0 d_\alpha^* e$, use $P_e^E = ([A_B, \zeta], d_\beta \zeta, 0)$ locally. Extending it by cutting off the potential gives a global exact gauge field. For $P_h^H = (0, b_h, \nu_h)$ extend $\ell_\alpha^{-1}h$ with constant coordinates in a smooth positive tangent frame, set $b_h = \ell_{\beta(s)}\nu_h$ on the seam, and extend smoothly into the negative side. A fixed cutoff gives an actual global matching field with

$$\|D_0^{\text{geo}}P_h^H\|_{C_B^jH_\Sigma^k} \leq C_{j,k}|h|.$$

It need not be homogeneous. Both profiles have zero base attaching value and preserve ordinary base divergence. Thus V^0 has all three values zero and $D_0^{\text{geo}}V^0 = -D_0^{\text{geo}}P_h^H$ on the local homogeneous region. All its fixed rescaled base and weighted surface norms remain

bounded by $C\mathfrak{b}$. The stream function exists because ordinary base divergence is zero; its seam trace is constant, taken to be zero. Its value and first base derivatives vanish at zero. Taylor's formula with the fixed higher rescaled estimates gives (176), also after surface differentiation.

The added stream coefficient $\psi * d\chi_-$ has rescaled size $Cq_0\mathfrak{b}$. Hence

$$\|\mathcal{C}_- V^0 - V^0\|_{X_{\text{phys}}} \leq Cr^{-\varepsilon} q_0 \mathfrak{b}.$$

The weak exterior derivative terms have an additional factor r/ℓ . In the scalar commutator the terms $\Delta\chi_- \psi$ and $d\chi_- d\psi$ have rescaled H^ε size $C\mathfrak{b}/\ell$. The base-connection bracket is smaller. In the mixed row use $(d\chi_-)(b^0 - *_\Sigma d_\beta \psi)$: b^0 is $O(q_0\mathfrak{b})$ and $d_\alpha \psi(r \cdot) = O(q_0^2 \mathfrak{b})$ by (176). The coefficient $\beta - \alpha = O(r)$ bounds its remainder. The positive row has the same estimate. Thus E_- has physical rescaled H^ε size $C\mathfrak{b}/\ell$. At this annulus $|\zeta| \asymp R_-^{-1}$, the cap source multiplier is $\rho f \asymp \rho R_-^2$, and the compact-base H^ε scale is $R_-^{\varepsilon-1}$. Their product proves (177).

The bounded source of P_h^H must be retained over the whole inner disc. On a compact-coordinate annulus of radius $v \geq R_-^{-1}$, its transformed rescaled coefficient has size $C\rho v^{-2}\mathfrak{b}$, with norm $C\rho v^{-1-\varepsilon}\mathfrak{b}$. The dyadic sum is dominated by $v \asymp R_-^{-1}$; the remaining fixed cap region contributes $C\rho\mathfrak{b}$. This proves (178). The true capped region at R_1^{-1} is not involved. $\square$

7.7. Conversion of complete sources and graph fields.

Lemma 7.11 (fractional coordinate conversion). *Let $z = c + \rho\Phi(\zeta)$ and $f_\infty = |d\Phi|_{g_B}$. For $R \leq c_1 R_1$ with $\rho R^2 \leq 1$, choose a cutoff χ_R zero for $|\zeta| \leq R^{-1}$ and one for $|\zeta| \geq 2R^{-1}$. The source conversions*

$$S_{\rho,R} F = \chi_R \rho f_\infty O F \circ (c + \rho\Phi), \quad T_{\rho,R} G = \left(\frac{\chi_R}{\rho f_\infty} O^{-1} G \right) \circ \Phi^{-1}((z - c)/\rho)$$

satisfy, for $0 \leq \varepsilon \leq 1/4$,

$$(180) \quad \|S_{\rho,R} F\|_{H_B^\varepsilon H_\Sigma^k} \leq CR^\varepsilon \|F\|_{H_{B,\text{phys}}^\varepsilon H_\Sigma^k}, \quad \|T_{\rho,R} G\|_{H_{B,\text{phys}}^\varepsilon H_\Sigma^k} \leq C\rho^{-\varepsilon} \|G\|_{H_B^\varepsilon H_\Sigma^k}.$$

For a physical field with strong $H_B^{1+\varepsilon} H_\Sigma^k$ coefficients and exterior surface derivatives in $H_B^\varepsilon H_\Sigma^k$, its cutoff cap conversion satisfies

$$(181) \quad \|C_\rho(\chi_R V)\|_{X_{\text{cap}}^{\text{raw}}} \leq C\varepsilon^{-1/2} R^\varepsilon \|V\|_{X_{\text{phys}}}.$$

Here cutoffs in the two coordinate systems are identified by pullback. The surface frame is the product frame and the positive frames are smooth up to the compactifying point. The source rotations O include the conformal phase. Inverse identities require nested cutoffs and data supported where those cutoffs are one.

Proof. The physical base volume is $\rho^2 f_\infty^2$ times the compact volume, so the uncut source maps are inverse L^2 isometries, up to the fixed bounded surface frames. Near infinity,

$$f_\infty \asymp |\zeta|^{-2}, \quad |d \log f_\infty| + |dO| \leq C|\zeta|^{-1}, \quad |df_\infty|/f_\infty^2 \leq C.$$

The forward H^1 estimate is

$$\|S_{\rho,R} F\|_{H^1} \leq C(R\|F\|_2 + \rho R^2\|F\|_{H^1}) \leq CR\|F\|_{H^1}.$$

For the reverse derivative, change volume first. The resulting compact L^2 expression is

$$\frac{1}{\rho f_\infty} (d(\chi_R O^{-1} G) - \chi_R O^{-1} G d \log f_\infty).$$

All coefficient factors divided by f_∞ are bounded, and $|d\chi_R|/f_\infty \leq C/R$. Hence its H^1 norm is at most $C\rho^{-1}\|G\|_{H^1}$, with no R loss. Interpolating the same maps with their L^2 bounds

proves (180). Fixed half-domain extensions are used at the geometric seam; zero extensions cross only the artificial boundaries where the smooth cutoffs vanish.

For the surface and positive coefficients the compact L^2 pullback norm is controlled by the physical C^0 norm, with constant $C\varepsilon^{-1/2}$. Derivatives of the pullback are the forward source transforms of physical base derivatives. On a compact annulus of radius v , conformal invariance of the two-dimensional Dirichlet integral gives

$$\|V(c + \rho\Phi(v\cdot))\|_{H^1} \leq C(\|V\|_{C^0} + \|d_B V\|_2) \leq C\varepsilon^{-1/2}\|V\|_{X_{\text{phys}}}.$$

For the cutoff derivative, its size v^{-1} cancels the L^2 scale v , leaving $v^{-\varepsilon}$. There is no rotating surface frame. Smooth cap and physical positive frames have derivative terms whose rescaled C^1 bound is $C(1 + \rho/v^2)$, bounded because $v \geq R^{-1}$ and $\rho R^2 \leq 1$. Their fractional contributions sum as $v^{1-\varepsilon}$. Thus no angular phase with derivative v^{-1} on every annulus is used for these strong coefficients.

The base coefficient is $f_\infty O_a a_{\text{phys}}/M_1$. Its differentiated physical field is a source transform with an additional factor f_∞/M_1 , bounded on the retained support. The endpoint calculation still gives constants C and CR . The derivatives of f_∞ , the cutoff and the base phase have annular H^ε bound

$$\frac{C\varepsilon^{-1/2}}{M_1 v^{2+\varepsilon}} \|V\|_{X_{\text{phys}}}.$$

Their geometric sum is at most $C\varepsilon^{-1/2} R^{2+\varepsilon} M_1^{-1} \|V\|$, which fits (181) since $R^2 \leq CM_1$. Finally the weak graph rows are source transforms of the physical exterior derivatives. A fixed smooth surface-dependent identification adds bounded zeroth-order terms with the same estimate. No surface divergence of b is introduced. $\square$

7.8. Matching transports and the geometric right inverse. On an overlap annulus $|\zeta| \asymp r$, $R_1^{-1} \lesssim r \lesssim R_1^{-1}$, put

$$(182) \quad q(r) = r + \rho/r + \delta, \quad \eta(r) = \delta + \rho^2/r^3.$$

In the fixed surface and positive frames, the cap, residual and preglued surface connections and positive values differ from the attaching values by $O(q(r))$. Their compact base coefficients and differences are $O(\eta(r))$. These bounds include every fixed rescaled base and surface derivative needed below. For the residual base coefficient this follows from $A_B(z) = O(|z|)$ and $z = c + \rho/\zeta$. The smooth interpolation has the same bounds by the chain rule, with affine interpolation in the base connection. For sufficiently small ρ , $\rho R_1^2 + \delta R_1 \leq c$, so $q(r) \leq Cr$.

Lemma 7.12 (paired matching transports). *There are paired domain and target maps T_{ji}^X, T_{ji}^Y between either model background i and the preglued background j , on the supports used below, with the following properties. They preserve the actual seam and negative normal condition, fix the base component and scalar target, and satisfy*

$$(183) \quad \|T_{ji}^X U\|_{X_{\text{cap}}^{\text{raw}}} \leq CR_1^\varepsilon \|U\|_{X_{\text{cap}}^{\text{raw}}}.$$

Their paired error $K_{ji} = \mathcal{A}_j T_{ji}^X - T_{ji}^Y \mathcal{A}_i$ is surface order zero and obeys

$$(184) \quad \|K_{ji} U(r\cdot)\|_{H_B^\varepsilon H_\Sigma^k} \leq C \|(a/f, b, \nu)(r\cdot)\|_{H_B^\varepsilon H_\Sigma^k}.$$

All assertions are on the retained Euclidean annuli. The maps are identity where the backgrounds agree.

Proof. Use the smooth local flat retraction of the surface-connection chart to define β^{flat} throughout each annulus. It equals β on the seam. In the fixed complex positive fiber V , extend the matching projector by

$$P = \begin{pmatrix} E_{\beta^{\text{flat}}} + \frac{1}{2}H_{\beta^{\text{flat}}} & \frac{1}{2}\ell_{\beta^{\text{flat}}} \\ \frac{1}{2}\ell_{\beta^{\text{flat}}}^* & \frac{1}{2}I \end{pmatrix}.$$

It is orthogonal and $JPJ^{-1} = I - P$ for the folded complex structure $J = (*_{\Sigma}, -J_V)$. Its interior extension is only an extension of the tangent projector; no interior matching is asserted. By (70), $P_j - P_i$ gains one surface derivative, with norm $Cq(r)$ and the same fixed rescaled derivative bounds, also one surface order lower. The finite blocks have smooth surface vectors. Set

$$A_{ji} = P_j P_i + (I - P_j)(I - P_i), \quad W_{ji} = A_{ji}(I - (P_j - P_i)^2)^{-1/2}.$$

The convergent power series and the projector identities show that W is unitary, commutes with J , and maps $\text{ran } P_i$ to $\text{ran } P_j$. Every term of $W - I$ contains $P_j - P_i$, so it has the same one-derivative gain and $Cq(r)$ bound. Apply W to (b, ν) in the domain and to the folded mixed and positive target rows, leaving the other components fixed. The cap map is used on the localized support $|w| \leq R_1/4$; its logarithm can be cut to zero nearer infinity, where that field vanishes. The residual map is used only for $|w| \geq R_-$, and is similarly extended where its input vanishes. These extensions remain complex unitary and have the same derivative bounds on their relevant supports.

The first compact base derivatives of W are bounded by C , and the second by CR_1 , since $q/r \leq C$ and $q/r^2 \leq CR_1$. Integer multiplier estimates and interpolation give $\|Wv\|_{H_B^{1+\varepsilon} H_{\Sigma}^k} \leq CR_1^{\varepsilon} \|v\|_{H_B^{1+\varepsilon} H_{\Sigma}^k}$. On H_B^{ε} the multiplier norm is C . The extra weak surface graph term contains $\rho f d_{\alpha} \pi_{\Sigma}(W - I)(b, \nu)$, an order-zero operator of size $C\rho/r$ and first compact base derivative $C\rho/r^2$. Both are bounded. This proves (183) without a surface divergence of b .

The normalized principal geometric rows on these annuli are

$$-f^{-1} \text{curl}_B a - \rho f *_{\Sigma} d_{\beta_i} b, \quad \bar{\partial}_B b - \rho \bar{\partial}_{\beta_i} a + [A_{B,i}^{0,1}, b], \quad \bar{\partial}_B \nu + \Omega_i \nu,$$

with the scalar bracket $f^{-1} * [A_{B,i} \wedge a]$ included. The positive connection coefficient Ω_i is $O(q/r)$. Pairing the domain and target maps cancels derivatives of the input in the base principal part. In the scalar row the surface difference has order zero and size $C\rho q/r^2$ on (b, ν) ; the base bracket difference has size $C\eta$ on a/f . The mixed rows contain $\bar{\partial}W$, of size Cq/r , base brackets of size $C\eta$, and $\rho(d_{\beta_j} - Wd_{\beta_i})a$. The last term is surface order zero because $(W - I)d_{\alpha}$ is order zero, using the gain one order lower; its size on a/f is $C\rho q/r^2$. Thus the right-hand coefficient before simplification is $C(q/r + \eta + \rho q/r^2)$, bounded under the stated assumptions. It and its first rescaled derivative are bounded, which proves (184) by interpolation. $\square$

Two consequences will be used in the construction. For the localized zero-attaching cap output $V = \mathcal{C}_1 U$, set

$$Z_* = C\varepsilon^{-2} R_0^8 (\|U\|_X + \|F_{\text{cap}}\|_Y) + C\varepsilon^{-2} (\rho + \delta) R_1^6 \|U\|_X.$$

Lemma 7.8 gives $\|(a_V/f, b_V, \nu_V)(r \cdot)\|_{H^{\varepsilon}} \leq CrZ_*$. On the finitely many retained annuli adjacent to r_1 , the rescaled proof of (144) gives the same bound, with the same restored terms. In particular the extra stream term is $f^{-1}\psi * d\chi = O(r_1)\psi(r_1 \cdot)$. The cap-to-preglued

difference is supported in $r \lesssim R_2^{-1}$. Using a smooth annular partition and summing $r^{2-\varepsilon}$ in (184), then using (180), gives

$$(185) \quad \|K_{\text{pre,cap}} V\|_{Y_\rho} \leq C \rho^{-\varepsilon} R_2^{-2+\varepsilon} Z_*.$$

For a residual field whose physical coefficient triple has rescaled $H_B^1 H_\Sigma^k$ bound $\mathfrak{b}$ on the relevant annuli, the corresponding sum is instead $\sum r^{1-\varepsilon}$, giving

$$(186) \quad \|K_{\text{pre,res}} V\|_{Y_\rho} \leq C \rho^{-\varepsilon} R_-^{-1+\varepsilon} \mathfrak{b}.$$

Both sums are uniform as ε decreases. The rescaled H^1 bound for an unmodified residual field follows from its fixed-base C^0 bound and conformal invariance of the Dirichlet integral, and is at most $C\varepsilon^{-1/2} \|V\|_{X_{\varepsilon,k}^{\text{res}}}$. Annular cutoffs are applied before extension, so these estimates are in the complete fractional source spaces.

Proposition 7.13 (geometric inverse on the complete source). *At the radii (157), the actual geometric linearization $D_\rho : X_\rho \rightarrow Y_\rho$ admits a right inverse*

$$(187) \quad D_\rho Q_\rho = I, \quad \|Q_\rho\| \leq CT^2.$$

The finite residual end parameters are retained. This is an inverse for geometric sources on the domain without a global Coulomb equation.

Proof. Let θ_0 be one on $|z - c| \leq \rho R_0$ and zero outside $|z - c| \leq 2\rho R_0$. For arbitrary $F \in Y_\rho$ put $F_{\text{in}} = \theta_0 F$ and $F_{\text{out}} = (1 - \theta_0)F$. Convert the inner source to G_{cap} , treating partition and conversion as the single cutoff source map of Lemma 7.11. Then

$$\|G_{\text{cap}}\|_{Y_{\text{cap}}} \leq CR_0^\varepsilon \|F\|_{Y_\rho}.$$

It vanishes near infinity on a half-disc of radius comparable to R_0^{-1} . The backgrounds agree on its support. Define $G_{\text{res}} = (T_{\text{pre,res}}^Y)^{-1} F_{\text{out}}$, with the zero Coulomb row inserted. This is a bounded source map. Indeed the source partition has norm $C(\rho R_0)^{-\varepsilon}$. On the outer source support the target map's difference from identity has physical coefficient size C/R_0 and first base derivative $C/(\rho R_0^2)$. Extend that difference by a cutoff inside the vanishing source region. Interpolation gives

$$\|((T^Y)^{-1} - I)(1 - \theta_0)F\|_{Y_{\varepsilon,k}^{\text{res}}} \leq C \rho^{-\varepsilon} R_0^{-1-\varepsilon} \|F\|_{Y_\rho}.$$

The integer surface source terms add no base derivative. The local product norm controls the lower isotropic term, proving $\|G_{\text{res}}\| \leq C\|F\|$. This source is zero on $D_{\rho R_0}$.

For the cap block take $U_{\text{cap}} = Q^0 G_{\text{cap}}$ with both attaching values zero, and localize by $\mathcal{C}_1$. After conversion and paired transport, call its output $P_{\text{cap}} F$. It is supported where $\theta_1 = 1$. Set

$$L_{\text{cut}} = C (\varepsilon^{-3} R_0^8 R_1^{-1+2\varepsilon} + \varepsilon^{-1} (\rho + \delta) R_1^{6+\varepsilon}).$$

The inverse and localization estimates, together with (185), give

$$(188) \quad \|D_\rho P_{\text{cap}} F - F_{\text{in}}\|_{Y_\rho} \leq C \rho^{-\varepsilon} (L_{\text{cut}} \|G_{\text{cap}}\|_Y + R_2^{-2+\varepsilon} Z_*).$$

Once the displayed localization error is small, its domain norm is at most $C\varepsilon^{-1} R_1^{2\varepsilon} R_0^\varepsilon \|F\|$. This smallness is checked below. No condition on the support of the new error is imposed on the eventual inverse.

Apply the fixed residual inverse to G_{res} and call the output U . Choose $\ell = c_0 \rho R_0$ so that $D_{2\ell}$ is contained in the homogeneous region. Lemma 7.10 applies with $\mathfrak{b} = C\varepsilon^{-1/2} \|F\|$ and gives U^G , the residual profiles $P = P_e^E + P_h^H$, and the stream function of $U^0 = U^G - P$.

Let $B = \mathcal{B}_\rho(e, h)$ be the true cap kernel field. Write $E = \|e\|_{H^k} + |h| \leq C\mathfrak{b}$, $r = \rho R_-$, and $\chi = \chi_-$. On the cutoff annulus put

$$V = \chi U^G + (\psi * d\chi, 0, 0) = \mathcal{C}_- U^0 + \chi P, \quad P_{\text{res}} F = T_{\text{pre, res}}^X V + (1 - \chi) B.$$

Inside D_r define it to be B , and outside D_{2r} it is the transported U^G . These definitions agree where the cutoffs are constant. The residual transport is used only where its input vanishes towards the inner core. The cap profile is supported where the pregluing is exactly the cap. Thus the actual seam, normal condition and all end parameters are preserved.

We record the source identity explicitly. Since $D_0^{\text{geo}} U^0 = G_{\text{res}} - D_0^{\text{geo}} P$ and $\chi G_{\text{res}} = G_{\text{res}}$,

$$D_0^{\text{geo}} V = G_{\text{res}} + E_- + [D_0^{\text{geo}}, \chi] P.$$

The terms $\chi D_0^{\text{geo}} P$ cancel, including the possibly nonzero source of P_h^H . The paired base symbols imply $T^Y [D_0^{\text{geo}}, \chi] P = [D_\rho, \chi] T^X P$. Since $D_\rho B = 0$ on the support of $1 - \chi$, we obtain

$$(189) \quad D_\rho P_{\text{res}} F - F_{\text{out}} = T^Y E_- + K_{\text{pre, res}} V + [D_\rho, \chi] (T^X P - B).$$

The first term is the stream commutator of (177), not the difference relative to the uncut source. The source of the harmonic residual profile has been canceled, not discarded.

The field V has rescaled physical $H_B^1 H_\Sigma^k$ size $C\mathfrak{b}$ on all relevant annuli. This holds for U by its C^0 bound and Dirichlet integral. For its gauge correction, the fixed rescaled homogeneous estimates at scale ℓ , including one extra surface order, give the same bound on smaller annuli. The extra stream coefficient on the r annulus has strong rescaled size $Cq_0\mathfrak{b}$ by (176). Hence (186) applies without another point-evaluation loss.

At the r annulus write $r_- = R_-^{-1}$ for its compact radius. The cap profile differs from its surface and positive attaching values by $Cr_- E$, and its physical base coefficient has size $C(\delta/\rho)r_-^2 E$. For the exact part this is the formula in Lemma 7.6; for the harmonic part it is the finite parameter bound (140). The residual profiles differ from their attaching values by $C(\rho/r_-)E$ and have base coefficient of that same size. The transport adds $Cq(r_-)E$ in the surface and positive fields. All these bounds hold in rescaled $H^{1+\varepsilon} H_\Sigma^k$. The commutator $[D_\rho, \chi]$ is $O(r_-^{-1})$ on the surface and positive coefficients and on the physical base coefficient. Compact scaling and source conversion therefore give

$$\|[D_\rho, \chi] (T^X P - B)\|_{Y_\rho} \leq C\rho^{-\varepsilon} \left(R_-^{-1+\varepsilon} + \rho R_-^{1+\varepsilon} + \frac{\delta}{\rho} R_-^{-2+\varepsilon} \right) E.$$

The intermediate δR_-^ε term is absorbed using $\delta R_1 \leq c$. Combining these estimates yields

$$(190) \quad \|D_\rho P_{\text{res}} F - F_{\text{out}}\|_{Y_\rho} \leq C\rho^{-\varepsilon} \varepsilon^{-1/2} \left(R_-^\varepsilon \frac{R_-}{R_0} + R_-^{-1+\varepsilon} + \rho R_-^{1+\varepsilon} + \frac{\delta}{\rho} R_-^{-2+\varepsilon} \right) \|F\|_{Y_\rho}.$$

We also check the complete domain bound. The outer factor of X_ρ sees only $(1 - \theta_1)U$, with norm at most $C\varepsilon^{-1/2}(\rho R_1)^{-\varepsilon} \|F\|$. On the raw cap factor, convert $\theta_1 U^G$ using (181), cut by χ , and apply (183). A scalar strong cutoff at radius R_-^{-1} has norm $C\varepsilon^{-1/2} R_-^\varepsilon$; its weak multiplier norm is smaller. Together with (179), this contribution is at most $C\varepsilon^{-3/2} \ell^{-\varepsilon} R_1^{2\varepsilon} R_-^\varepsilon \|F\|$, plus a lower-order term. The extra stream coefficient has strong raw norm at most

$$C \frac{R_-^2}{M_1} R_-^\varepsilon q_0 \mathfrak{b} \leq C R_-^\varepsilon q_0 \mathfrak{b},$$

and weak exterior derivative norm at most $C R_-^\varepsilon q_0^2 \mathfrak{b}$. For the latter, multiply its rescaled full surface derivative $q_0 \mathfrak{b}/\ell$ by ρR_-^2 and by the compact H^ε scale $R_-^{\varepsilon-1}$. The retained cap profile

has norm $C\varepsilon^{-1/2}R_-^\varepsilon E$ and lies entirely where $\theta_1 = 1$. Thus

$$(191) \quad \|P_{\text{res}}F\|_{X_\rho} \leq C\varepsilon^{-2}\rho^{-2\varepsilon}R_1^{4\varepsilon}\|F\|_{Y_\rho}.$$

Every operation was defined for arbitrary complete sources using the weak homogeneous estimates; no density assertion for a restricted source class is needed.

Set $P_\rho = P_{\text{cap}} + P_{\text{res}}$. To check the powers, write $R_- = T^a, R_0 = T^b, R_2 = T^c, R_1 = T^d$ temporarily. The nonexponential cap errors in (188) have exponents $3+8b-d$ and $3+8b-2c$. The residual errors have exponents $1/2+a-b$ and $1/2-a$. With $(a, b, c, d) = (4, 8, 40, 80)$ these are $-13, -13, -7/2, -7/2$. All factors $R_j^{C\varepsilon}$ and $\rho^{-C\varepsilon}$ are bounded. The fixed-order reconstruction estimate (158) makes every term containing ρ or δ/ρ exponentially small times a fixed power of T . It also verifies all inverse criteria and support separations. Therefore

$$\|D_\rho P_\rho - I\|_{Y_\rho \rightarrow Y_\rho} \leq CT^{-7/2} + Ce^{-T}T^{C_J} < \frac{1}{2}, \quad \|P_\rho\| \leq CT^2.$$

The Neumann series on the complete source space defines $Q_\rho = P_\rho(D_\rho P_\rho)^{-1}$ and proves (187). $\square$

7.9. The error of the exact-matching pregluing.

Lemma 7.14 (pregluing error). *Let $\mathcal{F}$ denote the geometric equation, with the target identifications defining Y_ρ . Its pregluing error satisfies*

$$(192) \quad \begin{aligned} e_\rho &:= \|\mathcal{F}(x_{\text{pre}})\|_{Y_\rho} \leq C\rho^{-\varepsilon} \left(R_2^{-1+\varepsilon} + \rho R_2^{1+\varepsilon} + \frac{\delta}{\rho} R_2^{-2+\varepsilon} \right) \\ &\leq CT^{-40} + Ce^{-T}T^{C_J}. \end{aligned}$$

It is supported on the single interpolation annulus.

Proof. Both backgrounds solve the exact same physical equations where they are used, and all perturbation outputs and dependencies are disjoint from the modification. Put $v = R_2^{-1}$ and rescale the transition by $\zeta = vZ$. The surface and positive differences have size $q = v + \rho/v + \delta \leq Cv$, and the compact base connection difference has size $\eta = \delta + \rho^2/v^3$. These are background coefficients A_B , not the normalized variation a . Interpolation in the smooth matching chart differs from affine interpolation of the surface fields by a remainder S of size Cq^2 in all fixed rescaled base and surface orders. Taylor’s integral formula contains two factors of the coordinate difference; differentiating retains two such factors because their fixed derivatives also have size $O(q)$. The positive interpolation has the same quadratic property.

The normalized scalar background equation is

$$(\rho f)^{-1} *_B F_B + \rho f *_\Sigma F_\beta.$$

Subtract the affine combination of the two exact equations. The base cutoff curvature has size $C\eta/v$, its quadratic bracket has size $C\eta^2$, and the surface curvature and chart terms have size Cq^2 . Thus the normalized rescaled scalar source is at most

$$C \left(\frac{v}{\rho} \eta + \frac{v^2}{\rho} \eta^2 + \frac{\rho}{v^2} q^2 \right) \leq C \left(\frac{\delta}{\rho} v + \frac{\rho}{v^2} \right).$$

For the quadratic base terms this uses $\delta^2 v^2/\rho$, $2\delta\rho/v$ and ρ^3/v^4 , absorbed using $\delta v \leq c$ and $\rho/v \leq c$. The surface term is $O(\rho)$. Compact H^ε scaling multiplies by $v^{1-\varepsilon}$ and gives $C((\delta/\rho)v^{2-\varepsilon} + \rho v^{-1-\varepsilon})$.

The compact mixed background equation is $\bar{\partial}_B \beta - \bar{\partial}_\beta A_B^{0,1} = 0$, with no additional ρ factor. Its affine cutoff term is Cq/v , its bracket difference is $C\eta q$, and the chart terms are $Cq^2/v + C\eta q^2$. The positive equation in a fixed smooth coordinate chart has the same leading

cutoff term and remainder Cq^2/v . Since $q \leq Cv$ and η is bounded, these rows have rescaled source norm C , hence compact norm $Cv^{1-\varepsilon}$. One rescaled derivative has the same bounds, so the fractional estimates are uniform by interpolation. A smooth cutoff before extension includes the fractional cross terms. Finally (180) contributes only $C\rho^{-\varepsilon}$. Substitution of the four radii and (158) proves the result. $\square$

7.10. Holonomy at the fractional graph order. We specify the primary perturbation class before selecting the regular residual solution. Enumerate DFL's cylinder functions by h_i , with original weights C_i . Let D_i dominate the fixed finite-family constants in the fractional estimates proved below. For each m , let $E_{i,m}$ dominate the defining loop, frame, coarea and class-function constants needed through order m on $W^{m,6}$, and through order three in the coarea exponents between three and six. Smooth-background bounds can be factored out of these constants. Set

$$(193) \quad C_i^\infty = \max \left(C_i, D_i, 2^i \max_{1 \leq m \leq i} E_{i,m} \right), \quad \sum_i C_i^\infty |a_i| < \infty.$$

For every fixed m the tail constants obey $E_{i,m} \leq 2^{-i} C_i^\infty$, and the head is finite. Thus the original DFL estimates and every fixed required higher estimate converge. This is an explicit strengthened choice of weights, not a consequence inferred from unspecified weights in [DFL26, Proposition 6.1]. Every finite cylinder direction remains available and their span is dense. DFL's cokernel-separation argument uses those finite directions, so its universal surjectivity and Sard–Smale argument apply in this weighted space as well. Finite cylinder additions preserve the condition.

Lemma 7.15 (fractional holonomy estimate). *For the finite secondary perturbations and primary series (193), at a fixed smooth background set*

$$R(v) = \mathcal{H}(A_0 + v) - \mathcal{H}(A_0) - D\mathcal{H}(A_0)v.$$

For compactly supported or weighted-decaying variations,

$$(194) \quad \|R(v) - R(w)\|_{Y_{\varepsilon,k}^{\text{res}}} \leq C\varepsilon^{-1} (\|v\|_{H_\omega^{1+\varepsilon}} + \|w\|_{H_\omega^{1+\varepsilon}}) \|v - w\|_{H_\omega^{1+\varepsilon}}.$$

For each fixed positive $\varepsilon \leq 1/4$, the perturbation map is C^1 from this connection space to the full residual source space. Constant finite end parameters are incorporated by exact end families, as in the next subsection.

Proof. For a secondary family $\gamma : S^1 \times D^4 \rightarrow X$, define $(Pv)(z)(t) = v_{\gamma(t,z)}(\partial_t \gamma)$, including a compact auxiliary output cutoff. In local bundle frames its path space is $\mathcal{E} = L^1(S^1; \mathfrak{su}(2))$; a smooth gluing matrix at the endpoints is included in the output function. Hölder in t and coarea for the full submersion give

$$\|Pv\|_{L_z^4 \mathcal{E}} \leq C\|v\|_{L_X^4}, \quad \|Pv\|_{W_z^{1,4} \mathcal{E}} \leq C\|v\|_{W_X^{1,4}}.$$

Differentiation hits v or fixed smooth loop coefficients. In particular $P : H^1 \rightarrow L^4 \mathcal{E}$ and $P : H^2 \rightarrow W^{1,4} \mathcal{E}$. Fubini gives an L^1 path for almost every output point. Coarea also makes the definition independent of the representative of v . No trace on each individual rough loop is asserted.

For a skew-Hermitian L^1 path a , parallel transport is the unique absolutely continuous solution of its integral equation and is unitary. Ordered Duhamel integrals, first for smooth

paths and then by L^1 approximation, give

$$\|D^m \text{Hol}(a)[h_1, \dots, h_m]\| \leq C_m \prod_j \|h_j\|_{L^1}.$$

The constants are independent of a . Smooth functions of unitary holonomy have the same fixed-order bounds by compactness of the unitary group. Write the local output as $f_z(a_0(z) + Pv(z))$. For $C_z(u) = Df_z(a_0(z) + u) - Df_z(a_0(z))$, the first three path derivatives give

$$\|C_z(u)\| \leq C\|u\|_{\mathcal{E}}, \quad \|C_x(u) - C_y(v)\| \leq C\|u - v\|_{\mathcal{E}} + C\|x - y\|\|v\|_{\mathcal{E}}.$$

The last term retains a factor of the variation because it is a difference of derivatives at the varying and reference paths.

For $\mathcal{E}$ -valued parameter functions use the raw difference seminorm

$$[g]_{\varepsilon,4,2}^2 = \int_{|h|<1} |h|^{-4-2\varepsilon} \|g(\cdot + h) - g\|_{L_z^4 \mathcal{E}}^2 dh.$$

For four-dimensional input frequency pieces v_j , the endpoint bounds give $\|Pv_j\|_{L^4} \leq C2^j\|v_j\|_2$ and $\|\nabla Pv_j\|_{L^4} \leq C2^{2j}\|v_j\|_2$. At $|h| \asymp 2^{-l}$, with $b_j = 2^{(1+\varepsilon)j}\|v_j\|_2$,

$$2^{\varepsilon l} \|\Delta_h Pv\|_{L^4} \leq C \sum_{j \leq l} 2^{-(1-\varepsilon)(l-j)} b_j + C \sum_{j > l} 2^{-\varepsilon(j-l)} b_j.$$

The convolution kernels have ℓ^1 norms C and $C\varepsilon^{-1}$. Young's inequality and the input Hilbert Sobolev norm give

$$\|Pv\|_{L^4 \mathcal{E}} + [Pv]_{\varepsilon,4,2} \leq C\varepsilon^{-1} \|v\|_{H^{1+\varepsilon}}.$$

This uses only norms and the triangle inequality in the path space. For the finite-dimensional output, Fourier integration of $|e^{i\xi \cdot h} - 1|^2 |h|^{-4-2\varepsilon}$ shows that $\|r\|_{H^\varepsilon} \leq C(\|r\|_2 + [r]_{\varepsilon,2,2})$ uniformly. Indeed adding the L^2 term to the lower bound $c(|\xi|^{2\varepsilon} - 1)/(2\varepsilon)$ controls the Bessel multiplier uniformly; low frequencies are covered by L^2 .

Taylor's formula writes the remainder difference as

$$R(v) - R(w) = \int_0^1 C_z(Pw + tP(v-w))P(v-w) dt.$$

In a translation difference of this expression one factor has a fractional difference and the other has only its L^4 norm. Hölder with exponents four and four therefore pays the ε^{-1} factor once. The explicit coefficient difference adds $|h|$ times a product, whose difference integral is bounded because $\varepsilon < 1$. This proves the unweighted estimate.

For a primary gradient the actual cylinder functions are smooth functions of integrated holonomy traces for immersed $\gamma_i : S^1 \times D^2 \rightarrow Y_0$ [DFL26, Section 6.1]. The gradient contains a local loop gradient multiplied by coefficients depending on all integrated traces. At an output point $\gamma_i(t, z)$, use the shifted whole loop $\tau \mapsto \gamma_i(t + \tau, z)$ for the local path. An output derivative in t then differentiates the pullback of the input. Finitely many immersion inverse charts and their Jacobians account for self-intersections. Include the global families in

$$\mathcal{E}_{\text{cyl}} = L^1(S^1) \oplus \bigoplus_i L^4(D^2; L^1(S^1)).$$

Their output dependence is only in the cylinder flow coordinate, apart from cutoffs. Hölder and coarea give the same two endpoint bounds. Products of at most three transverse path variations are integrable, so the integrated traces and the smooth coefficient functions have the bounded derivatives used above. Thus the proof applies to the full primary gradient. An L^1 norm in the transverse variable would not justify those products.

The fixed end weight is comparable between inputs and outputs on these loop families. Assign one weight to the factor with a difference, or to the corresponding differentiated factor; keep the other unweighted. Logarithmic derivatives of the weight contribute the same $|h|$ term. This proves the weighted estimate. Summation with (193) is legitimate in each bound. The perturbation outputs vanish where the principal surface operator and collar cutoff are supported. In $\Gamma_Y^j R$ the remaining terms are the scalar identity or smooth finite-rank operators bounded from L^2 : after the first finite-rank factor all subsequent surface derivatives act on smooth vectors. Thus the estimate is in the complete source norm, as asserted.

For differentiability fix $\varepsilon > 0$ and put

$$\sigma = \frac{3}{2}\varepsilon, \quad r = \frac{4}{1 + \varepsilon/2}, \quad q = \frac{4}{1 - \varepsilon/2}, \quad \frac{1}{r} + \frac{1}{q} = \frac{1}{2}.$$

Use the auxiliary primary path space $L_\tau^1 \oplus \bigoplus_i L_{y_i}^3 L_\tau^1$. Its derivatives through order three are bounded; both r and q exceed three, so the coarea endpoints hold for these two exponents. With $\kappa(t) = 4(1/2 - 1/t)$, input frequency pieces obey

$$\|Pv_j\|_{L^t} \leq C 2^{\kappa(t)j} \|v_j\|_2, \quad \|\Delta_h Pv_j\|_{L^t} \leq C \min(1, 2^j |h|) 2^{\kappa(t)j} \|v_j\|_2.$$

Since $\kappa(r) + \sigma = 1 + \varepsilon$, the preceding dyadic argument yields a σ -difference bound in L^r ; $\kappa(q) < 1 + \varepsilon$ gives the L^q bound. Applying Hölder to the candidate derivative remainder gives

$$\|(Df_z(Pv) - Df_z(0))Ph\|_{H^\sigma} \leq C_\varepsilon \|v\|_{H^{1+\varepsilon}} \|h\|_{H^{1+\varepsilon}}.$$

Taylor's remainder is $O(\|h\|^2)$ in L^2 by the L^4 coarea estimate and $O(\|h\|)$ in H^σ on a fixed input ball. Interpolation with $\varepsilon/\sigma = 2/3$ gives $O(\|h\|^{4/3})$ in H^ε . The derivative difference is $O(\|v - w\| \|h\|)$ in L^2 and $O(\|h\|)$ in H^σ , hence is operator-norm continuous into H^ε with modulus $C_{\varepsilon,R} \|v - w\|^{1/3}$. The same weighted and surface-source arguments apply. The constants for these coarea exponents were included in (193). This proves C^1 , separately from the quadratic Lipschitz estimate. $\square$

7.11. A complete nonlinear chart and the first correction.

Proposition 7.16 (nonlinear graph chart). *There is a chart Φ_ρ , with $\Phi_\rho(0) = x_{\text{pre}}$ and derivative I , from a ball in X_ρ to configurations satisfying the exact nonlinear matching, normal and outer Lagrangian conditions, with exact finite asymptotic representatives. Express the equation in the fixed target by parallel transport along its positive coordinate curves, and write*

$$F_\rho(U) = \mathcal{F}(\Phi_\rho(U)) = \mathcal{F}(x_{\text{pre}}) + D_\rho U + N_\rho(U).$$

For a fixed K_0 independent of ε and ρ , put

$$(195) \quad B_\rho = C\varepsilon^{-4} \rho^{-4\varepsilon} R_1^{K_0\varepsilon} (1 + \rho M_1^2 R_1^{K_0\varepsilon}) \leq CT^4.$$

The chart and its inverse are defined on a ball of radius c/B_ρ , in the finite-profile coordinates below, and

$$(196) \quad \|N_\rho(U) - N_\rho(V)\|_{Y_\rho} \leq B_\rho (\|U\|_{X_\rho} + \|V\|_{X_\rho}) \|U - V\|_{X_\rho}.$$

The ambient chart remainder satisfies the same quadratic difference bound. For each fixed positive scale the equation map is C^1 .

Proof. Use a smooth exponential-type family $e_\kappa(q, \nu)$ on the positive moduli space, with derivative I , adapted to the two outer Lagrangians and constant mixed ends as in [DFL26, Lemma 4.4]. The parameter κ is constant on the entire shrinking neighborhood. For a flat irreducible surface connection α , the surface chart of Proposition 6.13, with this positive coordinate, has a quadratic correction $S_{\alpha, \kappa}(b)$ satisfying

$$E_\alpha S_{\alpha, \kappa}(b) = 0, \quad \alpha + b + S_{\alpha, \kappa}(b) \text{ flat} \iff C_\alpha b = 0.$$

On the flat locus its class is $e_\kappa([\alpha], \ell_\alpha^{-1} H_\alpha b)$. The correction depends on the exact and harmonic coordinates; the coexact coordinate passes through linearly. If q is the flat shift determined by those coordinates, then

$$(197) \quad S_{\alpha, \kappa}(b) = H_\alpha S_{\alpha, \kappa}(b) - \frac{1}{2} d_\alpha^* G_\alpha^2 [q \wedge q].$$

It therefore gains one surface derivative, with quadratic difference bounds $H_\Sigma^k \rightarrow H_\Sigma^{k+1}$, $k \geq 2$.

For a definite choice of the background flat retraction, solve $F_{\beta - d_\beta^* \eta} = 0$ for the small two-form η . At a flat connection the derivative in η is $-d_\beta d_\beta^* : H^{k+1} \Omega^2 \rightarrow H^{k-1} \Omega^2$, an isomorphism by irreducibility. The implicit function theorem and uniqueness of the small solution give a smooth equivariant retraction $r(\beta)$ whose local definitions agree. Put $\alpha(z) = r(\beta_{\text{pre}}(z))$ on a seam collar. Choose its cutoff ψ to be one at the seam, using a fixed compact cap collar near the bubble and a fixed physical residual collar supported where $\varphi = 1$ outside. On the entire outer cap annulus both cutoffs are one. The same retraction and e_κ are used there, so these formulas agree. The background remains in the fixed retraction neighborhood on those collars. Its first compact base derivatives are bounded by C and its second by CR_1 ; outside the cap they have fixed bounds.

For the physical negative variation ξ , with surface part b , start with

$$\widehat{A}(U) = A_{\text{pre}} + \xi + \psi S_{\alpha, \kappa}(b), \quad u(U) = e_\kappa(u_{\text{pre}}, \nu).$$

The surface chart identities give exactly the nonlinear matching condition and its local converse. The normal base trace is unchanged; the adapted positive map gives exactly the outer Lagrangian conditions.

We next specify the ends. For a finite mixed-end tangent p , let $q_p = e_{\pm 1}(q, s_p)$ be its prescribed nearby Lagrangian value. Choose smooth perturbed-flat representatives B_p with derivative p by the local three-dimensional Coulomb slice, reparameterized by q_p . Prescribe their boundary representatives using the surface chart above. The original and prescribed boundary families have the same class and first derivative, so their small unitary boundary gauge is $I + O(|p|^2)$. Extend its logarithm through a fixed collar, with the covariant normal jet chosen to preserve zero normal variation, and then cut it off. Both the logarithm and prescribed jet are $O(|p|^2)$. Gauge covariance preserves the perturbed-flat equation and the derivative at zero. This is the end-representative construction of [DFL26, Section 4.1] with the specified boundary chart. If $\widehat{B}_p$ is the limiting connection of $\widehat{A}$ on a constant tangent profile, then $C_p = B_p - \widehat{B}_p$ is $O(|p|^2)$ with zero tangential boundary value and zero normal variation. With fixed end cutoffs $\tau_\pm$, define

$$(198) \quad A(U) = \widehat{A}(U) + \tau_+ C_{p_+} + \tau_- C_{p_-}.$$

This preserves the seam and normal conditions and has the exact end configurations $(B_{p_\pm}, q_{p_\pm})$.

For ambient configuration norms, retain p as a finite coordinate and subtract $\tau(B_p - B)$ before taking a weighted norm. Relative to the linear profile τp , the remaining chart

correction on each end is

$$(199) \quad \psi S_{\alpha, \kappa}(b) - \tau \psi_{\infty} S_{\alpha_{\infty}, \pm 1}(p_{\Sigma}).$$

It decays because the background converges exponentially and $b - p_{\Sigma}$ is in the weighted space. These ambient graph spaces are complete by the closed-distribution argument of Lemma 7.5, omitting the linear boundary constraints. An unsubtracted constant $B_p - B - p$ is never assigned a weighted-decay norm.

We prove the estimates for this chart. On the cap collar let $S(z, b) = S_{\alpha(z), \kappa(z)}(b)$. It is smooth from H_{Σ}^k to H_{Σ}^{k+1} , vanishes quadratically, and has first explicit base derivatives bounded by C and second by CL , $L \leq CR_1$, also after the fixed derivatives in b used below. A dyadic composition estimate avoids a full L loss. For low sums S_j and frequency pieces Δ_j , telescope

$$S(z, b) = \sum_{j \geq 0} m_j(z) \Delta_j b, \quad m_j = \int_0^1 D_b S(z, S_j b + t \Delta_j b) dt,$$

including the low-frequency term. On a small fixed C^0 ball,

$$\|m_j\|_{\infty} \leq C\|b\|_{\infty}, \quad \|Dm_j\|_{\infty} \leq C2^j\|b\|_{\infty}, \quad \|D^2m_j\|_{\infty} \leq C(2^{2j} + L)\|b\|_{\infty}.$$

Differences satisfy the same bounds with $\|b - c\|_{\infty}$. For output frequency $n > j$, one or two differentiations in its projection give

$$\|\Delta_n(m_j \Delta_j b)\|_2 \leq C\|b\|_{\infty} \|\Delta_j b\|_2 \min\{2^{-(n-j)}, (1 + L2^{-2j})2^{-2(n-j)}\}.$$

After the $H^{1+\varepsilon}$ weight, its convolution kernel is bounded by $C \min\{2^{\varepsilon(n-j)}, L2^{-(1-\varepsilon)(n-j)}\}$, whose sum is $C\varepsilon^{-1}L^{\varepsilon}$ by splitting at $n - j = \lceil \log_2 L \rceil$. For $n \leq j$ the kernel is $C2^{-(1+\varepsilon)(j-n)}$. Hilbert-valued square summation and the C^0 bound $C\varepsilon^{-1/2}\|b\|_{H^{1+\varepsilon}}$ give the coarser quadratic difference estimate

$$(200) \quad \|S(b) - S(c)\|_{H_B^{1+\varepsilon} H_{\Sigma}^{k+1}} \leq C\varepsilon^{-2} L^{K_1 \varepsilon} (\|b\|_{H_B^{1+\varepsilon} H_{\Sigma}^k} + \|c\|_{H_B^{1+\varepsilon} H_{\Sigma}^k}) \|b - c\|_{H_B^{1+\varepsilon} H_{\Sigma}^k}.$$

For a fixed physical patch, $L = 1$ and the second-derivative kernel alone is uniformly summable, leaving only the evaluation loss. The coarser fixed-patch constant $C\varepsilon^{-1}$ will suffice. Fixed coefficient multiplications are absorbed in K_1 . This argument uses Hilbert structure only for the input and output norms, not for the operator-valued coefficients.

On the cap, the quadratic a^2 term has coefficient ρ/f and bound

$$C\rho M_1^2 L^{\varepsilon} \varepsilon^{-1/2} \|a/M_1\|^2.$$

The ab term costs $C\rho M_1 \varepsilon^{-1/2}$. The surface derivative of S is controlled by (197) and (200); its scalar coefficient is ρf . Its base derivative and the positive nonlinear row use the strong/weak product estimate of Proposition 6.18. Thus the local cap equation remainder has quadratic difference constant

$$(201) \quad C\varepsilon^{-2} R_1^{K_1 \varepsilon} (1 + \rho M_1^2 R_1^{K_1 \varepsilon}).$$

There is no nonlinear surface derivative of a : its affine linear derivative was subtracted in N_{ρ} . No full additional derivative of the exact surface part is used. The background and target coefficients have the bounds just proved, including the normalized pregluing source on its transition. The same estimates therefore apply to the raw domain without a base Coulomb condition.

On the fixed four-dimensional residual body, curvature products obey $H^{1+\varepsilon} \times H^{1+\varepsilon} \rightarrow H^{\varepsilon}$ uniformly: interpolate $H^1 \times H^1 \rightarrow L^2$ and $H^2 \times H^2 \rightarrow H^1$, both consequences of $H^1 \rightarrow L^4$.

For an integer surface source order j , the principal part of Γ^j is $\chi^j \Lambda_\Sigma^j$. Elliptic order-zero comparison gives $\chi^j \nabla_\Sigma^j \xi \in H^1$ from the $\Gamma^j E^1$ input bounds, with lower powers and smooth finite-rank terms included. The comparison is normally constant, so this never divides by χ . In a quadratic product distribute $\chi^j = \chi^a \chi^{j-a}$ and use $H^1 \cdot H^1 \subset L^2$. The chart remainder is supported where $\varphi = 1$ and gains a surface derivative, so direct differentiation bounds its source and ambient graph norms. The positive equation uses the two-dimensional strong/weak estimates. Lemma 7.15 bounds the perturbations. Together these give constant $C\varepsilon^{-1}$ on the fixed residual pieces. In this boundedness calculation the chart remainder need not have linear boundary trace; no integration by parts against an adjoint with that trace is used.

At a nonzero end parameter, subtract the exact B_p family first. Products with a decaying field are smooth coefficient products. The pure parameter source has its linear term $D_\rho(\tau p)$ subtracted. Its quadratic remainder is supported at the fixed cutoff or decays like $C|p|^2 e^{-\lambda|\theta|}$, because the limiting configurations solve the exact equation and the background converges with all fixed derivatives. Choose the weight below λ . Differences have the same bound. Linear holonomy coefficients at the smooth end family use the differentiated coarea estimate; decaying remainders use Lemma 7.15. This proves the residual estimates including (199).

It remains to combine the two norms. Write $U_i = \theta_1 U$ and $U_o = (1 - \theta_1)U$. Their defining norms are bounded by $\|U\|_{X_\rho}$, and

$$\|(b, \nu)\|_{C_B^0 H_\Sigma^k} \leq C\varepsilon^{-1/2} \|U\|_{X_\rho}.$$

On a fixed outer compact annulus of radius R_1^{-1} ,

$$\|v(R_1^{-1} \cdot)\|_{H^{1+\varepsilon}} \leq C\varepsilon^{-1/2} \|v\|_{H^{1+\varepsilon}(B)}.$$

The value term uses C^0 , the derivative uses the unchanged Dirichlet integral, and its fractional part gains $R_1^{-\varepsilon}$. Cut U_i to an annular field U_i^a which agrees with it for $|w| \geq R_1/3$ and vanishes for $|w| \leq R_1/4$. It already vanishes beyond $R_1/2$. Since $f_\infty \asymp M_1$ there, conversion and physical scaling give

$$(202) \quad \|U_i^a\|_{H_{B, \text{phys}}^{1+\varepsilon} H_\Sigma^k} \leq C\varepsilon^{-1/2} (\rho R_1)^{-\varepsilon} \|U_i\|_{X_{\text{cap}}^{\text{raw}}}.$$

Zero extension at a radial boundary is bounded here because the value trace is zero and $1 + \varepsilon < 3/2$. At the straight seam use a fixed extension. Only the strong coefficient triple, not a full extra derivative of exact b , is asserted in this comparison. For a surface-gaining chart remainder the same calculation also controls its stronger residual surface norm.

The cross difference $N(U_i + U_o) - N(U_i) - N(U_o)$ is supported on this outer cap annulus. There U_i can be replaced by U_i^a : where the replacement changes it, U_o and its derivatives vanish. All nonlocal perturbation and end cross terms are zero because U_i vanishes on their outputs and dependencies. The local physical nonlinear estimate and (202) bound the cross difference with constant $C\varepsilon^{-2} \rho^{-2\varepsilon} R_1^{K_{2\varepsilon}}$. For the pure inner term use (201) and the source conversion $C\rho^{-\varepsilon}$; for the pure outer term use the fixed residual estimate. These give (196) with the coarser (195).

The same decomposition controls the ambient chart remainder. Its inner term has the full surface gain for the reverse annular comparison; its outer term transfers to the raw cap norm by (181). The cross term has that same surface gain, so its inner transfer costs at most a further $\varepsilon^{-1/2} R_1^\varepsilon$. The largest fractional loss is $\varepsilon^{-5/2} \rho^{-C\varepsilon}$ with $C \leq 3$, dominated by (195). The ambient near-identity map therefore has a contraction inverse on a ball of radius c/B_ρ , after fixing p and subtracting its exact profile. Recover p from the exact end value q_p . The

surface and positive inverse charts identify the exact nonlinear boundary conditions with the linear ones in X_ρ . This proves the chart assertion and its derivative I at zero. Finally, the fixed-scale surface and positive maps are smooth on their strong Banach algebras, curvature is a continuous quadratic map into the source, and the end families are smooth. The separate C^1 holonomy conclusion of Lemma 7.15 therefore gives $F_\rho \in C^1$. $\square$

Corollary 7.17 (first exact correction). *There is $U_\rho \in X_\rho$ with*

$$(203) \quad \mathcal{F}(\Phi_\rho(U_\rho)) = 0, \quad \|U_\rho\|_{X_\rho} \leq CT^{-38} + Ce^{-T}T^{C_J}.$$

The equation linearization in this graph chart has a right inverse of norm CT^2 at the corrected point.

Proof. Write $q_\rho = \|Q_\rho\|$ and $e_\rho = \|\mathcal{F}(x_{\text{pre}})\|$. The range of Q_ρ is closed because $Q_\rho D_\rho$ is a bounded projection. On its closed ball of radius $2q_\rho e_\rho$, define

$$\mathcal{T}(U) = -Q_\rho \mathcal{F}(x_{\text{pre}}) - Q_\rho N_\rho(U).$$

The inverse, error and nonlinear bounds give $B_\rho q_\rho^2 e_\rho = O(T^{-32})$ and $q_\rho e_\rho = o(B_\rho^{-1})$. Thus this ball lies in the chart, $\mathcal{T}$ preserves it, and its Lipschitz constant is at most $4B_\rho q_\rho^2 e_\rho < 1/2$. Completeness gives a fixed point, and applying $D_\rho Q_\rho = I$ proves its exact equation and the stated bound. Uniqueness at this stage is only within that inverse-range ball.

Since the equation is C^1 , the quadratic difference estimate implies $\|dN_\rho(U_\rho)\| \leq 2B_\rho \|U_\rho\|$. For $L_\rho = D_\rho + dN_\rho(U_\rho)$, the Neumann formula

$$(204) \quad \tilde{Q}_\rho = Q_\rho(I + dN_\rho(U_\rho)Q_\rho)^{-1}$$

is a right inverse with norm CT^2 , because its correction operator is $O(T^{-32})$. This conclusion concerns the geometric graph operator; the ordinary Fredholm quotient is addressed next. $\square$

7.12. Smooth representatives and the ordinary Fredholm problem.

Proposition 7.18 (ordinary regularity of the corrected solutions). *Any zero $x = \Phi_\rho(U)$ of the equation constructed in the preceding chart with $\|U\|_{X_\rho} \leq CT^{-37}$ represents a smooth mixed moduli solution at each fixed positive scale. It has finite analytical energy, the prescribed ordinary end classes, and the exact mixed end classes selected by its finite parameters. For a sufficiently small positive weight on the mixed ends only, its ordinary augmented Fredholm operator is onto and has real index four.*

Proof. We first establish smoothness and the ends. Fix $\rho > 0$ and $\varepsilon > 0$. The cap coefficient conversions are smooth and bounded at this fixed scale. For $k \geq 2$ the Fourier inequality

$$(1 + |\xi_B|^2 + |\xi_\Sigma|^2)^{1+\varepsilon} \leq C(1 + |\xi_B|^2)^{1+\varepsilon}(1 + |\xi_\Sigma|^2)^k$$

puts the negative strong cap coefficients in ordinary four-dimensional $H^{1+\varepsilon}$. The residual factor and ambient chart remainder have the same local regularity. Choose $p = 2 + \varepsilon/4$; then $2 < p < 4/(2 - \varepsilon)$ and $p < 2/(1 - \varepsilon)$. The negative connection and positive map are both locally $W^{1,p}$. The stronger collar continuity already gives the actual weak matching.

The local $W^{1,p}$ Coulomb gauge theorem described after [DFL24, Theorem 1] supplies a $W^{2,p}$ gauge relative to a smooth reference, with covariant Coulomb and zero normal boundary conditions. This use is at a fixed physical scale. The gauge is continuous and preserves the matching class. The mixed regularity theorem [DFL24, Theorem 1], equivalently [DFL26, Theorem 5.4], makes the unperturbed seam pair smooth there. Standard interior and totally real boundary regularity make the positive map smooth on its remaining domain.

In the perturbed negative interior, start with the pointwise bound for the holonomy perturbation, independent of the unitary connection [DFL26, Proposition 6.1(ii)]. The secondary forms have the same bound by unitarity and their compact defining data. In a local Coulomb gauge the first-order elliptic equation has a quadratic connection term and bounded forcing. If $2 < p < 4$,

$$a \in W^{1,p} \implies a \in L^{4p/(4-p)}, \quad a \wedge a \in L^{2p/(4-p)}, \quad 2p/(4-p) > p.$$

Iteration on smaller compact sets reaches $W^{1,6}$, choosing intermediate exponents to avoid the endpoint four. Only boundedness of the perturbation is used before this stage. The number of steps may depend on the fixed ε .

Once $p = 6$, differentiate the path ODE and use coarea to obtain $W^{1,6}$ regularity of the perturbation. At order m its Duhamel terms contain input derivatives of total order at most m . A factor of order m is in L^6 and the remaining factors are bounded; if there are several differentiated factors all have order at most $m - 1$. The integrated primary traces use Hölder in their transverse variable. Their fixed constants are precisely those included in (193). Thus every fixed differentiated series converges, and elliptic bootstrapping improves $W^{m,6}$ to $W^{m+1,6}$. Each compact output set is treated together with its full compact dependency neighborhood; for a primary series on a bounded flow interval this includes the whole fixed three-dimensional body on that interval. Local gauge transitions satisfy $dg = gA_j - A_i g$ and improve with the connection regularity. They are smooth at the end of the bootstrap, define a smooth bundle in the original topological class, and yield a global Sobolev gauge to a smooth representative. This is the perturbed interior and patching step discussed after [DFL26, Theorem 5.4]. No uniform bound for this gauge in X_ρ is claimed.

On compact subsets, H^1 and $H^1 \subset L^4$ give finite curvature and map energy. On a mixed end write $A = B_p + \xi$ using the exact end representative, where $\xi \in H_\omega^1$. Since B_p solves the perturbed-flat equation,

$$(205) \quad F_A + *_3 \nabla_A h = d_{B_p} \xi + \xi \wedge \xi + *_3 (\nabla_A h - \nabla_{B_p} h).$$

All terms on the right are L^2 , the last by the L^2 Lipschitz estimate of [DFL26, Proposition 6.1(ii)] or its Duhamel–coarea proof. Secondary perturbations vanish on far ends. The positive end has constant limit and L^2 derivative. The ordinary gauge end uses its fixed critical representative and the unweighted H^1 remainder in the same calculation; the ordinary symplectic end has finite energy by its fixed background and H^1 variation. Thus the analytical energy is finite. The unmodified F_{B_p} need not be L^2 on an infinite perturbed-flat cylinder, which is why (205) is required.

We can now apply [DFL26, Proposition 3.6]: local smoothness has supplied its initial local H^2 hypothesis and the required regularity of du . More explicitly, the small-energy tail estimates and temporal-gauge conclusion in [DFL26, Propositions 5.24, 5.27 and 5.29] give exponential convergence at every fixed order. Compact secondary perturbations do not change those tail equations. Choose

$$(206) \quad 0 < \omega_{\text{mod}} < \min(\omega, \delta_{\text{DFL}})$$

below the allowed decay rate on the compact end data. The positive mixed limit is q_p already: its original weighted base norm gives uniform convergence on unit translates and a negative gauge does not change it. The embedded three-manifold Lagrangian then fixes the negative limiting class as $[B_p]$. The positive ordinary limit is likewise prescribed. On the gauge end, translates where the original H^1 remainder tends to zero converge to its prescribed critical representative in the ordinary local quotient; its Hausdorff irreducible slice identifies the

class given by the end theorem. These arguments prove membership in the ordinary smooth configuration space with the stated end convention.

We next prove ordinary surjectivity. At any zero in the larger CT^{-37} ball, (196) gives $\|dN_\rho(U)Q_\rho\| \leq CT^{-31}$. Thus (204) remains a right inverse of size CT^2 there. Let $x = (A, u)$ now denote a smooth representative and consider

$$\mathcal{D}_x(\xi, \nu) = (d_A^* \xi, D_x^{\text{geo}}(\xi, \nu))$$

on the actual normal, matching and outer Lagrangian domain, with the finite tangent variables of the exact mixed end manifolds. The Fredholm and weak adjoint theorems [DFL26, Theorem 4.12], [DFL24, Theorem 5.9] identify its cokernel with the formal adjoint kernel in the positive-weight adjoint domain. In particular a cokernel vector (s, η) is smooth locally, globally L^2 , and positively weighted on mixed ends. The secondary linearization and adjoint have every fixed integer bound by differentiated coarea and are compact $H^1 \rightarrow L^2$ on the full dependency set. Thus the same weak regularity and end identification apply with that term. Retaining the finite end variables is essential to the stated adjoint weight.

For a compactly supported gauge potential ζ with covariant Neumann trace, $(d_A \zeta, 0)$ belongs to the actual domain and $D_x^{\text{geo}}(d_A \zeta, 0) = 0$ by covariance at the exact solution, including all perturbations. Pairing with the cokernel gives $\langle s, d_A^* d_A \zeta \rangle = 0$. Interior tests and the arbitrary boundary value of these Neumann tests imply $d_A^* d_A s = 0$ with covariant Neumann condition. Cutoff integration by parts, using the adjoint $\bar{H}^1$ and L^2 bounds, yields $d_A s = 0$. A parallel adjoint section restricts to a parallel section on a flat irreducible seam fiber, so $s = 0$.

For any smooth compact geometric source F in the original pregluing frames, apply the corrected graph inverse, the chart derivative, and the fixed smoothing gauge to obtain a variation V at x with

$$D_x^{\text{geo}} V = SF.$$

Here S is a bounded invertible source multiplication on weighted L^2 : negative unitary conjugation and positive target transport. The transformed variation is H^1 locally. Indeed its original physical regularity is $H^{1+\varepsilon}$, while the smoothing gauge is locally $W^{2,p}$ with $p > 2$; its derivative and the original variation have exponents greater than four, making the differentiated conjugation L^2 . Matching and normal traces are preserved. On mixed ends V has bounded L^2 size on unit strips, since before conjugation it is a fixed finite end profile plus a decaying remainder. It is L^2 on the other ends.

Choose an exhaustion cutoff χ_R common to the two sides of each seam, constant on all far primary loop dependencies at each flow coordinate, and one on the compact secondary dependencies. All holonomy commutators with it vanish. The compact variation $\chi_R V$ is in the ordinary first-order domain. The weak adjoint identity therefore gives

$$0 = \langle \eta, \chi_R SF \rangle + \langle \eta, [D_x^{\text{geo}}, \chi_R] V \rangle.$$

The remaining commutator is zeroth order with coefficient $Cd\chi_R$. Its pairing tends to zero by the positive adjoint weight and bounded unit-strip norm on mixed ends, and the two L^2 norms elsewhere. Thus η annihilates SF for every smooth compact F . Those images are dense in weighted L^2 because S is bounded and invertible, even if its gauge is initially Sobolev. Hence $\eta = 0$. This proves ordinary surjectivity without a uniform physical Coulomb projection.

Finally compute the index, retaining the relative seam framing. Choose a fixed positive-scale comparison circle outside the pregluing transition, where the pregluing agrees exactly with the residual. Join their mixed fillings to obtain a relative projective bundle P_{rel} over

$S^2 \times \Sigma_g$. On the positive disc pull back the universal projective bundle, choose flat surface representatives, and extend the negative base components across the seam by arbitrary smooth base components. The positive disc is contractible, so the lift exists with the prescribed seam identification. Since $F_\Sigma = 0$, the mixed curvatures are derivatives of the flat surface connection minus exact forms. Their derivatives are d_β -closed, and integration over Σ kills the exact terms in the wedge pairing. With the established orientation,

$$(207) \quad \int_{D^+ \times \Sigma} \text{tr}(F \wedge F) = 2 \int_{D^+} u^* \Omega.$$

This is the direct calculation of [DFL26, Lemma 5.10, equation (5.12)]. Initially piecewise smooth connections give the same integral by smoothing with the seam identification fixed. This calculation does not require the small-energy hypothesis of the annular argument in [DFL24, Lemma 4.67].

The outer connections coincide, so transgression vanishes and

$$E_{\text{top}}(x_{\text{pre}}) - E_{\text{top}}(x_0) = -\frac{1}{4} \langle p_1(P_{\text{rel}}), [S^2 \times \Sigma] \rangle.$$

Perturbation contributions cancel outside the comparison circle. Within the fixed attaching charts, contract the small residual filling to its central value and replace the cap's far tail by that same constant value. Perform these homotopies with the relative boundary identification fixed. The resulting closed projective bundle is $\text{ad } E$, so P_{rel} has its characteristic class. A positive-half trivialization has not been extended across the negative half. In particular a constant positive map in genus one does not remove this relative clutching class.

For the primitive extension (56), let f be the positive generator from S^2 and let $\lambda = c_1(\Lambda)$ be pulled back from Σ . Then $f^2 = \lambda^2 = 0$, $\langle f\lambda, [S^2 \times \Sigma] \rangle = 1$, and

$$(208) \quad c_1(E) = f + \lambda, \quad c_2(E) = f\lambda, \quad p_1(\text{ad } E) = c_1(E)^2 - 4c_2(E) = -2f\lambda.$$

Thus the relative topological energy is $1/2$, for either pencil rank. The configuration index formula [DFL26, Proposition 3.20, equation (3.21)] is

$$\text{ind } \mathcal{D}_x = 8E_{\text{top}}(x) + \epsilon(\beta) - \epsilon(\alpha).$$

The external end classes agree with the index-zero residual, so the preglued configuration has index four. The compact secondary term does not change that index.

The path $\Phi_\rho(tU)$ preserves the actual matching, outer conditions and exact end critical manifolds. Its continuous positive map and H_Σ^k -continuous seam data, together with the local surface gauge charts, give a continuous homotopy of the relative projective bundle. To use ordinary Fredholm invariance, smooth these continuous boundary data in the same charts, relative to smooth endpoints and retaining the end manifolds; extend the negative connection coefficients affinely. This gives a smooth configuration homotopy with the same clutching class. We have not applied a high-order operator homotopy theorem directly to a rough graph path. The smoothing gauge also preserves that class. Therefore the corrected smooth solution has index four and, by the surjectivity proved above,

$$(209) \quad \dim_{\mathbb{R}} \ker \mathcal{D}_x = 4.$$

The effective local slice divides by the central stabilizer $\{\pm 1\}$, as required for its free action. $\square$

The ordinary local slice and Fredholm implicit function theorem [DFL26, Propositions 3.18 and 4.6] now give a smooth four-dimensional moduli neighborhood. Equality of this dimension with the four expected gluing parameters does not establish that the constructed family parametrizes it. We next prove that assertion using four gauge-invariant holonomy functions.

7.13. Affine normalization and four holonomy functions. Use the holomorphic coordinate $w = -\theta + is$, whose negative half-plane has $\text{Im } w < 0$ and whose seam is real. This rotation must be retained in a later orientation comparison. Write a primitive extension pencil as

$$e(w) = vw + w_0, \quad [v, w_0] \in \mathbb{P}(V_L \oplus V_L), \quad V_L = H^1(\Sigma, \Lambda^{-1}L^2).$$

The maximal relative $\mathcal{O}(1)$ line identifies L ; these spaces have constant dimension g and form a smooth bundle over $\text{Pic}^0(\Sigma)$. Choose a smooth Hermitian metric, linear in its first argument. Stable attachment requires $v \neq 0$. Put

$$(210) \quad \kappa(E) = \frac{\langle w_0, v \rangle}{\|v\|^2}, \quad w_\perp = w_0 - \kappa v, \quad s(E) = \left((\text{Im } \kappa)^2 + \frac{\|w_\perp\|^2}{\|v\|^2} \right)^{1/2}.$$

They are invariant under common complex scaling. The condition $s = 0$ means a rank-one pencil with real basepoint $-\kappa$, excluded by seam admissibility. Thus they are smooth with $s > 0$ on the stated locus, including interior rank drop. Under pullback by $w \mapsto aw + b$, $a > 0$, $b \in \mathbb{R}$, the coefficients become $(av, w_0 + bv)$ and

$$\kappa \mapsto (\kappa + b)/a, \quad s \mapsto s/a.$$

There is a unique orbit representative in the slice

$$(211) \quad \text{Re } \kappa = 0, \quad s = 1,$$

obtained by $a = s(E)$, $b = -\text{Re } \kappa(E)$. The infinitesimal translation T_E and dilation D_E give, at this slice, the normal matrix

$$(212) \quad \begin{pmatrix} d\text{Re } \kappa(T_E) & d\text{Re } \kappa(D_E) \\ ds(T_E) & ds(D_E) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Both preserve evaluation at infinity. The regular attaching fiber has real dimension four, so its intersection with this slice is smooth of real dimension two.

Lemma 7.19 (holonomy coordinates for the limiting kernel). *Locally on the compact regular primitive family, four real linear combinations of traces of holonomy, supported on a fixed compact rescaled negative region, have independent differentials on the four-dimensional zero-attaching limiting kernel. They may be averaged over small real affine motions while retaining independence.*

Proof. First let A be a smooth irreducible unitary rank-two connection on a connected region. If a variation has zero first variation of every closed-loop trace, it is an infinitesimal gauge variation. To verify this, fix a basepoint and write H_γ for the based holonomy. Their complex span is $M_2(\mathbb{C})$: a nonscalar unitary supplies its two spectral projectors, and irreducibility supplies another unitary with nonzero off-diagonal entries. Multiplication by the projectors gives both off-diagonal matrix units. If $\sum c_\gamma H_\gamma = 0$, differentiation of traces of concatenated loops gives

$$\text{tr} \left(\sum c_\gamma \dot{H}_\gamma H_\eta \right) = -\text{tr} \left(\sum c_\gamma H_\gamma \dot{H}_\eta \right) = 0.$$

Nondegeneracy of the trace pairing makes $H_\gamma \mapsto \dot{H}_\gamma$ a well-defined derivation of the matrix algebra. It is inner: with matrix units, the derivation identity gives $D(X) = [K, X]$ for $K = \sum_i D(E_{i1})E_{1i}$. Preservation of the unitary involution makes $K + K^*$ scalar, so K may be taken skew-adjoint and trace-free. Subtract this basepoint conjugation. Now every based holonomy variation is zero. Differentiated parallel transport along a path from the basepoint defines a path-independent potential ζ , and differentiation along its final segment gives the

variation $d_A\zeta$. Smooth local path families give smoothness. Piecewise smooth loops can be smoothed without changing this conclusion. For fixed-determinant $U(2)$ connections take both real and imaginary trace parts.

Apply this fact in the based unitary trivialization of the limiting base-flat connection A_E^0 . Choose the determinant reference pulled back from Σ on the negative product. Its base component, as well as the trace-free base component, is zero. The surface $(0, 1)$ connection is holomorphic in w . A zero-attaching bundle tangent v has representative $(0, b_v, \nu_v)$ with $b_v^{0,1}$ holomorphic and $b_v(\infty) = 0$. Choose a relatively compact negative base region O close to a finite seam point and put $K = O \times \Sigma$. The surface zero-form gaps near the flat irreducible seam make the connection on K irreducible. If all its trace derivatives vanish, the preceding argument gives $(0, b_v) = d_{A_E^0}\zeta$ on K . The base equation makes ζ independent of w . The identity $b_v^{0,1} = d_{\beta_E(w)}^{0,1}\zeta$ extends to the whole negative half-plane by holomorphic uniqueness in surface Sobolev spaces. At infinity it gives $d_\alpha\zeta = 0$, so $\zeta = 0$ and $b_v = 0$. Matching makes ν_v zero on the seam. Stable fibers give a holomorphic extension of its classifying map and tangent bundle through a seam neighborhood. In local holomorphic frames, extension of ν_v by zero is weakly holomorphic across a boundary arc; uniqueness makes it zero everywhere. The limiting kernel/cohomology identification then gives $v = 0$.

Thus the real trace differentials separate this four-dimensional space. Select four real linear combinations spanning its dual; they remain independent on a smaller parameter patch. Averaging the loops over a small smooth affine neighborhood converges with its first finite-family derivative to the unaveraged functions, so a fixed sufficiently small average retains independence. A compact family is covered by finitely many such choices; no global frame of a possibly nontrivial kernel bundle is asserted. $\square$

Fix these four combinations, their loops and a smooth density $\chi(a, b)$ supported near $(1, 0)$ in $(0, \infty) \times \mathbb{R}$, with integral one. All moved loops stay in one fixed compact negative region. With real or imaginary parts and the selected combinations understood, define

$$(213) \quad \mathcal{O}_{c,\rho}(A) = \int \chi(a, b) \operatorname{Tr} \operatorname{Hol}_{(w,x) \mapsto (c+\rho(aw+b),x) \circ \gamma}(A), da db \in \mathbb{R}^4.$$

Write $\mathcal{O}_0(E)$ for their value at the limiting connection in rescaled coordinates. These are local moduli coordinates, not point-incidence conditions. For any signed smooth density σ use $\mathcal{O}^\sigma$ for the corresponding average. Changing variables in the compact affine integral gives the exact identities at a fixed input connection

$$(214) \quad \rho \partial_c \mathcal{O}_{c,\rho}^\chi = \mathcal{O}_{c,\rho}^{\sigma_T}, \quad \rho \partial_\rho \mathcal{O}_{c,\rho}^\chi = \mathcal{O}_{c,\rho}^{\sigma_D}, \quad \sigma_T = -\partial_b \chi, \quad \sigma_D = -\partial_a(a\chi) - \partial_b(b\chi).$$

The scale formula includes the two-dimensional Jacobian in (a, b) . These identities hold for continuous connections and iterate in the stated order for any fixed word of parameter vector fields. No input spatial derivative along a moving rough loop is needed.

Duhamel's formula bounds observation differences by $C\|A - A'\|_{C^0}$ and derivative differences by $C\|A - A'\|_{C^0}\|h\|_{C^0}$, also for these signed densities. On the fixed rescaled loop region the actual variation has coefficients $(\rho a/f, b)$, with f and f^{-1} bounded there. Therefore

$$(215) \quad \|(\rho a/f, b)\|_{C^0(K)} \leq C\varepsilon^{-1/2}(1 + \rho M_1)\|U\|_{X_\rho} \leq CT^{1/2}\|U\|_{X_\rho}.$$

In particular the linear observation norm is $CT^{1/2}$. Its ordinary connection quadratic remainder costs CT ; composition with the ambient chart remainder will be estimated below.

7.14. Finite bundle derivatives and the standard kernel.

Lemma 7.20 (finite pregluing derivatives). *Fix c, ρ and a smooth four-real-dimensional family $E(t)$ in its attaching fiber, with the actual attaching representative $\alpha(c)$ independent of t . Let $W_j = \partial_{t_j} x_{\text{pre}}$ in the actual linear domain and fixed target transport. Then*

$$(216) \quad \|W_j\|_{X_\rho} \leq C, \quad \|D_\rho W_j\|_{Y_\rho} \leq CT^{-40} + Ce^{-T} T^{C_J}.$$

On the fixed rescaled observation region W_j converges in C^0 to its zero-attaching limiting bundle representative.

Proof. Include one bundle derivative in the finite-order cap estimates. The derivative vanishes outside $|w| \leq 2R_2$, hence only the raw cap factor of X_ρ is used. With $r_2 = R_2^{-1}$, the surface and positive cap derivatives vanish at infinity and are $O(|\zeta|)$. Their tail cutoff has strong $H^{1+\varepsilon}$ norm $O(r_2^{1-\varepsilon})$. The differentiated matching interpolation remainder has two $O(r_2)$ difference factors, hence norm $O(r_2^{2-\varepsilon})$. Consequently the strong surface and positive norms are bounded. The normalized base derivative is $a_j = \rho^{-1} \chi \partial_{t_j} A_{B, \text{cap}}$ and satisfies

$$\|a_j/M_1\|_{H_B^{1+\varepsilon} H_\Sigma^k} \leq C \frac{\delta}{\rho M_1} R_2^\varepsilon, \quad \|\rho d_\alpha a_j\|_{H_B^\varepsilon H_\Sigma^k} \leq C \delta R_2^\varepsilon.$$

The cutoff derivative's size cancels its two-dimensional L^2 scale before the fractional factor is included. These quantities tend to zero. For the last weak row use the full surface smoothness of these finite parameter representatives. Extend f_∞ through the deleted tail with size CR_2^2 and derivative CR_2^3 . Its fractional multiplier has norm $CR_2^{2+\varepsilon}$, so $\|\rho f_\infty d_\alpha b_j\|_{H^\varepsilon H^k} \leq C \rho R_2^{2+\varepsilon}$. This tends to zero and imposes no new derivative on arbitrary graph inputs.

On the interpolation annulus, both the cap/residual surface difference and its bundle derivative are $O(q)$, and the base difference derivative is $O(\delta) \leq O(\eta)$, with the notation (182). Differentiate the affine cancellation of the exact cap and residual equations before estimating the error. It is still exact because the cap solves its equation for every t . The quadratic interpolation terms retain two difference factors after differentiation. The scalar error is therefore bounded by the same $C((v/\rho)\eta + (v^2/\rho)\eta^2 + (\rho/v^2)q^2)$, and the mixed and positive errors by the same quantities as in Lemma 7.14. Differentiated target frames multiply the already bounded error by bounded coefficients. This proves the second estimate in (216). On fixed rescaled compact sets the pregluing is the exact cap, and its metric and finite bundle derivative converge to their limiting base-flat representatives by the Dirichlet estimates. This proves C^0 convergence on the loops. No scale derivative has been used. $\square$

Lemma 7.21 (fixed-scale reduction of geometric kernel fields). *At a smooth representative of a corrected graph solution, every geometric kernel variation from the complete graph chart can be changed by an infinitesimal gauge into the ordinary augmented kernel, at a sufficiently small mixed-end weight. On the observation region this change preserves the observation differentials. The assertion is at fixed positive scale.*

Proof. On a mixed end take the smooth representative A^s in temporal gauge, with limit B_∞ , and write the original graph connection as $A^\circ = B_p + \xi$ with weighted H^1 remainder. The smoothing gauge satisfies $\partial_\theta g = -A_\theta^\circ g$, up to the action convention. Unitarity gives, for $s > t$,

$$\|g(s) - g(t)\|_{L^2(Y)} \leq C_\omega e^{-\omega t} \|e^{\omega \theta} A_\theta^\circ\|_{L^2([t, \infty) \times Y)}.$$

Thus it has an almost everywhere unitary L^2 limit g_∞ and weighted L^2 convergence for every smaller weight. Passing to the spatial gauge equation along slices where the original remainder tends to zero gives $d_Y g_\infty = B_\infty g_\infty - g_\infty B_p$ in distributions. Its smooth coefficients

and bounded solution imply smoothness by first-order bootstrapping. Subtracting the two spatial equations yields

$$(217) \quad \begin{aligned} d_Y(g - g_\infty) &= B_\infty(g - g_\infty) - (g - g_\infty)B_p \\ &\quad + (A_Y^s - B_\infty)g - g(A_Y^o - B_p). \end{aligned}$$

Together with the temporal equation this proves weighted H^1 convergence, without a supremum decay assertion for the gauge.

Conjugating a decaying H^1 variation preserves that regularity: the only rough differentiated product is bounded by $\|A^o - B_p\|_{L^4}\|V_0\|_{L^4}$, with a smaller weight. For a finite smooth tangent first subtract its conjugate by g_∞ , and use (217). The transformed field is therefore a finite smooth end tangent plus a weighted H^1 remainder. On the ordinary gauge end use the same equation with its unweighted H^1 bounds and smooth bounded limiting coefficients. The positive variation already has the corresponding end form.

At a mixed limit decompose its critical tangent as $v_\infty = h_\infty + d_{B_\infty}\zeta_\infty$ by the compact three-dimensional covariant Neumann Laplacian. This zero-form operator is invertible: a parallel section vanishes on the flat irreducible boundary fiber. Gauge covariance preserves the linearized critical equation, so h_∞ is exactly the harmonic critical tangent, with the same surface class and positive value. Subtract $d_A(\tau\zeta_\infty)$ globally. The normal trace and geometric equation remain unchanged, and the remaining field V' has precisely the allowed finite harmonic tangent. Its Coulomb source d_A^*V' is weighted L^2 on mixed ends and ordinary L^2 on the other ends.

Solve

$$(218) \quad d_A^*d_A\zeta = d_A^*V', \quad \nabla_n\zeta = 0.$$

At this fixed scale the operator is invertible from weighted H^2 to L^2 for a small positive mixed-end weight and zero ordinary end weight. Here are its Fredholm and kernel checks. The limiting compact cross-sectional zero-form Laplacians have positive gaps by irreducibility. On each product tail, the operator is $-\partial_\theta^2 + \Delta_B$; the separated Green kernels have indicial exponents $\pm\sqrt{\lambda_j}$ bounded away from zero. They give inverses at sufficiently small positive, negative or zero weights. Cut them off against a compact-core Neumann parametrix. Commutators are compact by Rellich, and the smoothly converging tail coefficients give arbitrarily small remaining $H^2 \rightarrow L^2$ error. This proves Fredholmness on the actual normal domain. At weight zero it is self-adjoint, and the energy identity and irreducibility kill its kernel; its index is zero. Weight deformation below the indicial gap keeps this index.

One can also eliminate the possible negative-weight formal cokernel directly. A weak cokernel section is smooth by local Neumann regularity. On a temporal tail put $f(\theta) = \|\zeta(\theta)\|_{L^2(Y)}$. The equation and boundary condition give, weakly also at zeros,

$$ff'' = \|\partial_\theta\zeta\|_2^2 - (f')^2 + \|d_{B(\theta)}\zeta\|_2^2 \geq \lambda f^2.$$

Compare on $[t_0, S]$ with $h'' = \lambda h$ having the same endpoint values. Negative weighted L^2 gives a sequence S with $f(S)e^{-\omega S}$ bounded. For $\omega < \sqrt{\lambda}$, the far endpoint contribution tends to zero, giving $f(\theta) \leq f(t_0)e^{-\sqrt{\lambda}(\theta-t_0)}$. Unit-strip elliptic estimates give decay of first derivatives as well. On the ordinary end they already have L^2 bounds. Exhaustion and integration by parts now give $d_A\zeta = 0$, hence zero. The positive-weight kernel is killed by the same energy identity. This verifies (218); no bound uniform in ρ has been asserted.

Subtracting $d_A\zeta$ gives an ordinary first-order augmented kernel element. Weak mixed regularity makes it a smooth kernel representative. The observations are unchanged also for the initially weak potential. On their compact region the original connection and variation

are continuous; the gauge is initially $W^{2,p}$, $p > 2$, hence continuous. Its first-order equation then makes it C^1 . The transformed variation is continuous, and its difference from the final smooth one is a continuous $d_A\zeta$. Since locally $\zeta \in H^2$ in dimension four, it is in every finite L^q ; the identity $d\zeta = d_A\zeta - [A, \zeta]$ gives $W^{1,q}$, then continuity and C^1 . The ordinary loop derivative of this gauge variation vanishes. The finite smooth end potentials do not change that local argument. $\square$

Corollary 7.22 (observations on the ordinary moduli tangent). *At any corrected zero of graph size CT^{-37} , the differential of $\mathcal{O}_{c,\rho}$ is an isomorphism from the ordinary four-dimensional moduli tangent to $\mathbb{R}^4$, for sufficiently small ρ on the chosen patch.*

Proof. Let $L = D_\rho + dN_\rho(U)$ and use its geometric right inverse $\tilde{Q}$ of size CT^2 . Put $K_j = (I - \tilde{Q}L)W_j$. Then $LK_j = 0$, and

$$\|K_j - W_j\|_{X_\rho} \leq CT^2(T^{-40} + T^4T^{-37}) + Ce^{-T}T^{C_J} \leq CT^{-31} + Ce^{-T}T^{C_J}.$$

By (215), the rescaled connection error is $CT^{-61/2}$. The chart derivative differs from the identity on the loops by its pointwise quadratic estimate; the corrected surface displacement is $O(T^{-73/2})$, and the cap metric error is exponentially small. Duhamel's derivative estimate therefore makes the observation matrix on $D\Phi_\rho(U)K_j$ converge to the invertible limiting matrix of Lemma 7.19. Lemma 7.21 identifies these as independent ordinary kernel classes with the same observations. Their dimension is four by (209), so they form a basis. These kernel vectors have not been asserted to be derivatives of a particular choice of corrected solution. $\square$

7.15. The observation-normalized exact solution.

Proposition 7.23 (normalized correction). *At fixed (c, ρ, E) in the chosen patch there is an exact solution $x_{c,\rho,E}^\mathcal{O} = \Phi_\rho(U^\mathcal{O})$ with*

$$(219) \quad \mathcal{O}_{c,\rho}(x_{c,\rho,E}^\mathcal{O}) = \mathcal{O}_0(E), \quad \|U^\mathcal{O}\|_{X_\rho} \leq CT^{-37}.$$

It is a regular ordinary moduli point of dimension four, and the observation values determine it uniquely in an ordinary moduli neighborhood at that positive scale.

Proof. Set $H_\rho(U) = \mathcal{O}_{c,\rho}(\Phi_\rho(U)) - \mathcal{O}_0(E)$ and augment the equation to $\mathbb{F}_\rho = (F_\rho, H_\rho) : X_\rho \rightarrow Y_\rho \oplus \mathbb{R}^4$, with the sum target norm. On the fixed rescaled loops the pregluing is the exact cap and differs from its limit by $C\delta$. Thus $|H_\rho(0)| \leq C\delta$ and $\mathfrak{e}_\rho = \|\mathbb{F}_\rho(0)\| \leq CT^{-40} + Ce^{-T}T^{C_J}$.

Write $D = D_\rho$, $Q = Q_\rho$, and $\ell = dH_\rho(0)$, so $\|\ell\| \leq CT^{1/2}$. For the four finite derivatives put $Z_j = (I - QD)W_j$ and $Zt = \sum t_j Z_j$. Then $DZ = 0$, $\|Z\| \leq C$, and $\|Z_j - W_j\| \leq CT^{-38}$. The matrix $M = \ell Z$ differs from the invertible limiting observation matrix by $O(T^{-75/2}) + O(e^{-T}T^{C_J})$, and hence $\|M^{-1}\| \leq C$. The space

$$\mathcal{B}_\rho = \text{im } Q \oplus Z\mathbb{R}^4$$

is a closed direct sum in the actual complete domain: QD is a bounded projection, and applying D and then M proves the intersection is zero. Its coordinate inverse is $V \mapsto (DV, M^{-1}\ell(V - QDV))$. On this space $\mathbb{A}_\rho = (D, \ell)$ is an isomorphism, with explicit inverse

$$(220) \quad \mathbb{Q}_\rho(f, h) = Qf + ZM^{-1}(h - \ell Qf), \quad \|\mathbb{Q}_\rho(f, h)\| \leq CT^{5/2}\|f\| + C|h|.$$

In particular $\mathfrak{q}_\rho = \|\mathbb{Q}_\rho\| \leq CT^3$. This is a specified complement, not an inverse on the whole geometric domain with its gauge kernel.

The ambient chart remainder has quadratic constant CT^4 . Restriction to the loops costs $CT^{1/2}$, so composition in the observation remainder costs at most $CT^{9/2}$. Its ordinary connection quadratic term costs only CT . Duhamel's higher derivatives and the chart difference estimate give, on the complete ball of radius cT^{-5} ,

$$\|\mathbb{N}_\rho(U) - \mathbb{N}_\rho(V)\| \leq CT^5(\|U\| + \|V\|)\|U - V\|, \quad \mathbb{F}_\rho(U) = \mathbb{F}_\rho(0) + \mathbb{A}_\rho U + \mathbb{N}_\rho(U).$$

The equation is C^1 by the preceding proof; the observations are C^1 by continuous restriction to C^0 and the smooth path ODE maps. The finite end corrections lie outside the loop region.

On the closed $\mathcal{B}_\rho$ ball of radius $2\bar{q}_\rho \bar{\epsilon}_\rho$ use the contraction $U \mapsto -\mathbb{Q}_\rho(\mathbb{F}_\rho(0) + \mathbb{N}_\rho(U))$. Fixed upper bounds permit this radius even if the actual initial error vanishes. Its size is CT^{-37} and its Lipschitz constant is bounded by $CT^{5+6-40} = CT^{-29} < 1/2$. The ball lies in the chart. Completeness and the isomorphism (220) give the exact augmented zero, unique within that ball. This choice may differ from the first correction, which used only $\text{im } Q$. Proposition 7.18 and Corollary 7.22 apply at its CT^{-37} size. The ordinary inverse function theorem makes $\mathcal{O}_{c,\rho}$ a coordinate map near this point, proving the stated local moduli uniqueness. This does not assert that its neighborhood contains all solutions near a broken configuration. $\square$

7.16. Continuity of the normalized solutions at positive scale. For parameter dependence we fix the optional finite cap normalization. Let $K_{\rho,E}$ be the four-dimensional cap kernel with both attaching values zero, and let Π_K be its projection for the surface and positive L^2 inner product. The Gram matrix is uniformly invertible and its smooth basis has bounded graph norm by Proposition 6.17. Replace Q^0 in the preceding construction by

$$(221) \quad Q_{\rho,E}^{\text{can}} = (I - \Pi_K)Q_{\rho,E}^0.$$

Its equation, ordinary base gauge, normal trace and zero attaching values are unchanged, with bound $C\varepsilon^{-1}R_1^\varepsilon$. It is now the unique output of its equation with $\Pi_K U = 0$: a difference is a kernel vector annihilated by its kernel projection. In particular it is independent of the temporary spectral truncation and, on common inputs, of the permitted exponent ε . Use this choice in P_ρ , Q_ρ and (220) from now on. All preceding bounds and identities persist. This specifies the selected solution; it need not agree with a choice that omitted the finite projection.

Proposition 7.24 (positive-scale continuity). *With (221), the normalized solutions $[x_{c,\rho,E}^\mathcal{O}]$ are continuous in the ordinary moduli topology on each chosen local parameter patch with $\rho > 0$.*

Proof. First allow the simultaneous range

$$(222) \quad (2T)^{-1} \leq \varepsilon \leq 2/T.$$

All preceding estimates hold there with the same powers of T : $\varepsilon^{-1} \asymp T$, $\rho^{-\varepsilon} \leq e^2$, and $R_j^{C\varepsilon}$ is bounded. The cap inverse is the same on common inputs by uniqueness, and the augmented fixed residual inverse is the same by its index-zero uniqueness. The formulas for cutoffs, profiles, based gauge potentials and paired transports contain no exponent. Thus P_ρ is the same distributional operator, and its Neumann series has identical terms and converges in every norm in (222). The same is true of Q_ρ , Z , M and $\mathbb{Q}_\rho$. The nonlinear chart and almost everywhere holonomy formulas are also independent of the exponent. Starting the augmented iteration at zero gives identical iterates in the highest space and every lower space in this interval. Their limits agree as distributions. Hence the chosen solution is independent of this auxiliary exponent.

Fix $\lambda_0 = (c_0, \rho_0, E_0)$ with $\rho_0 > 0$ and use one fixed $\varepsilon_* > 0$ in every nearby exponent window. At fixed positive scale the cap metric depends smoothly on the finite parameters: its Dirichlet linearization is elliptic and invertible, so the usual implicit function theorem applies. The normalized base gauge and its finite attaching conditions have the same property. Polar matching maps identify the nearby cap domains, preserve the ordinary base gauge and total zero attaching value, and gain one surface derivative in their differences. Thus the coefficients and finite Gram projection give a continuous family of bounded isomorphisms $(\mathcal{A}_{\text{cap}}, \Pi_K)$ in fixed-exponent operator charts. The inverse identity gives continuous, in fact smooth, local dependence of (221). This uses no derivative of Sobolev rescaling on a rough input.

We identify the global spaces next. Let $h_\lambda = \rho R_1(\rho)$. Choose smooth physical base diffeomorphisms supported inside the fixed perturbation-free collar, preserving the seam, its normal lines, and both sides, which align the θ_1 interfaces with those at λ_0 . Near the interface use translation to c and dilation h_λ/h_{λ_0} . On the smaller observation region use translation and the physical scale ratio ρ/ρ_0 , so that all observation loops become fixed. Radial interpolation between these separated regions and the identity gives diffeomorphisms near λ_0 ; even normal cutoffs preserve normal lines. Use the same map on the positive side and the actual polar transports to identify matching domains. The ends, perturbation dependencies and finite end coordinates stay fixed.

If the based residual representative of Lemma 7.3 changes with the center, include its smooth unitary gauge in these identifications. It is supported in a fixed collar and is identity at the ends and dependency regions. Covariance conjugates its augmented residual operator and inverse to the single fixed residual problem. Thus references to the fixed residual inverse here concern a fixed gauge class with these explicit smooth representative identifications, not an assumption that the point-normalized representative is independent of c .

These maps give locally uniformly bounded identifications with fixed complete spaces X_*, Y_* . Aligning θ_1 aligns its two unequal-regularity factors. On the raw cap factor, base pullback preserves separate Sobolev orders and does not change exterior surface differentiation into a full gradient. Changing d_α adds only a zeroth-order coefficient, and $d_\alpha(W - I)b$ is order zero. On the overlap and residual factor the full additional surface derivative is already present. The support of the coordinate change lies strictly where $\chi = \varphi = 1$, so the transition of the surface regularizer and all weighted ends stay fixed. At this positive scale all retained coefficient maps and their inverses are smooth and finite. Their pullbacks are strongly continuous, by approximation in the ambient localized graph spaces and the local uniform bounds. One may first regularize in the surface, commuting its elliptic regularizer past the exterior derivatives, and then regularize in the base. The lower-order commutators are controlled by the strong norms. No operator-norm derivative of a rough pullback is asserted.

In these charts $D_\lambda : X_* \rightarrow Y_*$ is continuous in operator norm. Its transported rows have smoothly varying coefficients of controlled base derivatives and the exterior derivatives $d_\alpha a, d_\alpha b$; the polar error is surface order zero. The residual factor controls the stronger local derivatives, and all nonlocal perturbation regions are unchanged. This is an operator coefficient assertion after transport, distinct from norm continuity of raw pullback.

The constructed P_λ is strongly continuous. Source cutoffs and conversions have that property; the fixed residual inverse is bounded and the canonical cap inverse has the local dependence just established. The only additional issue is residual evaluation on a moving homogeneous disc. Near λ_0 all such discs contain a common smaller positive-size disc. After the fixed residual identification, differences of the inverse outputs solve the same homogeneous equation there. Lemma 7.9 upgrades their graph convergence to every fixed local base and surface order on a smaller disc. Moving point values and derivatives in the based gauge

are therefore continuous. Dilation turns its moving Dirichlet and Neumann potential problems into fixed-disc problems with continuously varying normalized coefficients. The exact and harmonic profiles then vary continuously. The cap output has a strict support margin $|w| \leq R_1/4$ inside $\theta_1 = 1$, so it never acquires a stronger exterior norm when the interface moves. The residual field already has that norm where it crosses the interface.

For completeness, the density used in this strong-continuity argument can be checked in the actual source norm. Truncate weighted ends by flow cutoffs, which commute with all surface and finite-rank terms; fractional truncations converge by integrable tails. Regularize next in Σ by a fixed elliptic resolvent J_η , which commutes with χ . Expanding Γ^j gives $\chi^j \Lambda_{\Sigma, \theta}^j$ plus lower surface-order and finite-rank terms. Since the principal symbol is independent of θ , its commutator with J_η has at most $j - 1$ surface orders and is controlled by lower Γ norms. The commutators are uniformly bounded and tend strongly to zero, first on smooth surface fields and then by the lower-order bounds. This preserves $\varphi F \in H_B^{\varepsilon*} H_\Sigma^k$. After surface regularization all surface orders are finite; base and interior smoothing then converge in the source norm, using bounded extension across the seam below its trace threshold. A diagonal choice gives smooth compact source approximation. For the ambient domain, a base derivative of these commutators adds $\chi^{j-1} d\chi$ times at most $j - 1$ surface derivatives, again controlled by the lower norms. There is no division by χ . This also justifies the ambient approximation used above.

Put $R_\lambda = D_\lambda P_\lambda - I$. It is strongly continuous. Its norm in an arbitrary equivalent common norm need not be below $1/2$, but its powers satisfy

$$\|R_\lambda^n\|_{Y_* \rightarrow Y_*} \leq C_{\lambda_0} 2^{-n}.$$

Conjugate the whole power to the intrinsic Y_λ norm; the two identification bounds occur once, not once per factor. The Neumann series therefore converges locally uniformly in operator norm and is strongly continuous, as is Q_λ . The finite W_j and their sources are continuous. The observation functional ℓ_λ is norm-continuous because its loops are fixed in the chosen chart. Consequently Z_λ , M_λ^{-1} and (220) are strongly continuous.

The transported nonlinear remainders are jointly continuous in λ, U . On local regions use the fixed-exponent product and surface-gaining composition estimates; on the unchanged dependency regions use Lemma 7.15 and the fixed summability class. Exact end subtraction uses one smooth finite family, and observations use continuous C^0 restriction. Starting the augmented iteration at zero thus gives continuous X_* -valued iterates. The intrinsic contraction estimate implies

$$\|U_{\lambda, n} - U_\lambda^O\|_{X_*} \leq C_{\lambda_0} 2^{-n}.$$

Uniform convergence proves graph continuity, including the finite end coordinates, without requiring a contraction constant below one in the particular common norm.

We now pass to ordinary moduli continuity, which requires control of the ends as well as compact sets. For a sequence $\lambda_i \rightarrow \lambda_0$, graph convergence gives physical $W^{1,p}$ convergence on compact sets for a fixed $p > 2$ and positive C^0 convergence. On each mixed end subtract its own exact representative B_{p_i} ; the finite parameters converge, and the remainders converge in weighted H^1 . On ordinary ends there is ordinary H^1 convergence. Formula (205), the L^2 holonomy Lipschitz estimate and $H^1 \subset L^4$ show that the analytical energy densities are uniformly integrable with uniformly small tails. For the ordinary gauge end use the fixed residual solution and its integrable perturbed curvature; the positive chart gives the same conclusion for map energy.

On compact sets the $W^{1,p}$ bound gives an L^p curvature bound because $a^2 \in L^{2p/(4-p)}$ with exponent larger than p , taking $p < 4$. Hence energy cannot concentrate at an interior point

or on a shrinking matching neighborhood. The positive $W^{1,p}$ bound excludes positive and boundary concentration as well. The compactness theorem [DFL26, Theorem 5.2], with the all-order perturbation class and empty concentration sets, gives subsequential smooth convergence on compact sets modulo gauge. It includes the full compact secondary dependency sets.

This smooth limit represents the graph limit. For the gauges relating the original A_i to the smoothly convergent A'_i , the equation $dg_i = g_i A'_i - A_i g_i$ and unitarity give local $W^{1,q}$ bounds, $q = 4p/(4-p) > 4$. Differentiating gives $W^{2,p}$ bounds since $q/2 > p$. Compact embedding and a diagonal subsequence give a limiting unitary gauge relating the smooth limit to A_0 . The positive limit is already u_0 . Thus no continuous prior choice of smoothing gauges was needed.

On a mixed tail the topological integral required for the small-tail theorem is also uniformly small. Write $A = B_p + \xi$ and note that $F_{B_p} \wedge F_{B_p} = 0$ because it is pulled back from a three-manifold. Expanding leaves $F_{B_p} \wedge d_{B_p} \xi$, $F_{B_p} \wedge \xi^2$, $d_{B_p} \xi \wedge d_{B_p} \xi$, $d_{B_p} \xi \wedge \xi^2$ and ξ^4 . With weighted H^1 bound K , weighted Cauchy–Schwarz bounds the first absolute integral by $Ce^{-\omega R}K$, and weighted $H^1 \subset L^4$ bounds the others by $Ce^{-\omega R}(K^2 + K^3 + K^4)$, after decreasing the weight. Positive area is bounded by energy. The uniform tail therefore meets the fixed threshold of [DFL26, Proposition 5.29]. Temporal gauges give uniform smooth exponential decay to the individual critical limits. The argument of [DFL26, equation (5.36)] promotes compact convergence to weighted convergence after subtracting those limits: split at a large R , use compact convergence before R , and bound the weighted tail in each fixed order by $C_l e^{-cR}$. The limiting classes are the original p_i classes, fixed by the positive limits and embedded critical Lagrangians, and converge to p_0 .

Uniformly small ordinary analytical tails exclude a nonconstant trajectory or strip escaping to infinity: translating a compact piece of such a limit would have positive energy in a uniformly vanishing tail. Global H^1 control relative to the residual also fixes the limiting critical classes. To make the ordinary Sobolev conclusion explicit, choose cross-sectional gauge slices near the fixed nondegenerate critical points, or the corresponding strip coordinates. Their self-adjoint first-order Hessians have a fixed spectral gap. The inverse estimate and a small-neighborhood absorption bound the H^1 displacement by the L^2 gradient. The local action is quadratic in the displacement, so remaining energy satisfies $E'(t) \leq -cE(t)$ along the gradient equation. Every sufficiently far strip stays in that neighborhood: otherwise compactness of a translated sequence, uniform small tails, and the fixed class would give a zero-energy limit with a nonzero critical displacement. Unit-strip elliptic estimates upgrade uniform exponential energy decay to every fixed smooth order. Splitting into compact and tail parts gives the ordinary unweighted Sobolev convergence. This is the no-energy-loss case of the ordinary-end discussion in [DFL26, Section 5]; its later index-at-most-one exclusion is not used here.

The smooth limit is related to the smooth representative of x_0 by an allowed gauge: on temporal tails their relating gauge has zero temporal derivative and the required limiting cross-sectional value. After the prescribed ordinary-end identifications that value is a stabilizer. Its spatial equation gives smoothness. Every subsequence therefore has a further subsequence converging to $[x_0]$ in the ordinary moduli topology, at any fixed finite Sobolev order and a sufficiently small common mixed-end weight. The effective local slice makes this topology metrizable. The full sequence converges, proving the proposition. $\square$

7.17. Smoothness and nonsingularity of the parameter map. Evaluation at infinity is a submersion on the regular primitive patch, and the two affine normal functions have independent differentials inside its fibers by (212). The finite-dimensional submersion theorem therefore gives a local smooth family $E = E(c, t_1, t_2)$ with

$$\mathrm{ev}_\infty E(c, t) = [\alpha(c)], \quad \mathrm{Re} \kappa(E) = 0, \quad s(E) = 1.$$

Use the based surface representative $\alpha(c)$ in this family. It is independent of the scale. The t derivatives span the two-dimensional normalized fiber tangent. Shrink the coordinate neighborhood so that the fixed four observations above work throughout, and take constants on its compact closure. Define

$$(223) \quad \mathrm{Glue}_{\mathrm{loc}}(c, \rho, t) = [x_{c, \rho, E(c, t)}^{\mathcal{O}}].$$

Completion of the proof of Theorem 7.2. At every positive scale the image point is an ordinary regular four-dimensional moduli point, and $d_x \mathcal{O}_{c, \rho}$ is an isomorphism by Corollary 7.22. On its ordinary moduli neighborhood consider

$$G(c, \rho, t, x) = \mathcal{O}_{c, \rho}(x) - \mathcal{O}_0(E(c, t)).$$

It is jointly smooth. Choose high Sobolev slice orders for any fixed number of moving-loop derivatives; elliptic regularity identifies these finite-dimensional moduli charts. The limiting Dirichlet metric is a smooth finite family as well. The implicit function theorem gives a unique smooth zero branch near each parameter point. Proposition 7.24 puts the selected solutions in that same neighborhood, and their defining observation equations identify them with this branch. Thus (223) is smooth for $\rho > 0$.

We identify its differential by affine equivariance of the limiting reconstruction. For $g(w) = aw + b$, holomorphic pullback preserves the trace-free base-flat equation and pulls its stable boundary projectively flat metric to the corresponding metric of g^*E . Dirichlet uniqueness therefore identifies the projective zero-scale metric of g^*E with $g^*H_0(E)$, and hence

$$[A_{g^*E}^0] = [g^*A_E^0].$$

The determinant reference was chosen pulled back from Σ on the negative product, so it is invariant under this affine pullback. Locally the projective gauge lifts to a determinant-preserving unitary identification. Thus this equality applies to the actual trace functions, with no central base-curvature contribution. The positive classifying map has the same equivariance.

Let $\hat{A}_\lambda$ be the normalized solution pulled back by $(w, x) \mapsto (c + \rho w, x)$ to the loop region. The graph correction and (215) give

$$(224) \quad \|\hat{A}_\lambda - A_{E(c, t)}^0\|_{C^0(K)} \leq CT^{-73/2} + Ce^{-T}T^{C_J}.$$

The chart's pointwise quadratic term is smaller, and the pregluing is exactly the cap there. Apply Duhamel's value estimate with σ_T, σ_D in (214). Equivariance and the same change of variables in the limiting average imply

$$\mathcal{O}_0^{\sigma_T}(A_E^0) = d\mathcal{O}_0(E)T_E, \quad \mathcal{O}_0^{\sigma_D}(A_E^0) = d\mathcal{O}_0(E)D_E.$$

These are left compositions on the loops: translation changes (a, b) to $(a, b + h)$ and dilation to $(e^h a, e^h b)$, which fixes both signs and the two-dimensional Jacobian.

Put $L_\lambda = d_x \mathcal{O}_{c, \rho}$ on the ordinary moduli tangent. Differentiate the exact normalization $\mathcal{O}_{c, \rho}(\mathrm{Glue}_{\mathrm{loc}}(c, \rho, t)) = \mathcal{O}_0(E(c, t))$, now justified by smoothness. On the parameter frame

$(\rho\partial_c, \rho\partial_\rho, \partial_{t_1}, \partial_{t_2})$ the columns of $L_\lambda d\text{Glue}_{\text{loc}}$ are

$$(225) \quad \begin{aligned} J_c &= \rho d\mathcal{O}_0(E)\partial_c E - \mathcal{O}_{c,\rho}^{\sigma_T}(x_\lambda^\mathcal{O}), \\ J_\rho &= -\mathcal{O}_{c,\rho}^{\sigma_D}(x_\lambda^\mathcal{O}), \\ J_{t_1} &= d\mathcal{O}_0(E)\partial_{t_1} E, \quad J_{t_2} = d\mathcal{O}_0(E)\partial_{t_2} E. \end{aligned}$$

There is no scale derivative of E . Its bounded c derivative is multiplied by ρ . By (224), these columns differ by at most $C\rho + CT^{-73/2} + Ce^{-T}T^{C_J}$ from

$$d\mathcal{O}_0(E)(-T_E, -D_E, \partial_{t_1} E, \partial_{t_2} E).$$

This limiting matrix is invertible by (212) and Lemma 7.19, uniformly on the smaller compact patch. Thus (225) is invertible for every sufficiently small positive scale. Since L_λ is an isomorphism, $d\text{Glue}_{\text{loc}}$ is invertible. The displayed parameter frame is a basis whenever $\rho > 0$, so the ordinary inverse function theorem gives the claimed local diffeomorphism. Its correction is CT^{-37} by (219), and its topological component is the one computed in (208). This completes the local theorem. $\square$

The two minus signs in the limiting parameter matrix do not by themselves give the determinant-line orientation comparison. The coordinate rotation, ordered gluing parameters and coherent orientations must also be retained. Proposition 10.3 identifies the derivative with an explicit analytic kernel map and computes the finite complex sign. Proposition 10.14 then establishes the excision comparison, and Section 12 identifies the coherent signs. The four holonomy functions serve as coordinates; the point-incidence class is constructed in Section 9.2. Theorem 8.16 gives injectivity on a whole small parameter product. The hypotheses for converse gluing are stated at the start of Section 8.

7.18. The constructed broken limit. We now identify the limit of the local construction. Throughout this subsection the parameters stay in a compact subset of one sufficiently small regular seam-admissible patch from Theorem 7.2. The residual $x_0 = (A_0, u_0)$ is fixed. Write $x_\rho = (A_\rho, u_\rho)$ for the observation-normalized exact solution, x_{pre} for its pregluing, and

$$s_\rho = T^{-37}, \quad T = \log(1/\rho).$$

The notation $x_E^0 = (A_E^0, u_E)$ denotes the smooth compactified base-flat reconstruction of the primitive bundle E . In its prescribed unitary frame the negative base component is zero. $\hat{}$ denotes physical pullback under $z = c + \rho w$.

Convergence of holomorphic bundles on compact sets means convergence of their holomorphic transition functions on a finite product cover, after bundle identifications. In particular this notion retains an unstable surface fiber and is stronger than convergence of classifying maps on the stable locus alone. No insertion points are marked in the following result.

Theorem 7.25 (limit of the constructed family). *Suppose $\rho_\nu \rightarrow 0$, $c_\nu \rightarrow c_*$, and $E_\nu \rightarrow E_*$ in the above patch, with its prescribed affine normalization and attaching data. Then the constructed solutions have the following limits.*

- (1) *Away from $\{c_*\} \times \Sigma_g$, the residual limit is x_0 , smoothly on compact sets modulo gauge. On the mixed ends there is convergence in every prescribed weighted Sobolev order after subtraction of the individual critical limits, for one sufficiently small positive weight. On the ordinary ends the corresponding convergence is unweighted.*
- (2) *On every fixed rescaled compact set, including the seam, the pair $\hat{x}_{\rho_\nu}$ converges modulo local unitary gauges to $x_{E_*}^0$ in every prescribed finite differentiability order. The*

doubled holomorphic projective bundles converge to $\mathbb{P}(E_*)$ in the sense just defined, including near an interior unstable fiber.

- (3) The whole neck has vanishing limiting energy and surface/map diameter. The rescaled normalized energy measures converge in total variation to the smooth measure of $x_{E_*}^0$, of mass $\frac{1}{2}$. The component at infinity has the prescribed attaching identification and extends holomorphically as $\mathbb{P}(E_*)$, with no additional modification.

Consequently Glue_{loc} extends continuously at scale zero, on its own image, by the prescribed residual-plus-primitive configuration. This assertion gives neither a neighborhood of that image nor a converse or a differentiable collar at scale zero.

7.18.1. Compatibility of the surface completions.

Lemma 7.26. *The observation-normalized construction is independent of the integer surface completion. For every fixed $k \geq 2$, its correction satisfies*

$$(226) \quad \|U_\rho^\mathcal{O}\|_{X_\rho^{(k)}} \leq C_k T^{-37}$$

for $0 < \rho < \rho_k$. The threshold may depend on k .

Proof. Keep the canonical cap inverse, its finite L^2 kernel projection, the four observations, cutoffs, paired transports, and the operators Γ_X, Γ_Y fixed before choosing a completion. In particular their finite end projections and added scalar shifts do not depend on k . The exact cap metric is also independent of the derivative order used to estimate it. For any fixed k , its error is bounded by $C_k \rho^2 T^{N_k}$ for a finite N_k . All four radius powers in (157) remain unchanged. Factors $R_1^{C_k \varepsilon}$ are bounded for each fixed k , and the exponential factor ρ^2 absorbs every fixed additional power of T . Thus all the preceding estimates hold at this surface order.

The higher-order cap solution lies in the lower-order domain. The cap inverse is unique after base gauge, zero total attaching value, and the fixed kernel projection; its outputs therefore agree. The augmented residual inverse is already unique in its ordinary weighted H^1 domain, including its finite end variables. Every term in the geometric parametrix and its Neumann series is consequently independent of the completion. The surface chart is defined by the same uniquely solvable small elliptic equations, and the end profiles are the same exact critical representatives. Starting nonlinear iteration at zero gives identical iterates at each order. Their limits agree as distributions. This proves (226) without comparing contraction-ball radii or assuming uniqueness in a larger moduli neighborhood. No single threshold valid for all k is asserted. $\square$

7.18.2. *Energy on the whole neck.* Fix a paired physical disc $D(c, r_*)$ contained in the unperturbed Euclidean matching collar, uniformly for the allowed centers. For the paired annulus $A(c; a, b)$ let $\mathfrak{E}_\rho(a, b)$ be the analytical energy: the negative integral of $|F_{A_\rho}|^2$ and the positive integral of $|du_\rho|^2$, with the fixed conventions of [DFL26, equation (3.5)]. Perturbations vanish on this disc.

Proposition 7.27. *There is $\eta_\rho \rightarrow 0$, uniform on the parameter patch and independent of subsequently chosen R, r , such that*

$$(227) \quad \mathfrak{E}_\rho(R\rho, r) \leq CR^{-2} + Cr^2 + \eta_\rho, \quad 2 \leq R \leq R_2/2, \quad 4\rho R_2 \leq r \leq r_*.$$

In the actual attaching surface frames, for every fixed $k \geq 2$,

$$(228) \quad \sup_{A(c; R\rho, r)^-} \|\beta_\rho - \alpha(c)\|_{H_\Sigma^k} + \sup_{A(c; R\rho, r)^+} d(u_\rho, [\alpha(c)]) \\ \leq C_k(R^{-1} + r) + C_k T^{-73/2} + o_\rho(1).$$

The final error is an exponential times a fixed power of T .

Proof. The physical product metric in a compact cap coordinate is $(\rho f)^2 g_B + g_\Sigma$, with $1 \leq f \leq M_1$. Its curvature norm retains the three weights

$$(229) \quad \|F\|_{L^2_{\text{phys}}}^2 = \|(\rho f)^{-1} F_{BB}\|_2^2 + \|F_{B\Sigma}\|_2^2 + \|\rho f F_{\Sigma\Sigma}\|_2^2,$$

using orthonormal component conventions. The mixed integral, like the positive Dirichlet integral, is conformally invariant in the base. For a normalized cap base variation ρa , its physical coefficient is a/f . Change of variables and the two-dimensional H^1 embedding give

$$(230) \quad \|a/f\|_{L^4_{\text{phys}}} \leq C\rho^{1/2} M_1 \|a/M_1\|_{H_B^1 H_\Sigma^k}.$$

Indeed the fourth power on the left is $\rho^2 \int f^{-2} |a|^4$, and $f \geq 1$.

Write the actual ambient displacement as $\xi = A_\rho - A_{\text{pre}}$, including the nonlinear surface correction, and $u_\rho = e(u_{\text{pre}}, \nu)$. Its norm is at most $C(s_\rho + T^4 s_\rho^2) \leq C s_\rho$. Localize it as $\xi_i = \theta_1 \xi = (\rho a_i, b_i)$ and $\xi_o = (1 - \theta_1) \xi$. The complete norm gives

$$(231) \quad \begin{aligned} & \|(a_i/M_1, b_i)\|_{H_B^{1+\varepsilon} H_\Sigma^k} + \|\rho d_\alpha a_i\|_{L_B^2 H_\Sigma^k} + \|\rho f d_\alpha b_i\|_{L_B^2 H_\Sigma^k} \leq C s_\rho, \\ & \|\xi_o\|_{H_{\text{phys}}^1} + \|\xi_o\|_{H_{B,\text{phys}}^{1+\varepsilon} H_\Sigma^k} \leq C s_\rho. \end{aligned}$$

All physical norms in this proof are on the fixed collar or the indicated localized support. Its cutoff functions are one on a slightly larger collar; finite end variables contribute only bounded smooth fields here. Both localized fields already include their differentiated cutoffs. Differentiating their sum requires no additional estimate of $(d\theta_1)\xi$ in an unlocalized norm. The two-dimensional evaluation estimate yields

$$(232) \quad \|b_i\|_{C_B^0 H_\Sigma^k} + \|\xi_o\|_{C_{B,\text{phys}}^0 H_\Sigma^k} \leq C\varepsilon^{-1/2} s_\rho = CT^{-73/2}.$$

The positive displacement has the same localized estimates.

On the inner support the preglued surface coefficients and their fixed surface derivatives are bounded, while its compact base coefficient is bounded by $C(\delta + \rho^2 R_1^3) = o(1)$. For the residual contribution, the based physical coefficient is $O(|z - c|)$ and conversion multiplies it by ρf , giving $O(\rho^2 |\zeta|^{-3})$. The cap estimate and interpolation give the other contributions. On the outer support the pregluing is exactly the smooth residual.

The surface curvature difference is

$$d_{\beta_{\text{pre}}}(b_i + b_o) + (b_i + b_o) \wedge (b_i + b_o).$$

The differentiated inner term is bounded by $C s_\rho + C \rho M_1 s_\rho$ in physical L^2 , including the reference change in d_α . Its square costs $C \rho M_1 \varepsilon^{-1/2} s_\rho^2$. The outer derivative and square cost $C(s_\rho + s_\rho^2)$; the cross term costs $C\varepsilon^{-1/2} s_\rho^2$. Since $\rho M_1 \rightarrow 0$, this proves

$$(233) \quad \|F_{\Sigma\Sigma}(A_\rho) - F_{\Sigma\Sigma}(A_{\text{pre}})\|_{L^2_{\text{phys}}} \leq C s_\rho.$$

Only the exterior surface derivative in (231) is used.

For mixed curvature the differentiated inner terms are $d_B b_i - \rho d_\alpha a_i$; the remaining linear terms contain either the small compact base coefficient times b_i or a bounded surface coefficient times ρa_i . Their norms are $O(s_\rho)$. The inner quadratic term costs $C \rho M_1 \varepsilon^{-1/2} s_\rho^2$, and the outer terms cost $C(s_\rho + s_\rho^2)$. The two cross terms, in physical coefficients, are $(a_o)_{\text{phys}} b_i$ and $(a_i/f) b_o$. Equations (230)–(232) bound them by $C\varepsilon^{-1/2} s_\rho^2$ and $C\rho^{1/2} M_1 s_\rho^2$, respectively. Therefore the mixed curvature difference also has physical L^2 norm at most $C s_\rho$.

The base curvature must use the exact scalar equation:

$$(234) \quad (\rho f)^{-1} *_B F_{BB}(A_\rho) = -\rho f *_\Sigma F_{\Sigma\Sigma}(A_\rho).$$

The pregluing scalar error has physical L^2 norm $O(T^{-40})$. Subtract its equation from (234) and use (233). This proves the full estimate

$$(235) \quad \|F_{A_\rho} - F_{A_{\text{pre}}}\|_{L^2_{\text{phys}}(D(c, r_*))} \leq C s_\rho.$$

This estimate concerns exact zeros. Directly differentiating a general raw base field would instead lose M_1 : an interior supported field $a = M_1 s_\rho a_0$, $b = 0$, has raw graph size $O(s_\rho)$, whereas its normalized base curvature has size $M_1 s_\rho$. Such a field does not satisfy the scalar equation.

The first compact derivatives of u_{pre} are bounded on the inner support. They are bounded for the cap; in the residual region they cost $O(\rho f)$; and on the transition $\zeta = vZ$, $v = R_2^{-1}$, the values differ by $O(v + \rho/v + \delta) = O(v)$ while the cutoff derivative costs v^{-1} . On the outer support use the fixed residual derivatives. In smooth ambient target coordinates,

$$|du_\rho - du_{\text{pre}}| \leq C(|\nabla \nu| + |\nu||du_{\text{pre}}|).$$

Apply the localized estimates and conformal invariance to obtain

$$(236) \quad \|du_\rho - du_{\text{pre}}\|_2 \leq C s_\rho, \quad \sup d(u_\rho, u_{\text{pre}}) \leq CT^{-73/2}.$$

The product in the derivative estimate uses the L^2 norm of ν , so it has no extra evaluation loss.

It remains to integrate the pregluing itself. Divide the annulus into the exact cap $|w| \leq R_2$, the interpolation annulus, and the exact residual $|z - c| \geq 2\rho R_2$. Smoothness of the limiting cap at $\zeta = 0$ gives

$$(237) \quad \int_{|w| \geq R} |d_B \beta_E^0|^2 + \int_{|w| \geq R} |du_E|^2 \leq CR^{-2}.$$

The cap mixed curvature converges in compact C^0 to $d_B \beta_E^0$, with error $O(\delta)$. Its weighted surface curvature has norm at most $C\rho M_1$, as does its base curvature by exact ASD. Thus its annular contribution is $CR^{-2} + o(1)$.

On $\zeta = vZ$ the surface and positive deviations from attachment are $O(v)$ with bounded first compact derivatives. Their mixed and positive energies are $O(v^2)$. The weighted surface curvature norm is $O(\rho/v)$; scalar base curvature is bounded by this term and the pregluing error. The transition energy is therefore at most

$$Cv^2 + C(\rho/v)^2 + CT^{-80} + o(1),$$

where the last term is exponential times a fixed power of T . The smooth residual contributes at most Cr^2 . This proves the pregluing version of (227) and a uniform bound for its energy on $D(c, r_*)$. Equations (235) and (236) transfer both assertions to the exact solution. They also give L^1 convergence of the two analytical energy densities on this fixed collar, by factoring the difference of squares.

Finally the cap surface and positive values differ from attachment by $C/R + C\delta$, the transition by $O(v + \rho/v + \delta)$, and the residual by Cr . Add (232) and (236) to prove (228). There is no sum over a growing number of annuli in either argument. $\square$

Corollary 7.28. *On $|z - c| \leq \rho R_1/2$, pull the normalized topological energy measure back to the rescaled base and extend it by zero over the omitted neighborhood of infinity. These*

measures converge in total variation to

$$(238) \quad \mu_E^0 = \frac{1}{8\pi^2} \int_{\Sigma} \text{tr}(F_{A_E^0} \wedge F_{A_E^0}) + \frac{1}{4\pi^2} u_E^* \Omega, \quad \mu_E^0(\mathbb{P}^1) = \frac{1}{2},$$

where the two terms are placed on their respective hemispheres. For $E_\nu \rightarrow E_*$ the limit is $\mu_{E_*}^0$.

Proof. On every fixed rescaled compact set the pregluing is the cap and $\theta_1 = 1$. The compact base displacement is ρa_i , whose $H_B^1 H_\Sigma^k$ norm is $O(\rho M_1 s_\rho)$; the surface and positive displacements have size $O(s_\rho)$. Consequently the coefficients converge in these first-order norms to x_E^0 . On the whole retained disc, the pregluing cap mixed curvature converges in compact L^2 to the limiting mixed curvature. The omitted tail has energy $O(R_2^{-2})$, the transition energy tends to zero, and the residual part has energy $O((\rho R_1)^2)$. Its weighted base and surface curvatures tend to zero, by exact ASD off the transition and the vanishing scalar error on it.

Wedge products are continuous from $L^2 \times L^2$ to L^1 . Thus (235) also gives convergence of the negative Chern–Weil densities. The L^1 norm of a top-degree form is unchanged under pull-back, so no rescaling factor remains. Equation (236) and smoothness of Ω give convergence of the positive area forms. This proves total-variation convergence, including the omitted region at infinity. The same argument before integrating over Σ excludes concentration at an interior point of a rescaled product chart.

The limiting measure is positive: its negative mixed curvature has the ASD mixed-type relation, its base curvature is zero, and its positive map is holomorphic. Sew the negative connection to a flat surface family on the positive disc, retaining the prescribed seam identification. The flat-family formula [DFL26, Lemma 5.10 and equation (5.12)] identifies its positive Chern–Weil integral with twice the symplectic area. The resulting adjoint bundle is $\text{ad } E$. Since $p_1(\text{ad } E) = -2$, the normalization in (238) has mass $-\frac{1}{4}(-2) = \frac{1}{2}$. An interior rank-drop fiber causes no singularity in this smooth negative reconstruction. In genus one a constant positive map does not trivialize the seam clutching bundle. $\square$

7.18.3. Regularity on fixed rescaled domains.

Lemma 7.29. *On nested fixed rescaled compact sets $K \subset K'$, including a seam segment, there are local unitary base gauges such that, for every fixed m, k ,*

$$(239) \quad \|g_\rho^* \hat{A}_\rho - A_E^0\|_{W_B^{m,4} H_\Sigma^k(K)} + \|\hat{u}_\rho - u_E\|_{W^{m,4}(K^+)} \longrightarrow 0.$$

The gauges are close to the identity in $H_B^2 H_\Sigma^k$ on their fixed defining domains. Each fixed order has its own small-scale threshold.

Proof. On a fixed rescaled domain the pregluing is the cap and $\theta_1 = 1$. Lemma 7.26 and the cap estimates give, with $\hat{A}_\rho = A_{B,\rho} + \beta_\rho$,

$$(240) \quad \|A_{B,\rho}\|_{H_B^1 H_\Sigma^k} \leq C_k \rho M_1 s_\rho + C_k \rho^2 T^{N_k} = o(1),$$

$$\|\beta_\rho - \beta_E^0\|_{H_B^1 H_\Sigma^k} + \|\hat{u}_\rho - u_E\|_{H^1} = o(1).$$

The coefficient differences also tend to zero in $C_B^0 H_\Sigma^k$, using the evaluation loss $\varepsilon^{-1/2} = T^{1/2}$.

Choose a fixed smooth negative base domain D with any artificial boundary separated from K . It may have a straight seam segment. The ordinary zero-mean Neumann problem with one-form source a has

$$\|\phi\|_{H_B^2 H_\Sigma^k} \leq C_k \|a\|_{H_B^1 H_\Sigma^k}, \quad \Delta_B \phi = -d_B^* a, \quad \partial_n \phi = -a(n).$$

The two signs are understood through the corresponding variational problem; the compatibility condition is the divergence theorem. The scalar inverse acts coefficientwise on the Hilbert surface space and commutes with surface derivatives. For $g = e^\phi$ write $g^*a = a + d_B\phi + R(a, \phi)$. Two-dimensional multiplication gives

$$\|R(a, \phi) - R(a, \psi)\|_{H_B^1 H_\Sigma^k} \leq C_k(\|a\|_{H^1} + \|\phi\|_{H^2} + \|\psi\|_{H^2})\|\phi - \psi\|_{H^2}.$$

Here $H_B^2 \cdot H_B^1 \rightarrow H_B^1$ and $H_B^1 \cdot H_B^1 \rightarrow L_B^2$ control, in particular, the differentiated term $d\phi d\phi$. The entire remainder is in H^1 , so its normal trace is controlled as well. Applying the Neumann inverse to $a + R(a, \phi)$ is a contraction. For $a = A_{B,\rho}$ this gives

$$(241) \quad d_B^*(g_\rho^*a) = 0, \quad (g_\rho^*a)(n) = 0, \quad \|g_\rho - I\|_{H_B^2 H_\Sigma^k} \leq C_k \|a\|_{H_B^1 H_\Sigma^k}.$$

Its logarithm has zero mean. Identical iterations identify the gauges at different surface orders. On the seam $a(n) = 0$ originally, so the new normal condition implies $\partial_n g_\rho = 0$. The matching class is preserved and the positive map is unchanged. The transformed surface connection contains $g_\rho^{-1} d_\Sigma g_\rho$; use one additional surface order in (241). Hence, writing the transformed pair as $a_\rho + \beta_E^0 + b_\rho$ and $\hat{u}_\rho$, its differences are still small in the H^1 and C^0 norms of (240).

Fold the positive half-disc across the seam. The limiting linear boundary condition is the smooth real subspace

$$C_{\beta_E^0} b = 0, \quad H_{\beta_E^0} b = \ell\nu.$$

Its local complex straightening, followed by complex-conjugate reflection, gives a Cauchy–Green parametrix for the limiting operator L_E . It is bounded from $L_B^p H_\Sigma^k$ to $W_B^{1,p} H_\Sigma^k$ for $p = 2, 4$, with the same integral formula. The scalar singular-integral estimate extends to the Hilbert space by random signs and square-function inequalities. Smooth straightening adds bounded zeroth-order coefficients; the local remainders improve base regularity. No attaching-point evaluation is required in this local estimate.

Use the nonlinear matching chart of Proposition 6.13, centered at the limiting pair. Its derivative is the identity and its negative correction gains one surface derivative. The difference coordinate v_ρ satisfies the fixed linear boundary condition, is small in $H_B^1 H_\Sigma^k$ and $C_B^0 H_\Sigma^k$, and solves

$$(242) \quad L_E v_\rho = B(z, v_\rho) D_B v_\rho + f_\rho, \quad \|B(z, v_\rho)\|_{L^\infty \mathcal{L}(H_\Sigma^k)} = o(1).$$

The source f_ρ contains $d_\Sigma a_\rho$, bounded coefficients times a_ρ, v_ρ , and their zeroth-order products, but no derivative of v_ρ . It vanishes at the limiting pair. The higher surface orders and $H_B^1 \subset L_B^4$ give $\|f_\rho\|_{L_B^4 H_\Sigma^k} = o(1)$. Localizing with a fixed cutoff and applying the parametrix R gives

$$(243) \quad (I - RBD_B)(\chi v_\rho) = h_\rho, \quad \|h_\rho\|_{W_B^{1,4} H_\Sigma^k} = o(1).$$

The right side includes the source, cutoff commutators applied to $v_\rho \in L^4$, and regularizing remainders. For small ρ the operator RBD_B has norm less than $\frac{1}{2}$ on both $W^{1,2}$ and $W^{1,4}$. Its inverse is the same Neumann series on common inputs. The given weak $W^{1,2}$ solution therefore belongs to $W^{1,4}$ on the smaller domain and tends to zero there. Interior patches use the same argument without reflection. No fixed $p > 2$ estimate has been inferred from the shrinking fractional exponent.

The base field now satisfies

$$(244) \quad d_B^* a_\rho = 0, \quad a_\rho(n) = 0, \quad d_B a_\rho = -a_\rho \wedge a_\rho - \rho^2 f_\infty^2 *_B *_\Sigma F_\Sigma(\beta_E^0 + b_\rho).$$

The conformal factor and all fixed derivatives are bounded on the chosen rescaled compact sets. The local base Hodge estimate is

$$\|a_\rho\|_{W_B^{m+1,p}H_\Sigma^k(K)} \leq C(\|d_B a_\rho\|_{W_B^{m,p}H_\Sigma^k(K')} + \|a_\rho\|_{L_B^p H_\Sigma^k(K')}).$$

At $m = 0, p = 4$, the square of a_ρ is small in L^4 by $H_B^1 \subset L_B^8$, and the surface term is $O(\rho^2)$ using the just proved estimate at a higher surface order. Hence $a_\rho \rightarrow 0$ in $W^{1,4}$. Since this space is a two-dimensional algebra, the same equation gives $a_\rho \rightarrow 0$ in $W^{2,4}$.

Differentiate (242) tangentially at the seam. The principal coefficient remains small in L^∞ . The new product $(D_B B)D_B v_\rho$ is small in L^2 , using two L^4 factors; derivatives of f_ρ are controlled in the same space. Tangential difference quotients justify applying the $p = 2$ estimate before the new derivative is known: each nonzero quotient is already $W^{1,2}$, its new source is uniformly bounded in L^2 , and the resulting estimate passes to the weak derivative. Straightening has made tangential differentiation preserve the real boundary condition. The normal principal coefficient is invertible, so the equation recovers normal derivatives. This proves $v_\rho \rightarrow 0$ in $W^{2,2}$.

Now $D_B v_\rho$ belongs to every finite L_B^q . Use $q = 8$ for both factors in $(D_B B)D_B v_\rho$ and apply the $p = 4$ estimate to obtain $v_\rho \rightarrow 0$ in $W^{2,4}$. For $m \geq 2$, assume both fields converge in $W^{m,4}$ at each required fixed surface order. Equation (244) first gives $a_\rho \rightarrow 0$ in $W^{m+1,4}$. After m derivatives of (242), the only unestimated derivative is $BD_B D^m v_\rho$. All other terms lie in L^4 by two-dimensional Sobolev multiplication and the improved base estimate. Absorb this small principal term and use difference quotients as before. This gives $v_\rho \rightarrow 0$ in $W^{m+1,4}$ and proves the induction. Each step uses finitely many higher surface orders already supplied by Lemma 7.26. It requires no uniform four-dimensional elliptic constant as $\rho \rightarrow 0$. $\square$

7.18.4. The residual component and its ends.

Lemma 7.30. *The residual and end conclusions of Theorem 7.25 hold for the constructed family. On a fixed exterior matching collar the gauges identifying the smooth limit with x_0 can be chosen to converge to the identity in C^0 .*

Proof. Undo the smooth center-dependent based gauge before comparison with A_0 . Outside $|z - c| \leq \rho R_1/2$ the pregluing equals the residual and the graph norm is its residual factor. Thus on every fixed negative compact set K disjoint from $\{c_*\} \times \Sigma$,

$$(245) \quad \|A_\rho - A_0\|_{H^1(K)} \leq C_K s_\rho.$$

On a fixed exterior matching collar, at each fixed surface order,

$$(246) \quad \|A_\rho - A_0\|_{C_B^0 H_\Sigma^k} \leq C_k T^{1/2} s_\rho = o(1).$$

The positive maps converge in H^1 and C^0 in their construction charts. All these statements include the nonlinear ambient correction.

On the mixed ends write $A_\rho = B_{p_\rho} + \xi_\rho$ and $A_0 = B_0 + \xi_0$, subtracting the individual exact critical representatives. With v_ρ denoting the positive displacement from its own q_{p_ρ} , the graph estimate gives

$$(247) \quad |p_\rho| + \|\xi_\rho - \xi_0\|_{H_\omega^1} + \|v_\rho - v_0\|_{H_\omega^1} \leq C s_\rho.$$

On the ordinary ends the analogous estimate is unweighted, with fixed ordinary critical classes. In particular all these remainders have uniformly small tails.

Strong physical H^1 convergence gives strong L^2 curvature convergence on fixed exterior sets, since in four dimensions

$$F_{A_0+a} - F_{A_0} = d_{A_0}a + a \wedge a, \quad \|F_{A_0+a} - F_{A_0}\|_2 \leq C(\|a\|_{H^1} + \|a\|_{H^1}^2).$$

The holonomy gradients converge on their compact outputs by their L^2 Lipschitz estimates; their entire dependency neighborhoods are outside the modified collar. The finite secondary perturbations have the same property. Smooth positive charts give convergence of the positive derivatives in L^2 . Hence analytical energy densities are uniformly integrable on fixed exterior sets.

On a mixed end exact criticality is needed to cancel the constant term:

$$(248) \quad \begin{aligned} F_{A_\rho} + *_3 \nabla_{A_\rho} h &= d_{B_{p\rho}} \xi_\rho + \xi_\rho \wedge \xi_\rho \\ &\quad + *_3 (\nabla_{B_{p\rho} + \xi_\rho} h - \nabla_{B_{p\rho}} h). \end{aligned}$$

The weighted H^1 embedding and the L^2 Lipschitz estimate control this expression in weighted L^2 , after decreasing the positive weight. On an ordinary end compare with the fixed residual's integrable perturbed curvature and use its unweighted H^1 displacement. Together with the positive estimates this proves

$$(249) \quad \lim_{L \rightarrow \infty} \limsup_{\rho \rightarrow 0} \mathfrak{E}(x_\rho; \text{all ends beyond } L) = 0.$$

The unmodified critical curvature has not been assigned an infinite cylinder L^2 norm. Proposition 7.27 now gives a uniform global analytical energy bound.

Apply the standard-perturbation compactness theorem [DFL26, Theorem 5.2] to the smooth representatives. The interior and positive concentration sets are empty and the seam set is contained in $\{c_*\}$. Its smooth conclusion off that set uses the all-order primary summability fixed before selecting the residual, and the complete exterior dependency sets. The uniform integrability above supplies absence of exterior concentration without a uniform $p > 2$ curvature estimate.

We identify its smooth exterior limit A' at the critical H^1 order. The relating unitary gauges satisfy

$$(250) \quad dg_\rho = g_\rho A'_\rho - A_\rho g_\rho, \quad A'_\rho \longrightarrow A' \text{ smoothly.}$$

Unitarity and (245) give a local $W^{1,4}$ bound for g_ρ . Rellich compactness gives a subsequence converging in every finite L^q and almost everywhere to a unitary g . The right side of (250) converges strongly in L^4 : use $A_\rho \rightarrow A_0$ in L^4 and dominated convergence against the fixed smooth coefficients. Thus the gauges converge strongly in $W^{1,4}$. Differentiation gives strong $W^{2,2}$ convergence, using products of two L^4 terms and bounded gauges times the strongly convergent L^2 derivatives of A_ρ . The limiting equation $dg = gA' - A_0g$ has smooth coefficients and bounded g ; it first gives $W^{1,\infty}$ and then smoothness by differentiation. Hence the residual limit is x_0 modulo a smooth gauge. On an exterior matching collar, (246) bounds the coefficients in (250) uniformly. After composition with the limiting gauge, the relating gauges tend to I in L^2 with uniformly bounded first derivatives, and therefore in C^0 . This last assertion uses the extra collar estimate, not a false embedding of $W^{2,2}$ into C^0 in four dimensions.

For mixed-end convergence at higher orders, we also check the topological small-tail hypothesis of DFL's decay theorem. Since B_p is pulled back from three dimensions, $F_{B_p} \wedge F_{B_p} = 0$. The other curvature-square terms have the forms

$$F_{B_p} \wedge d_{B_p} \xi, \quad F_{B_p} \wedge \xi^2, \quad d_{B_p} \xi \wedge d_{B_p} \xi, \quad d_{B_p} \xi \wedge \xi^2, \quad \xi^4.$$

If K bounds the weighted H^1 remainders, their absolute integrals past L are at most $Ce^{-cL}(K + K^2 + K^3 + K^4)$, by weighted Cauchy–Schwarz and H^1 multiplication. The constants are uniform for p near zero. Positive area is bounded by derivative energy, and the perturbation terms are controlled by (248). Thus all required small-tail thresholds hold uniformly after increasing L .

In temporal gauge, DFL’s estimates [DFL26, Propositions 5.24 and 5.29] give uniform exponential decay. The higher-order argument retains the initial decay rate at each derivative order, while shrinking the allowed energy threshold. The preceding tail estimate supplies that threshold at each fixed order. Choose one weight $\omega_* > 0$ below this common rate. The critical classes are the prescribed p_ρ classes: their positive values q_{p_ρ} are fixed by the end chart, and the critical Lagrangians are embedded. Split an end at large L , use smooth convergence before L , and integrate the uniform exponential estimate after it. This gives convergence in every prescribed $H_{\omega_*}^l$ order after subtracting individual critical limits, as in [DFL26, equation (5.36)].

On ordinary ends (249) excludes a translated nonconstant trajectory or strip, since a compact piece of such a limit would have positive energy. The global H^1 displacement fixes its zero-energy limiting critical class. In its ordinary Coulomb slice, or the chart at the transverse Lagrangian intersection, the self-adjoint critical Hessian has a spectral gap. Its inverse and absorption of the quadratic remainder give

$$\|b\|_{H^1} \leq C \|\text{grad } \mathcal{A}(b)\|_2, \quad |\mathcal{A}(b) - \mathcal{A}(0)| \leq C \|b\|_{H^1}^2.$$

The family stays in this neighborhood on every sufficiently far tail: otherwise a translated first-exit strip has energy tending to zero, converges locally to a constant critical solution, and has displacement tending to zero in L^2 by the global comparison. Local elliptic compactness in the fixed critical slice contradicts the nonzero exit displacement. Along the flow toward that critical point, the remaining energy therefore satisfies $e'(t) \leq -ce(t)$.

Uniform unit-strip elliptic estimates follow by compactness near this critical solution, as in the proof of DFL’s Proposition 5.27. On the far end the equation and perturbations are translation invariant, so the temporal derivative solves the homogeneous linearized equation. Its high Sobolev norms on smaller strips are bounded by its L^2 norm, already exponentially decaying. Integrating the temporal derivative gives the corresponding displacement bound. Splitting the end at large L now proves every prescribed unweighted H^l convergence. These ends retain weight zero. No index-at-most-one exclusion is used in this proof. $\square$

7.18.5. Holomorphic bundle convergence. We first make the stable surface slice quantitative. Let V be a stable rank-two surface bundle with fixed determinant and put

$$D = \bar{\partial}_V : \Omega^0(\text{End}_0 V) \longrightarrow \Omega^{0,1}(\text{End}_0 V), \quad H = \ker D^*.$$

Stability gives $\ker D = 0$. Hodge decomposition on the compact curve therefore makes

$$(251) \quad H^{k+1}(\text{End}_0 V) \oplus H \longrightarrow H^k(\Omega^{0,1} \text{End}_0 V), \quad (\eta, h) \longmapsto D\eta + h$$

an isomorphism, for $k \geq 2$. The nonlinear map

$$(\eta, h) \longmapsto e^{-\eta}(\bar{\partial}_V + h)e^\eta - \bar{\partial}_V$$

is holomorphic between these Banach spaces and has derivative (251). The complex implicit-function theorem gives unique small parameters satisfying

$$(252) \quad \|\eta(b)\|_{H^{k+1}} + |h(b)| \leq C_k \|b\|_{H^k}.$$

Every surface $(0, 1)$ connection is integrable, since the surface has complex dimension one. Shrink the slice so its members are stable and distinct h represent distinct isomorphism

classes. This is the local slice for the effective complex gauge action on stable bundles [Owe01, Section 4, pp. 769–770]. The inverse maps agree at higher orders, have smooth parameter dependence, and are uniformly bounded on a compact stable reference family. The gauge is chosen near I , fixing the choice between the central lifts $\pm I$.

Consider now an integrable product Dolbeault operator whose surface part $D + b(z)$ stays in this neighborhood. The slice parameters are $\eta(z), h(z)$. Integrability says that $\partial_{\bar{z}}b$ is an infinitesimal complex gauge variation. The derivative of h annihilates that variation, since its value is constant along a sufficiently small gauge orbit. The slice map is holomorphic, so the chain rule gives $\partial_{\bar{z}}h = 0$. Apply $g = e^\eta$ to the full product operator. Its surface part is $D + h(z)$ and its base part has trace-free coefficient γ . Integrability then gives

$$(253) \quad D\gamma + [h(z), \gamma] = 0.$$

Stability implies $\gamma = 0$. Thus the slice identifies the complete projective holomorphic family with the pullback slice family, including its base coefficient. The central part is a base-line factor, removed by projectivization. Fiberwise surface isomorphisms alone would not supply this identification.

Lemma 7.31. *The finite-scale doubled projective bundles of the constructed family converge to $\mathbb{P}(E_*)$ on every fixed rescaled compact set, with the prescribed attaching identification. The assertion holds through an interior unstable fiber and identifies the stable extension at infinity.*

Proof. The small gauges of Lemma 7.29 are compatible on overlaps. Indeed their relative gauges h_ν tend uniformly to I and obey $dh_\nu = h_\nu A'_\nu - A_\nu h_\nu$, where both connections converge to the same limiting coefficients. This gives convergence of dh_ν to zero; successive differentiation gives every fixed higher-order convergence. The cocycle identities are exact before taking a limit and limit to the identity.

On a fixed rescaled seam segment, finitely many stable reference fibers supply the preceding slices. By Lemma 7.29 and smooth dependence of the slice inverse, the negative surface isomorphisms and classifying maps converge in every fixed finite order. The positive classifying maps converge to u_{E_*} . Matching identifies their continuous values on the seam. In a complex chart their union is holomorphic by Morera’s theorem. Uniform convergence on a paired neighborhood and Cauchy estimates give all finite orders of holomorphic convergence on smaller neighborhoods. Pulling back the slice family supplies holomorphic bundle frames on both halves; equation (253) identifies the full negative product operator. The slice and frame transition functions converge to those of E_* . The same argument applies on positive interior compact sets.

An unstable surface fiber in the negative interior requires a different argument. Choose a product polydisc $D_z \times D_\zeta$ and a holomorphic frame of E_* on a slightly larger polydisc. In the smooth unitary identifications already obtained, the varying Dolbeault operators have coefficients

$$(254) \quad \bar{\partial} + B_{z,\nu} d\bar{z} + B_{\zeta,\nu} d\bar{\zeta}, \quad B_{z,\nu}, B_{\zeta,\nu} \longrightarrow 0$$

at every prescribed finite order. Use a fixed mixed Hölder norm $C_z^{m,\alpha}(C_\zeta^{m,\alpha})$, $0 < \alpha < 1$, controlled by a sufficiently high ordinary smooth norm. The Cauchy–Green inverse T_z on a fixed disc is bounded in this parameter-dependent norm: the usual kernel-difference estimate holds for functions with values in the Banach space $C_\zeta^{m,\alpha}$. It commutes with parameter derivatives. On nested fixed discs the contraction

$$(255) \quad g_\nu = I - T_z(B_{z,\nu}g_\nu), \quad k_\nu = I - T_\zeta(C_{\zeta,\nu}k_\nu)$$

constructs an invertible $g_\nu \rightarrow I$ first. In its frame the z coefficient vanishes and the remaining coefficient $C_{\zeta,\nu}$ is holomorphic in z by integrability. It tends to zero, using one additional coefficient order for $\partial_{\bar{z}}g_\nu$. The second contraction in (255) has iterates holomorphic in z , since T_ζ acts on a fixed disc and commutes with $\partial_{\bar{z}}$. Thus $g_\nu k_\nu$ gives a holomorphic frame converging to the chosen limiting frame. Higher orders follow in the successively higher fixed norms. No stability of the surface fiber is used here. On a finite product cover, the resulting holomorphic transition matrices converge, including on overlaps with the stable seam neighborhoods. Fixed-determinant central lifts may be used locally; equivalently this is a projective frame construction.

It remains to check attachment on the whole neck. Put $\alpha_\nu = \alpha(c_\nu)$. Equation (228) places every negative surface operator and positive class on $A(c_\nu; R\rho_\nu, r)$ in one fixed stable slice, for R large, r small, and then ν large. Apply (252) there to obtain

$$(256) \quad \sup_{A^-} \|\eta_\nu\|_{H_\Sigma^{k+1}} + \sup_A |h_\nu - h(c_\nu)| \leq C_k(R^{-1} + r) + o_\nu(1).$$

The first term is in the slice centered at α_ν , or has its small reference gauge subtracted in a fixed slice. The negative classifying map is holomorphic, and it glues to the positive map across both seam intervals. Consequently the doubled family on the full annulus is the pullback of that one slice family. The slice map and its small logarithm are single-valued in the actual attaching frame, even though the annulus is not simply connected. Hence no additional projective transition arises from unrelated gauge choices on different arcs. This argument controls values and surface derivatives; it does not assume uniform physical base derivative bounds on the shrinking annulus.

The limit on $\mathbb{C} \times \Sigma$ is now the restriction of the specified primitive bundle E_* . Its fibers near infinity are stable, and its classifying coordinate tends to $[\alpha(c_*)]$. Let $R \rightarrow \infty$ and $r \rightarrow 0$ in (256) after taking the sequence limit. This identifies precisely its original attaching value and stable family at infinity. Only the limiting bundle is extended. A finite annulus has not been replaced by a punctured disc, and no holomorphic cap for each finite-scale residual variation is asserted.

The isomorphisms used here are inherited from the chosen fixed-determinant rank-two families and small complex gauges. Two such isomorphisms to the same stable family differ by a scalar, so agree projectively. This does not assert that every abstract projective automorphism lifts to the fixed rank-two bundle. Base-line twists have no adjoint effect. On the compact sphere the normalization

$$\det E = \pi_\Sigma^* \Lambda \otimes \pi_B^* \mathcal{O}(1)$$

removes their remaining degree ambiguity: a base twist of degree d changes the determinant degree by $2d$, and a degree-zero line on $\mathbb{P}^1$ is trivial. Thus the normalized rank-two limit is E_* as well. $\square$

Completion of the proof of Theorem 7.25. Lemma 7.30 identifies the residual and all ends. Lemma 7.29 supplies every fixed finite order on rescaled compact sets, and Lemma 7.31 identifies the doubled bundle, including an interior unstable fiber. The whole-neck estimate (227) gives

$$\lim_{r \downarrow 0} \lim_{R \rightarrow \infty} \limsup_{\nu \rightarrow \infty} \mathfrak{E}_{\rho_\nu}(R\rho_\nu, r) = 0.$$

Equation (228) gives the corresponding diameter assertion. Corollary 7.28 retains all the half-unit mass. At rank drop the modification remains in the limiting holomorphic bundle; a stable-map sphere describing its algebraic degeneration is not an additional analytic component. The stable whole-neck family inserts no modification at infinity.

The component gauges retain the same attaching frame. For completeness, choose successively larger fixed rescaled domains and thresholds at which each finite list of estimates holds, and let these domains grow by a diagonal choice. Decrease the thresholds to keep their physical diameters tending to zero and their supports inside the cap. The small base gauge can be extended as the identity by cutting off its logarithm near the artificial boundary. On a regular exterior matching collar, Lemma 7.30 gives relating gauges close to I . These may likewise be cut off toward the neck. In relative normal boundary gauge both original and transformed seam normal coefficients vanish, so their relating gauges have zero normal derivative there; cutoffs constant in that direction preserve it. For each retained compact truncation the two component regions are disjoint for small scale, and these Sobolev bundle identifications extend across the discarded annulus. No estimate for their derivatives on that entire annulus is needed. The gauges at different fixed orders agree where both constructions apply.

These are the stated analytic, energy, and holomorphic convergence properties of the broken configuration. They are uniform for each finite collection of compact domains and derivative orders and on the compact parameter patch. Such collections give a countable exhaustion defining this convergence topology, so the local construction extends continuously at scale zero. Constant unmarked components with only their attaching node are omitted; insertion ratios and their possible marked components are not part of this theorem. No assertion of converse entry or collar uniqueness has entered the proof. $\square$

8. CONVERSE GLUING ON REGULAR PRIMITIVE STRATA

8.1. Stable families and one elementary modification. We first state the converse with explicit hypotheses on the doubled families and their exterior representatives. We group them into four conditions, retaining the fixed data throughout this section. Section 8.13 derives the technical exterior and end conditions from convergence and energy exhaustion.

(C1) Residual and convergence. The pair $x_0 = (A_0, u_0)$ is a fixed smooth irreducible regular residual of index zero. The primary perturbations satisfy the all-order summability used above. The product matching neighborhood is disjoint from all perturbation supports and dependency regions. We retain the fixed-determinant lifts and use the real seam coordinate in Section 7.

Let x_ν be smooth mixed solutions converging to x_0 on compact sets away from one seam point.

(C2) Doubled families and outer framings. On shrinking doubled discs suppose Lemma 4.1 and the coherent framing and capping convention of Proposition 4.3 apply, with $n(E_\nu) = 1$. The outer classifying loops and an annulus about them lie in a shrinking stable chart about the residual attaching class. Their caps in that chart have area tending to zero. On the negative outer arcs there are actual unitary surface representatives converging to the residual attaching connection in every prescribed fixed surface order. These are data on the chosen growing rescaled circles, not a consequence of convergence on a fixed rescaled circle.

(C3) Family type and regular attaching data. After a subsequence, fix one of the following alternatives:

- (1) $d_\nu = 0$, $\ell_\nu = 1$. There is one Langton step with quotient of degree zero. Its point is in the open negative half-plane. Translate by its physical real part and divide by the absolute value of its physical imaginary part, so its normalized position is $-i$.
- (2) $g = 2$, $d_\nu = 1$, $\ell_\nu = 0$. All finite surface fibers are stable. The limiting image line obtained below corresponds to $L_* \in \text{Pic}^0(\Sigma)$ with $H^0(\Sigma, \Lambda L_*^{-2}) = 0$. Normalize the comparison pencil by (210), as in Proposition 8.3 below, and assume its limit is seam-admissible.

For each ν , write E_ν^0 for the normalized comparison primitive associated with x_ν : the inverse elementary transform with the same modification data in the first case, and the degree-one extension pencil in the second. These constructions are given in Propositions 8.2 and 8.3. In both cases assume the normalized comparison data lie in a relatively compact regular seam-admissible patch with one observation system. Write their physical centers and scales as c_ν, ρ_ν , and put $w = (z - c_\nu)/\rho_\nu$. The retained doubled discs contain $D_{2R_\nu} \times \Sigma$, with fixed additional radius margins, where

$$(257) \quad T_\nu = \log(1/\rho_\nu), \quad R_\nu \geq c\rho_\nu^{-1/2}, \quad r_\nu = \rho_\nu R_\nu \longrightarrow 0.$$

All finite-family hypotheses must hold on these retained discs. The enlargement argument below explains when coherent charge-one outer data preserve the second alternative.

(C4) Exterior and end estimates. In normal gauges, convergence to the based residual representative holds on

$$r_\nu/32 \leq |z - c_\nu| \leq r_\nu/8$$

with physical $C_B^1 C_\Sigma^k$ connection error at most r_ν , for each fixed k and sufficiently large ν . The other required finite physical derivatives have their exterior convergence bounds. There is no other concentration on those compact sets. On the ends assume weighted smooth convergence after subtracting each solution's own exact critical representative, retaining its finite critical parameters. The weight is fixed below the residual asymptotic gaps and the given convergence rate. Smooth convergence off the atom allows a sufficiently slow diagonal choice of exterior radii. The bundle and charge comparison on an enlargement, and the end convergence under exhaustion of the global defect, are proved in Propositions 8.17 and 8.18 below.

Theorem 8.1 (converse for stable families and one elementary modification). *Suppose conditions (C1)–(C4) hold. Then x_ν eventually belongs, modulo the allowed gauge group, to the local gluing family of Theorem 7.2. Write (q, t_1, t_2) for local coordinates on the normalized primitive space, with q its attaching class. Let t_ν be the last two coordinates of E_ν^0 . There are unique nearby solutions $(\hat{c}_\nu, \hat{\rho}_\nu, \hat{t}_\nu)$ of the four observation equations such that*

$$(258) \quad [x_\nu] = \text{Glue}_{\text{loc}}(\hat{c}_\nu, \hat{\rho}_\nu, \hat{t}_\nu),$$

$$\left| \frac{\hat{c}_\nu - c_\nu}{\rho_\nu} \right| + \left| \log \frac{\hat{\rho}_\nu}{\rho_\nu} \right| + |\hat{t}_\nu - t_\nu| \leq CT_\nu^{-79/2}.$$

The original solution has a representative in the full gluing chart at these parameters with coordinate norm at most CT_ν^{-38} . The second alternative includes stable finite families approaching negative rank drop.

The theorem does not classify every index-four degeneration. In particular it excludes positive rank-drop limits and special genus-two lines, and it retains the exterior and end hypotheses just stated. We first construct the comparison bundles, then prove common analytic estimates, and finally prove full-chart entry and uniqueness.

8.2. A single elementary modification.

Proposition 8.2. *In the first alternative the stable replacements have image tending uniformly to the attaching class. After a subsequence the modification sublines converge smoothly. The doubled bundles converge holomorphically on fixed compact products, including the modification fiber, to the primitive inverse transform determined by those limiting data. There is a comparison primitive E_ν^0 with the same modification point and subline, and an identification of its base-holomorphic structure with that of E_ν .*

Proof. The stable-reduction map f_ν has area tending to zero, since $d_\nu = 0$ and its cap has vanishing area. Small area and the outer image condition imply uniform image convergence. Indeed, a holomorphic curve through the center of a target coordinate ball, with no domain boundary mapping into that ball, has area at least $c\delta^2$ in a ball of radius δ . In Euclidean coordinates, Stokes and coarea give $2A(s) \leq sA'(s)$ at regular radii; the density at a nonconstant branch is a positive integer multiple of π . Uniform comparison with the Kähler metric gives the assertion. A point a fixed distance from the attaching class would provide such a ball missing the outer boundary, contradicting small area.

Use a stable complex slice about the attaching bundle V_* . Its coordinate $t_\nu(w)$ is holomorphic and satisfies

$$(259) \quad \|t_\nu\|_\infty \leq \eta_\nu \rightarrow 0, \quad \|\partial_w^j t_\nu\|_\infty \leq C_j \eta_\nu R_\nu^{-j}$$

on smaller retained discs. The slice pullback represents the stable replacement up to a line from the disc, which is holomorphically trivial. Its trace-free base operator is $\bar{\partial}_w$: the remaining base coefficient is a holomorphic trace-free endomorphism of a stable surface bundle, hence zero.

Let $a = -i$, $V_\nu = F_\nu|_{\{a\} \times \Sigma}$, and let $G_\nu \subset V_\nu$ be the degree-zero subline supplied by the Langton inclusion. After taking a subsequence, the classes of G_ν converge in the compact Picard torus. Normalize their inclusions to have L^2 norm one. Uniform surface Dolbeault estimates give a smoothly convergent subsequence of those maps. Its nonzero limit $G \rightarrow V_*$ has no zero: saturation at a nonzero effective divisor would give a line of degree at least one in the stable degree-one rank-two bundle. Thus the orthogonal projections onto G_ν , in the NS metrics of V_ν , converge in every surface order.

Writing $t = w - a$, the inverse-transform lattice is generated by $e_1/t, e_2$ when e_1 spans G_ν at $t = 0$. If the stable transition matrix is

$$Q_\nu = \begin{pmatrix} q_{11} & q_{12} \\ tc_{21} & q_{22} \end{pmatrix},$$

the actual transition matrix is

$$(260) \quad \begin{pmatrix} t & 0 \\ 0 & 1 \end{pmatrix} Q_\nu \begin{pmatrix} t^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} q_{11} & tq_{12} \\ c_{21} & q_{22} \end{pmatrix}.$$

Division of the vanishing entry by t is controlled by its base derivative on smaller charts. These matrices converge with nonzero determinant and prove locally free holomorphic convergence through the divisor.

Replace the stable family by the constant family V_ν and keep G_ν and a . Its inverse transform E_ν^0 has extension

$$0 \longrightarrow \mathcal{O}(1) \boxtimes G_\nu \longrightarrow E_\nu^0 \longrightarrow \mathcal{O} \boxtimes \Lambda G_\nu^{-1} \longrightarrow 0,$$

with coefficient pair $e(w) = (w - a)v$. The extension compactifies directly on $\mathbb{P}^1 \times \Sigma$; it has $c_2 = 1$ and $p_1(\text{ad } E_\nu^0) = -2$. The same smooth splitting and lattice $e_1/t, e_2$ identify

the actual and comparison base-holomorphic structures. The surface operators differ. Since $t_\nu(w) - t_\nu(a) = (w - a)k_\nu(w)$, this difference in the divisor frame has entries

$$\begin{pmatrix} (w - a)k_{11} & (w - a)^2k_{12} \\ k_{21} & (w - a)k_{22} \end{pmatrix}, \quad \|\partial_w^j k_\nu\|_{C_\Sigma^m} \leq C_{j,m} \eta_\nu R_\nu^{-1-j}.$$

It is small and nonsingular on the fixed divisor chart. At large radius use the stable frame, rather than this polynomial bound. The current rank-one pencil already satisfies (211). $\square$

8.3. Degree-one stable families in genus two.

Proposition 8.3. *In the second alternative, the classifying image approaches a line in M_2 . Near a regular limiting line, the actual family has slowly varying Jacobian and extension coordinates. Their values at a unique projective pole define a comparison primitive E_ν^0 without division by its transverse coefficient. The affine normalization produces c_ν, ρ_ν with $c_\nu, \rho_\nu \rightarrow 0$ and a compact normalized primitive family. On a seam-admissible subsequence the actual and comparison base-holomorphic structures are identified uniformly through negative rank drop.*

Proof. Use the realization of M_2 as the intersection of two quadrics in $\mathbb{P}^5$, with hyperplane class the primitive class [Muñ99b, Example 24]. Initially write the physical domain as D_{8r_ν} , with outer annulus $r_\nu \leq |z| \leq 8r_\nu$ mapping close to $p = [1 : 0 : \dots : 0]$. Let $\epsilon_\nu \rightarrow 0$ bound its five affine coordinates. The section Z_0 of the pullback of $\mathcal{O}(1)$ has exactly one simple zero $a_\nu \in D_{r_\nu}$: its relative zero count is the capped degree one, using Z_0 as a nonvanishing trivialization on the small-chart cap. A holomorphic lift, divided by the nowhere-zero function $s_0/(z - a_\nu)$, gives

$$(261) \quad f_\nu(z) = [(z - a_\nu)(1, h_\nu(z)) + (0, v_\nu)], \quad v_\nu \in \mathbb{C}^5 \setminus \{0\}.$$

The residue formula on $|z| = 7r_\nu$, the maximum principle, and Cauchy's estimate give

$$|v_\nu| \leq C r_\nu \epsilon_\nu, \quad \|h_\nu\|_{D_{6r_\nu}} \leq C \epsilon_\nu, \quad \|\partial_z^j h_\nu\|_{D_{4r_\nu}} \leq C_j \epsilon_\nu r_\nu^{-j}.$$

Put $p_0 = (1, 0, \dots, 0)$ and $u_\nu = v_\nu/|v_\nu|$. The two summands in $(z - a_\nu)p_0 + (0, v_\nu)$ are orthogonal, and its perturbation $(z - a_\nu)(0, h_\nu)$ has relative norm at most $C\epsilon_\nu$. Hence the projective distance to the line $\mathbb{P}\text{span}\{p_0, (0, u_\nu)\}$ is uniformly $O(\epsilon_\nu)$, independently of $|v_\nu|$.

For each defining quadric Q_j , with polar form B_j , evaluation and differentiation of (261) at a_ν give

$$Q_j(0, v_\nu) = 0, \quad B_j((0, v_\nu), (1, h_\nu(a_\nu))) = 0.$$

After $u_\nu \rightarrow u_*$ these exact identities and $Q_j(p_0) = 0$ show that $\ell_* = \mathbb{P}\text{span}\{p_0, (0, u_*)\}$ lies in M_2 . The image of the whole smaller disc tends uniformly to ℓ_* . The only division by $|v_\nu|$ here is in the exact identity identifying a direction.

Let $N = \mathbb{P}(\mathcal{V}) \rightarrow \text{Pic}^0(\Sigma)$, $\mathcal{V}_L = H^1(\Sigma, L^2\Lambda^{-1})$, and let $i : N \rightarrow M_2$ send an extension to its middle bundle. The primitive classification identifies $\ell_* = i(N_{L_*})$ [Muñ99b, Propositions 2–3 and 5]. The relative $\mathcal{O}(1)$ subline recovers L_* uniquely: push the classified product bundle twisted by $\mathcal{O}(-1)$ to Σ . At a pair $L \subset V$, the infinitesimal subline variations with fixed V are $H^0(\Sigma, \Lambda L^{-2})$, by the graph description of a subbundle. Stable automorphisms are scalar. Thus

$$(262) \quad \ker di = H^0(\Sigma, \Lambda L^{-2}).$$

The source and target have complex dimension three. Our regularity hypothesis makes di invertible along the whole compact fiber N_{L_*} . It is injective on that fiber: the extension tensored by L_*^{-1} gives $h^0(VL_*^{-1}) = 1$, so the subline is unique. It is then injective on

a neighborhood of the fiber. Otherwise pairs of distinct points with the same image in shrinking neighborhoods would limit to the same fiber point, contradicting local invertibility there. Consequently, for a small ball B about L_* , $i : N|_B \rightarrow U$ is biholomorphic. The actual maps lift to $N|_B$, with Jacobian coordinate $L_\nu(z) \rightarrow L_*$ uniformly.

Trivialize $\mathcal{V}$ on B . The pullback of the projective hyperplane bundle has degree one on each fiber $\mathbb{P}^1$, so its coordinate sections are a holomorphic rank-two matrix $A(L) : \mathbb{C}^2 \rightarrow \mathbb{C}^6$, after trivializing a base line. Choose holomorphic extension vectors $e_1(L), e_2(L)$ on which its first row is $1, 0$, and choose $e_1(L_*)$ to represent p . The extension coordinate has the same unique simple pole:

$$(263) \quad q_\nu(z) = \frac{b_\nu}{z - a_\nu} + k_\nu(z), \quad b_\nu \neq 0.$$

The inverse i^{-1} and the outer data make q_ν small on the outer annulus. The residue and Cauchy estimates give, on smaller discs and for $\eta_\nu \rightarrow 0$,

$$|b_\nu| \leq C\eta_\nu r_\nu, \quad \|L_\nu - L_*\| + \|k_\nu\| \leq C\eta_\nu, \quad \|\partial_z^j L_\nu\| + \|\partial_z^j k_\nu\| \leq C_j \eta_\nu r_\nu^{-j} \quad (j \geq 1).$$

The projective residue in (261) is b_ν times the last five components of $A(L_\nu(a_\nu))e_2$; their norm is bounded above and below.

Put $L_\nu^0 = L_\nu(a_\nu)$, $k_\nu^0 = k_\nu(a_\nu)$, and $v = e_1(L_\nu^0) + k_\nu^0 e_2(L_\nu^0)$. The current comparison pencil is

$$(264) \quad e_\nu^0(z) = (z - a_\nu)v + b_\nu e_2(L_\nu^0).$$

To identify the actual bundle, choose the local Poincaré family on one fixed smooth line bundle, with line Dolbeault operator holomorphic in L . Choose representatives $\psi_j(L)$ of $e_j(L)$ holomorphic in L and uniformly smooth in Σ . Two fixed $(0, 1)$ -forms spanning the negative-degree H^1 at L_* remain a basis nearby; a holomorphic change of basis gives these representatives. All $(0, 1)$ -forms are closed on a curve, H^0 is zero, and $h^1 = 2$ is constant. The actual operator has base part $\bar{\partial}_z$ and surface part

$$(265) \quad \begin{pmatrix} \bar{\partial}_{L_\nu(z)} & (z - a_\nu)\psi_1(L_\nu(z)) + [b_\nu + (z - a_\nu)k_\nu(z)]\psi_2(L_\nu(z)) \\ 0 & \bar{\partial}_{\Lambda L_\nu(z)^{-1}} \end{pmatrix}.$$

Its coefficients are holomorphic in z , so it is integrable. Its stable fibers have precisely the lifted extension classes. The fine-moduli property identifies it projectively with the original doubled bundle; the remaining line from the disc is trivial. Replacing L_ν, k_ν by their current values at a_ν gives E_ν^0 with the same base operator.

For $e_\nu^0(z) = zv + w_0$ set

$$(266) \quad c_\nu = -\operatorname{Re} \frac{\langle w_0, v \rangle}{\|v\|^2}, \quad \rho_\nu = s(e_\nu^0) > 0.$$

The prescribed metric in (210) gives uniformly

$$\rho_\nu \asymp (|\operatorname{Im} a_\nu|^2 + |b_\nu|^2)^{1/2}, \quad |c_\nu - \operatorname{Re} a_\nu| \leq C|b_\nu|.$$

Indeed the projection of e_2 perpendicular to v is bounded below, and $\kappa + a_\nu = O(b_\nu)$. In $w = (z - c_\nu)/\rho_\nu$, divide the pencil by its common scalar ρ_ν and put $A_\nu = (a_\nu - c_\nu)/\rho_\nu$, $B_\nu = b_\nu/\rho_\nu$. These coefficients are bounded and satisfy the affine slice exactly. The slope remains nonzero. If the limit has $B = 0$, normalization forces $A = \pm i$; $A = -i$ is the negative rank-drop limit, whereas $A = i$ is the excluded positive one.

On the normalized domains the slow parameters $t_\nu = (L_\nu, k_\nu)$ satisfy

$$(267) \quad |t_\nu(w) - t_\nu(A_\nu)| \leq C\eta_\nu |w - A_\nu|/R_\nu, \quad |\partial_w^j t_\nu| \leq C_j \eta_\nu R_\nu^{-j}.$$

Near the bounded A_ν the operator difference in the extension frame is nonsingular, with all surface derivatives $O(\eta_\nu/R_\nu)$. On the outer region use the common $\mathcal{O}(1)$ frame at infinity, conjugating by $\text{diag}(w^{-1}, 1)$. Its off-diagonal entry is

$$(268) \quad (1 - A_\nu/w)[\psi_1(L_\nu(w)) + k_\nu(w)\psi_2(L_\nu(w))] + (B_\nu/w)\psi_2(L_\nu(w)).$$

The whole-disc surface-operator error is $O(\eta_\nu)$. On a logarithmic disc it and its fixed derivatives are η_ν/R_ν times a fixed polynomial in the radius. No division by B_ν occurs. The common base operators and transitions identify the base-holomorphic structures, and the converging integrable coefficients give holomorphic bundle convergence on fixed compact products, including negative rank drop. $\square$

Remark 8.4 (enlarging the stable-family disc). Suppose a larger shrinking doubled disc contains the original one, has the same coherent charge one, and has an outer cap of vanishing area about the same attaching class. Its stable-reduction map restricts to the original stable map and has area at least $1/2 + o(1)$. Its integer degree is therefore at least one for large ν . The identity $d + \ell = 1$ gives $d = 1, \ell = 0$. The intervening annulus has area tending to zero and both boundary images near the attaching point; the target-ball monotonicity argument gives uniform image smallness on that annulus. The inverse extension-family branch extends, and its pole, residue, and current values at the pole are unchanged. Thus center and scale are fixed before enlargement. When these coherent outer data are available, slow diagonal radii at least $\sqrt{\rho_\nu}$ give (257), after fixed margins and recentering. Boundary values on one preassigned disc alone do not justify this enlargement.

8.4. Comparison metrics on expanding half-discs. Let H_ν be the actual projective HYM metric. We use metrics on the same actual holomorphic bundle, rather than compare two surface operators as if they were holomorphically identical.

Lemma 8.5. *In either sector there is a metric K_ν on the actual negative bundle with zero trace-free base curvature and uniformly bounded surface curvature in every fixed surface order. In a base-parallel frame it is independent of w and has attaching NS connection α_ν . The actual Dolbeault operator in that frame is*

$$(269) \quad \bar{\partial}_w + \bar{\partial}_{\alpha_\nu} + c_\nu(w, x), \quad \bar{\partial}_w c_\nu = 0.$$

The coefficients have uniformly bounded fixed surface derivatives on the growing half-disc and uniformly bounded fixed mixed derivatives on its unit neighborhoods away from the artificial outer edge. For $s_\nu = \log(K_\nu^{-1}H_\nu)$, determinant normalized, the boundary values satisfy

$$(270) \quad \begin{aligned} \|s_\nu(x, \cdot)\|_{C_\Sigma^k} &\leq C_k \eta_\nu (1 + |x|)/R_\nu \quad (x \in [-R_\nu, R_\nu]), \\ \sup_{\substack{|w|=R_\nu \\ \text{Im } w \leq 0}} \|s_\nu(w, \cdot)\|_{C_\Sigma^k} &\leq C_k (b_{k,\nu} + \eta_\nu + R_\nu^{-1}). \end{aligned}$$

Here $b_{k,\nu} \rightarrow 0$. On every logarithmic real interval the seam trace has the analogous small bounds for each fixed tangential and surface derivative, with a fixed polynomial in that interval's radius.

Proof. In the one-modification case let P_ν be the NS-orthogonal projection onto $G_\nu \subset V_\nu$ and set

$$(271) \quad S_\nu(w) = (I - P_\nu) + \frac{w - a}{w - \bar{a}} P_\nu, \quad K_\nu = S_\nu^* H_{\text{NS}, \nu}.$$

On the inverse-transform frame $e_1/(w - a), e_2$, the two factors of S_ν are $1/(w - \bar{a}), 1$. Thus it is invertible on the actual bundle throughout the closed negative half-plane. It is holomorphic

in the base and smooth in the surface. Consequently the base Chern curvature of K_ν is zero. On the seam $|(w - a)/(w - \bar{a})| = 1$; at infinity $S_\nu = I + O(1/|w|)$. For E_ν^0 this is its unique base-flat Dirichlet metric. For the actual bundle the retained surface-operator error in the divisor and stable frames proves all required surface-curvature bounds.

In the stable-family case transfer the base-flat Dirichlet metric of E_ν^0 by the base-holomorphic identification in Proposition 8.3. The compact Dirichlet construction and its parameter dependence give uniform smooth metrics on the closed negative hemisphere. All boundary fibers, including infinity, stay in a compact stable family; an unstable limiting interior fiber is irrelevant to the base Dirichlet problem. The two uniform extension charts and (268) bound the surface error. This proves

$$(272) \quad F_{K_\nu, BB}^\perp = 0, \quad \sup |\nabla_\Sigma^j F_{K_\nu, \Sigma\Sigma}^\perp| \leq C_j.$$

Parallel transport for the flat projective base connection from infinity identifies the bundle with the pullback of its attaching surface bundle. This identification is base-holomorphic and unitary for K_ν , and smooth in the compact family. It gives (269) in both cases. The extension charts on a fixed core and at infinity, together with the slow-parameter Cauchy bounds, verify the stated coefficient bounds.

On the seam both metrics are NS metrics for nearby surface operators. The trace-free NS metric linearization has zero kernel on the compact stable family, so its surface implicit-function theorem has uniform parameter and surface bounds. The difference of the actual and current parameters vanishes at a or A_ν and is bounded by $C\eta_\nu(1 + |x|)/R_\nu$. This proves the first line of (270) and its local tangential versions. For the first sector use also the unitary $S_\nu(x)$ on the seam; for the second use the common infinity extension chart at large x .

On the outer negative semicircle the stipulated unitary connections are close to the residual attaching connection. A stable complex slice supplies a close isomorphism to their holomorphic representative. The comparison frame at infinity is in that same small stable neighborhood and its metric differs from its attaching NS metric by $O(R_\nu^{-1})$. These maps compare the actual metrics there in all fixed surface orders. To compare their identifications with the chosen global one, observe that both are isomorphisms of the same stable rank-two surface bundle. Their ratio is scalar and disappears from the projective metric ratio. This proves the second line of (270). No norm of the scalar, or determinant square root on an annulus, is needed. $\square$

Set

$$U_\nu = \log \frac{\text{tr}(K_\nu^{-1} H_\nu)}{2\sqrt{\det(K_\nu^{-1} H_\nu)}}.$$

For determinant-one ratio with eigenvalues $e^\lambda, e^{-\lambda}$, $U_\nu = \log \cosh \lambda$. The Chern trace inequality on the actual holomorphic bundle and (272) give

$$(273) \quad (\Delta_w + \rho_\nu^2 \Delta_\Sigma) U_\nu \leq C\rho_\nu^2,$$

with nonnegative-spectrum scalar Laplacians. The boundary estimate and the barrier $C\rho_\nu^2(R_\nu^2 - |w|^2)$ imply

$$\sup U_\nu \leq C(b_{0,\nu} + \eta_\nu + R_\nu^{-1} + r_\nu^2) \rightarrow 0.$$

In particular $s_\nu \rightarrow 0$ uniformly on the entire growing half-disc.

A sharper seam barrier supplies the rate needed near the core. Write $w = x - it$, $t \geq 0$, choose fixed $A_0 \geq 1$, and put

$$(274) \quad \begin{aligned} P_R(w) &= -\frac{2}{\pi} \operatorname{Im} \log \frac{R+w}{R-w}, \\ J_R(w) &= x^2 - t^2 + Rt + A_0^2, \quad V_R(w) = \frac{1}{2}t(R-t). \end{aligned}$$

The function P_R is harmonic, zero on the diameter and one on the semicircle; $P_R \leq Ct/R$ on $D_{R/2}^-$. The function J_R is nonnegative harmonic with seam value $x^2 + A_0^2$, and $V_R \geq 0$, $\Delta_w V_R = 1$. The first boundary estimate gives $U(x) \leq C\eta^2(x^2 + A_0^2)/R^2$. If m_R is the bounded outer discrepancy, comparison with $C\eta^2 R^{-2} J_R + m_R P_R + C\rho^2 V_R$ yields

$$(275) \quad \sup_{D_L^- \times \Sigma} U \leq C(L/R + \rho^2 RL), \quad \|s\|_{L^\infty(D_L^- \times \Sigma)} \leq C(L/R + \rho^2 RL)^{1/2} \leq CL^{1/2} \rho^{1/4}$$

for $1 \leq L \leq R/2$. At the two corners continuity and the seam estimate give the required boundary upper limits. The last inequality uses (257) and $r \leq 1$. No convergence rate of the exterior data occurs in this bound.

8.5. Surface derivatives and the inner connection norm. The following derivative argument keeps surface orders unweighted; rescaled four-dimensional elliptic estimates alone would lose them.

Lemma 8.6. *For every fixed k , all surface derivatives of s_ν through order k tend uniformly to zero on the growing half-disc. For every fixed $A > 0$ and finite m , on $|w| \leq T_\nu^A$ the actual solution can be put in a unitary normal axial gauge with*

$$(276) \quad \begin{aligned} \|\widehat{\beta}_\nu - \beta_\nu^0\|_{C^m} + \|u_\nu - u_\nu^0\|_{C^m} &\leq C_{A,m} \rho_\nu^{1/10}, \\ \|\widehat{A}_{B,\nu}\|_{C^m} &\leq C_{A,m} \rho_\nu^2 T_\nu^A. \end{aligned}$$

The base component in the second line is expressed in w . Map differences use the exponential chart along the compact primitive family. The localized ambient raw cap difference from its exact physical cap is $o(T_\nu^{-N})$ for every fixed N .

Proof. In the frame (269) write $H = K^0 h$, and let ∇ be the attaching flat adjoint connection and the fixed surface tensor connection. Put

$$q = h^{-1} \nabla_\Sigma h, \quad \mathcal{D}_\xi v = \nabla_\xi v + \frac{1}{2}[q(\xi), v], \quad v_1 = q, \quad v_{j+1} = \mathcal{D}_\Sigma v_j.$$

This connection preserves the real H -self-adjoint bundle. Let A be the actual Chern connection, $d_{A,\rho} = (d_{A,B}, \rho \nabla_{A,\Sigma})$, and $L_{A,\rho} = \nabla_{A,\rho}^* \nabla_{A,\rho} / 2$. The differentiated metric equations have the estimate

$$(277) \quad |L_{A,\rho} v_k| \leq C_k \rho^2 (1 + |v_k|) + C_k \sum_{\substack{1 \leq i, j \leq k \\ i+j \leq k}} |d_{A,\rho} v_i| |d_{A,\rho} v_j|.$$

Here the constants use bounded lower surface derivatives and the bounded surface coefficients of c , but no separate bound for a base derivative of the unknown metric.

We justify this form. Under unitary metric transport a metric velocity $v = H^{-1} \dot{H}$ changes the connection by $(\partial_A - \bar{\partial}_A)v/2$. A change γ of the Dolbeault operator adds $\gamma - \gamma^{*H}$. For surface differentiation γ has only surface components. The variation of L in a skew connection direction Θ is, in parallel tangent frames,

$$\dot{L}w = - \sum_\mu \lambda_\mu [\Theta_\mu, \nabla_{A,\mu} w] - \frac{1}{2} \left[\sum_\mu \lambda_\mu \nabla_{A,\mu} \Theta_\mu, w \right], \quad \lambda_B = 1, \quad \lambda_\Sigma = \rho^2.$$

Here is the second variation and the structure preserved by further differentiation. Write $\mathbf{m} = (i\Lambda_\rho F_A)_0 = 0$, and use s, t for surface derivative indices. Put

$$K(v)_\mu = -\frac{i}{2}J_\mu^\nu \nabla_{A,\nu} v, \quad C_s = \gamma_s - \gamma_s^{*H}, \quad \gamma_s = \nabla_s c.$$

The reference variation C_s has only surface components. On endomorphism coefficients the transported connection variation is

$$[\mathcal{D}_s, \nabla_{A,\mu}] = \text{ad } B_{s,\mu}, \quad B_s = K(v_s) + C_s.$$

On tensor-valued fields this commutator also contains the fixed surface tensor-curvature action in a surface slot. It vanishes in a base slot. The two identities

$$(i\Lambda_\rho d_A K(v))_0 = L_{A,\rho} v, \quad \text{div}_{A,\rho} K(v) = -\frac{1}{2}[\mathbf{m}, v]$$

give the first differentiated equation $L_{A,\rho} v_t + \rho^2 S_t = 0$, where

$$S_t = (i\Lambda_\Sigma d_A C_t)_0 + S_t^{\text{geom}}, \quad |S_t^{\text{geom}}| \leq C|v_1|.$$

The term S_t^{geom} is the fixed surface tensor-curvature contribution. Thus $|S_t| \leq C(1 + |v_1|)$. For $v_{st} = \mathcal{D}_s v_t$, the second differentiated equation is

$$\begin{aligned} L_{A,\rho} v_{st} &= \sum_{\mu} \lambda_{\mu} [K(v_s)_{\mu}, \nabla_{A,\mu} v_t] \\ (278) \quad &+ \rho^2 \sum_{\mu \in T\Sigma} [C_{s,\mu}, \nabla_{A,\mu} v_t] + \frac{\rho^2}{2} [\text{div}_{A,\Sigma} C_s, v_t] \\ &- \rho^2 \mathcal{D}_s S_t + \rho^2 R_{st}^{\text{geom}}. \end{aligned}$$

The omitted metric divergence is a commutator with $\mathbf{m}$ and vanishes. The tensor R_{st}^{geom} contains only fixed surface geometry and surface metric jets. Moreover,

$$\begin{aligned} \mathcal{D}_s C_t &= \nabla_s \gamma_t - (\nabla_s \gamma_t)^{*H} + \frac{1}{2}[v_s, \gamma_t + \gamma_t^{*H}], \\ \mathcal{D}_s S_t &= (i\Lambda_\Sigma (d_A \mathcal{D}_s C_t + [B_s \wedge C_t]))_0 + \text{surface geometry terms}. \end{aligned}$$

Only the surface restriction of B_s occurs in the second line. The surface Chern formula is

$$A_\Sigma = \alpha + \frac{1}{2}q - \frac{i}{2}q \circ J_\Sigma + c - c^{*H}, \quad \nabla_{A,\Sigma} v_j = v_{j+1} + [-\frac{i}{2}q \circ J_\Sigma + c - c^{*H}, v_j].$$

It bounds all reference terms in (278) by $C\rho^2(1 + |v_2|)$ once v_1 is bounded. The first sum in that equation pairs two base gradients with weight one or two surface gradients with weight ρ^2 . It is bounded by $C|d_{A,\rho} v_1|^2$. This proves (277) at order two.

For the induction, retain the stronger form $L_{A,\rho} v_k = \rho^2 R_k + P_k$. The expression R_k has no base derivative of the unknown metric and is at most linear in v_k , with coefficients controlled by lower surface jets and the reference data. Each term of P_k is bilinear in $\nabla_A v_i, \nabla_A v_j$, where $i, j < k$ and $i + j \leq k$; the derivative indices are contracted within the same product factor. Base pairs have weight one and surface pairs have weight ρ^2 .

Apply one more ordered surface derivative. Variation of L adds the paired product $K(v_1)\nabla_A v_k$ and ρ^2 -weighted reference terms. In differentiating P_k use

$$\mathcal{D}_s (\nabla_{A,\mu} v_i) = \nabla_{A,\mu} v_{si} + [K(v_s)_\mu + C_{s,\mu}, v_i] + R_{s\mu}^\Sigma v_i.$$

Here $R_{s\mu}^\Sigma$ is the fixed tensor-curvature action and is zero for a base index μ . The first term raises one derivative index by one. The $K(v_s)$ term replaces one gradient by a gradient of v_1 times a bounded lower jet, preserving the paired contraction. The C_s term occurs only in a surface slot: its partner is therefore also a surface gradient and the whole term retains ρ^2 .

The surface Chern formula absorbs it in R_{k+1} . Differentiating coefficients and R_k introduces only surface jets, with v_{k+1} at most linear.

Fixed tensor-curvature commutators also have the surface factor ρ^2 . Ordered differentiation requires no interchange of parameter indices. If they are interchanged, the parameter curvature $-[q \wedge q]/4$ acts on $\mathfrak{m}$ or a lower differentiated equation, preserving the same form. Thus the induction closes before taking norms and proves (277) at every order.

Let G be the scalar Dirichlet Green operator of $\Delta_B + \rho^2 \Delta_\Sigma$ on $D_R^- \times \Sigma$. It is positive, $\|G1\|_\infty \leq CR^2$, and

$$\|G\sqrt{fg}\|_\infty \leq \|Gf\|_\infty^{1/2} \|Gg\|_\infty^{1/2} \quad (f, g \geq 0).$$

These assertions follow from the base barrier and Cauchy–Schwarz in the positive Green measure. Put $M_k = \sup |v_k|$, $\beta_k = \sup_\partial |v_k|$, $I_k = \|G|d_{A,\rho}v_k|^2\|_\infty$. Kato’s inequality and the identity

$$(\Delta_B + \rho^2 \Delta_\Sigma)|v_k|^2 = 4\langle L_{A,\rho}v_k, v_k \rangle - 2|d_{A,\rho}v_k|^2$$

give

$$(279) \quad \begin{aligned} M_k &\leq \beta_k + C_k r^2(1 + M_k) + C_k \sum \sqrt{I_i I_j}, \\ I_k &\leq \tfrac{1}{2}\beta_k^2 + C_k M_k(r^2(1 + M_k) + \sum \sqrt{I_i I_j}). \end{aligned}$$

The sums have the indices in (277) and are empty for $k = 1$. Since $r \rightarrow 0$ and the boundary derivatives tend to zero, induction gives $M_k \leq C_k B_k$ and $I_k \leq C_k B_k^2$, where $B_k = \max_{j \leq k} \beta_j + r^2$. Ordinary derivatives of s are polynomial expressions in the v_j and bounded h, h^{-1} . Zeroth order is also controlled: on the compact irreducible attaching family $\|s\|_{C_\Sigma^0} \leq C\|\nabla_\alpha s\|_{C_\Sigma^0}$, since otherwise a normalized sequence would limit to a nonzero parallel trace-free section. This proves the first assertion.

On $L = T^A$, interpolate (275) with the uniform C_Σ^{2k} bound to obtain $\|s\|_{C_\Sigma^k} \leq C_k L^{1/4} \rho^{1/8}$. The logarithmic metric equation in the base-parallel frame is

$$(280) \quad \Delta_w s = Q(s)(d_w s, d_w s) + \rho^2 f.$$

Every fixed surface derivative of f is bounded using two further surface orders; f has no base derivative of h . On translated unit discs and seam half-discs, the two-dimensional interpolation estimate

$$\|d_w s\|_{L^{2p}}^2 \leq C\|s\|_\infty \|d_w^2 s\|_{L^p} + C\|s\|_\infty^2, \quad p > 2,$$

absorbs the quadratic term in the localized Laplace estimate. At the seam subtract the known small Dirichlet trace. Surface differentiation first gives the same estimate at every prescribed surface order; further base differentiation then uses the already obtained orders in f and the two-dimensional $W^{2,p}$ algebra. The coefficient and seam-jet bounds are uniform on the unit cover. Taking suprema, rather than summing its cardinality, proves mixed derivative bounds by a logarithmic polynomial times $\rho^{1/8}$. The actual and comparison surface operators differ by η/R times such a polynomial. Unitary metric transport by $h^{1/2}$ therefore gives the surface comparison. In the positive hemisphere the primitive family is uniformly stable, and its classifying map is smooth in w and the slow parameters; the same argument gives the positive comparison.

Finally use parallel transport along the seam from its center, then along normal segments, with the convention $A^g = gAg^{-1} - dg g^{-1}$. The gauge differs from the initial one by a

logarithmic polynomial times $\rho^{1/8}$, at every fixed order. In $w = x - it$ it satisfies

$$(281) \quad A_t^g = 0, \quad A_x^g(x, 0) = 0, \quad A_x^g(x, t) = \int_0^t F_{A^g, tx}(x, \tau) d\tau, \quad F_{A^g, tx} = \rho^2 \mathcal{C}(F_{A^g, \Sigma\Sigma}),$$

where $\mathcal{C}$ is the fixed signed contraction. All fixed surface and unit base derivatives of the last curvature are bounded. Differentiating the integral proves the base estimate in (276). The strict exponent gap from $1/8$ to $1/10$ absorbs every fixed logarithmic polynomial. The gauge preserves the actual matching condition.

The exact physical caps have error $\rho^2 T^J$ from this same compact base-flat family, at every prescribed finite order. Apply the same axial construction on their Euclidean regions. For their difference from the actual solution, $\Delta A_w = O(\rho^2 T^J)$ and the surface/map difference is $O(\rho^{1/10} T^J)$. The raw cap base coefficient is $a = (f_\infty/\rho) O_a \Delta A_w \circ \Phi$, as in the source conversion above. Thus a/M_1 is a positive power of ρ times a polynomial in T , even after its physical ρ^{-1} conversion. The other strong terms and the weak terms $\rho d_\alpha a$, $\rho f_\infty d_\alpha b$ have the same property. Fixed compact-coordinate derivatives of the inner cutoff and frame conversions cost only powers of T on $|w| \leq T^{81}$. Bounding these exponential errors at two integer base orders proves their ambient raw norm is $o(T^{-N})$ for every fixed N . $\square$

8.6. The critical seam trace and the larger axial gauge. Set $Z = w/R = X - iY$. We need an $H^{3/2}$ bound for the seam logarithm after this dilation; small pointwise values alone do not give this trace estimate.

Lemma 8.7. *On every fixed smaller real Z interval and for each fixed surface order k ,*

$$(282) \quad \|s(R \cdot)\|_{H_X^{3/2} H_\Sigma^k} \leq C_k \eta.$$

In the base-parallel frame write $c = c^0 + e$, where c^0 is the comparison-primitive coefficient. On a fixed smaller negative Z half-disc,

$$(283) \quad \begin{aligned} \|e\|_{L_Z^\infty C_\Sigma^k} + \|d_Z e\|_{L_Z^2 H_\Sigma^k} &\leq C_k \eta, \\ \|c^0\|_{L_Z^\infty C_\Sigma^k} + \|d_Z c^0\|_{L_Z^2 H_\Sigma^k} &\leq C_k. \end{aligned}$$

The integrals of $|d_Z e|_{C_\Sigma^k}$ and $|d_Z c^0|_{C_\Sigma^k}$ along every vertical segment of a smaller fixed rectangle are bounded by $C_k \eta$ and C_k , respectively.

Proof. In either case the seam metric logarithm is a smooth function $F_E(x, d)$, with $F_E(x, 0) = 0$, of the compact comparison primitive E and the difference $d(x)$ of its slow parameters. It is smooth also at $x = \infty$ in the coordinate $1/x$, with uniformly bounded parameter and surface derivatives. For the first sector this follows from the NS family and conjugation by $S(x)$ in (271). For the second it follows from the common extension frames and the compact stable seam family. Only a function on the compact real boundary is being extended at infinity.

The Taylor integral gives $F_E(x, d) = Q_E(x, d)d$, with Q_E real-linear in its vector argument. The exact divided difference is $d(w) = (w - A)a(w)$, with $|\partial_w^j a| \leq C_j \eta R^{-1-j}$ and bounded A ($A = -i$ in the first sector). In real coordinates let M_A be multiplication by A . Taylor expansion at infinity gives

$$(284) \quad Q_E(x, d)(xI - M_A) = xQ_\infty(d) + Q_0(d) + K_E(x, d), \quad |\partial_x^j \partial_d^\gamma \nabla_\Sigma^k K_E| \leq C_{j, \gamma, k} (1 + |x|)^{-1-j}.$$

The two ends of the real line have the same expansion in $1/x$. The bound on a fixed interval follows by smoothness. Put $b(X) = Ra(RX)$. Both b and $d(RX)$ have all fixed X derivatives

$O(\eta)$ on the retained interval. The seam logarithm is the sum

$$XQ_\infty(d(RX))b(X) + R^{-1}Q_0(d(RX))b(X) + R^{-1}K_E(RX, d(RX))b(X).$$

The first two terms have $C_X^2 C_\Sigma^k$ size $O(\eta)$. For the last, insert a fixed outer cutoff. Change of variables and (284) give

$$\|R^{-1}K_E(R \cdot, d(R \cdot))\|_{H_X^1 H_\Sigma^k} \leq C_k R^{-1/2}, \quad \|R^{-1}K_E(R \cdot, d(R \cdot))\|_{H_X^2 H_\Sigma^k} \leq C_k R^{1/2}.$$

The largest second derivative is $R\partial_x^2 K_E$; derivatives of $d(RX)$ are bounded and all parameter derivatives of K_E have the same decay. Multiplication by b costs $C_k \eta$, taking further surface orders where needed. Interpolation between these fixed Hilbert norms proves (282). A higher uniform trace order is not asserted.

For the first sector the surface error in the global S frame has the four blocks

$$e(w) = \begin{pmatrix} (w-a)k_{11} & (w-a)^2(w-\bar{a})^{-1}k_{12} \\ (w-\bar{a})k_{21} & (w-a)k_{22} \end{pmatrix}.$$

Polynomial division of $(w-a)^2/(w-\bar{a})$ and the slow derivatives of k give $|e| \leq C\eta(1+|w|)/R$, $|d_w e| \leq C\eta/R$, in every fixed surface order. The pole is outside the closed negative half-plane. In the second sector write $e = G_E(w, d(w))$, where G_E is smooth on the compact negative hemisphere, $G_E(w, 0) = 0$, and its fixed-parameter base derivative is $O((1+|w|)^{-2})$. These facts follow from (265) and (268) in the fixed primitive base-parallel frame. The same bounds for $e, d_w e$ follow. In both cases $|d_w c^0|_{C_\Sigma^k} \leq C_k(1+|w|)^{-2}$. The Dirichlet integral is conformally invariant under $w = RZ$, and the vertical integral of this last decay is uniformly bounded after the substitution $t = RY$. These observations prove all coefficient and segment assertions. $\square$

Let $d_{k,\nu}$ be the supremum of the surface derivatives of s_ν through order k . They tend to zero by Lemma 8.6. We write $\tau_{k,\nu} \rightarrow 0$ for a finite sum of η_ν, r_ν^2 , and sufficiently many of these $d_{j,\nu}$; this asserts no logarithmic rate. On fixed nested Z half-discs the metric equation is

$$\Delta_Z s = Q(s)(d_Z s, d_Z s) + r^2 f.$$

Use the fixed trace extension $H^{3/2} \rightarrow H^2$ and $\|d_Z s\|_4^2 \leq C\|s\|_\infty \|d_Z^2 s\|_2 + C\|s\|_\infty^2$ to absorb its quadratic term. Differentiating in Σ first, the highest new first-derivative term has the small coefficient $d_Z s$ in L^4 ; all other products use preceding orders. The bounded surface derivatives of f and the small boundary trace close the finite induction. Thus

$$(285) \quad \|s(R \cdot)\|_{H_Z^2 H_\Sigma^k} \leq C_k \tau_{k,\nu}.$$

Pointwise surface uses of this estimate take two further surface orders. It does not commute a surface supremum with a base L^2 norm for free.

After unitary metric transport, A'_Z is a smooth expression in $s, d_Z s$, so $\|A'_Z\|_{H_Z^1 H_\Sigma^k} \leq C_k \tau_{k,\nu}$. For an H^1 function on a fixed rectangle the L^2 trace on each parallel line segment of a smaller rectangle is uniformly bounded by its H^1 norm: apply the one-dimensional H^1 estimate with values in the other variable's L^2 space. The same statement holds at the real boundary. Consequently the seam and normal transport integrals of A'_Z are small, in every prescribed surface order. The axial gauge g based at the seam center satisfies $\|g - I\|_{L_Z^\infty C_\Sigma^k} \leq C_k \tau_{k,\nu}$.

The curvature identity now gives $A_Z^g = O(r^2)$. To differentiate its axial integral, the vertical integral of $d_Z F_{\Sigma\Sigma}$ is uniformly bounded: differentiate the surface Chern formula, use (285) and the segment trace for $d_Z s$, and use (283) for $d_Z c$. The first-order gauge equation gives

the integrals of $d_Z g$ after the A_Z^g bound is known. It therefore also bounds the differentiated conjugation of surface curvature. Differentiating the axial integral proves

$$(286) \quad \begin{aligned} & \|A_Z^g\|_{L_Z^\infty C_\Sigma^k} + \|d_Z A_Z^g\|_{L_Z^\infty C_\Sigma^k} \leq C_k r^2, \\ & \|\beta^g - \beta^0\|_{L_Z^\infty C_\Sigma^k} + \|\beta^g - \beta^0\|_{H_Z^1 H_\Sigma^k} \leq C_k \tau_{k,\nu}. \end{aligned}$$

For the second line, before g the difference consists of e , surface derivatives of h , and small metric factors times bounded reference coefficients. Differentiating in Z uses (283), (285), and the finite Dirichlet norm of c^0 . After g use the differentiated first-order gauge equation and the small factor $g - I$ multiplying $d_Z \beta^0$. This does not require a uniform H^1 multiplier bound for the rapidly varying reference. In physical coordinates, the first line says $A_B^{\text{phys}} = O(r)$, $d_z A_B^{\text{phys}} = O(1)$ and its physical H^1 norm on the shrinking disc is $O(r)$.

On the positive side the difference from the comparison primitive has the same C^0 and H_Z^1 smallness. In the stable-family case its parameter variation is smooth on the compact positive hemisphere; in the first case it is the slow stable slice. In both cases smoothness of the primitive at infinity gives

$$(287) \quad |\nabla_w^j (\beta^0 - \alpha_\nu, u^0 - p_\nu^0)|_{C_\Sigma^k} \leq C_{j,k} |w|^{-1-j}, \quad p_\nu^0 = [\alpha_\nu], \quad |w| \geq C_*.$$

For the positive map use the attaching chart and omit the surface norm. The map need not be constant in the second sector.

8.7. Removing the primitive core.

Proposition 8.8. *Put $C_\nu = T_\nu^{40}$ and $h_\nu = \rho_\nu C_\nu$. There are smooth mixed configurations y_ν , equal to (α_ν, p_ν^0) on $|z - c_\nu| \leq h_\nu$ and equal to x_ν outside $2h_\nu$, which preserve exact matching, zero normal trace, and all ends. Their geometric equation source has complete residual norm $O(T_\nu^{-40})$ at $\varepsilon_\nu = 1/T_\nu$. They have global representatives converging strongly to x_0 in the critical residual ambient norms and in C^0 after each fixed surface derivative on the collar. Their background divergence, with its exact end limit subtracted, tends to zero in every required fixed L^p norm.*

Proof. On the annulus $|w| \sim C_\nu$, set $v = w/C_\nu$. Equations (276) and (287) give, in every fixed annular and surface order,

$$(288) \quad \|(\beta - \alpha_\nu, u - p_\nu^0)\| \leq q_\nu, \quad \|A_v\| \leq p_\nu, \quad q_\nu = C(T_\nu^{-40} + \rho_\nu^{1/10} T_\nu^J), \quad p_\nu = C\rho_\nu^2 T_\nu^J.$$

The exponent J may increase with the fixed derivative count. In the exact nonlinear matching chart about the flat pair, interpolate all surface and positive coordinates by a fixed radial cutoff $\chi(v)$, zero inside radius one and one outside radius two. Interpolate the base one-form by the same cutoff. The common harmonic coordinates on the seam remain equal, and the coexact seam coordinate remains zero. The resulting surface connection is $\alpha_\nu + \chi(\beta - \alpha_\nu) + \mathcal{R}$, with all fixed derivatives of $\mathcal{R}$ bounded by Cq_ν^2 . There is the same quadratic remainder on the positive side.

The exact input equations cancel. In physical coefficients the scalar-row source is bounded by $C[h_\nu^{-2}(p_\nu + p_\nu^2) + q_\nu^2]$, and the mixed row by $Ch_\nu^{-1}(q_\nu + p_\nu q_\nu + q_\nu^2)$. The positive row obeys the latter bound without the p_ν term. For a coefficient of fixed-annulus $H_v^1 H_\Sigma^k$ norm K , physical dilation and interpolation give

$$\|f((z - c)/h)\|_{H_B^\varepsilon H_\Sigma^k} \leq Ch^{1-\varepsilon} K, \quad 0 \leq \varepsilon \leq 1/4.$$

The annular support margins permit extension by zero. All residual source rows, including their surface operators, are controlled by this local norm, with one further fixed surface order

when needed. Therefore

$$(289) \quad \|\mathcal{F}(y_\nu)\|_{Y_{\varepsilon\nu,k}^{\text{res}}} \leq C_k [h_\nu^{-1-\varepsilon\nu}(p_\nu + p_\nu^2) + h_\nu^{1-\varepsilon\nu}q_\nu^2 + h_\nu^{-\varepsilon\nu}(q_\nu + p_\nu q_\nu + q_\nu^2)] \leq C_k T_\nu^{-40}.$$

The source is identified by the current attaching surface frame and the bounded positive target frame on its support. Here $h_\nu^{-\varepsilon\nu}$ is bounded and p_ν/h_ν is exponentially small. The perturbation dependency assumption ensures there is no source elsewhere.

The gauge in (286) agrees with the inner axial gauge by uniqueness of their seam-then-normal transport. On the transition its base coefficient is $O(r^2 C_\nu/R)$ in Z , so a derivative of the removal cutoff preserves its $O(r^2)$ derivative bound. The reference's Dirichlet tail outside $|w| \geq C_\nu$ is $O(C_\nu^{-1})$. Together with (286) this gives, on a fixed smaller physical relative disc,

$$(290) \quad \begin{aligned} & \|A_{y,B}^{\text{phys}}\|_\infty \leq C_k r, \quad \|d_z A_{y,B}^{\text{phys}}\|_\infty \leq C_k, \quad \|A_{y,B}^{\text{phys}}\|_{H_B^1 H_\Sigma^k} \leq C_k r, \\ & \|\beta_y - \alpha_\nu\|_{H_B^1 H_\Sigma^k} + \|u_y - p_\nu^0\|_{H_B^1} \leq C_k(\tau_{k,\nu} + T_\nu^{-40} + r), \\ & \|\beta_y - \alpha_\nu\|_{C_B^0 C_\Sigma^k} + \|u_y - p_\nu^0\|_\infty \leq C_k(\tau_{k,\nu} + T_\nu^{-40}). \end{aligned}$$

Suprema of base fields in the first line include the displayed surface order. On the shrinking transition, estimate derivatives of the cutoff using (288); one must not apply an unbounded cutoff multiplier to the qualitative error $\tau_{k,\nu}$.

On the exterior overlap compare this axial representative with the stipulated representative converging to the based residual. If $A_2 = A_1^G$, the gauge identities are

$$d_\Sigma G = G\beta_1 - \beta_2 G, \quad d_B G = G A_{1,B} - A_{2,B} G.$$

Unitarity and the first identity give bounded surface derivatives. The physical base bounds and their first derivatives give $d_B G = O(r)$, $d_B^2 G = O(1)$, in all fixed surface orders. Fix one base value G_0 on the connected overlap. Integration along paths of length Cr gives

$$H = G G_0^{-1}, \quad H - I = O(r^2), \quad d_B H = O(r), \quad d_B^2 H = O(1).$$

First retain G_0 as a constant surface gauge on the inner representative, then cut off only $\log H$ on the overlap. The cutoff gauge has base derivative $O(r)$ and second derivative $O(1)$; its physical H^1 contribution tends to zero. Choose it normal-constant on the seam. Undo the common based residual gauge using its fixed-collar extension. This proves global critical and C^0 convergence.

The background divergence uses first derivatives of the base component, which are bounded on a region of area $O(r^2)$, and surface derivatives of the surface difference, which tend uniformly to zero. Thus it tends to zero in each fixed finite L^p norm on compact sets. The joined configurations equal the original ones on all ends. If $B_{p,e}$ is the exact critical family, define

$$(291) \quad e_e(p) = d_{B_{0,e}}^*(B_{p,e} - B_{0,e}), \quad \sigma_p = \sum_e \chi_e e_e(p).$$

Since $B'_0 = p$ is co-closed, $e_e(p) = O(|p|^2)$. Subtracting σ_{p_ν} gives the asserted weighted divergence convergence. Its unsubtracted nondecaying value need not belong to a weighted source space. $\square$

8.8. A fixed gauge and the quantitative residual estimate.

Lemma 8.9. *Let $a_\nu = A_{y_\nu} - A_0$ in the preceding global representatives and $f_\nu = d_{A_0}^* a_\nu - \sigma_{p_\nu}$. There are smooth gauges $g_\nu = \exp \phi_\nu$, with zero covariant normal derivative and tending to the identity on all ends, such that*

$$d_{A_0}^*(A_{y_\nu}^{g_\nu} - A_0) = \sigma_{p_\nu}, \quad \|\phi_\nu\|_{W_\omega^{2,2} \cap W_\omega^{2,6}} \rightarrow 0.$$

The same convergence holds after every fixed surface derivative on a smaller fixed matching collar. The gauges preserve critical convergence, all exact critical profiles, and the geometric source estimate (289).

Proof. The fixed operator $P = d_{A_0}^* d_{A_0}$ with covariant Neumann condition has a positive global L^2 gap. On each end cross-section irreducibility excludes a parallel zero-form and gives a tangential Poincaré inequality. It controls the far tails of a putative normalized sequence with derivative tending to zero. Rellich compactness on the remaining fixed compact part would then give a nonzero parallel zero-form for A_0 , a contradiction. For $q = 2, 6$, integration against $|v|^{q-2}v$ gives

$$\int \langle Pv, |v|^{q-2}v \rangle \geq \frac{4(q-1)}{q^2} \|d_{A_0}(|v|^{q/2-1}v)\|_2^2 \geq \frac{4(q-1)\lambda}{q^2} \|v\|_q^q.$$

Fixed Neumann estimates therefore give $\|v\|_{W^{2,q}} \leq C_q \|Pv\|_q$. For compact smooth forcing, start with the L^2 inverse, use fixed local regularity, and insert end cutoffs in this inequality; density gives the agreeing L^2 and L^6 inverses. Conjugation by a sufficiently small fixed weight $e^{\omega t_0}$ changes P by a small first-order operator and leaves its Neumann domain unchanged. A Neumann series yields

$$Q_N = P^{-1} : L_\omega^2 \cap L_\omega^6 \longrightarrow W_{\omega,N}^{2,2} \cap W_{\omega,N}^{2,6}.$$

The notation denotes the one bounded inverse on the intersection.

For the gauge action used above set $M_\phi = e^{\text{ad } \phi}$, $J_\phi = \int_0^1 e^{s \text{ad } \phi} ds$. Then $a^g = M_\phi a - J_\phi d_{A_0} \phi$ and

$$\begin{aligned} d_{A_0}^* a^g - \sigma_p &= f - P\phi + N(\phi), \\ (292) \quad N(\phi) &= (M_\phi - I)(f + \sigma_p) - (\nabla_{A_0} M_\phi) \lrcorner a \\ &\quad - (J_\phi - I)P\phi + (\nabla_{A_0} J_\phi) \lrcorner d_{A_0} \phi. \end{aligned}$$

The only derivative of a is its known divergence. In $Y_g = L_\omega^2 \cap L_\omega^6$ and $Z_g = W_{\omega,N}^{2,2} \cap W_{\omega,N}^{2,6}$, the fixed $W^{2,6} \rightarrow C^1$ embedding gives

$$\|N(\phi) - N(\psi)\|_{Y_g} \leq C(\|a\|_\infty + \|\sigma_p\|_\infty + \|f\|_{Y_g} + \|\phi\|_{Z_g} + \|\psi\|_{Z_g}) \|\phi - \psi\|_{Z_g}.$$

The factor σ_p is used only in its unweighted supremum; a decaying ϕ supplies the weighted factor. The contraction $\phi = Q_N(f + N(\phi))$ on the ball $2\|Q_N\|\|f\|_{Y_g}$ proves the claim at order zero. Normal trace follows directly from the exact gauge formula.

For surface differentiation first rearrange the equation as

$$P\phi = J_\phi^{-1} \{M_\phi f + (M_\phi - I)\sigma_p - (\nabla M_\phi) \lrcorner a + (\nabla J_\phi) \lrcorner d_{A_0} \phi\}.$$

Its principal operator is fixed and its right side contains only first derivatives of ϕ . After k surface derivatives, the highest new field and its first derivative are controlled by the preceding surface level's two physical derivatives. The commutator with P has that same preceding-order control. The remaining terms use the small surface derivatives of a, f , and bounded derivatives of σ_p with a decaying ϕ factor. Fixed local Neumann estimates on nested collars give the finite induction. Smoothness for each ν follows from this semilinear equation and ordinary regularity. The gauges tend to identity on weighted ends by embedding; on unweighted ordinary ends their unit-region $W^{2,6}$ norms tend to zero and give the same conclusion. The exact formula, the derivative bounds, and covariance preserve critical convergence and the localized geometric source bound. $\square$

Proposition 8.10. *Let V_ν be the inverse nonlinear residual coordinate of the gauged y_ν about x_0 , with all finite end variables. For every fixed $k \geq 2$,*

$$(293) \quad \|V_\nu\|_{X_{1/T_\nu, k}^{\text{res}}} \leq C_k T_\nu^{-40}.$$

The same ambient residual bound holds after subtracting the exact critical profiles. The attaching class and the central flat surface representative satisfy errors $O(T_\nu^{-79/2})$ relative to the fixed residual at c_ν .

Proof. The proved C^0 surface and map convergence permits the inverse nonlinear matching chart; no critical Sobolev embedding into C^0 is assumed. Its coordinate V has small critical norm, and all its required C^0 surface coefficient norms tend to zero. Let their finite sum, together with $|p|$, be $s_m(V)$. The residual inverse in Lemma 7.4 also holds at $\varepsilon = 0$: its initial first-order inverse, its integer surface commutators, and the reflected Bessel multiplier proof all include this endpoint on the same domain.

Write $A_{\Phi_0(V)} - A_0 = \xi + C(V)$ for its connection component. The nonlinear correction C on the collar is surface-valued, quadratic, and gains one surface derivative; its end value is $B_p - B_0 - p$. Define

$$N_{\text{geom}}(V) = \mathcal{F}(\Phi_0(V)) - D_{\text{geom}}V, \quad R_C(V) = \sigma_p - d_{A_0}^* C(V).$$

For sufficiently many fixed surface supremum orders,

$$(294) \quad \|N_{\text{geom}}(V)\|_{Y_{0,k}^{\text{res}}} + \|R_C(V)\|_{Y_{0,k}^{\text{res}}} \leq C_k s_m(V) \|V\|_{X_{0,k}^{\text{res}}}.$$

Here each source is placed only in its geometric or Coulomb row. To verify the estimate, distribute the surface weights χ^j in the Γ norms between the two factors of each product. Put one factor and its fixed surface derivatives in the small supremum norm and the other in weighted L^2 . Surface ellipticity of the principal part $\chi\Lambda_\Sigma$ and the smooth finite-rank end corrections give the required weighted derivative bounds. There is no division by χ . Where the chart correction is supported, the unweighted $H_B^1 H_\Sigma^k$ and $L_B^2 H_\Sigma^{k+1}$ norms are available. One base derivative of its surface correction has form $DS(b)d_B b + (\partial_B S)(b)$, with the second term quadratic. The Coulomb row takes only a surface derivative of this correction. The positive remainder has a small coefficient multiplying $d_B v$, plus quadratic terms with fixed background derivatives. For holonomy, the second Duhamel expansion, one supremum insertion, and submersion coarea give $\|\mathcal{H}(A_0 + a) - \mathcal{H}(A_0) - D\mathcal{H}_{A_0} a\|_2 \leq C\|a\|_\infty \|a\|_2$. The prescribed primary series sums these bounds, and its dependencies miss the surface-localized collar.

To check decay, choose a smooth linear extension V_p of the finite end tangent and write $V = V_p + W$, with W decaying. The exact critical geometric equation implies $N_{\text{geom}}(V_p) = O(|p|^2)$ with exponential decay before taking the weighted norm. Also $\sigma_p - d_{A_0}^* C(V_p)$ has the same bound: its two nondecaying limits cancel exactly. Subtracting these expressions from those at $V_p + W$ leaves a decaying W factor and a small coefficient. For the chart derivative, split $DS(b_p + b_W)d_B b_W$ from $[DS(b_p + b_W) - DS(b_p)]d_B b_p$; the latter uses the smooth $O(|p|)$ derivative of the finite profile. This proves (294) with exact ends retained.

The actual augmented equation is

$$DV_\nu = (\mathcal{F}(y_\nu^{g\nu}) - N_{\text{geom}}(V_\nu), R_C(V_\nu)).$$

The fixed endpoint inverse and (289) give $\|V_\nu\|_{X_{0,k}^{\text{res}}} \leq C_k T_\nu^{-40} + C'_k s_m(V_\nu) \|V_\nu\|_{X_{0,k}^{\text{res}}}$. Absorb the final coefficient. No factor T_ν multiplies the qualitative smallness in this endpoint argument.

For fractional order a polynomial second-order bound suffices. Before core removal, the coefficient and seam-jet bounds and (280) give uniformly bounded fixed w derivatives on unit discs throughout a smaller retained region. Integrated norms cost powers of $R \leq \rho^{-1}$. The axial identity and its first-order gauge equation, differentiated successively, retain $A_w^{\text{ax}} = O(\rho^2 R)$ and bounded fixed mixed derivatives of the gauge and surface fields. This is a finite induction using unitary transport and the curvature integral, not exponential growth with the path length. Physical differentiation costs fixed powers of ρ^{-1} . The two cutoff radii satisfy $h \geq \rho$ and $r \geq c\sqrt{\rho}$, so their differentiated contributions are also polynomial. The exterior joining gauge has the same property by its first-order equation.

One further derivative of the fixed gauge equation gives a polynomial H^3 bound: its principal operator is fixed, its coefficients have small C^1 norms, and the differentiated forcing and connection use the just established bounds. The surface-order version is obtained first, before the extra base derivative. Differentiating the inverse surface chart twice then gives

$$\|V_\nu\|_{E_\omega^2} + \|\varphi V_\nu\|_{H_B^2 H_\Sigma^k; \omega, \text{as}} \leq C_k \rho_\nu^{-J_k}.$$

The suffix means exact profile subtraction with the finite coordinates added. The chart is a surface chart; no four-dimensional H^2 algebra estimate is used. Interpolation in each complete residual component yields

$$(T^{-40})^{1-1/T} (\rho^{-J_k})^{1/T} = T^{-40} \exp(J_k + 40 \log T/T).$$

It proves (293), including the Γ terms, the next-surface-order weak term, and all finite end variables. The argument works at any exponent comparable to $1/T$.

Finally a normalized average on a half-disc of radius h has H^{-1} norm at most $C\sqrt{1 + \log(1/h)}$. Indeed its Fourier transform is $\widehat{\eta}(h\xi)$; integrating $(1 + |\xi|^2)^{-1}$ up to $|\xi| = 1/h$ gives the logarithm, and the smooth Fourier tail is bounded. Apply a fixed half-plane extension and outer cutoff. The erased positive map is exactly p_ν^0 on that disc. Averaging its critical $O(T_\nu^{-40})$ difference from the residual gives

$$(295) \quad d(p_\nu^0, u_0(c_\nu)) \leq CT_\nu^{-79/2}.$$

The variation of u_0 across the disc is exponentially smaller. For the surface representative after the fixed gauge, its oscillation on the erased disc is $O(h_\nu)$ in every fixed surface order, since the gauge has bounded first derivatives. The same Hilbert-valued average therefore gives $\|\beta_{y_\nu^0}(c_\nu) - \alpha_0(c_\nu)\|_{H_\Sigma^k} \leq C_k T_\nu^{-79/2}$. This representative is flat. The constant surface factor from the exterior join is retained in this comparison. $\square$

8.9. Uniqueness in the full small chart.

Proposition 8.11. *Fix the parameters of a sufficiently small regular gluing chart. Let U_* be its normalized correction and let $U \in X_\rho$ solve the same geometric equation and the same four observation equations. If $\|U\|_{X_\rho} \leq T^{-9}$, then $\Phi_\rho(U)$ and $\Phi_\rho(U_*)$ represent the same ordinary mixed moduli point.*

Proof. Write the augmented equation of Proposition 7.23 as

$$\mathbb{F}(U) = \mathbb{F}(0) + \mathbb{A}U + \mathbb{N}(U), \quad \mathbb{A} = (D, \ell), \quad \mathcal{B} = \text{im } Q \oplus Z\mathbb{R}^4.$$

Its inverse on $\mathcal{B}$, denoted by $\mathbb{Q}$, satisfies $\mathbb{A}\mathbb{Q} = I$ and $\|\mathbb{Q}\| \leq CT^3$. The geometric inverse has norm CT^2 , $\|\ell\| \leq CT^{1/2}$, and

$$(296) \quad \|\mathbb{N}(U) - \mathbb{N}(V)\| \leq CT^5(\|U\| + \|V\|)\|U - V\|, \quad \|U\|, \|V\| \leq c_0 T^{-5}.$$

The equation is C^1 , and its geometric part has quadratic constant CT^4 . The constructed correction belongs to $\mathcal{B}$ and has norm at most CT^{-37} .

Set $V = U - U_*$, $\delta = \|U\| + \|U_*\|$, and decompose $V = w + b$ by $b = \mathbb{Q}\mathbb{A}V$. The endpoint equations give

$$b = -\mathbb{Q}(\mathbb{N}(U) - \mathbb{N}(U_*)), \quad \|b\| \leq CT^8\delta^2, \quad w \in \ker \mathbb{A}, \quad \|w\| \leq \delta + CT^8\delta^2.$$

For $0 \leq t \leq 1$, solve in $\mathcal{B}$

$$(297) \quad v(t) = -\mathbb{Q}\{\mathbb{N}(U_* + tw + v(t)) - \mathbb{N}(U_*)\}.$$

The ball of radius $C_1T^8\delta^2$ is invariant for C_1 large, and the contraction constant is at most $CT^8\delta = o(1)$. The solution depends continuously on t , has $v(0) = 0$, $v(1) = b$, and gives an exact path $U(t) = U_* + tw + v(t)$ with the four observations fixed and $\sup_t \|U(t)\| \leq 4\delta$. Only the difference estimate at the original origin is used here.

We check ordinary regularity at every point U_0 of this path. For $L = D + dN_{\text{geo}}(U_0)$ the formula

$$\tilde{Q}_{U_0} = Q(I + dN_{\text{geo}}(U_0)Q)^{-1}, \quad \|\tilde{Q}_{U_0}\| \leq CT^2$$

is valid because $\|dN_{\text{geo}}(U_0)Q\| \leq CT^6\delta = o(1)$. The fixed-scale regularity argument in Proposition 7.18 gives a smooth finite-energy representative with its exact critical limits. The small seam chart retains an irreducible flat fiber. For an ordinary augmented cokernel pair (s, η) , testing compact covariant-Neumann gauge variations gives $d_A^*d_A s = 0$, hence $s = 0$ by irreducibility and the end conditions. Pair η with $\tilde{Q}_{U_0}$ applied to compact smooth geometric sources. The end cutoffs can be chosen constant on each far primary dependency family and equal to one on all secondary dependencies. Their remaining commutators tend to zero against the positively weighted adjoint solution; a graph variation has only the allowed finite critical tangent and a weighted remainder. Density gives $\eta = 0$. The index computation of the local gluing theorem still applies: the bundle, positive relative class, and end components have not changed. Thus the ordinary moduli space is smooth of dimension four at every U_0 .

Let W_j be the four fixed-attaching pregluing derivatives used for (225). They satisfy $\|W_j\| \leq C$, $\|DW_j\| \leq CT^{-40}$, and $(\ell W_j)_j = M_0(E) + o(1)$ with uniformly invertible M_0 . Set $K_j = (I - \tilde{Q}_{U_0}L)W_j$. Then $LK_j = 0$ and $\|K_j - W_j\| \leq C(T^{-38} + T^6\delta)$. For $H(U) = \mathcal{O}_{c,\rho}(\Phi_\rho(U)) - \mathcal{O}_0(E)$, the observation part of (296) gives

$$(dH(U_0)K_j)_j = M_0(E) + o(1) + O(T^{13/2}\delta).$$

This matrix is invertible. The fixed-scale reduction in Lemma 7.21 transfers these vectors to the ordinary augmented kernel. It retains the critical tangent and solves the decaying covariant Neumann equation after removing its limiting infinitesimal gauge. Irreducibility and the end spectral gaps justify that solve at this positive scale. Gauge invariance retains the displayed matrix. The observations are therefore local coordinates on the ordinary moduli space at every point of the path.

Finally, graph continuity of this exact path implies continuity of its ordinary moduli classes by the argument in Proposition 7.24. Indeed, at fixed ρ it gives local physical $W^{1,p}$ convergence for some $p > 2$, positive C^0 convergence, convergence of the finite end coordinates, and weighted H^1 convergence after exact profile subtraction. These imply uniform integrability of energy on compact sets and uniformly small end tails. Compactness has no concentration, the comparison gauge equation identifies every smooth limit with the graph limit, and the end gaps give the ordinary weighted convergence. This argument only uses exactness and graph convergence. The continuous path has constant observations, hence is locally constant in the ordinary moduli space. Its endpoint classes agree. $\square$

8.10. Exact parameters and compatible inner and outer gauges. We now complete the proof of Theorem 8.1. Write $\Delta_\nu = T_\nu^{-79/2}$. Evaluation and the two affine normal functions have jointly surjective differential, by (212). Choose local primitive coordinates (q, t_1, t_2) on the normalized slice, with q the attaching class. Let $\bar{E}_\nu = E(c_\nu, t_\nu)$ have the fixed residual attaching value. Equation (295) gives $d(E_\nu^0, \bar{E}_\nu) \leq C\Delta_\nu$. This adjustment is allowed to leave the rank-one locus.

Lemma 8.12. *The parameters can be changed by (258) so that the original solution satisfies*

$$(298) \quad \mathcal{O}_{\hat{c}_\nu, \hat{\rho}_\nu}(x_\nu) = \mathcal{O}_0(E(\hat{c}_\nu, \hat{t}_\nu)).$$

The solution of these four parameter equations is unique in a fixed sufficiently small box in scaled center, log scale, and primitive coordinates.

Proof. Set

$$J_\nu(c, \rho, t) = \mathcal{O}_{c, \rho}(x_\nu) - \mathcal{O}_0(E(c, t)), \quad (c, \rho, t) = (c_\nu + \rho_\nu b, \rho_\nu e^h, t_\nu + \tau).$$

The inner comparison on the fixed observation-loop region and the attaching estimate give $|J_\nu(0)| \leq C\Delta_\nu$. The affine averaging formula (214) differentiates only the fixed smooth averaging densities. Their supports stay in one compact loop region on a small fixed box in (b, h) . The inner comparison therefore controls these differentiated observations without differentiating an incoming gauge with respect to the trial parameters. At the box center the derivative converges to the observation matrix on

$$T_E, \quad D_E, \quad -\partial_{t_1} E, \quad -\partial_{t_2} E.$$

The c derivative of $E(c, t)$ contributes $O(\rho_\nu)$. Affine equivariance of the limiting Dirichlet metric, as in the proof of (225), identifies the first two columns. These four vectors form a basis of the fixed-attaching tangent space, so the matrix M_ν is uniformly invertible.

Repeated changes of variables again differentiate only the densities. On the box, $\|dJ_\nu - M_\nu\| \leq C(\text{box radius}) + o(1)$. The map $z \mapsto z - M_\nu^{-1} J_\nu(z)$ is a contraction on a ball of radius $C\Delta_\nu$. It supplies the stated parameters. Integrating $dJ_\nu - M_\nu$ on segments proves injectivity on the smaller fixed convex box as well. $\square$

Proposition 8.13. *In one allowed gauge the original solution lies in the full configuration chart at the parameters of Lemma 8.12, with $\|U_\nu\|_{X_{\hat{\rho}_\nu}} \leq CT_\nu^{-38}$.*

Proof. We suppress ν and repeat the interpolation of Proposition 8.10 at $\varepsilon = 1/\hat{T}$. Since $\hat{T}/T \rightarrow 1$, the factors $\rho^{-J/\hat{T}}$ remain bounded. The core-erased solution y in the fixed residual gauge still has complete residual norm CT^{-40} . Use the residual frame based at $\hat{c}$ and let $\ell = \hat{\rho}\hat{T}^{81}$. It contains the entire support of the inner gluing cutoff. Write $y = y_{\text{ax}}^G$, where both representatives have zero normal base trace. The axial estimate through the erased annulus gives $A_{y_{\text{ax}}, B} = O(\rho T^J)$ on this disc. In the fixed based frame $A_{0, B} = O(\ell)$. The Hilbert-valued base embedding, of norm $C\sqrt{T}$ at this exponent, gives $\|A_{y, B} - A_{0, B}\|_{C_B^0 C_\Sigma^m} \leq C_m \Delta$. The gauge equation consequently implies

$$(299) \quad d_B G = G A_{y_{\text{ax}}, B} - A_{y, B} G, \quad \|d_B G\|_{C_B^0 C_\Sigma^m} \leq C_m d_T, \quad d_T = \Delta + \rho T^J m.$$

The global joining construction has uniformly bounded surface derivatives of G and bounded first base derivatives.

The localized $H_B^{1+\varepsilon} H_\Sigma^m$ norm of $d_B G$ is also bounded by $C_m d_T$. To verify this assertion, localization of the residual estimate costs $C\sqrt{T} \ell^{-\varepsilon}$. In differentiating a product fG , the term $(df)G$ uses the H^ε multiplier bound for uniformly C^1 functions. Substitute (299) in the other

term $f dG$; the resulting products of base coefficients use their supremum and H^ε bounds. The axial term is exponentially small with all required fixed derivatives. Background terms localized to this disc are $O(\ell^{1-\varepsilon})$. Additional surface derivatives use the same equation at the next fixed surface orders. All factors $\ell^{-\varepsilon}$ are bounded.

Let $G_c = G(\hat{c})$ and, on this disc, define

$$q(z) = \log(G(z)G_c^{-1}), \quad H(z) = \exp(-\chi(z)q(z)),$$

where χ is radial, one on the half-radius disc and zero near its outer circle. Then $\|q\|_{C_\Sigma^m} \leq C\ell d_T$, $\|dq\|_{C_\Sigma^m} \leq C d_T$, and $\|D_B^2 q\|_{H_B^\varepsilon H_\Sigma^m} \leq C d_T$. The normal derivative vanishes on the seam. Extending H by the identity gives an allowed gauge with $HG = G_c$ on the inner disc. The order of these factors follows from $(A^G)^H = A^{HG}$.

The cost of this change in the complete residual norm is at most CT^{-39} . For the largest cutoff term, $D_B^2 \chi q$ has size $C\ell^{-1} d_T$ on area $O(\ell^2)$, hence L^2 norm $C d_T$ and fractional factor $\ell^{-\varepsilon}$. Terms with one differentiated cutoff use dq ; the remaining terms use the preceding fractional estimate. Allowing one additional $\sqrt{T}$ factor gives $C\sqrt{T}\Delta = CT^{-39}$. Surface terms have the additional factor ℓ . The same product estimates apply to conjugating $y - x_0$. Thus, after this change,

$$(300) \quad \|y' - x_0\|_{\text{res}, 1+\varepsilon; \text{amb}} \leq CT^{-39}, \quad x' = y' \text{ outside the old removal disc.}$$

On the inner disc x' is the original axial solution in the constant surface frame G_c . All ends are unchanged.

Since $\hat{c}$ lies inside the old erased core, its flat surface representative is $\gamma = G_c \alpha_\nu G_c^{-1} - d_\Sigma G_c G_c^{-1}$. Before the last gauge change, the CT^{-40} residual estimate gives $\|\gamma - \alpha_0(\hat{c})\|_{C_\Sigma^m} \leq C_m \Delta$. The fixed irreducible surface slice expresses γ in the chosen flat family by a surface gauge k with $\|k - I\|_{C_\Sigma^m} \leq C_m \Delta$. The comparison primitive in the new coordinate is the pullback of E_ν^0 by $w = b + a\hat{w}$, with $|b| + |a - 1| \leq C\Delta$. It differs from $\hat{E} = E(\hat{c}, \hat{t})$ by $C\Delta$ in the full finite-dimensional family.

Use the compact smooth family of limiting base-flat representatives in base-parallel frames fixed at infinity. The remaining identification of frames is precisely the base-constant surface gauge k . Hence the limiting surface connections and positive maps differ in every fixed compact norm by $C_m \Delta$, while both base connection coefficients vanish. The exact small-metric caps in axial gauge have base coefficient $O(\hat{\rho}^2 \hat{T}^J)$ in rescaled coordinates, or $O(\hat{\rho} \hat{T}^J)$ physically. Their surface and map errors retain $C_m \Delta + O(\hat{\rho}^2 \hat{T}^J)$. It follows that the inner ambient norm satisfies

$$(301) \quad \|\theta_1(x' - x_{\text{cap}, \hat{E}})\|_{\text{cap}, \text{amb}} \leq C\sqrt{T}\Delta + \rho^{1/20} T^J.$$

For the $C\Delta$ term this is the fractional cutoff estimate on a smooth compact surface/map family difference; its cost is $C\varepsilon^{-1/2} R_1^\varepsilon$. The physical base conversion includes ρ^{-1} , but its numerator is $O(\rho^2 T^J)$. Thus it is included in the last term of (301). No such conversion is applied to the $C\Delta$ surface difference.

The pregluing cap need not originally use axial gauge. Apply its inverse axial gauge to both x' and y' . Its logarithm and required rescaled derivatives are $O(\rho^2 T^J)$. A radial extension to a slightly larger disc, followed by a cutoff, preserves the zero normal derivative on the straight seam rays. The resulting physical base change has H^1 norm ρT^J and H^2 norm T^J ; interpolation gives $\rho^{1-\varepsilon} T^J$ in $H^{1+\varepsilon}$. All needed surface orders satisfy the same estimates. Consequently (300) and (301) persist in the actual pregluing representative, and $x' = y'$ off the removal disc.

The primitive tail in (287) and the smooth residual attaching expansion give

$$\|\theta_1(x_{\text{cap},\widehat{E}} - x_{\text{pre}})\|_{\text{cap,amb}} \leq CT^{-40} + \rho^{1/20}T^J.$$

Indeed the strong tail bound is $CR_2^{-1+\varepsilon}$; the residual variation and based base coefficient are ρT^J . The nonlinear matching interpolation is quadratic in $R_2^{-1} + \rho R_2 + \rho^2 T^J$, with fractional factor R_2^ε . Weak rows retain their additional physical factors. On the support of $1 - \theta_1$ the old removed core is absent. There $x' = y'$ and $x_{\text{pre}} = x_0$. The residual cutoff costs $C\sqrt{T}$ in (300), giving $CT^{-77/2}$. These estimates yield

$$\|x' - x_{\text{pre}}\|_{\widehat{X}_{\widehat{\rho}}} \leq CT^{-38}.$$

Both configurations satisfy exact matching and zero normal trace, with the same ordinary outer conditions and individually subtracted exact end profiles. The inverse nonlinear chart of Proposition 7.16 applies: its quadratic constant is $C\widehat{T}^4$, and its pointwise chart condition follows from the $C\sqrt{T}$ embedding. It gives $U = \Phi_{\widehat{\rho}}^{-1}(x')$ with norm at most CT^{-38} . The inverse retains actual linear matching, normal trace, outer conditions, and finite end tangents. Thus U belongs to the full domain $X_{\widehat{\rho}}$. $\square$

Completion of the proof of Theorem 8.1. Propositions 8.2 and 8.3 supply the comparison primitive models in the two sectors. The metric, connection, core-removal, and fixed-gauge estimates prove Proposition 8.10. Lemma 8.12 solves the observations, and Proposition 8.13 puts the original solution in the full chart at those parameters. The observations are gauge invariant, so they still hold there. Its norm CT^{-38} is smaller than T^{-9} . Proposition 8.11 identifies its ordinary moduli class with the normalized constructed solution, proving the theorem. $\square$

8.11. Energy and diameter in the converse sectors.

Corollary 8.14. *Retain the hypotheses of Theorem 8.1. Suppose in addition that the normalized analytical energy measures near the attaching point satisfy*

$$\mu_\nu \rightharpoonup \mu_0 + \tfrac{1}{2}\delta_{c_*}, \quad \mu_0(\{c_*\}) = 0.$$

Pull back $\mu_\nu|_{D(c_\nu, r_\nu)}$ by $w \mapsto c_\nu + \rho_\nu w$ and extend it by zero; denote the resulting measure by ν_ν . If $E_\nu^0 \rightarrow E_$, then $\|\nu_\nu - \mu_{E_*}\|_{\text{TV}} \rightarrow 0$, where μ_{E_*} is the smooth primitive measure of mass $\frac{1}{2}$. In particular, for all retained L ,*

$$\mu_\nu(D(c_\nu, r_\nu) \setminus D(c_\nu, L\rho_\nu)) \leq CL^{-2} + o(1).$$

In the unitary reference frames used above, the surface connection and positive map on the whole retained annulus have distance at most $C_k L^{-1} + o(1)$ from the attaching pair, in every fixed surface order.

Proof. The local connection estimates include the seam and any interior unstable fiber. In rescaled coordinates the three negative curvature contributions to physical energy have weights ρ^{-2} , 1, and ρ^2 on F_{BB} , $F_{B\Sigma}$, and $F_{\Sigma\Sigma}$ respectively. The scalar ASD equation gives $F_{BB} = O(\rho^2 F_{\Sigma\Sigma})$, with bounded surface curvature on compact sets. Thus both pure contributions vanish locally, and the mixed contribution converges. Positive energy is conformally invariant. This proves local total variation convergence to the primitive measure. The flat-family Chern–Weil identity used in Theorem 7.25 gives its mass $-\frac{1}{4}p_1(\text{ad } E_*) = \frac{1}{2}$. The derivative of the tail (287) gives $\mu_{E_*}(\mathbb{C} \setminus D_L) \leq CL^{-2}$.

The additional weak energy hypothesis bounds the limiting mass in the shrinking physical disc above by $\frac{1}{2}$: first enclose it in a fixed continuity disc around c_* and then shrink that

disc. Local convergence bounds the liminf below by $\mu_{E_*}(D_L)$ for every fixed L . Consequently the total masses tend to $\frac{1}{2}$. Splitting into D_L and its complement now proves global total variation convergence and the simultaneous tail bound. For the diameter assertion, the actual surface connection in its reference unitary frame is a smooth expression in c , $h - I$, and one surface derivative of h . On the entire exterior half-annulus their estimates above give $C_k(L^{-1} + \eta_\nu) + o(1)$. The positive reference map has the same $1/w$ tail and the actual map differs from it by $o(1)$ there. This proves the assertion without selecting intermediate circles. The analytical defect was imposed separately from the algebraic charge identity. $\square$

8.12. Injectivity and transitions between regular charts. The next result concerns the constructed charts themselves. All charts use the same fixed residual, determinant and clutching conventions, and the same Hermitian metric in (210). They may use different local primitive coordinates and different sets of admissible observations. Primitive isomorphisms below retain the inherited attaching data. Each chart is restricted to a relatively compact subset of a slightly larger regular seam-admissible chart. Denote these local gluing maps by Glue_i .

Lemma 8.15 (comparison of two rescalings). *Suppose two constructed sequences represent the same ordinary moduli point at each ν :*

$$\text{Glue}_i(c_\nu, \rho_\nu, t_\nu) = \text{Glue}_j(c'_\nu, \rho'_\nu, t'_\nu), \quad \rho_\nu, \rho'_\nu \longrightarrow 0.$$

Assume their remaining parameters stay in the compact subsets just specified. Then

$$(302) \quad \frac{c'_\nu - c_\nu}{\rho_\nu} \longrightarrow 0, \quad \frac{\rho'_\nu}{\rho_\nu} \longrightarrow 1.$$

Their normalized primitive parameters approach the same bundle: any convergent subsequences have isomorphic limits with the prescribed attaching data.

Proof. Pass to a subsequence on which the centers and primitive parameters converge. The energy defect in Theorem 7.25 first identifies the two limiting physical centers. Choose a physical disc about that common center, with radius tending to zero sufficiently slowly to contain both centers and every fixed multiple of both core radii. Restrict the common analytical energy measure to this disc. In either rescaling it converges in total variation to the corresponding primitive measure of mass $\frac{1}{2}$. Indeed local convergence gives the lower mass bound, whereas the weak physical defect gives the upper bound, exactly as in Corollary 8.14.

Write $a_\nu = \rho'_\nu / \rho_\nu$ and $b_\nu = (c'_\nu - c_\nu) / \rho_\nu$. If $a_\nu \rightarrow 0$, a fixed disc in the second coordinate is a disc of radius $a_\nu L$ in the first. Choose L so that it contains, for instance, more than $\frac{3}{8}$ of the second limiting mass. Total variation convergence to the first smooth measure implies that the masses of all discs of radius $a_\nu L$ tend uniformly to zero, regardless of their centers. To see this uniformity, first discard a small tail of the limiting integrable density and then use absolute continuity of its integral on a compact set. This is a contradiction. Interchanging the two descriptions rules out $a_\nu \rightarrow \infty$. With a_ν bounded above and below, $|b_\nu| \rightarrow \infty$ would send that same positive mass outside every fixed first-coordinate disc, contrary to tightness. After a further subsequence, $a_\nu \rightarrow a > 0$ and $b_\nu \rightarrow b \in \mathbb{R}$.

The two limiting pairs are isomorphic after pullback by $g(w) = aw + b$. Here it is necessary to identify the connections, rather than just their energy densities. The rescaled smooth convergence of Theorem 7.25 holds on every fixed negative product chart, including seam segments. The gauge relating the two exact representatives is unitary. In their convergent local frames it satisfies $dh_\nu = h_\nu A'_\nu - A_\nu h_\nu$. Its values lie in a compact group, and this equation bounds every fixed derivative on compact sets. A diagonal subsequence gives a

smooth unitary isomorphism of the limiting negative connections on the whole half-plane. The local transition identities pass to the limit. In the limiting base-parallel frames the base components both vanish, so the base part of this gauge equation says that the isomorphism is independent of w . In particular it extends smoothly to the negative compactifying point. The positive maps agree after the same affine change because they came from the same solution.

The limiting negative isomorphism is holomorphic for the product Dolbeault operators and includes an unstable surface fiber. On the stable seam and positive side, the universal surface family identifies the two holomorphic projective bundles. Scalar uniqueness makes that identification agree with the negative one on their overlap. Near infinity both families are stable and extend in the fixed attaching slice. The same universal-family identification extends there, with no modification. These are the surface-slice and extension arguments used in the proof of Theorem 7.25. Thus $E'_* \simeq g^*E_*$ in the primitive family, with the common determinant and attaching conventions. The local primitive charts retain the same lift of this projective identification.

Both bundles satisfy $\operatorname{Re} \kappa = 0$ and $s = 1$. The affine transformation law immediately gives $a = 1$, $b = 0$. Every convergent subsequence of the bounded relative parameters has these limits, proving (302) and the asserted primitive identification. $\square$

Theorem 8.16 (injectivity and boundary limits of regular charts). *After reducing its upper scale bound, each regular chart Glue_i above is injective on its entire parameter product and is a smooth open embedding at positive scale. Adjoin its normalized primitive configurations at scale zero. The extended map is a homeomorphism onto its own image in the unmarked broken-convergence topology of Theorem 7.25.*

On an existing overlap of two positive-scale images the coordinate change is smooth. Along its closure at regular boundary parameters in the specified compact patches, it extends continuously by the natural identification of normalized primitive bundles, and satisfies (302). These assertions neither give coverage of a full neighborhood of the boundary stratum nor assert differentiability at scale zero.

Proof. Suppose that a single chart is noninjective for every upper scale bound. There are distinct parameter pairs with the same image and both scales tending to zero. By Lemma 8.15, their relative center and log scale tend to zero and their primitive coordinates have the same limit. They therefore lie in one of the fixed scaled boxes used in Lemma 8.12. For the common constructed solution, the observation comparison (224) and differentiation of the averaging densities give the same uniform derivative estimate for J as in that lemma. Both parameter pairs solve $J = 0$, by the definition of the normalized construction. Injectivity of J on the fixed convex box contradicts their distinctness. Hence some positive upper bound gives injectivity throughout the product. The local diffeomorphism of Theorem 7.2 then makes its positive-scale restriction an open embedding.

Continuity at zero is Theorem 7.25. For inverse continuity, consider a sequence in this image converging to one of its broken configurations, and keep the boundary parameters in a smaller compact patch. Its scales must tend to zero: a subsequence with a positive limiting scale would have a smooth ordinary limit and no half-unit atom. Any subsequential limit of the remaining parameters has the prescribed physical center by the energy defect. The rescaling comparison in the preceding lemma also applies to its given broken convergence: the two rescalings have the same full primitive mass, local smooth limits, and the same compatible attaching identifications. It therefore gives the same normalized primitive bundle. Compactness of the parameter closure now proves convergence of all parameters. At a positive-scale point the inverse is already smooth; a sequence with scales tending to zero

cannot converge there because of its atom. This proves inverse continuity on the image, with the usual sequential broken topology on these fixed compact patches.

The transition on a positive overlap is the composition of one open embedding with the inverse of the other, so it is smooth. Consider any sequence in this overlap whose first parameters tend to a regular boundary point and whose second parameters remain in the specified compact patch. A positive limiting second scale would again contradict the energy defect. Lemma 8.15 identifies every subsequential limit of its second parameters, proving the continuous boundary extension and the relative scale limits. Coordinate cocycles hold on positive overlaps by equality of the moduli points and on their regular boundary closures by continuity. This argument applies only where such overlaps accumulate; it does not prove that two image sets cover a common neighborhood of every shared boundary point. $\square$

8.13. Deriving the outer and end hypotheses. We now derive the technical exterior hypotheses of the converse from physical convergence. Retain the fixed smooth irreducible regular index-zero residual x_0 , determinant lifts, perturbations, and integrable product collar of this section. Suppose x_ν are smooth mixed solutions converging to x_0 , modulo the allowed gauges, smoothly on every compact set off one seam point c_* . The convergence includes all fixed physical and surface derivative orders. Assume that on the unperturbed matching neighborhood their normalized analytical energy measures satisfy

$$(303) \quad \mu_\nu \rightharpoonup \mu_0 + \frac{1}{2}\delta_{c_*}, \quad \mu_0(\{c_*\}) = 0,$$

where μ_0 is the smooth residual measure there. No shrinking radius, outer framing, or convergence norm on an infinite end is prescribed in these assumptions.

Proposition 8.17 (outer data from physical convergence). *Under these assumptions, the finite doubles can be taken on one fixed neighborhood of c_* . On sufficiently slowly shrinking outer circles they have coherent relative charge one, stable outer fibers, caps of vanishing area in a fixed fine-moduli chart, and the actual unitary outer representatives required by Theorem 8.1. After a subsequence their finite-family splitting is $(d_\nu, \ell_\nu) = (0, 1)$ or $(1, 0)$. The first alternative has one degree-zero Langton quotient in the open negative half-disc; it is the only alternative in genus one.*

In the first alternative, or in the second when $g = 2$ and the limiting image line is regular, extract the current primitive by the holomorphic constructions in Propositions 8.2 and 8.3. Its physical center and scale satisfy $c_\nu \rightarrow c_$ and $\rho_\nu \rightarrow 0$, with $\rho_\nu > 0$. The outer circles can then be enlarged so that all the retained-domain and exterior-annulus assumptions of Theorem 8.1 hold, including (257), without changing that comparison primitive. Regularity and seam-admissibility of its normalized limit remain separate algebraic restrictions.*

Proof. Translate the seam coordinate so $c_* = 0$. The residual double has stable fibers near zero, by openness of stability at its seam fiber. Choose a fixed disc D on which all these fibers are stable and its classifying map f_0 lies in a contractible fine-moduli coordinate chart U about $p = f_0(0)$. Lemma 4.1 applies to each actual smooth solution on this same disc. It requires no uniform width of the stable seam collar as ν varies. On every compact punctured annulus, physical convergence and the stable complex slice used above make the actual surface fibers stable for large ν . They give convergence of the classifying maps to f_0 in every fixed physical derivative order.

We first identify the relative charge of the double. On the negative half its underlying bundle is the bundle of the actual connection. On the positive half it is the pullback of the universal stable bundle. Equip the latter with the smooth family of Narasimhan–Seshadri metrics. At the seam, the projective holomorphic isomorphism is unique and is unitary: both

metrics are projectively flat, and uniqueness of the determinant-normalized stable metric identifies them. Scalar automorphisms disappear on the adjoint bundle. The projective seam transition of the double is therefore the actual unitary matching transition.

Inserting the flat cylinder in the connection-valued filling implements this same transition; collapsing its collar recovers the underlying projective bundle of the double. This is the annular bundle construction in [DFL24, Lemma 4.67, equations (4.68)–(4.69)]. For each finite ν , smoothness of the solution at the seam origin and a stable-family trivialization extend this comparison across the center. We use the branch fixed by this actual filling, without applying the small-energy branch choice in the removable-singularity argument to the concentrating sequence. The normalization of the positive contribution is [DFL26, Lemma 5.10].

The outer identification must also be retained. In the physical convergence gauge on a fixed stable annulus, the complex slice normalizations for the actual and residual surface operators differ by gauges tending to the identity. Their polar decompositions give a homotopy between the universal-family outer identification and the physical convergence identification. This is a statement about their ratio in the common convergence frame. Neither identification is assumed to extend over the original disc. In particular any large component of the convergence gauge occurs in both relative bundles. Since the residual doubled disc lies in U , it is homotopic there to the constant cap at p . Thus the capped adjoint bundle of the actual double has precisely the Pontryagin number of $P_{\nu,r}$ in Theorem 3.2. At a sufficiently small fixed regular radius, that theorem and (303) give

$$(304) \quad n(E_\nu) = -\frac{1}{2}p_1(P_{\nu,r}) = 1$$

for all sufficiently large ν .

Choose a smaller fixed outer radius r_0 within D . A diagonal choice $s_\nu \downarrow 0$ gives convergence gauges on $s_\nu \leq |z| \leq r_0$ with every prescribed finite derivative error at most s_ν^2 . Explicitly, choose radii $s_j \downarrow 0$ and thresholds N_j so convergence through order j on the corresponding fixed annulus has this bound for $\nu \geq N_j$, then set $s_\nu = s_j$ between consecutive thresholds. All actual fibers on this annulus are stable and their classifying maps lie in U .

For an outer circle of radius comparable to $r \geq Cs_\nu$ and contained in D_{Cs_ν} , its classifying loop is $O(r)$ from p and has length $O(r)$. The first assertion uses $f_0(z) - p = O(|z|)$; the second uses the uniformly bounded physical first derivative. Contract the loop to p in U . Its radial and angular derivatives are both $O(r)$ in local coordinates, so its signed normalized area satisfies

$$(305) \quad |\text{Area}(\text{cap}_{\nu,r})| \leq Cr^2.$$

The derivative estimate is needed here; small image alone does not bound the area of a rapidly winding loop.

Between two such circles surrounding D_{s_ν} , retain the actual annular restriction and the same contraction in U . The annulus with its two caps is a sphere in a contractible chart, whose universal adjoint pullback has zero Pontryagin number. Consequently (304) holds on every retained circle. The same argument allows displaced circles surrounding D_{s_ν} . There is no new Chern–Simons branch when a circle is moved.

Apply Lemma 4.2 and use Proposition 4.3. On a shrinking retained disc its stable replacement has

$$1 = d_\nu + \ell_\nu, \quad \frac{d_\nu}{2} = \text{Area}(f_\nu) + \text{Area}(\text{cap}_{\nu,r}).$$

The interior area is nonnegative, the cap area tends to zero, and d_ν is an integer. Thus $d_\nu \geq 0$ for large ν , giving the two alternatives. If $\ell_\nu = 1$, positivity of every modification

charge gives exactly one step with degree-zero quotient. Its point is negative, since every positive and seam fiber is stable, and it is inside D_{s_ν} . If $\ell_\nu = 0$, there is no unstable fiber on the retained disc. For $g = 1$, the classifying map is constant, so $d_\nu = 0$.

The holomorphic part of the current model construction can now be performed before imposing a lower bound on the retained radius. In the first alternative the stable replacement has area tending to zero and small outer image. The monotonicity argument in Proposition 8.2 makes its whole image tend to p and gives compactness of the modification subline. If a_ν is the modification point, set $c_\nu = \operatorname{Re} a_\nu$ and $\rho_\nu = -\operatorname{Im} a_\nu > 0$. Both tend to zero.

In genus two with $d_\nu = 1$, choose the projective hyperplane avoiding p used in Proposition 8.3. Its unique intersection with the classifying disc is a simple pole a_ν inside D_{s_ν} . The residue calculation and exact quadratic identities there extract its limiting image line. If that line is regular, the unique inverse extension neighborhood gives $L_\nu(z)$ and

$$q_\nu(z) = \frac{b_\nu}{z - a_\nu} + k_\nu(z).$$

Freezing the two slow parameters at a_ν and applying (210) gives the current center and scale, with $c_\nu \rightarrow 0$ and $\rho_\nu \rightarrow 0$, $\rho_\nu > 0$. These holomorphic steps use only the charge and small outer data already obtained.

Choose a sufficiently large fixed constant C_0 and put

$$(306) \quad r_\nu = C_0 \max\{s_\nu, \sqrt{\rho_\nu}, |c_\nu|\}, \quad R_\nu = r_\nu / \rho_\nu.$$

Then $r_\nu \rightarrow 0$ and $R_\nu \geq C_0 \rho_\nu^{-1/2}$. All fixed radius margins lie inside the original doubling domain for large ν . Taking C_0 large enough puts the physical annulus $r_\nu/32 \leq |z - c_\nu| \leq r_\nu/8$ outside D_{s_ν} . Its physical $C_B^1 C_\Sigma^k$ error is at most $s_\nu^2 \leq r_\nu$ for every fixed k and large ν . The smooth uniformly bounded gauge family basing the residual at c_ν preserves this bound after adjusting the diagonal accuracy by a fixed factor. The actual outer surface representatives and positive maps tend to the attaching pair, and (305) applies.

No new modification appears on enlargement, since the whole added annulus has stable fibers. In the stable genus-two case, its added image is near p , while the inner image is near the limiting line. Both lie in the same inverse extension neighborhood. Uniqueness of that inverse makes L_ν, q_ν , their pole and residue, and the frozen data agree on the overlap. In the modification case the stable replacement and its modification data agree by restriction. The current primitive and its normalization are therefore unchanged. This proves all the claimed outer and radius hypotheses. $\square$

For complete end convergence one also needs the global energy budget. In addition to (303), assume the prescribed ordinary ends agree with those of x_0 and

$$(307) \quad \mathcal{E}(x_\nu) - \mathcal{E}(x_0) \rightarrow \frac{1}{2}.$$

Proposition 3.1 and Corollary 3.3 supply this situation for the index-four primitive degeneration under consideration.

Proposition 8.18 (end convergence from exhaustion of the defect). *Under the physical hypotheses above, including (307), there are allowed gauges in which the mixed ends converge in every fixed weighted Sobolev order after subtracting each solution's exact critical representative. Their finite critical parameters converge to those of x_0 . The weight can be fixed positive below the common exponential decay rate and the residual asymptotic gaps. On the ordinary ends the corresponding convergence holds in every fixed unweighted order.*

Proof. Let K_R be a compact truncation containing the matching atom and every secondary perturbation support, with all cylindrical ends starting beyond R . Smooth convergence on the compact complement of a small matching disc gives convergence of the topological integrals there, including the compact perturbation terms. On that unperturbed disc, topological energy equals normalized analytical energy. Equation (303) therefore gives, for each sufficiently large fixed R ,

$$\mathcal{E}(x_\nu; K_R) \longrightarrow \mathcal{E}(x_0; K_R) + \frac{1}{2}.$$

Subtract this from (307), using the same restriction convention for the perturbed energy on the cylinders. The topological energy of an exact solution on a far end is nonnegative: its gauge term is the square of the perturbed gradient and its symplectic term is the holomorphic energy. Hence

$$(308) \quad \lim_{R \rightarrow \infty} \limsup_{\nu \rightarrow \infty} \sum_e \mathcal{E}(x_\nu; e_{[R, \infty)}) = 0.$$

This also bounds every tail separately. Compact secondary terms have already been handled by convergence on K_R . No unmodified curvature-square integral of a nonflat critical representative is used on an infinite cylinder.

For each finite derivative order l , choose a fixed tail start R_l so that (308) meets the topological small-energy threshold in [DFL26, Proposition 5.29] for all large ν . Its proof retains the positive decay rate from [DFL26, Proposition 5.24], while decreasing the threshold at higher orders. In temporal gauges, patched to the compact exterior gauges as in [DFL26, Section 5.4], there are exact critical mixed pairs (B_ν, q_ν) such that, on far unit strips,

$$(309) \quad \|A_\nu - B_\nu\|_{C^l(e_t)} + \|u_\nu - q_\nu\|_{C^l(e_t)} \leq C_l e^{-\delta_0(t-R_l)}.$$

All time derivatives are supplied by the same equation and unit-strip estimates. A common rate does not require a single threshold for all l .

Compactness of the critical family gives a convergent subsequence of its classes. Anchor temporal gauges at a fixed cross-section and use a smooth critical slice. For any fixed large time t , compact convergence identifies the limits of the actual sections there. The distance from each section to its own critical limit is bounded by (309). Letting first ν and then t tend to infinity shows that the limiting critical pair is (B_0, q_0) . In the chosen smooth family of exact representatives (B_p, q_p) , embeddedness of the critical Lagrangian therefore gives $p_\nu \rightarrow 0$.

Subtract these exact representatives, retaining their full dependence on p_ν . For a weight ω below δ_0 , compact convergence and the integrable exponential majorant give, in smooth end coordinates,

$$(310) \quad \begin{aligned} & \| (A_\nu - B_{p_\nu}) - (A_0 - B_0) \|_{H_\omega^l(e)} \\ & + \| (u_\nu - q_{p_\nu}) - (u_0 - q_0) \|_{H_\omega^l(e)} \longrightarrow 0. \end{aligned}$$

Indeed, split the end at a fixed large time, use smooth convergence on the finite strip, and then make the uniformly dominated tail small. This is the argument of [DFL26, equation (5.36)], now with its small-tail premise supplied by (308). Decrease the weight also below the residual asymptotic gaps.

On an ordinary end the prescribed critical point is isolated in its ordinary slice and has a fixed spectral gap. Choose a small slice neighborhood whose closure contains no other critical point. Compact convergence places a sufficiently late fixed section in a smaller neighborhood. A sufficiently small remaining energy prevents a first exit from the chosen neighborhood. Otherwise, for a sequence of remaining energies tending to zero, translate unit strips to the

first exit. Ordinary local compactness and regularity give a constant critical limit. Its central section lies on the fixed boundary of the slice neighborhood, which contains no critical point, a contradiction. This argument takes place in the local gauge quotient and gives a positive exit-energy threshold met by (308).

In this neighborhood, the inverse Hessian and quadratic action estimates are

$$\|b\|_{H^1} \leq C \|\text{grad } \mathcal{A}(b)\|_2, \quad |\mathcal{A}(b) - \mathcal{A}(0)| \leq C \|b\|_{H^1}^2.$$

The exact limit is the prescribed critical point, so the remaining energy satisfies $e'(t) \leq -ce(t)$. Uniform exponential energy decay follows. The temporal derivative solves the homogeneous linearized equation on the translation-invariant far end. Uniform elliptic estimates on unit strips, the all-order primary perturbation bounds, and integration of that derivative give decay at every fixed derivative order. Splitting into a finite strip and a tail proves the required ordinary-end convergence, where the complete domain uses weight zero. These end arguments use the estimates preceding the low-index compactness conclusion in [DFL26, Section 5.4]; they do not apply that index-at-most-one conclusion to the present sequence. $\square$

Theorem 8.19 (coverage for sequences with regular algebraic limits). *Retain the fixed residual and equation data of (C1). Assume the following:*

- (1) *The solutions converge smoothly to x_0 on every compact set off c_* , in every fixed physical and surface derivative order. The local and global energy defects are both one-half, as in (303) and (307).*
- (2) *The finite-family reduction has one degree-zero elementary modification, or $g = 2$ and the finite families are stable with regular limiting image line.*
- (3) *The normalized comparison primitives supplied by Proposition 8.17 converge to a regular seam-admissible primitive E_* .*

Every sufficiently late $[x_\nu]$ then belongs to any fixed sufficiently small regular gluing chart whose primitive patch contains E_ in its interior and whose center interval contains c_* . Its observation-corrected parameters satisfy (258), and its full chart representative has norm $O(T_\nu^{-38})$. Representation in that chart is unique modulo the allowed gauge group. The retained rescaled energy measures converge in total variation to the half-unit primitive measure, with the whole-neck energy and diameter conclusions of Corollary 8.14.*

Proof. Proposition 8.17 supplies the finite doubles, coherent charge, caps, actual outer representatives, exterior annulus, and retained radii. Proposition 8.18 supplies the complete end convergence. The algebraic limit puts the current data in a compact subset of a slightly larger regular seam-admissible coordinate patch with one observation system. All hypotheses of Theorem 8.1 now hold. That theorem gives exact representation and the parameter and norm estimates. The corrected primitive parameters tend to those of E_* , so the representation lies in the prescribed smaller patch for large ν . Theorem 8.16 gives whole-patch uniqueness, and Corollary 8.14 gives the energy and diameter conclusions. $\square$

The two finite-family alternatives can alternate along a sequence. If their normalized data tend to the same regular primitive, and the stable-family subsequences satisfy the regular-line condition, apply the theorem to each alternative. Both are eventually represented in the same chart. Thus continuity of the finite-family splitting is not assumed at negative rank drop.

This theorem supplies sequential coverage in the stated regular sectors. Neither its physical hypotheses nor the extraction of the algebraic data uses membership in a gluing image. If two

primitive coordinate patches contain the same limit, each such incoming sequence is eventually in both images. Their coordinate changes have the boundary limits in Theorem 8.16. It remains to establish which other algebraic limits can approach these boundary points. In particular, at negative rank drop a constant positive classifying map does not determine the modification subline or the limiting image line of stable finite families. Boundary data must retain the full reduced primitive bundle and its attaching identification. The theorem does not identify a topology recording only the residual and the energy atom with the required compactification.

Corollary 8.20 (very stable attaching values). *In genus two, the separate regular-image-line hypothesis in the stable-family alternative of Theorem 8.19 may be replaced by very stability of the residual attaching bundle $e_0(c_*)$. All its other hypotheses on physical convergence and the normalized primitive limit are retained. Special extension points cannot approach a compact set of very stable attaching values in either finite-family alternative.*

Proof. The projective-pole part of Proposition 8.3 does not use regularity of the limiting line. It supplies a line $\ell_* = i(N_{L_*})$ through $p = e_0(c_*)$. If L_* were special, the definition of B would give $p \in B$, contrary to Proposition 5.3. Thus $H^0(\Sigma, \Lambda L_*^{-2}) = 0$, as required for the remaining inverse and comparison argument. For a one-modification primitive limit, use its retained attaching coefficient $[e_\infty]$ over p ; it too has nonspecial subline parameter by Proposition 5.4. That proposition and compactness also give the uniform exclusion. No subline is recovered from a constant positive map in this argument. $\square$

The corollary excludes special lines under a condition on the residual value. Proposition 9.2 supplies that condition on the whole residual seam after a sufficiently small finite perturbation enlargement. The corollary alone retains the restrictions on positive rank drop and convergence of the full normalized primitive data. Theorem 8.26 below derives the normalized limit at finite centers for the chosen genus-two point cutdown and for genus-one sequences before incidence.

8.14. Positive sphere limits and the insertion. Continue with the stable degree-one families of Proposition 8.3, before restricting to a seam-admissible normalized limit. In the rotated coordinate the seam is $\text{Im } z = 0$ and the positive side has $\text{Im } z > 0$. The actual lift is

$$(311) \quad f_\nu(z) = i(L_\nu(z), [z - a_\nu : b_\nu + (z - a_\nu)k_\nu(z)]), \quad b_\nu \neq 0.$$

On retained smaller discs the pole and Cauchy estimates give

$$(312) \quad \begin{aligned} |a_\nu| &< r_\nu, & |b_\nu| &\leq C\eta_\nu r_\nu, & |L_\nu - L_*| + |k_\nu| &\leq C\eta_\nu, \\ |\partial_z^j L_\nu| + |\partial_z^j k_\nu| &\leq C_j \eta_\nu r_\nu^{-j}, & \eta_\nu &\rightarrow 0. \end{aligned}$$

The limiting line is $i(N_{L_*})$ and contains the residual attaching value $p = i(L_*, [1 : 0])$.

Proposition 8.21 (positive sphere limit in the stable-family sector). *Assume the physical half-unit concentration and the stable degree-one holomorphic hypotheses above, with $p \notin B$. Write $a_\nu = c'_\nu + id_\nu$, and suppose $d_\nu > 0$ and $|b_\nu|/d_\nu \rightarrow 0$. Then the positive maps at scale $|b_\nu|$ converge to a single degree-one sphere through p , which exhausts the concentrating energy. At scale d_ν the classifying maps converge to p away from the sphere point, including on the seam. The sphere operator and its node evaluation are regular.*

For a positive insertion z_ν^{mark} in the retained disc, a limit $[z_\nu^{\text{mark}} - a_\nu : b_\nu] \rightarrow [R : B']$ gives marked value $i(L_, [R : B'])$. The frozen holomorphic model has node equation $r\xi = b$, where $r = z - a$; in a principal coordinate normalized by the distance d to the seam its coefficient is b/d .*

Proof. Positive rank drop after (266) has $d_\nu/\rho_\nu \rightarrow 1$ and $b_\nu/\rho_\nu \rightarrow 0$, so it satisfies the stated ratio. Here $c'_\nu = \operatorname{Re} a_\nu$ differs from the normalized center by $O(|b_\nu|)$; they need not coincide. Set $z = a_\nu + b_\nu \zeta$. The actual lift becomes

$$(313) \quad f_\nu(a_\nu + b_\nu \zeta) = i(L_\nu(a_\nu + b_\nu \zeta), [\zeta : 1 + \zeta k_\nu(a_\nu + b_\nu \zeta)]) \longrightarrow F(\zeta) = i(L_*, [\zeta : 1]).$$

Every fixed compact ζ -set lies on the positive side for large ν , since $d_\nu/|b_\nu| \rightarrow \infty$, and in the retained disc since $|b_\nu|/r_\nu \rightarrow 0$. The slow bounds give convergence in every fixed C^k norm. Homogeneous coordinates include $\zeta = 0$. The limit extends to a degree-one sphere with $F(\infty) = p$ and energy $1/2$. The phase of b_ν is retained in this conformal coordinate.

At scale d_ν , set $z = c'_\nu + d_\nu w$. The affine extension coordinate is

$$\frac{b_\nu/d_\nu}{w - i} + k_\nu(c'_\nu + d_\nu w).$$

Since $d_\nu < r_\nu$, the slow bounds give C^k convergence to p on compact sets away from $w = i$, including the seam. On the negative side this concerns the classifying map, not high-order convergence of the unitary connection. For $|z| \leq 2r_\nu$ and $|z - a_\nu| \geq R|b_\nu|$, the same affine coordinate gives

$$(314) \quad d(f_\nu(z), p) \leq C(R^{-1} + \eta_\nu).$$

This proves vanishing map-image diameter between scales.

Let μ_ν be the nonnegative normalized energy measure, pushed to the base on the gauge side. The energy on $B_{R|b_\nu|}(a_\nu)$ converges to that of $F|_{|\zeta| \leq R}$, which increases to $1/2$. Positivity and the exhausted half-unit atom imply

$$(315) \quad \lim_{R \rightarrow \infty} \lim_{\delta \downarrow 0} \limsup_{\nu \rightarrow \infty} \mu_\nu(D_\delta \setminus B_{R|b_\nu|}(a_\nu)) = 0.$$

The smooth residual energy on D_δ tends to zero. Thus no second concentrating component or additional neck energy remains. The sphere and quasimap basepoint record the same charge-one degeneration.

Very stability gives $H^0(\Sigma, \Lambda L_*^{-2}) = 0$. The regular extension chart identifies a neighborhood of N_{L_*} with a neighborhood of its image. On this fiber the tangent sequence is

$$0 \longrightarrow \mathcal{O}(2) \longrightarrow TN|_{N_{L_*}} \longrightarrow T_{L_*} \operatorname{Pic}^0(\Sigma) \otimes \mathcal{O} \longrightarrow 0.$$

Its quotient is $\mathcal{O}^{\oplus 2}$, and $H^1(\mathcal{O}(2)) = 0$ splits the sequence. Consequently

$$(316) \quad F^*TM_2 \cong \mathcal{O}(2) \oplus \mathcal{O} \oplus \mathcal{O}, \quad H^1(F^*TM_2) = H^1(F^*TM_2(-\infty)) = 0.$$

These vanishings give sphere regularity and surjective evaluation of its kernel at the node. Domain automorphisms fixing the node have zero node value, so removing them preserves surjectivity. For a regular residual and a node at a fixed positive interior point, solve the residual and sphere equation rows, then correct their matching row with a sphere kernel vector. This proves regularity of that linear nodal pair, but not an inverse estimate uniform as the node approaches the seam.

Compactness of $\mathbb{P}^1$ supplies the stated subsequential limit of $[z_\nu^{\text{mark}} - a_\nu : b_\nu]$. The slow bounds in homogeneous coordinates give its marked value. For $B' \neq 0$ the sphere coordinate is R/B' ; for $B' = 0$ the value is p . In particular the path $z^{\text{mark}} = it$ has sphere coordinate

$$(317) \quad \frac{-\operatorname{Re} a_\nu + i(t_\nu - d_\nu)}{b_\nu}.$$

Both real positions must be retained.

For frozen L, k , evaluation $i(L, [r : b + kr])$ extends over $\mathrm{Bl}_{(0,0)} \mathbb{C}^2$. Its charts $r = b\zeta$ and $b = r\xi$ have evaluations

$$i(L, [\zeta : 1 + k\zeta]), \quad i(L, [1 : \xi + k]).$$

The second chart has the exact equations

$$(318) \quad r\xi = b, \quad (w - i)\xi = b/d.$$

At $b = 0$ the branches $\xi = 0$ and $r = 0$ are the principal component and sphere, with common node value $i(L, [1 : k])$. A mark colliding with the node requires a constant three-pointed component in marked stable reduction; its value remains p and it adds no energy. The ratio $\lambda = d/|b|$ specifies only the modulus $|b/d| = \lambda^{-1}$, not the complex phase. These holomorphic model identities do not identify corrected mixed-chart transitions with a nodal chart. $\square$

On the positive side DFL’s evaluated mixed bundle is $u^*\mathbb{F}_q$, where $\mathbb{F}_q$ is the universal adjoint bundle at the surface point q [DFL26, equation (7.9)]. Its pullbacks by the two evaluation charts extend the positive model over the sphere and node. Actual and frozen evaluations are uniformly close by (312); short-geodesic transport in a fixed connection gives bundle identifications on their retained domains. This does not imply that arbitrary configuration-dependent incidence sections have the required limits.

Proposition 8.22 (a symplectic representative avoiding primitive lines). *Let $\mathcal{C}$ be a compact smooth residual family of real dimension at most one with $e_x(0) \notin B$ for every $x \in \mathcal{C}$. There is an arbitrarily small change of the symplectic section pair defining the point representative such that it is independent along every primitive line through these values, at the surface point q . For this choice, no sequence satisfying the stable degree-one holomorphic hypotheses of Proposition 8.3, with atom at the interpolation crossing and residual in $\mathcal{C}$, can lie in the point incidence if its insertion is in the retained positive doubling disc.*

In particular, the positive sphere of Proposition 8.21 cannot carry the insertion. For parameterized section pairs the choice can be made relative to a prescribed parameter collar where this independence already holds.

Proof. Pull back the cover of Proposition 5.4 along $p(x) = e_x(0)$:

$$Q = \{(x, L, [v]) : i(L, [v]) = p(x)\} \longrightarrow \mathcal{C}.$$

Let $W \rightarrow Q$ have projective fiber $\mathbb{P}(\mathcal{V}_L)$. It is smooth and compact, of real dimension $\dim \mathcal{C} + 2 \leq 3$, and evaluates by

$$(319) \quad \Psi : W \longrightarrow M_2 \times \Sigma, \quad (x, L, [v]; [w]) \longmapsto (i(L, [w]), q).$$

This image can have overlaps and need not be a submanifold. Other target factors are carried at the corresponding residual values. A line can pass through B because a different subline there is special; it need not lie entirely in $M_2 \setminus B$.

At every point of W , global section variations supported near its image vary the two values through all of $\mathrm{Hom}(\mathbb{C}^2, \mathbb{C}^3)$. Finitely many suffice over compact W . Parametric Sard makes the pulled-back pair transverse to the rank-one and rank-zero strata, of real codimensions four and twelve, including on ∂W . Both inverse images are empty. For nonempty W , compactness in a fixed Hermitian metric gives

$$(320) \quad \min_{w \in W} |\sigma_1(\Psi(w)) \wedge \sigma_2(\Psi(w))| > 0.$$

For empty W the assertion is vacuous. This open condition allows sufficiently small further genericity changes and does not change the characteristic class or coorientation. For relative

data, retain the independent pair on the prescribed collar and multiply the remaining finite variations by parameter cutoffs. Independence is open near the compact fixed part; its complement projects outside the collar, where the same finite-cover argument applies.

Choose these as the symplectic sections s_i^S before choosing DFL's mixed interpolating sections. Their positive restriction is exactly $\sigma_i(u(z), q)$ [DFL26, Section 7.2, equation (7.9)]. The image extraction in Proposition 8.3 places every retained marked value at distance tending to zero from a limiting line through $p(x)$. Very stability places that line among the four fibers defining W . A subsequence of marked values therefore limits to $\Psi(W)$. The margin (320) and continuity contradict dependence of the two actual positive section values. Compactness gives the same conclusion for parameterized pairs.

A mark entering the smaller sphere scale has $|z_\nu^{\text{mark}} - a_\nu| = O(|b_\nu|)$, so it is automatically in the retained positive disc. Escaping sphere coordinates are also excluded while the insertion remains in that disc. The proof uses the whole limiting line and applies as well to the positive part of a full-rank stable-family primitive. $\square$

For a negative insertion, DFL's sections on the compact gauge part can depend on the connection on that whole part [DFL26, Section 7.2, condition (iii)]. The following convergence result controls this dependency.

Proposition 8.23 (negative connection at positive rank drop). *Retain the physical and stable finite-family hypotheses of Proposition 8.21. In the real affine normalization write $z = c_\nu + \rho_\nu w$ and $e_\nu^0(w) = (w - A_\nu)v_\nu + B_\nu e_{2,\nu}$, with $A_\nu \rightarrow i$ and $B_\nu \rightarrow 0$. The original negative connections admit representatives converging to the residual connection in $C^0(K) \cap H^1(K)$ on every fixed compact gauge region K , including its seam boundary. On the product collar convergence holds in $C_B^0 C_\Sigma^k \cap H_B^1 H_\Sigma^k$ for each fixed finite k . The local identifications are normal along the seam and join to the prescribed exterior representatives. No negative core is removed.*

Proof. For $\text{Im } A \geq 1/2$, trivialize the degree-one base line on the closed negative hemisphere by the section whose only zero is A . The extension class in this frame is

$$(321) \quad v + \frac{B}{w - A} e_2.$$

Its expression at infinity is $v + B\zeta(1 - A\zeta)^{-1} e_2$. These two charts form a smooth compact family through $B = 0$, where the negative bundle is the pullback of the stable attaching extension. Its NS boundary metric and base-flat Dirichlet metric depend smoothly on the parameters. At $B = 0$, Dirichlet uniqueness identifies this metric with the constant attaching NS metric. In the base-parallel unitary frame, normalized at infinity, its surface connection therefore satisfies

$$(322) \quad |\nabla_w^j(\beta^0 - \alpha)|_{C_\Sigma^k} \leq C_{j,k} |B| (1 + |w|)^{-1-j}, \quad j \geq 0.$$

The estimate on a bounded affine region follows from smooth parameter dependence; at infinity the difference vanishes and has a smooth Taylor expansion. The trace-free base coefficient is zero. This construction concerns only the negative hemisphere. The positive classifying map still has a basepoint at $B = 0$.

Choose the coherent enlarged radius as in Proposition 8.17, with $R_\nu \geq c\rho_\nu^{-1/2}$ and $r_\nu = \rho_\nu R_\nu \rightarrow 0$. Transfer the base-flat metric to the actual negative extension frames. The slow-parameter bounds (267) still hold: their divided difference is taken on the doubled disc, which contains the positive point A_ν . The negative frames in (321) are uniform, so the reference-curvature and NS boundary arguments of Lemma 8.5 apply unchanged on this side. In

particular the seam metric logarithm has size $C_k \eta_\nu (1 + |x|)/R_\nu$ in C_Σ^k , and the actual outer metric error tends to zero. The scalar comparison and Lemma 8.6 give smallness of the metric logarithm in every fixed surface order. They use no positive interior metric or map estimate.

For completeness, the larger-disc estimate also extends to this family. Its seam NS logarithm is $F_E(x, d)$ with $F_E(x, 0) = 0$, smooth on the compactified real line through $B = 0$. Writing $d = (x - A)a(x)$ gives the expansion (284), uniformly in this neighborhood. The concentrated remainder has H^1 norm $O(R^{-1/2})$ and H^2 norm $O(R^{1/2})$ after dilation, so interpolation gives the same $H^{3/2}$ trace bound as Lemma 8.7. The actual surface-operator error has size $C_k \eta (1 + |w|)/R$ and base derivative $C_k \eta/R$. The frozen coefficient has derivative $C_k |B|(1 + |w|)^{-2}$ by the compact negative family. Thus the coefficient and segment estimates used in (285) and (286) hold here as well.

Set $Z = w/R$. Those estimates, on fixed smaller relative half-discs, give for some $\tau_{k,\nu} \rightarrow 0$

$$\|\beta_\nu^g - \beta_\nu^0\|_{L_Z^\infty C_\Sigma^k \cap H_Z^1 H_\Sigma^k} \leq C_k \tau_{k,\nu}, \quad \|A_{Z,\nu}^g\|_{C_Z^1 C_\Sigma^k} \leq C_k r_\nu^2.$$

The axial transport is close to the base-parallel identification by the H_Z^1 segment trace estimate for its base coefficient. This does not infer transport bounds from the metric supremum alone. Combine these inequalities with (322) and the conformally invariant base Dirichlet integral. In physical coordinates, $A_z = r^{-1} A_Z$ and $d_z A_z = r^{-2} d_Z A_Z$, hence

$$(323) \quad \begin{aligned} \|\beta_\nu^g - \alpha_\nu\|_{L_z^\infty C_\Sigma^k} + \|\beta_\nu^g - \alpha_\nu\|_{H_z^1 H_\Sigma^k} &\leq C_k (|B_\nu| + \tau_{k,\nu} + r_\nu), \\ \|A_{z,\nu}^g\|_{L_z^\infty C_\Sigma^k} + \|A_{z,\nu}^g\|_{H_z^1 H_\Sigma^k} &\leq C_k r_\nu, \quad \|d_z A_{z,\nu}^g\|_{L_z^\infty C_\Sigma^k} \leq C_k. \end{aligned}$$

No positive-map comparison or core-removal step is used.

Join to the exterior gauge by the connection argument in the proof of Proposition 8.8. On its shrinking overlap both representatives have bounded surface jets, physical base coefficient $O(r)$ and bounded first physical derivative. Their transition satisfies $d_B G = O(r)$ and $d_B^2 G = O(1)$. Retain its base-constant surface factor G_0 in the inner identification. Then $GG_0^{-1} - I = O(r^2)$, and cutting off its small logarithm contributes $O(r)$ to the physical base coefficient. The transformed attaching connection converges to the residual one by comparison at an overlap anchor. The radial cutoff preserves the normal seam condition. Equation (323), the shrinking area and exterior smooth convergence prove the asserted compact C^0 and H^1 convergence. The same argument retains every fixed surface order and leaves the exterior end representatives unchanged. $\square$

This result concerns the original negative connection on the whole compact dependency region. It is not convergence of the bubbling mixed pair in its original configuration norm. Nor does it give all higher physical Sobolev norms: a profile of amplitude B_ν on scale ρ_ν has j th base derivative of L^2 size $|B_\nu| \rho_\nu^{1-j}$. For $j \geq 2$ the stated hypotheses need not make that expression small. We next choose sections continuous in the proved topology.

Proposition 8.24 (compact negative insertion intervals). *Let C be a compact smooth residual family of real dimension at most one, with irreducible restrictions to a common compact gauge region. Fix a compact interval $I \subset (-\infty, 0)$ on the negative insertion path and a surface point q . The compact gauge-side interpolating pair may be chosen so that no point-cut sequence with insertion in I has a positive-rank-drop limit as in Proposition 8.23 with residual in C . This choice preserves the prescribed symplectic pair and the far negative gauge pullback, after increasing its finite compact cutoff if necessary.*

Proof. Take a connected compact region X_c containing the path segment and all holonomy dependencies to be chosen. In the fixed-determinant matrix convention, normalize the determinant of a based loop holonomy by one fixed square root of its fixed determinant holonomy. The resulting $SU(2)$ matrix $H_\gamma(A)$ gives the gauge-equivariant vector

$$v_\gamma(A) = \frac{1}{2}(H_\gamma(A) - H_\gamma(A)^*) \in \mathfrak{su}(E_{y_0}).$$

Irreducibility supplies two loops with noncommuting holonomies. Their vectors v_1, v_2 are not parallel, so $v_1, v_2, [v_1, v_2]$ span the real adjoint fiber. Transport these vectors along fixed families of paths to points in small spatial charts, multiply by spatial cutoffs, and extend by zero. These are sections of the universal adjoint bundle over $\mathcal{B}^*(X_c) \times X_c$, defined even where their values fail to span. Compactness supplies finitely many complexified sections $E_1, \dots, E_N$ spanning at every point $([A_x|_{X_c}], y(t, q))$, $(x, t) \in C \times I$.

For two rows of complex coefficients set

$$(324) \quad s_i^a = \sum_{j=1}^N a_{ij} E_j, \quad i = 1, 2.$$

Coefficient variation spans the whole pair fiber $\text{Hom}(\mathbb{C}^2, \mathbb{C}^3)$. Parametric Sard on $C \times I$ and its boundary strata therefore makes the pair transverse to ranks one and zero. Their real codimensions are four and twelve, whereas $\dim(C \times I) \leq 2$. For generic a the pair is independent everywhere, with a positive wedge minimum if the parameter space is nonempty.

To impose this choice on the actual mixed interpolation, increase the finite cutoff T to contain I , the spatial supports and all loops in X_T . Let χ be a spatial cutoff equal to one near $y(I, q)$ and supported strictly inside the negative compact region. Replace the old pair by

$$\tilde{s}_i = \chi s_i^a + (1 - \chi) s_i^{\text{old}}.$$

This factors through $A|_{X_T}$, as required by DFL's condition (iii), and preserves the seam, positive restriction and pullback for $t \leq -T$ [DFL26, Section 7.2, equations (7.9)–(7.10)]. It is exactly (324) on a neighborhood of the specified insertion segment.

Parallel transport has, in common unitary representatives, the estimate

$$(325) \quad \|U_{A'}(\gamma) - U_A(\gamma)\| \leq C \text{length}(\gamma) \|A' - A\|_{C^0(X_T)}.$$

It follows by variation of constants; the two transport factors in its integral have norm one. This estimate is uniform for the finitely many compact path families. Matrix operations and finite sums retain the continuity, and their ODE formulas give the required differentiability at sufficiently high Sobolev regularity. Proposition 8.23 thus identifies the actual section limits with their residual values, also when $t_\nu \rightarrow t_0$ in I . Their positive wedge minimum contradicts the point incidence. $\square$

The preceding proposition chooses the pair for a specified compact negative interval. To make one choice for the full path, we retain the actual prescriptions at its two ordinary ends and its seam.

Proposition 8.25 (exclusion of the positive rank-drop end). *Let C be a compact smooth residual family of real dimension at most one, with uniform ordinary-end convergence, a common compact perturbation support, and prescribed ends in finite sets of irreducible non-degenerate gauge and transverse symplectic critical points. Assume the compact gauge restrictions and all sufficiently far incoming slices are irreducible, and the residual values at the insertion crossing are very stable. There is one choice of symplectic pair, gauge pair and mixed interpolating pair satisfying DFL's prescriptions for which no point-cut sequence*

with residual in C , the same ordinary ends, and an exhausted global half-unit defect has a positive-rank-drop limit at a finite seam center in the regular stable finite-family sector of Proposition 8.23. This holds for every sequence of insertion parameters in $\mathbb{R}$, including limits at the seam and at either infinity.

Proof. Choose the symplectic pair independent both on the compact line family W of (319) and on all residual positive insertion evaluations $(u_x(t, 0), q)$, $x \in C$, $t \geq 0$, and their critical limits. Finitely many global section variations span the pair fiber on the compact target. Apply parametric Sard to W , to $C \times [0, n]$ for each positive integer n , and to the finite critical set, including their boundary strata. Their dimensions are at most three, two and zero, below the rank-one codimension four. The countable union of exceptional parameter sets still has measure zero. Uniform ordinary-end convergence makes the completed residual evaluation image compact. Thus one pair σ^S has a positive wedge minimum on this image and on W . No differentiability at compactified infinity is required.

For large fixed T , the completed image of the incoming restrictions $[A_x(t)]$, $x \in C$, $t \leq -T$, is a compact subset of $\mathcal{B}^*(Y_\#, E_\#)$. It includes the prescribed critical limits. Use the holonomy frames in the preceding proof on $Y_\#$ to choose finitely many sections spanning the evaluated fiber on this compact image. The same countable Sard argument on $C \times [-n, -T]$ and the critical set gives a finite holonomy pair σ^G independent there, with a positive wedge minimum. It defines sections of the universal bundle on $\mathcal{B}^*(Y_\#, E_\#) \times Y_\#$ even where its constituent frames cease to span.

Near the negative seam define the pair by normal transport:

$$(326) \quad s_i^\partial(A, t, q) = P_A(y(0, q), y(t, q)) \sigma_i^S([\beta(0, 0)], q), \quad -2\epsilon \leq t \leq 0.$$

The seam connection is flat and DFL identifies its evaluated bundle with the symplectic universal bundle. This is a gauge-equivariant formula with the same wedge norm as the seam pair. It agrees with the positive prescription at zero.

We check its compact-dependency requirement on the configuration space. On the stable open set of unitary surface connections, the normalized NS metric and its canonical unitary transport define a smooth equivariant retraction $r(\beta)$ to the projectively flat connections. It is the identity on flat connections, by uniqueness of the NS metric. The surface implicit-function theorem used in the metric comparisons gives its local smoothness. Pull σ^S back by r on the same fixed Hermitian adjoint fibers. In the ordinary surface slice quotient choose a smooth cutoff equal to one near the compact flat moduli space, with support contained in the stable open set, and extend the product by zero. Finitely many small slice neighborhoods with closures in this open set give such a cutoff. This extends the seam section to arbitrary surface connections. Restriction from $A|_{X_T}$ followed by normal transport extends (326) to the compact universal bundle. Work at a sufficiently high Sobolev order for the restrictions and the finite differentiability used here. On mixed configurations the surface cutoff equals one. Thus the formula depends only on $A|_{X_T}$ and is continuous in the compact connection and seam-surface convergence of Proposition 8.23.

On the far incoming end use the exact slice pullback of σ^G . Near the seam use (326). A partition in t gives a background pair on a tube around the compact middle of the negative insertion path, equal to these prescriptions near its two ends. Both are independent on the residual family there. Add finitely many holonomy variations supported in the remaining middle, retaining smaller fixed collars. Their pair values span by the preceding proposition. Relative parametric Sard on the compact parameter space C times the middle interval, of dimension at most two, makes the pair independent everywhere; sufficiently small variations

preserve the margins on the collars. Extend from this tube by a spatial cutoff and an existing interpolating pair. The new and old boundary values agree with σ^S at the seam and σ^G on the far end, so the extension preserves both prescriptions. Each term factors through $A|_{X_T}$. This produces one allowed mixed pair, independent on the entire residual insertion path, whose negative evaluations have the proved continuity.

We next treat positive insertions approaching the atom but outside the originally retained disc. Suppose the atom is at the interpolation crossing, and write $p = u_0(0, 0)$. Proposition 8.23 and exact matching give uniform convergence of seam classes on a fixed compact seam interval. On a small fixed positive outer semicircle the original smooth convergence holds. For fixed large R , the sphere circle $|z - a_\nu| = R|b_\nu|$ has image at distance at most $C/R + o(1)$ from p , by (314). All boundary images of $\Omega_{\nu, \delta, R} = D_\delta^+ \setminus B_{R|b_\nu|}(a_\nu)$ are consequently near p . Its energy tends to zero in the iterated limit of (315). Interior target-ball monotonicity gives

$$(327) \quad \lim_{R \rightarrow \infty} \lim_{\delta \downarrow 0} \limsup_{\nu \rightarrow \infty} \sup_{z \in \Omega_{\nu, \delta, R}} d(u_\nu(z), p) = 0.$$

Indeed, a value a fixed distance ε from p would give a target ball of radius $\varepsilon/4$ missing the whole boundary image, once R is large and δ small. Its preimage component has compact closure in the domain. Monotonicity would force energy at least $c\varepsilon^2$, contradicting exhaustion. This uses the seam and both circular boundary controls, not energy smallness alone.

For a positive mark tending to zero, a bounded sphere ratio $(z_\nu - a_\nu)/b_\nu$ has its limit on the line in W . An unbounded ratio has value tending to p by (327), even outside the initial retained disc. Both are excluded by σ^S . If the atom is at a different finite seam center, a mark near the crossing lies away from the atom and smooth convergence identifies its residual value.

Finally pass to a subsequence of hypothetical cut points with $t_\nu \rightarrow t_0$ in the extended real line. A finite negative t_0 uses compact negative connection convergence and the chosen continuous formulas. A negative approach to zero uses the unchanged wedge norm in (326). Positive approaches to zero were just treated, and finite positive t_0 use smooth convergence away from the seam atom. For $t_\nu \rightarrow \pm\infty$, Proposition 8.18 gives uniform ordinary-end convergence of the actual sequence to its prescribed critical representatives. The finite holonomy pair σ^G or the smooth pair σ^S therefore has its chosen nonzero critical wedge limit. Every case contradicts incidence. $\square$

The proposition chooses the end and mixed representatives; it does not apply to arbitrary prescribed pairs. Relative fixed parameter collars require independence and compatible section limits already on those collars. The sections represent the same universal characteristic class. Their proper relative count and compatible transversality are constructed in the subsequent incidence and boundary argument. The nonconstant negative primitives do not satisfy Proposition 8.23, so their contribution is not excluded. Positive rank drop remains an unmarked degeneration. The proposition excludes it from this chosen point cutdown in the stated sector, so an ambient nodal collar is not needed for that exclusion.

8.15. Normalized compactness at finite centers.

Theorem 8.26 (finite-center coverage in genera one and two). *Fix the equation and a finite regular index-zero residual set C in a fixed energy and ordinary-end range. Let J be a compact interval of finite seam centers containing the insertion crossing. Suppose smooth mixed solutions converge to $x_0 \in C$ smoothly modulo gauge off $c_* \in J$, with the local and global half-unit defects (303) and (307), and the same ordinary ends. In genus one, their actual normalized*

primitive data have a regular seam-admissible limit after passing to a subsequence. In genus two the same conclusion holds for the point cutdown chosen in Proposition 8.25, provided the residual attaching values on $C \times J$ are very stable.

Every sufficiently late member of this subsequence is represented in each sufficiently small regular gluing chart whose primitive patch contains the limit in its interior. The representation has the estimates of Theorem 8.19 and is unique in the patch modulo the permitted gauge and real affine actions. Existence, regularity and seam admissibility of the normalized limit are conclusions here.

Proof. For a primitive pencil $e(w) = vw + w_0$ retain its nonzero attaching coefficient $[v]$ and subline L . In the affine slice (210), write $\kappa = i\tau$, $\tau \in \mathbb{R}$, and let $\xi : \mathbb{C}v \rightarrow v^\perp$ send v to $w_\perp$. Common complex scaling changes neither τ nor ξ . The slice equations are exactly

$$(328) \quad \tau^2 + \|\xi\|^2 = 1, \quad e(w) = v(w + i\tau) + \xi(v), \quad \xi \in \text{Hom}(\mathbb{C}v, v^\perp).$$

In genus two this is the unit two-sphere in a real rank-three bundle over the attaching extension space. Over a compact set of very stable values the latter is a compact four-sheeted cover by Proposition 5.4. Thus the normalized coefficient completion is compact.

If $\xi \neq 0$, the pencil has independent coefficients and no basepoint. If $\xi = 0$, its common zero is $-i\tau$, with $\tau = \pm 1$. These are precisely negative and positive rank drop. A real common zero would require both $\xi = 0$ and $\tau = 0$, contradicting the unit norm. A zero at infinity would require $v = 0$. There is no additional seam-basepoint or coefficient-escape stratum in this normalized single-pencil completion.

Proposition 8.17 gives exactly the finite-family alternatives $(d, \ell) = (0, 1)$ and $(1, 0)$. In the one-modification alternative, compactness of its degree-zero subline follows from the proof of Proposition 8.2: normalize the inclusion into the converging stable replacement, pass to a Picard limit, and apply elliptic compactness. The nonzero limiting inclusion cannot have zeros, since its saturation would have degree at least one in a stable rank-two degree-one bundle. It remains a subbundle. The actual center $\text{Re } a_\nu$ and scale $-\text{Im } a_\nu$ put the modification point at $-i$ exactly. This gives a converging negative-rank-drop primitive, retaining its full subline data. In genus one this is the only finite-family alternative, and $\dim_{\mathbb{C}} V_L = 1$ leaves no perpendicular coefficient.

In the stable genus-two alternative, the projective-pole part of Proposition 8.3 extracts a line through the residual value before assuming its regularity. Very stability makes its subline parameter nonspecial by Corollary 8.20. The regular inverse therefore supplies the actual current pencil. Equation (328) gives a convergent normalized subsequence. Its full-rank and negative-rank-drop limits are seam admissible. The only remaining limit is positive rank drop, excluded for these actual cut points by Proposition 8.25. Very stability similarly makes the subline nonspecial in the genus-two one-modification alternative.

The obtained seam-admissible limits are regular. Proposition 6.4 gives $H^2(\mathbb{P}^1 \times \Sigma, \text{End}_0 E) = 0$. By Proposition 6.17, the complex attaching-evaluation cokernel has dimension $g - 2 + h^0(\Lambda L^{-2})$. It vanishes in genus one, where the degree-one line has one section, and in genus two for the nonspecial sublines just obtained. These computations include negative rank drop. Thus primitive evaluation is onto and the fiber product with the fixed regular index-zero residual is regular.

The automatic enlargement and end results now supply all other hypotheses of Theorem 8.19. Apply that theorem to obtain the asserted exact chart representation, estimates and uniqueness. Alternating finite-family alternatives can be treated on separate subsequences; when their full limits agree, both enter the same chart. The full attaching data are retained even when the positive classifying map is constant. $\square$

The point-cut hypothesis excludes the positive pole in genus two. Section 11 treats centers escaping down mixed ends and proves compactness of the remaining cut. The relative incidence construction in Section 9.2 determines the local counts on the finite-center neighborhoods.

On the complement of the positive pole in (328), stereographic coordinates give the smooth bundle identification

$$(329) \quad \zeta = \frac{\xi}{1 + \tau}, \quad \tau = \frac{1 - \|\zeta\|^2}{1 + \|\zeta\|^2}, \quad \xi = \frac{2\zeta}{1 + \|\zeta\|^2}.$$

Thus at a fixed very stable genus-two attaching value the four seam-admissible normalized primitive branches are four planes. Their origins are negative rank drop, and their added points at infinity are positive rank drop. Over varying attaching data the planes form the complex line bundle $\text{Hom}(\mathbb{C}v, v^\perp)$ on the four-sheeted extension cover. This is a smooth identification determined by the chosen metric; it does not identify the oriented relative incidence class or a holomorphic structure on the real affine quotient.

8.16. Coverage of the point-cut ends. The local construction gives smooth parameters at every sufficiently small positive scale on compact regular seam-admissible patches. Theorem 7.25 identifies their full broken limit. Theorems 8.1 and 8.16 give converse entry, injectivity, and transition limits on the stated regular sectors. Theorem 8.19 derives the necessary outer and end hypotheses for physical sequences with a regular reduced limit.

For the point cut, the regular-limit restriction is removed by the subsequent evaluation and incidence arguments. Very stable attaching values exclude special genus-two lines. The finite-center theorem covers the genus-one and genus-two alternatives, and the point-incidence sections exclude positive sphere escape. Propositions 11.1 and 11.4 complete the exhaustion for arbitrary sequences in the chosen cut. Theorem 11.5 then uses these finitely many regular neighborhoods. An ambient differentiable collar through every uncut primitive and positive nodal configuration is not required.

9. RESIDUAL TRANSVERSALITY AND RELATIVE INCIDENCE

We choose residual evaluation and incidence data so that the primitive ends of the point cut lie in the regular neighborhoods constructed above. The relative seam calculations give local degrees eight and four. Section 10 compares their coordinate orientations with the actual mixed gluing maps. Section 11 excludes the other bubbling alternatives and proves the signed boundary identity. Section 12 fixes its coherent orientation signs and proves the integral point theorem.

9.1. Evaluation transversality. Let $\mathcal{P}$ consist of standard secondary holonomy perturbations and domain-dependent almost-complex variations, with output and dependency regions disjoint from the fixed matching neighborhood. We specify the holonomy convention. On DFL's permitted gauge region X_c , take submersions $\gamma : S^1 \times B \rightarrow X_c$ with $\gamma(1, b) = b$ and set

$$(330) \quad V_{\gamma, \omega}(A) = \pi_{\text{su}(2)}(\omega \otimes \text{Hol}_\gamma(\tilde{A})), \quad \omega \in \Omega_c^{2,+}(B; \mathbb{C}).$$

Here $\tilde{A}$ is a unitary lift with fixed determinant connection, and $\pi_{\text{su}(2)}$ is the real orthogonal projection. Real and imaginary coefficients give independent real directions. This is the convention of [Kro05, Section 3(i)]; it includes DFL's displayed real-form multiples of projected holonomy. We allow this standard enlargement explicitly. Projection commutes with gauge transformations, $|V_{\gamma, \omega}(A)| \leq C|\omega|$, and both support and dependency remain outside the

collar. Kronheimer’s interior compactness theorem [Kro05, Proposition 11], together with DFL’s unchanged matching and end analysis, gives index-zero compactness for a fixed finite family of these directions. Their coefficient norms can be made small enough to retain the nonnegative-energy bound in [DFL26, Section 6.2]. No assertion about a smaller coefficient family is needed below.

Let $\mathcal{S}$ be a stratum in an algebraic Whitney stratification for which i_g and the restrictions of ev_∞ to the full-rank and Segre strata are stratified submersions. On the universal residual moduli space consider

$$(331) \quad \text{Ev}_\gamma : \mathcal{M}_0^{\text{univ}}(\alpha, \beta) \times \mathbb{R}_s \longrightarrow M_g, \quad \text{Ev}_\gamma([A, u], \eta, s) = [A|_{(s,0) \times \Sigma_g}].$$

The matching condition gives the same value from u .

The required genericity concerns evaluation of actual residual solutions. It does not follow from the dimension of an arbitrary path in M_g . We prove the needed evaluation statement using two consecutive Sobolev realizations of the mixed operator.

At a smooth regular index-zero residual use DFL’s $l = 2$ realization of the full gauge-fixed operator, denoted by $D : \mathcal{E}^2 \rightarrow \mathcal{Y}^1$. The domain retains linear matching, zero normal trace, outer boundary conditions, and the exact end terms; the target is the unrestricted weighted H^1 sum of equation fields. DFL’s configuration and Fredholm theory allow this choice [DFL26, Definition 3.9 and Proposition 3.10]. Regularity and the gauge splitting make D invertible with the stated end convention. For a fixed seam point s , let $K = d\text{Ev}_\gamma|_{\dot{s}=0}$. This is bounded on $\mathcal{E}^2$, since H^2 point evaluation on the two-dimensional map domain is bounded. It is onto on smooth configuration variations: prescribe a harmonic matching trace locally and extend it to both sides with compact support. The connection restriction gives the same harmonic class.

Let Z_s be the evaluation surface $\{s\} \times \Sigma$ in the boundary of the gauge domain, together with its corresponding point on the boundary of the map domain. Write P for variation of the equation in all allowed smooth secondary directions. The adjoint equation for evaluation has a distributional source: an annihilator consists of $\lambda \in (\mathcal{Y}^1)^*$ and $\nu \in T_{\text{Ev}_\gamma(s)}^* M_g$ satisfying

$$(332) \quad \langle \lambda, D\xi \rangle + \nu(K\xi) = 0, \quad \langle \lambda, P\dot{\eta} \rangle = 0$$

for all domain variations and all allowed perturbation directions. We will show that λ is supported on Z_s , including possible distributions on the boundary, and then that it vanishes.

Proposition 9.1 (residual seam evaluation). *At every smooth regular index-zero residual and fixed finite seam point, the operator*

$$(333) \quad (\xi, \dot{\eta}) \longmapsto (D\xi + P\dot{\eta}, K\xi)$$

is surjective. Finitely many directions of (330) suffice at the fixed residual and seam point. In a fixed finite energy and end range and on a compact seam interval I , a finite-dimensional enlargement gives generic residual paths avoiding B in genus two on I . For $g \geq 3$ the analogous conclusion is avoidance of $i_g(N_g)$. These statements use only the index-zero compactness theorem.

Proof. We first regularize the adjoint away from Z_s . Denote the given realization by $D_2 : \mathcal{E}^2 \rightarrow \mathcal{Y}^1$ and the realization with the same matching and exact end conditions by $D_1 : \mathcal{E}^1 \rightarrow \mathcal{Y}^0$. DFL prove Fredholmness starting at $k = 1$, with regularity at all higher orders [DFL26, Theorem 4.12 and Lemma 4.34]. Finite secondary holonomy derivatives are compact $\mathcal{E}^1 \rightarrow \mathcal{Y}^0$. Indeed, at the fixed smooth connection the derivative of parallel transport has the

form

$$(334) \quad D \operatorname{Hol}_\gamma|_A(a)(b) = - \int_0^1 U_A(1, t; b) a_{\gamma(t, b)}(\partial_t \gamma) U_A(t, 0; b) dt.$$

Pullback by the submersion, smooth multiplication, and integration in t are bounded on H^j for $j = 0, 1$; compact support and Rellich give compactness into L^2 . The fixed cylinder Hessians have the same local bounds, with the longitudinal derivative retained. These are linear estimates at a smooth connection, not assertions of nonlinear smoothness on H^1 . If $D_1 \xi = 0$, the higher-order estimate gives $\xi \in \mathcal{E}^2$, so $\xi = 0$. The closed image of D_1 contains the dense subspace $\mathcal{Y}^1$ because D_2 is onto. Thus D_1 is invertible and

$$(335) \quad \|D_1^{-1} f\|_{\mathcal{E}^1} \leq C \|f\|_{\mathcal{Y}^0}, \quad D_1^{-1} f = D_2^{-1} f \quad (f \in \mathcal{Y}^1).$$

Let χ be a smooth compact cutoff zero near Z_s , constant in the surface variable at the seam and equal on its two sides. Multiplication by χ preserves the matching, normal trace, and outer conditions and removes the exact end terms. Such cutoffs can equal one near every point off Z_s ; near another seam point they are chosen on the base and include the whole surface factor. For $f \in \mathcal{Y}^1$, the first adjoint identity applied to $\chi D_2^{-1} f$ gives

$$(336) \quad \begin{aligned} (\chi \lambda)(f) &= -\lambda([D, \chi] D_2^{-1} f), \\ |(\chi \lambda)(f)| &\leq C_\chi \|\lambda\|_{(\mathcal{Y}^1)^*} \|f\|_{\mathcal{Y}^0}. \end{aligned}$$

Here $K(\chi D_2^{-1} f) = 0$. The differential commutator has order zero; the nonlocal commutators also map $\mathcal{E}^1 \rightarrow \mathcal{Y}^1$, by (334). They need not be differential operators. Equation (335) proves the bound. Density and the L^2 representation theorem now identify $\chi \lambda$ with an L^2 field, with the dual weight. This is an identity on the unrestricted H^1 target, including its boundary traces, so no additional boundary functional is omitted.

On a seam segment disjoint from Z_s , variations supported in the collar have no perturbation output: the dependency regions also avoid the collar. The weak adjoint equation there is homogeneous. The L^2 field just obtained satisfies [DFL24, Theorem 5.9(ii)], which gives H^1 regularity and the adjoint matching condition. The higher-order part of that theorem, applied to the formal adjoint, gives smoothness up to the segment. This proves the negative-order boundary regularity needed here without asserting L^2 regularity at Z_s itself.

Write $\lambda = (\phi, \beta, z)$ for its scalar gauge, self-dual, and map components, initially as functionals. For any scalar H^1 target f , put $\xi = D_2^{-1}(f, 0, 0)$. Its physical equation derivative is zero. Since the residual has zero kernel on the gauge quotient, ξ is an infinitesimal gauge variation; use the Coulomb chart and gauge splitting in [DFL26, Propositions 3.10 and 4.6]. Evaluation is gauge invariant, so $K\xi = 0$. The adjoint identity implies $\phi(f) = 0$, hence $\phi = 0$. This step uses no integration by parts at the evaluation surface.

The restriction of A to X_c is irreducible, as in [DFL26, Section 6.2]. Locally in X_c , choose finitely many based loops whose holonomies span the complex endomorphism algebra, and thicken them to submersions. Their holonomies remain a smooth spanning family on a smaller output ball. Varying the complex self-dual coefficients in (330) then produces every smooth compactly supported adjoint self-dual section on that ball. This is the local spanning argument in [Kro05, Lemma 13]. The perturbation annihilation therefore gives $\beta|_{X_c} = 0$ as a distribution.

For a gauge variation compactly supported in the interior, the fixed and secondary holonomy terms have output in X_c and are annihilated by β . The first adjoint identity reduces to

$$(337) \quad d_A^{+,*} \beta = 0 \quad \text{in } \operatorname{int} X.$$

On self-dual forms this operator has injective principal symbol; $d_A^+ d_A^{+,*}$ has scalar Laplace principal part, up to the factor $1/2$. Distributional elliptic regularity makes β smooth in the interior. Unique continuation for this scalar-principal system [Aro57, Remark 3] and its vanishing on X_c give $\beta = 0$ throughout the gauge interior. Every connected gauge component meets the permitted region. The continuation is applied after eliminating the nonlocal terms.

Choose a seam arc away from s . The smooth adjoint matching condition there gives zero trace for z , since the gauge field is zero. In the map interior $D_u^* z = 0$. In a unitary trivialization this has Cauchy–Riemann principal part. Extend z by zero across the arc and extend the smooth coefficients. Its zero trace removes the boundary distribution, so the extension solves the homogeneous equation weakly. Interior regularity and unique continuation make it zero, first near the arc and then throughout the connected map interior. This applies also when u is constant; no almost-complex variation is required. All interior components of λ vanish. Equation (336) excludes boundary functionals at every point off Z_s . Thus $\text{supp } \lambda \subset Z_s$.

Finally, a continuous functional on the unrestricted H^1 target supported on Z_s is zero. Choose a smooth cutoff χ_ϵ in half-disc coordinates transverse to Z_s , zero for $r \leq \epsilon^2$ and one for $r \geq \epsilon$, with

$$(338) \quad |\nabla \chi_\epsilon| \leq \frac{C}{r|\log \epsilon|}, \quad \int |\nabla \chi_\epsilon|^2 \leq \frac{C}{|\log \epsilon|}.$$

For every smooth compactly supported target section ψ , $\chi_\epsilon \psi \rightarrow \psi$ in H^1 . The L^2 error and the term $(1 - \chi_\epsilon) \nabla \psi$ tend to zero by the shrinking tube volume; the remaining term is controlled by (338). Integrating over Σ leaves the same estimate on the gauge side. A finite partition of unity handles the bundles, and the weights are bounded on this compact neighborhood. The functional vanishes on every $\chi_\epsilon \psi$, hence on ψ , and is zero by density.

We have proved that an annihilator of (333) has $\lambda = 0$. Then $\nu K = 0$, and surjectivity of configuration evaluation gives $\nu = 0$. There is no closed-range issue: D is invertible and the remaining factor is finite dimensional. Equivalently, the values of $KD^{-1}P$ span that factor, so finitely many directions suffice. Their solution variations are smooth by mixed elliptic regularity. Thus the same finite-dimensional derivative is obtained in the higher Sobolev realizations used to construct smooth universal moduli spaces.

For the final assertion, the regular index-zero residual moduli space in the given range is compact and discrete, hence finite [DFL26, Proposition 3.26]. Its product with I is compact. Choose a finite union of the local separating directions. Surjectivity persists for small parameters; parameterized low-index compactness excludes additional residuals outside these neighborhoods. The perturbations remain in a fixed finite smooth family and can be chosen small enough to preserve DFL’s nonnegative-energy bound. Parametric Sard applied to a Whitney stratification of B , including the endpoint restrictions on I , makes each residual path transverse to every stratum. Every stratum has real codimension at least two by Proposition 5.3, so a transverse one-dimensional path misses it. Equation (37) gives the same argument for $g \geq 3$. $\square$

We next extend avoidance to the noncompact seam. Put $L_+ = L(Y, E)$ and $L_- = L(Y', E')$, retaining the fixed primary perturbations. The seam parameter s here is denoted by θ in [DFL26]. The two mixed-end limits are

$$(339) \quad q_\pm(x) = \lim_{s \rightarrow \pm\infty} e_x(s) \in L_\pm, \quad e_x(s) = u(0, s).$$

They can vary independently in the configuration space with fixed Floer generators α, β ; see [DFL26, Proposition 3.6(iii)] and the following configuration definition]. Write $Z_2 = B$ and $Z_g = i_g(N_g)$ for $g \geq 3$. These are proper closed complex algebraic subsets of M_g .

Proposition 9.2 (avoidance on the whole residual seam). *At a smooth regular index-zero residual, let $K_\partial = d(q_+, q_-)$ be the configuration endpoint derivative. Then*

$$(340) \quad (\xi, \dot{\eta}) \longmapsto (D\xi + P\dot{\eta}, K_\partial \xi) \quad \text{in } \mathcal{Y}^1 \oplus T_{q_+} L_+ \oplus T_{q_-} L_-$$

is onto, using finitely many of the permitted holonomy directions. In a fixed finite energy and end range, an arbitrarily small finite-dimensional enlargement can be chosen so that every index-zero residual seam avoids Z_g , including its two mixed-end limits. For the resulting perturbation, the finite union of these completed seam images is a compact subset of $M_g \setminus Z_g$. This avoidance persists under sufficiently small changes in any subsequent fixed finite permitted family.

Proof. The exact critical end coefficients in DFL’s domain give bounded maps

$$K_\partial : \mathcal{E}^j \longrightarrow T_{q_+} L_+ \oplus T_{q_-} L_-, \quad j = 1, 2.$$

Their norms are explicit in [DFL26, equation (4.5)]. Boundary restriction identifies the perturbed harmonic end spaces with the tangent spaces of the embedded Lagrangians. The end charts in [DFL26, Section 4.1] vary the two representatives independently. Extend their matching variations on the respective ends and cut them off towards the compact part; the chart trivializations give the decaying corrections at the actual residual. A smooth infinitesimal gauge transformation whose scalar vanishes on the boundary removes the normal component without changing the tangential matching trace or the endpoint classes; its end correction decays since the exact harmonic representatives have zero normal trace. Thus K_∂ is onto on smooth configuration variations. It vanishes on compactly supported variations.

Suppose $\lambda \in (\mathcal{Y}^1)^*$ and $\nu \in (T_{q_+} L_+ \oplus T_{q_-} L_-)^*$ annihilate (340). The compatible lower-order inverse in (335) gives

$$(341) \quad \lambda(f) = -\nu(K_\partial D_1^{-1} f), \quad |\lambda(f)| \leq C|\nu| \|f\|_{\mathcal{Y}^0}, \quad f \in \mathcal{Y}^1.$$

Consequently the entire functional is represented by a weighted L^2 field. The weight is dual to that of $\mathcal{Y}^0$ and is equivalent to the ordinary weight on compact sets. We have not imposed a vanishing adjoint end coefficient: the endpoint source is still present in this identity.

The separation argument in Proposition 9.1 now applies without a finite evaluation set. More explicitly, write $\lambda = (\phi, \beta, z)$. For a scalar target f , $D_2^{-1}(f, 0, 0)$ has zero physical derivative, hence is infinitesimally gauge by regularity and index zero. Gauge invariance of the endpoint classes gives $\phi(f) = 0$. Holonomy annihilation gives $\beta|_{X_c} = 0$. Since all holonomy outputs lie in X_c , compact interior connection tests give $d_A^{+,*} \beta = 0$; elliptic regularity and unique continuation give $\beta = 0$ throughout the gauge interior.

Every compact domain test has zero endpoint derivative. Near a finite seam arc the adjoint equation is therefore homogeneous. Equation (341) supplies its L^2 hypothesis in [DFL24, Theorem 5.9(ii)], and higher regularity gives the smooth adjoint matching condition. The zero gauge trace forces the map adjoint trace to vanish. Zero extension across the arc and Cauchy–Riemann unique continuation give $z = 0$ in the connected map interior. This includes a constant residual map. Thus the representing L^2 field is zero almost everywhere, so $\lambda = 0$. There is no additional boundary or infinity functional on the unrestricted weighted L^2 target. Surjectivity of K_∂ gives $\nu = 0$. Invertibility of D_2 reduces the range question to the finite endpoint target; hence finitely many directions suffice. Their solution variations are smooth by mixed regularity, as in Proposition 9.1.

We use an elementary consequence of the Lagrangian condition: if L is a smooth Lagrangian in a Kähler manifold M of complex dimension n , and $Z \subset M$ is a proper closed complex analytic subset, then $L \setminus Z$ is open dense. To prove density, suppose a nonempty open subset of L lies in Z . In a coordinate neighborhood use a finite complex stratification and its closed filtration by dimension. Remove the closed lower-dimensional part whenever the open subset meets its complement; otherwise descend the filtration. This produces a nonempty open subset of L contained in a single smooth complex stratum S . Then $TL \subset TS$ is an isotropic subspace of the symplectic space $(TS, \omega|_S)$. Its dimension is n , whereas an isotropic subspace of TS has dimension at most $\dim_{\mathbb{C}} S < n$, a contradiction. Closedness gives openness. No real analyticity of L or transversality of L to Z is required. Applying this with the standard complex structure J_* gives the open dense set

$$(342) \quad (L_+ \setminus Z_g) \times (L_- \setminus Z_g).$$

Start at the given regular perturbation η_0 . The residual set in the specified range is finite. Choose the finite union of the endpoint directions just obtained. For small parameters, each residual continues uniquely by the implicit function theorem, and each of its endpoint maps is a submersion. Fixed-family index-zero compactness excludes additional residuals, as in Proposition 9.1. The inverse image of (342) under each endpoint map is open dense, since a submersion is open. The finite intersection of these sets is open dense. Choose a parameter η_1 in it, arbitrarily close to η_0 . Joint submersion for the endpoints of distinct residuals is unnecessary.

Write $\mathbb{R} = [-\infty, +\infty]$. The completed seam map $\bar{e}_x$ depends continuously on x as a map to $C^0(\mathbb{R}, M_g)$ in a high-order weighted configuration chart. Indeed, the exact end coefficients control $q_{\pm}(x)$. The weighted H^2 norm of the decaying map remainder controls its supremum on each unit half-strip, with a uniform Sobolev constant after translation. A bounded set in an end chart therefore satisfies, for $t \geq t_0$,

$$(343) \quad \sup_{s \in [t, t+1]} d(e_x(s), q_+(x)) \leq Ce^{-\delta t},$$

and the analogous negative-end estimate. The reference solution converges exponentially by [DFL26, Proposition 5.29]; smooth end chart constants are absorbed in C . Compactly supported secondary perturbations do not change the equation sufficiently far out on either end. Ordinary H^2 embedding treats the compact part.

There is consequently a common S such that the tails $|s| \geq S$ of all residuals at η_1 have positive distance from Z_g . Continuity of the completed seam maps gives configuration neighborhoods on which these same tails retain a smaller positive margin. These are neighborhoods in configuration space, chosen before the next finite family.

Apply Proposition 9.1 to $[-S, S]$ at η_1 and adjoin its finitely many directions. Shrink the parameter ball so that all continued residuals stay in the preceding neighborhoods; compactness again excludes additional residuals. Generic parameters make every residual path on $[-S, S]$ transverse to every stratum of Z_g , also at the interval endpoints. The strata have real codimension at least two, so every path avoids them. The tails already avoid Z_g . The union of the two finite families has compact support and dependency disjoint from the fixed matching collar, and its coefficients can retain the required energy bound.

For the final perturbation η_2 , the finite union of the completed seam images is compact and disjoint from the closed set Z_g . If both are nonempty, this gives

$$(344) \quad \inf_{\substack{x \in \mathcal{M}_0^{\eta_2} \\ s \in \mathbb{R}}} d(\bar{e}_x(s), Z_g) > 0.$$

The same configuration continuity, regularity and index-zero compactness preserve avoidance under sufficiently small changes in a subsequent fixed finite family. This proves the assertion. $\square$

In genus two, Proposition 5.4 now gives uniform local inverse bounds over all the completed residual seam images. Corollary 8.20 excludes special limiting lines at every finite attaching center, retaining the other hypotheses on the full normalized primitive. The separation also holds at the limiting mixed-end values. It does not give a gluing chart for a center escaping down an end.

The relative parameter problem remains distinct. Density of the complement in a Lagrangian need not permit paths between prescribed endpoints: locally, for $L = \mathbb{R}^n \subset \mathbb{C}^n$ and $Z = \{z_1 = 0\}$, a path in L changing the sign of its first coordinate must meet Z . Moreover, seam evaluation for a one-parameter residual family has a two-dimensional domain, so transversality to real codimension two does not imply avoidance. These observations limit the density argument; they assert no obstruction for a particular DFL family. When $g = 1$, the target is a point and evaluation has no role; the excess-incidence problem remains separate. Regular parameterized residual problems. The evaluation argument also applies when the linearized operator has a kernel. We keep the primary perturbations and Floer generators fixed. Let P be a finite-dimensional parameter manifold for the secondary equation data, with compact output and dependency away from the matching collar. At a solution over $p \in P$, write the full parameterized derivative as

$$(345) \quad T_j : \mathcal{V}^j = \mathcal{E}^j \oplus T_p P \longrightarrow \mathcal{Y}^{j-1}, \quad T_j(\xi, \tau) = D_j \xi + R\tau, \quad j = 1, 2.$$

The parameter columns $R\tau$ are smooth equation fields. Regularity of the parameterized problem means that T_2 is onto; it does not require D_2 to be onto.

Proposition 9.3 (evaluation for a regular parameterized problem). *Suppose T_2 in (345) is onto. Let K be joint configuration evaluation at finitely many distinct finite seam points, optionally including either or both mixed-end values. Then*

$$(346) \quad (v, \dot{\eta}) \longmapsto (T_2 v + P_x \dot{\eta}, K v)$$

is onto, using finitely many permitted additional holonomy directions. Its finite-point targets are tangent spaces to M_g and its endpoint targets are tangent spaces to $L_{\pm}$. Here P_x denotes equation variation, not the parameter manifold.

Proof. The operators T_j are finite-rank extensions of the Fredholm operators D_j . If $T_1(\xi, \tau) = 0$, mixed regularity applied to $D_1 \xi = -R\tau$ gives $\xi \in \mathcal{E}^2$, and then smoothness. This uses the linear holonomy bounds in (334) when applying the unperturbed regularity theorem. Thus $H = \ker T_1 = \ker T_2$ is finite dimensional. The closed image of T_1 contains the dense subspace $\mathcal{Y}^1$, since T_2 is onto, so T_1 is onto.

Choose continuous linear observations $\ell : \mathcal{V}^1 \rightarrow \mathbb{R}^{\dim H}$ restricting to an isomorphism on H . Compactly supported field observations together with observations of τ suffice: they separate every nonzero smooth vector, hence finitely many separate H . The operators (T_j, ℓ) are isomorphisms. Their right inverses with zero observation are

$$(347) \quad Q_j f = (T_j, \ell)^{-1}(f, 0), \quad T_j Q_j = \text{id}, \quad Q_2 f = Q_1 f \quad (f \in \mathcal{Y}^1), \quad \|Q_1 f\|_{\mathcal{V}^1} \leq C \|f\|_{\mathcal{Y}^0}.$$

This construction also gives local smooth choices with uniform bounds on compact regular families.

An annihilator (λ, ν) of (346) satisfies

$$\lambda(T_2 v) + \nu(Kv) = 0, \quad \lambda(P_x \dot{\eta}) = 0.$$

In particular νK vanishes on H . Let Z be the finite union of evaluation surfaces and map boundary points. For a compact domain-preserving cutoff χ zero near Z , put $C_\chi(\xi, \tau) = (\chi\xi, 0)$. Then $KC_\chi = 0$, including its endpoint components, and

$$(348) \quad (TC_\chi - \chi T)(\xi, \tau) = [D, \chi]\xi - \chi R\tau.$$

Both terms map $\mathcal{V}^1$ to $\mathcal{Y}^1$. Consequently

$$(\chi\lambda)(f) = -\lambda((TC_\chi - \chi T)Q_2 f), \quad |(\chi\lambda)(f)| \leq C_\chi \|\lambda\|_{(\mathcal{Y}^1)^*} \|f\|_{\mathcal{Y}^0}.$$

This is the local L^2 representation used in Proposition 9.1. If only endpoints are evaluated, K is bounded on $\mathcal{V}^1$ and $\lambda(f) = -\nu K Q_1 f$ gives the global representation directly.

The scalar gauge argument must retain the kernel. For a scalar target f , put $v = Q_2(f, 0, 0)$. Its combined physical linearization is zero. Coulomb splitting gives $v = v_C + v_G$, where v_G is infinitesimally gauge with zero parameter component and v_C has zero gauge row. Gauge equivariance gives zero physical rows for $T_2 v_G$, so $T_2 v_C = 0$. Therefore $\nu K v_C = 0$ and $K v_G = 0$. The annihilator identity gives $\lambda(f, 0, 0) = 0$. Thus the scalar component vanishes without asserting that every physical kernel vector is gauge.

The separation and continuation proof of Proposition 9.1 now applies to tests with parameter component zero. Holonomy annihilation gives $\beta|_{X_c} = 0$; removing all holonomy outputs gives $d_A^{+,*}\beta = 0$ in the interior, hence $\beta = 0$. The local L^2 adjoint theorem supplies matching regularity on an arc off Z . Zero gauge trace, zero extension and Cauchy–Riemann continuation give zero map adjoint. The parameter derivative of the almost complex structure, if present, is a column of R and does not occur in these tests. Boundary functionals off Z are excluded by the local L^2 representation. The logarithmic capacity argument (338), with finitely many cutoffs, removes the remaining source on Z . Compact smooth tests up to the boundary are dense in the weighted $\mathcal{Y}^1$ target; it has no exact critical coefficients and supports no separate functional at infinity. Thus $\lambda = 0$.

Configuration evaluation K is onto. Prescribe its finite values using disjoint compact matching variations. Prescribe the exact end coefficients independently and correct their finite values by these compact variations. Hence $\nu = 0$. Since T_2 is onto, subtracting its right inverse reduces the range question to the finite evaluation factor, including $K(H)$; its range is a finite-dimensional subspace and is closed. Trivial annihilator proves surjectivity, and finitely many added directions suffice. Higher regularity identifies the same derivatives in smooth universal moduli charts. $\square$

Corollary 9.4 (relative evaluation on a compact regular family). *Let $\mathcal{C}$ be a compact regular moduli space of parameterized residuals over a compact parameter manifold P , with boundary coming from ∂P and with regular restricted problems there. Let A be a prescribed closed collar of ∂P . Suppose all mixed-end values of $\mathcal{C}$ avoid Z_g , and seam evaluation is transverse to a complex Whitney stratification of Z_g over A , including the boundary restriction. Then arbitrarily small finite secondary perturbations depending smoothly on p , unchanged over A , give transversality on the whole seam of the continued family $\mathcal{C}_a$. The mixed-end values and both sufficiently long tails still avoid Z_g . This concerns the deformation in a neighborhood of $\mathcal{C}$.*

If $\dim \mathcal{C} = 1$, the exceptional inverse image is finite and interior for $g = 2, 3$, and empty for $g \geq 4$.

Proof. A compact regular zero set has a uniform implicit-function neighborhood. Choose a smooth complement to its tangent bundle in the configuration and parameter space. The equation derivative on that complement is an isomorphism; the implicit function theorem solves for a small normal section uniformly over $\mathcal{C}$. Thus its continued zero sets are compact manifolds $\mathcal{C}_a$ diffeomorphic to $\mathcal{C}$. They agree over A , where the equation is unchanged. The construction at ∂P uses the restricted charts and is fixed on its collar. This is a local deformation assertion, not a compactness assertion for additional components of the full moduli space.

Weighted H^2 embedding and the exact end coefficients make completed seam evaluation continuous on $\mathcal{C} \times \mathbb{R}$. Compactness and the endpoint hypothesis give a common S and configuration neighborhoods on which the tails $|s| \geq S$ stay away from Z_g . For a point over $p \notin A$ and $s \in [-S, S]$, multiply the directions in Proposition 9.3 by a smooth parameter function supported outside A and equal to one near p . The evaluation derivative there is unchanged. Transversality on the prescribed collar extends to an open neighborhood of it and persists on a smaller such neighborhood for small deformations. This uses compactness and Whitney condition (a). The remaining compact set projects outside A . A finite cover of that set supplies finitely many directions making the universal evaluation a submersion there. The fixed collar remains transverse. Whitney condition (a) and compactness preserve this simultaneous transversality near its boundary.

Parametric Sard for the finitely many strata gives the assertion on $\mathcal{C}_a \times [-S, S]$, including its boundary faces. The tails retain avoidance. All spatial output and dependency remain compact and away from the matching collar, and the parameter coefficients vanish on A .

If $\dim \mathcal{C} = 1$, the evaluation domain has dimension two. For $g = 2, 3$, the top exceptional strata have real codimension two and their inverse images have dimension zero. Lower complex strata have real codimension at least four and are missed. The fixed boundary restriction has dimension one and also misses Z_g . The closed exceptional set and tail avoidance make the inverse image compact, hence finite and interior. For $g \geq 4$ its minimal real codimension is at least four, so the whole inverse image is empty. $\square$

The endpoint hypothesis holds automatically for small continuations around the perturbation of Proposition 9.2. In any further fixed finite family, take a sufficiently small parameter ball U . The index-zero residuals are finitely many smooth branches $x_j(a)$, and their completed seam images stay away from Z_g . Smooth compact parameter families in this ball therefore satisfy the corollary; their entire residual seams already avoid Z_g . The point-interpolation parameter t in (7) varies the incidence data, not the residual equation. Its residual evaluation images are unchanged, for all t .

In genus two, pull back the degree-four cover in Proposition 5.4 along $U \times \overline{\mathbb{R}} \rightarrow M_2 \setminus B$ for each residual branch. This contractible domain gives four continuous branches after labeling one fiber, smooth at finite seam points. A smaller closed ball gives uniform local inverse bounds by compactness. This is a labeling on the specified parameter domain, not on all of $M_2 \setminus B$. No differentiability in a compactified seam coordinate or at bubble scale zero follows.

9.2. Seam sections and a relative local problem. The following local constructions use the proved finite-center comparisons and the chosen interpolating sections. Their limiting relative class retains the actual universal fiber at the seam. The subsequent determinant comparison and global boundary argument identify this local count with its contribution to the point homotopy.

Proposition 9.5 (relative seam-section model). *Fix a regular residual $x_0 = (A_0, u_0)$ in the setting of Theorem 8.26. Use the section construction of Proposition 8.25, with an independent pair at the point M_1 in genus one. Its compact-middle holonomy loops and paths may be chosen in the interior gauge region. There is a fixed compact negative insertion interval I such that every finite-center point-cut degeneration has its center at the interpolation crossing, bounded ratio $s = -c/\rho$, and insertion limit in I .*

Let P be the elliptic Jacobian in genus one, or one of the four normalized two-spheres in genus two. In a fixed attaching unitary frame put

$$(349) \quad S(E, s) = \tilde{\sigma}^S(\beta_E(s), q), \quad S_\infty = \tilde{\sigma}^S(\alpha, q),$$

where β_E is the base-flat negative Dirichlet surface connection and the tilde denotes the equivariant universal surface section. The actual section pair converges, in the residual point frame, to a pair of the form

$$(350) \quad H(E, s, t) = R(t) + b(t)(S(E, s) - S_\infty) \quad \text{on } P \times S^1 \times I.$$

Here R is the independent residual pair, $b = 0$ near the far end of I , and $b = 1$, $R = S_\infty$ near its seam end. The pair defines a relative positive- p_1 degeneracy class with compact support away from the independent boundary data. Allowed compact section variations make its model degeneracy locus an oriented compact zero-manifold. On compact regular gluing patches the convergence is uniform in C^0 after $c = -\rho s$. At a regular positive scale, local intersection numbers equal the model local degrees when oriented by these gluing coordinates.

Proof. Noncommuting loops and their connecting paths can be moved slightly into the interior, retaining the finite spanning property used in Proposition 8.24. They then have a fixed positive distance from the seam. On a tube around a compact middle interval the mixed pair is

$$a(t)\sigma^G(A_t) + b(t)P_A(0, t)\tilde{\sigma}^S(\beta(0, 0), q) + K(A, t),$$

where the first term is supported in an incoming product neighborhood and K is a finite interior holonomy sum. The two end collars have the prescribed gauge pair and the exact seam-transport pair respectively.

Choose a flat attaching representative α on a fixed Hermitian bundle V . The negative Dirichlet family is smooth on P , including negative rank drop; in genus two it also extends through the positive pole by Proposition 8.23. At that pole it is constant α . Its seam connection is smooth on the compactified real line, with value α at infinity and uniform $O((1 + |s|)^{-1})$ decay. Thus (349) is a smooth independent pair in the fixed fiber $(\text{ad } V)_q \otimes \mathbb{C}$. It equals S_∞ at seam infinity and, in genus two, on the positive-pole circle. Its independence uses the entire line in genus two and the unique class in genus one. The equivariant fiber identification is part of the formula.

For an actual sequence with compact regular primitive limit E and $-c_\nu/\rho_\nu \rightarrow s$, the inner connection comparison gives $\beta_\nu(0, 0) \rightarrow \beta_E(s)$ in every fixed surface order. Retain the joining gauge's constant surface factor and the convergent attaching representative. This compares the full point fiber; a residual stabilizer acts trivially on its adjoint. Normal transport to $t \in I$ converges to the residual transport: the physical base coefficient on the shrinking disc is $O(r)$ by (286), also after joining, so its segment integral is $O(r^2)$. On the rest of the segment use the diagonal exterior convergence and the unitary transport ODE. The fixed interior holonomies converge to their residual values. Transporting the point fiber back by $P_{A_0}(0, t)$ now gives (350). On a compact gluing patch the evaluated bundle on I is identified with the residual bundle outside the pregluing disc; the nonlinear correction is on that same bundle. This gives a continuous point-fiber identification in which the estimates are uniform

on the patch. The converse parameter estimate identifies current and observation-normalized ratios in this limit; no derivative at $\rho = 0$ is used.

If the finite atom center differs from the crossing, every dependency just listed is separated from it. The limiting pair is then the independent residual pair. If the center tends to zero but $|c_\nu|/\rho_\nu \rightarrow \infty$, enlarge the outer radius to include the crossing in a smaller relative disc. The original negative comparison in (286) and the uniform primitive tail give $\beta_\nu(0,0) \rightarrow \alpha$. The same transport and holonomy argument again gives the residual pair. The positive pole is excluded by Proposition 8.25. These arguments prove center and normalized-parameter localization. The short negative seam collar is independent by exact transport. Positive marked values near zero lie near the limiting line inside the enlarged retained disc, by the actual line estimate, or near the residual value in the diagonal exterior region. In genus one the stable replacement is uniformly constant in this limit. Finite marks away from zero use smooth convergence; insertion infinity uses the actual uniform end convergence. Thus the remaining insertion lies in the compact middle I .

On $Z = P \times S^1 \times I$, the pair H is independent near both t boundaries and on $P \times \{\infty\} \times I$; in genus two it is also independent on the positive-pole cylinder. Compactness gives a closed neighborhood A of these subsets with independent values. The rank-one degeneracy condition defines its relative class in $H^4(Z, A; \mathbb{Z})$ with DFL's coorientation.

To make the model transverse, take finitely many surface universal sections spanning the point fiber along the compact line image, or at M_1 . Extend them by the NS retraction used in (326) and transport them normally. Multiply by a t cutoff vanishing on the fixed end collars and positive on the possible degeneracy region. Two independent rows of complex coefficients span the pair fiber at every such model point. Small coefficients preserve the margins on A and the residual pair. Parametric Sard gives a transverse rank-one locus of dimension zero and no rank-zero points. These variations are allowed compact mixed sections and preserve the two end pairs. They may be chosen generic simultaneously on countably many regular positive-scale gluing charts.

Equation (350) describes the initial representative. These added terms have their corresponding seam-section limits by the same convergence argument. They give a transverse representative of the same relative class, since the whole small homotopy stays independent on A .

Finally take a small four-ball about an isolated transverse model zero, with independent pair on its boundary. Uniform C^0 convergence makes the straight boundary homotopy to the actual small-scale pair independent. At a regular scale its intersection number in the ball equals the model degree. This uses the chosen local coordinate orientation; comparison with the physical determinant orientation remains separate. $\square$

Polar orthonormalization of the seam pair, followed by its determinant-one third column, defines a map $\Gamma : P \times S^1 \rightarrow SU(3)$, constant on $P \times \{\infty\}$. In genus two it is also constant on the positive-pole circle and therefore factors through $S^2 \wedge S^1 \cong S^3$. The fibration $SU(2) \rightarrow SU(3) \rightarrow S^5$ gives $\pi_3(SU(3)) \cong \mathbb{Z}$. The usual complex generator is represented by the pair $(e_1, z_1 e_2 + z_2 e_3)$ on a four-ball, with one transverse rank-one degeneracy. Thus the relative number above is the obstruction of this specific seam frame, with DFL's coorientation sign. Equivalently, clutch trivial rank-three bundles over $P \times D^2$ by Γ to obtain $V_\Gamma \rightarrow P \times S^2$. Then

$$(351) \quad \#D_H = \pm \langle c_2(V_\Gamma), [P \times S^2] \rangle.$$

The sign records factor order, boundary orientation and DFL’s opposite c_2 convention. The far insertion boundary has a constant pair and contributes zero. This identifies an actual relative boundary obstruction; it does not assign multiplicity one to a genus-two branch merely because that branch is a plane.

Proposition 9.6 (genus-one seam obstruction). *For the genus-one relative seam model in Proposition 9.5,*

$$|\langle c_2(V_\Gamma), [J \times S^2] \rangle| = 8.$$

This is the absolute integral relative obstruction. Its sign in the physical DFL boundary formula requires the remaining orientation comparison.

Proof. Let V be the fixed stable rank-two degree-one elliptic attaching bundle and $\mathcal{P}$ the normalized Poincaré line over $\Sigma \times J$. For every $L \in J$, $H^1(VL^{-1}) = 0$ by duality and stability of the negative-slope bundle V^*L . Riemann–Roch gives $h^0(VL^{-1}) = 1$. Thus $T = \pi_*(V \otimes \mathcal{P}^{-1})$ is a line bundle, and evaluation gives the universal subbundle

$$G = \mathcal{P} \otimes \pi^*T \subset \text{pr}_\Sigma^* V.$$

It is nowhere zero: a zero of a fiber inclusion would give a positive-degree saturated subline in V . With $\int_J \theta = 1$, the usual Poincaré normalization has $\pi_*(c_1(\mathcal{P})^2) = -2\theta$. Since $\text{Td}(\Sigma) = 1$ and $\text{ch}(V) = 2 + \omega_\Sigma$, the direct Grothendieck–Riemann–Roch calculation is

$$(352) \quad \text{ch}(T) = \pi_*((2 + \omega_\Sigma)e^{-c_1(\mathcal{P})}) = 1 - 2\theta, \quad c_1(G_q) = -2\theta.$$

The restriction $\mathcal{P}_q$ is topologically trivial. In particular $L \mapsto G_{L,q} \in \mathbb{P}(V_q)$ has degree two. No torsion quotient of the retained Jacobian subline parameter is taken.

Let P_L be orthogonal projection onto G_L in the NS metric. For the normalized modification point $-i$, the actual base-flat Dirichlet identification is

$$(353) \quad S_L(w) = (I - P_L) + \frac{w + i}{w - i} P_L, \quad K_L = S_L^* H_{\text{NS}}.$$

It is nonsingular in the inverse-transform bundle frame, as in (271). On the real seam it is unitary. Hence in its base-parallel target frame $\beta_L(s) = \alpha^{S_L(s)}$, and equivariance gives the pair $\text{Ad}_{S_L(s,q)} \sigma_0$. Polar orthonormalization commutes with this unitary action, so Γ is its complexified adjoint clutching map, up to a constant change of frame.

Choose the sphere orientation so the scalar clutching factor gives $\mathcal{O}(1)$, with positive class h . The rank-two point clutching bundle is

$$U = (G_q \boxtimes \mathcal{O}(1)) \oplus (G_q^\perp \boxtimes \mathcal{O}).$$

For $g = c_1(G_q) = -2\theta$, its complement has class $-g$, and $g^2 = h^2 = 0$. Therefore

$$(354) \quad c_1(U) = h, \quad c_2(U) = -gh, \quad c_2(\text{End}_0 U) = 4c_2(U) - c_1(U)^2 = 8\theta h.$$

The bundle V_Γ is $\text{End}_0 U$ by the actual seam formula. Integration gives eight in this clutching orientation. Reversing the seam orientation reverses its sign, so the asserted absolute value is eight. The physical relation $s = -c/\rho$, DFL’s coorientation and the mixed determinant orientation have not been combined here. $\square$

Proposition 9.7 (genus-two seam obstruction). *Fix a very stable genus-two attaching value and one of its four normalized coefficient spheres P . Orient P by the complex normal extension coordinate ξ at its positive pole, and orient the base sphere by its complex coordinate w . With clutching convention $x_- = \Gamma x_+$,*

$$(355) \quad \langle c_2(V_\Gamma), [P \times S^2] \rangle = 1.$$

This convention is intrinsic on all four branches, so their total is four. In particular each branch has absolute relative obstruction one. The physical DFL orientation is not asserted to be this product orientation.

Proof. Fix the attaching subline L , put $Q = \Lambda L^{-1}$, and choose an orthonormal basis v, v_2 of $H^1(LQ^{-1})$ with v representing the attaching extension. The actual normalized family has

$$\tau^2 + |\xi|^2 = 1, \quad e_{\tau, \xi}(w) = (w + i\tau)v + \xi v_2.$$

Fixed surface cocycles for v, v_2 construct the smooth parameter family

$$(356) \quad 0 \longrightarrow \mathcal{O}_w(1) \boxtimes L \longrightarrow \mathcal{E} \longrightarrow \mathrm{pr}_\Sigma^* Q \longrightarrow 0 \quad \text{over } P \times \mathbb{P}_w^1 \times \Sigma.$$

Its coefficient at infinity is the fixed v . There is no additional line from P in this family.

Every nonzero surface extension is stable. Its only split fiber in the positive region $P \times D_+$ is $z_+ = (\tau = -1, \xi = 0, w = i)$; the other zero lies in the negative region. The positive classifying map is $[w + i\tau : \xi]$ into the fixed extension line. Near z_+ , use the coordinates $r = w + i\tau(\xi), \xi$. They preserve the chosen product orientation, and the extension class is exactly $rv + \xi v_2$. Blow up this point in these coordinates:

$$\pi : \widetilde{M} \longrightarrow M = P \times \mathbb{P}_w^1, \quad D = \pi^{-1}(z_+).$$

The normal line of D is $\mathcal{O}_D(-1)$, so $D^2 = -1$. The negative region and seam are unchanged.

The pulled-back extension splits on $D \times \Sigma$. Take its elementary transform using projection to the L summand there. In a blowup chart write $r = tR, \xi = tX$. The original upper-right surface transition coefficient is $t(Rv_{ij} + Xv_{2,ij})$. The new frame (te_L, e_Q) replaces it by $Rv_{ij} + Xv_{2,ij}$. This proves local freeness and the exact sequence

$$(357) \quad 0 \longrightarrow (\pi^* \mathcal{O}_w(1) \otimes \mathcal{O}_{\widetilde{M}}(-D)) \boxtimes L \longrightarrow \widehat{\mathcal{E}} \longrightarrow \mathrm{pr}_\Sigma^* Q \longrightarrow 0.$$

On D the class is $[R : X]$, hence is stable. The positive classifying map therefore extends over its resolved region as a map $\widehat{f}$ to the same extension line. The adjoint of this stable family is the pullback of the universal adjoint; any base-line ambiguity has disappeared.

The chosen symplectic pair is independent on the whole extension line. Its polar completion is a global complex frame of the universal adjoint point bundle there. Pull it back by $\widehat{f}$ to frame $\mathrm{End}_0 \widehat{\mathcal{E}}_q$ on the resolved positive region. This is a complex frame, not a real $SO(3)$ frame: the exceptional restriction has nontrivial real w_2 , while its complexification $\mathcal{O}(1) \oplus \mathcal{O} \oplus \mathcal{O}(-1)$ is trivial.

On the negative region the family is unchanged, and its base-flat Dirichlet metric gives the attaching point frame used in (349). On the seam the coordinates of the positive frame in the negative frame are exactly Γ . In the chosen positive frame the complex bundle descends to a trivial bundle on $P \times D_+$. Thus, with the stated clutching convention,

$$(358) \quad \mathrm{End}_0 \widehat{\mathcal{E}}_q \cong \pi^* V_\Gamma.$$

This is a complex-bundle comparison. It neither descends the real adjoint across D nor constructs an analytic positive nodal collar.

Put $h = \pi^* c_1(\mathcal{O}_w(1))$ and $d = \mathrm{PD}[D]$. At the fixed point q , the spaces L_q, Q_q are constant. Equation (357) gives $c_1(\widehat{\mathcal{E}}_q) = h - d$ and $c_2(\widehat{\mathcal{E}}_q) = 0$. Furthermore $h^2 = 0$, $hd = 0$ since the base projection is constant on D , and $\int_{\widetilde{M}} d^2 = -1$. The degree-one blowdown and (358) now give

$$\langle c_2(V_\Gamma), [P \times S^2] \rangle = \int_{\widetilde{M}} (4c_2(\widehat{\mathcal{E}}_q) - c_1(\widehat{\mathcal{E}}_q)^2) = - \int_{\widetilde{M}} (h - d)^2 = 1.$$

A change of complex extension frame preserves the positive-pole orientation. Changing the independent pair changes its completed frame by a map $S^2 \rightarrow SU(3)$, which is null-homotopic. Thus the number is independent of these choices and applies with the same convention on all four branches. $\square$

The compact section variations can also be chosen transverse on the negative-pole cylinder $\{(\tau, \xi) = (1, 0)\} \times S^1 \times I$. Its dimension is two, below the rank-one codimension four, and the same coefficient variations span. Its cutdown is therefore empty, with an independent neighborhood by compactness. For this refinement all genus-two model cut points lie in the full-rank primitive locus away from both poles. The actual section limits exclude finite-center cut sequences approaching the negative pole as well. Genus one retains its entire rank-drop Jacobian family.

The orientation in Proposition 9.7 is opposite to the complex stereographic coordinate $\zeta = \xi/(1 + \tau)$ at the negative pole. Together with $s = -c/\rho$ and DFL’s positive- p_1 coorientation, this must still be compared with the physical mixed determinant. The following proposition assembles the local degrees into an actual truncation at the covered finite-center primitive ends.

Proposition 9.8 (finite-center truncation). *Retain the fixed equation, residual and physical hypotheses of Theorem 8.26, with the transverse section system above. There are finitely many disjoint positive-scale gluing neighborhoods of the model zeros, with independent side boundaries, which exhaust the stated finite-center primitive cut degenerations. They admit common regular scale truncations. Write $n(x_0) = \pm 1$ for the sign of the residual. With outward normal pointing toward the removed smaller scales, their total artificial boundary count is*

$$(359) \quad B_{\text{finite}}(x_0) = \begin{cases} 8n(x_0)\eta_1(x_0), & g = 1, \\ n(x_0) \sum_{j=1}^4 \eta_{2,j}(x_0), & g = 2. \end{cases}$$

Here η_1 and each $\eta_{2,j}$ are constant signs on the corresponding connected primitive branch. They compare the physical DFL orientation with the ordered local parameters (P, c, ρ) , after factoring out $n(x_0)$. Their values and agreement across branches and residuals are not asserted.

Proof. Orient $P = J$ complexly in genus one and use the positive-pole orientation in genus two. With complex base orientation and clutching convention $x_- = \Gamma x_+$, put $k(P) = \langle c_2(V_\Gamma), [P \times S^2] \rangle$. Then $k(J) = 8$ and $k(P_j) = 1$. For genus one this sign is fixed by the actual inverse transform at $-i$: its determinant transition is the Blaschke factor and its base determinant is $\mathcal{O}(1)$.

Give the model the orientation $o_Z = o_P \wedge ds \wedge dt$. Cap its constant far insertion boundary inside the negative hemisphere, with t increasing toward the seam. The oriented seam boundary of that hemisphere has decreasing s , so the annular orientation is $ds \wedge dt$. The independent positive frame has transition Γ on the seam. The ordinary complex degeneracy count is therefore $k(P)$, and DFL’s positive- p_1 count is $-k(P)$.

First check orientation compatibility of local gluing charts. A smaller compact common regular primitive patch lies in both constructed images at all sufficiently small scales: otherwise a sequence of constructed solutions with its proved regular primitive limit would contradict the physical regular-sector converse. This does not use a positive nodal collar. The transition

estimates give

$$\theta' = \psi(\theta) + o(1), \quad c' - c = \rho o(1), \quad \rho'/\rho = 1 + o(1).$$

In $c = \delta b$, $\rho = \delta e^h$ they converge uniformly on compact sets to $(\psi(\theta), b, h)$ as $\delta \rightarrow 0$. This limiting map preserves $o_P \wedge db \wedge dh$. The boundary homotopy on a small four-ball shows that the smooth positive-scale transition has degree +1. Thus it preserves $o_P \wedge dc \wedge d\rho$; no derivative limit at zero is required.

On any such chart define η by

$$(360) \quad o_{\text{DFL}, \mathcal{M}} = n(x_0) \eta o_P \wedge dc \wedge d\rho.$$

Positive-scale regularity makes this sign locally constant, and the overlap argument makes it well-defined on the connected primitive branch. This gives the signs in the statement. It does not identify different branches.

Choose disjoint closed four-balls B_a around the finitely many transverse model zeros in coordinates (θ, s, t) . Their boundaries have independent pair. Define the actual neighborhoods by

$$(\theta, s, t, \rho) \mapsto (\text{Gl}_a(\theta, -\rho s, \rho), t), \quad 0 < \rho \leq \epsilon.$$

They are embeddings for positive scale. Uniform section convergence makes their side boundaries independent for small ϵ . They are mutually disjoint after shrinking that bound: a sequence of common configurations would force the same full primitive limit, scale ratio tending to one, center difference $o(\rho)$, and the same insertion limit, contradicting disjointness of the closed model balls.

Choose a common regular ϵ for the scale functions on the finitely many one-dimensional cuts. The slices are compact zero-manifolds, and their local degrees equal the model degrees by the independent boundary homotopy. Conversely every stated finite-center cut degeneration limits to a model zero by Proposition 9.5 and enters its neighborhood by the physical converse. This exhausts these degenerations, without asserting compactness of the remainder through other possible strata.

Finally, using (360) gives

$$\begin{aligned} o_{\mathcal{M} \times \mathbb{R}_t} &= n(x_0) \eta o_P \wedge dc \wedge d\rho \wedge dt \\ &= -n(x_0) \eta \rho o_P \wedge ds \wedge d\rho \wedge dt \\ &= n(x_0) \eta \rho d\rho \wedge o_Z. \end{aligned}$$

The outward normal of the remaining moduli space points toward decreasing ρ , giving $-n(x_0) \eta o_Z$. The incidence has even real codimension four, so it commutes with this boundary operation without another sign. Multiplying by the DFL model count $-k(P)$ gives $n(x_0) \eta k(P)$. Sum over the balls and branches to obtain (359). $\square$

For comparison, the finite-dimensional complex orientation of the fixed-attaching bubble fiber pulls back to $-o_J \wedge dc \wedge d\rho$ in genus one: its local coordinates are (L, κ) with $\kappa = -c + i\rho$. In genus two the physical coefficients are $\kappa = -c + i\rho\tau$, $\chi = \rho\xi$. At the negative pole, $d\tau = 0$ and the chosen o_P is opposite to the complex ξ orientation. The two reversals cancel, giving $o_P \wedge dc \wedge d\rho$. These coordinate orientations are compared with the mixed Fredholm orientation in Section 10. The truncation uses independent side boundaries and regular positive-scale slices; it requires no differentiable collar at scale zero. Section 11 supplies the compactness and regularity needed for the global boundary identity.

10. ORIENTATIONS OF THE PRIMITIVE GLUING MAPS

We first construct determinant-line excision at positive scale. The main analytic step identifies this excision with the kernel map obtained by differentiating the corrected gluing family. We then compare the compact primitive orientation with DFL’s coherent orientation, first on the selected genus-two component and then with an ordinary closed insertion. The simultaneous normalization on both matching components is established in Section 12.

10.1. Excision of the mixed determinant line. The full-Coulomb realization used here places gauge fixing in the target. Its ordinary source imposes the normal trace and canonical matching conditions, but does not impose the Coulomb equation on every input field. This distinction permits the following excision.

Proposition 10.1 (positive-scale determinant excision). *Fix the regular low-genus gluing data of Proposition 9.8, with strictly positive scale ρ and attaching value $p = e_0(c)$. Let D_{gl} be the ordinary mixed operator at the corrected solution and D_0 the residual operator, with their prescribed weighted domains. On each sufficiently small compact regular attaching patch there is a continuous comparison of orientation double covers*

$$(361) \quad \text{Or}(\det D_{\text{gl}} \otimes \det T_p M_g) \cong \text{Or}(\det D_0 \otimes \det H^1(\mathbb{P}^1 \times \Sigma_g, \text{End}_0 E)).$$

It is defined by the configuration paths and cutoff rotation below, and is unchanged under homotopies of these choices within the fixed attaching chart and retained bundle identification. It includes negative rank drop. No extension to $\rho = 0$, identification with the gluing-map derivative, or compatibility with DFL’s coherent orientation is asserted.

Proof. Write C_E for the compact reconstructed mixed operator at the positive scale and C_α for the constant compact operator at the flat attaching representative α . The compact index and regularity results give

$$\ker C_\alpha = T_p M_g, \quad \text{coker } C_\alpha = 0, \quad \ker C_E = H^1(\text{End}_0 E), \quad \text{coker } C_E = 0.$$

Multiplying the nonlinear-chart correction by a parameter in $[0, 1]$ first gives a path in the graph configuration topology. As in the relative-index argument, smooth its positive and seam data in the matching charts relative to smooth endpoint representatives, retain the exact end critical manifolds, and extend the negative connection coefficients affinely. Use this smooth configuration path to join the corrected solution to the prescribed pregluing. It need not consist of solutions. Along it the ordinary Fredholm domains vary continuously: use the bounded local domain identifications of [DFL24, Proposition 5.27], the retained asymptotic gaps, and the established low-order holonomy bounds. Orientability of the global mixed determinant line [DFL26, Section 4.7] makes its orientation transport independent of the chosen smooth configuration path with these endpoints. We do not apply ordinary operator transport directly to the rough graph path.

In the common gluing annulus use the nonlinear matching chart about α . Let z_E, z_0 denote the primitive and residual coordinate vectors. Replace the original interpolation by $az_E + bz_0$, where $a, b \geq 0$, $a + b \leq 1$, a decreases from one to zero before b increases from zero to one, and both vanish on an intervening open annulus. Convex interpolation of these coordinate formulas gives a path in the same chart. The seam equations are exactly $v = 0$ and equality of the harmonic coordinates on the two sides, so they are preserved. Interpolate base components affinely in the retained bundle frame; their zero normal component remains zero.

Simultaneously flatten the residual inside the outer transition and the compact primitive toward its attaching cap. The latter exterior remains in the chart: in $\zeta = 1/w$, the base-flat model has deviation $O(|\zeta|)$ from the attaching value, and the capped reconstruction adds the vanishing fixed-coordinate error from the pregluing estimates. Thus its surface and map coordinates are small on the whole exterior of the chosen large affine radius. The retained bundle identification is unchanged. Choose equal product metrics on the common annulus. All paths use smooth positive metrics and mixed configurations.

Denote the flattened operators by $D_{\text{gl}}^f, D_0^f, C_E^f$. Exchanging the inner mixed discs of $D_{\text{gl}}^f \oplus C_\alpha$ gives $D_0^f \oplus C_E^f$. Their differential expressions and canonical matching domains agree in the common constant annulus. The equation's perturbation supports and dependencies are outside this local unperturbed neighborhood and remain on the global exterior.

For the source rotation use

$$R(\vartheta) = \begin{pmatrix} \cos \vartheta & -\sin \vartheta \\ \sin \vartheta & \cos \vartheta \end{pmatrix},$$

where ϑ is a radial base function, zero on the outer edge and $\pi/2$ on the inner edge. Apply the same rotation to the two gauge fields and the two positive map fields. It is independent of the surface coordinate, and its two seam traces agree. For the actual linearized boundary conditions,

$$(362) \quad \iota_{\mathbf{n}}(Ra) = R\iota_{\mathbf{n}}a = 0, \quad d_\alpha(Ra_\Sigma) = 0, \quad \Pi_\alpha(Ra_\Sigma) = \kappa_\alpha(R\nu).$$

These are trace-space identities for the full mixed domain. They do not require the cutoff field to solve the gauge-fixing row. Extend the maps by the identity on the exteriors and signed interchange on the interiors. The resulting bounded invertible source and target maps U_X, U_Y preserve the global weights and finite exact end variables.

The equal differential expressions imply

$$(363) \quad U_Y(D_{\text{gl}}^f \oplus C_\alpha)U_X^{-1} = D_0^f \oplus C_E^f + K,$$

where K is a zero-order commutator supported in the common annulus, on its four- and two-dimensional components. Rellich compactness makes it compact from the ordinary H^1 mixed domain to the L^2 target. Hence $D_0^f \oplus C_E^f + tK$ is Fredholm for every $0 \leq t \leq 1$. Neither smallness nor surjectivity along this path is needed.

To transport orientations through possible kernel jumps, choose a finite-dimensional target stabilization F on a compact parameter neighborhood such that $(D_t, j_t) : X \oplus F \rightarrow Y$ is onto for every parameter. Cokernel representatives at one point, openness of surjectivity, and a finite cover provide it. Its kernels W_t form a vector bundle, and

$$0 \longrightarrow \ker D_t \longrightarrow W_t \longrightarrow F \longrightarrow \text{coker } D_t \longrightarrow 0$$

identifies $\det D_t$ with $\det W_t \otimes (\det F)^*$. This constructs the orientation double cover continuously through kernel jumps. Enlarging F adds cancelling factors; homotopies give the same endpoint orientation transport.

Apply this construction to (363) and undo the configuration and metric paths. It gives

$$\text{Or}(\det D_{\text{gl}} \otimes \det C_\alpha) \cong \text{Or}(\det D_0 \otimes \det C_E).$$

The compact kernel identifications prove (361). All factors have even real index, so reordering the displayed factors adds no graded sign. Homotopies of the radial cutoffs, small chart interpolations and positive metrics prove the asserted choice compatibility. Smooth locally free bundle frames and stable seam fibers make the same argument valid through negative rank drop. $\square$

Attaching evaluation on these regular compact kernels is onto. Its exact sequence identifies the right-hand side of (361), after cancelling $\det T_p M_g$, with $\det D_0 \otimes \det_{\mathbb{R}} \ker d\text{ev}_{\infty}$. This constructs an excision comparison for the ordinary mixed Fredholm operators. To compare it with the derivative of gluing, we control the kernels along the compact commutator path. Propositions 10.11 and 10.14 provide this control. The coherent orientation convention is specified next.

10.2. The coherent orientation normalization. DFL’s gauge orientation system includes a choice of orientation for a path p_0 from a reference generator to itself whose index is $8k + 4$. Reference paths from the other generators, cylinder composition and the standard orientations for instantons on S^4 then determine the system [DFL26, Section 2.4, pp. 21–22]. This choice affects the primitive signs even after the model clutching numbers have been fixed.

Proposition 10.2 (change of coherent orientation normalization). *Keep the reference-path orientations and standard S^4 instanton orientations fixed, and reverse only the chosen orientation of p_0 . For the actual finite-center gluing families of Proposition 9.8, the signs relative to their residuals change by*

$$(364) \quad \eta'_1(x_0) = -\eta_1(x_0), \quad \eta'_{2,j}(x_0) = -\eta_{2,j}(x_0).$$

The seam clutching numbers are unchanged. In particular, this normalization change cannot alter the ratio of two branch signs or two residual-relative signs. Changing only the reference-path orientations, with the orientation of p_0 fixed, leaves every η unchanged.

Proof. In DFL’s extension recipe, a path expression with m copies of p_0 , allowing inverse paths, changes orientation by $(-1)^m$. An S^4 insertion has index divisible by eight and retains its fixed orientation. The reference-path contributions cancel when comparing two paths with the same endpoints. If their indices differ by four modulo eight, their numbers of p_0 factors have opposite parity, so their orientation-change factors have ratio -1 .

DFL orient a mixed configuration by deforming it to a symplectically constant pair and gluing a gauge cylinder to the constant mixed solution [DFL26, Section 4.7, pp. 59–60]. The latter operator has zero kernel and cokernel and its determinant has the natural real orientation. The glued and residual mixed configurations have the same ordinary generators and indices differing by four. Their gauge cylinders in this prescription have the same index difference, because the constant mixed factor has index zero.

Let $\chi_{\text{gl}}, \chi_0 \in \{1, -1\}$ be the changes of mixed orientation at these configurations. The preceding parity calculation gives $\chi_{\text{gl}}/\chi_0 = -1$. The residual sign becomes $n'(x_0) = \chi_0 n(x_0)$. Applying this to the defining equation (360) gives $\eta' = (\chi_{\text{gl}}/\chi_0)\eta = -\eta$. The geometric bundles and their complex Chern numbers do not depend on the coherent orientation choice. Finally, if the reference-path orientations change by signs $\epsilon(\alpha)$, the composition recipe multiplies the orientation of a path from α to β by $\epsilon(\alpha)\epsilon(\beta)$. The mixed prescription has the same endpoint factor. It cancels between the glued configuration and its residual, proving the last assertion. $\square$

For a selected surface component of genus at least two, DFL fix the free choice by gluing its positive generator sphere to a constant strip and using mixed shifting [DFL26, Remark 4.76]. Their scope is explicit: if both components have nontrivial π_2 , they guarantee sphere-gluing compatibility for one chosen summand [DFL26, Remark 2.39, pp. 16–17]. Simultaneous compatibility for both components does not follow from that remark. A genus-one component has $\pi_2(M_1) = 0$, so a torus-only splitting has no generator sphere for this prescription. In

that case the orientation of p_0 , or an independently specified cylinder convention, must be retained as gauge orientation data. One cannot choose a new normalization separately for each residual, branch or marked point in a single coherent problem. Nor does fixing it identify the actual gluing orientation with the complex primitive orientation.

10.3. Comparison of the analytic kernel map with determinant-line excision. Let

$$K_E = \ker(d\text{ev}_\infty : H^1(\text{End}_0 E) \longrightarrow T_p M_g)$$

on the regular low-genus attaching patch. It has complex dimension two. Write $W_\rho(v)$ for the derivative of the actual pregluing in $v \in K_E$, keeping c, ρ fixed and using all four real fixed-attaching directions before taking the affine slice. Let $L = dF_\rho(U)$ at the observation-normalized correction, and let $\tilde{Q}$ be its geometric right inverse. Define

$$(365) \quad \mathcal{T}_\rho(v) = [D\Phi_\rho(U)(I - \tilde{Q}L)W_\rho(v)] \in \ker D_{\text{gl}}.$$

The brackets denote the ordinary full-Coulomb representative given by Lemma 7.21. The vector before reduction is a genuine geometric kernel vector, and gauge reduction preserves its observations and the exact finite end variables.

Proposition 10.3 (orientation of the analytic kernel map). *For sufficiently small positive scale on each compact regular attaching patch, $\mathcal{T}_\rho$ is an isomorphism. Let o_{K_E} be its source's complex orientation and orient P as in Proposition 9.8. Then*

$$(366) \quad d\text{Gl}(o_P \wedge dc \wedge d\rho) = \sigma_g \mathcal{T}_\rho(o_{K_E}), \quad \sigma_1 = -1, \quad \sigma_2 = +1.$$

This compares the derivative with the analytic map (365); it does not identify that map with the excision transport of Proposition 10.1.

Proof. Set $M_E = d\mathcal{O}_0(E)|_{K_E}$ and $\ell_\rho = d\mathcal{O}_{c,\rho}|_{\ker D_{\text{gl}}}$. These are isomorphisms to $\mathbb{R}^4$ by Lemma 7.19 and Corollary 7.22. The finite derivative estimate and the corrected right inverse give

$$\|(I - \tilde{Q}L)W_\rho(v) - W_\rho(v)\|_{X_\rho} \leq CT^{-31}|v|, \quad T = \log(1/\rho).$$

Restriction to the rescaled loop region costs $T^{1/2}$. The cap derivative and the connection there converge to their base-flat limits. Duhamel's derivative formula therefore gives

$$(367) \quad M_E^{-1} \ell_\rho \mathcal{T}_\rho = I + o(1)$$

uniformly on the smaller compact patch. This proves invertibility and shows that $\mathcal{T}_\rho$ has the same orientation as $\ell_\rho^{-1} M_E$: the straight homotopy of the matrix in (367) to I is invertible once its error norm is less than one. Observations are used here only on smooth finite-dimensional kernels, with their gauge-invariance already proved.

On the parameter frame

$$(\rho \partial_c, \rho \partial_\rho, \partial_{\theta_1}, \partial_{\theta_2}),$$

define B_ρ to have columns $(-T_E, -D_E, \partial_{\theta_1} E, \partial_{\theta_2} E)$ in K_E . It is an isomorphism by the affine normalization. Equation (225), retaining $\rho d\mathcal{O}_0(E) \partial_c E$, gives $\ell_\rho d\text{Gl} = M_E B_\rho + o(1)$ in this frame. Thus

$$\mathcal{T}_\rho^{-1} d\text{Gl} B_\rho^{-1} = I + o(1).$$

It is homotopic through invertible maps to the identity, so $d\text{Gl}$ and $\mathcal{T}_\rho B_\rho$ have the same orientation.

It remains to compute the sign of B_ρ . In genus one, the fixed-attaching holomorphic coordinates are (L, κ) . At $\kappa = i$, the vectors $(-T_E, -D_E)$ have κ coordinates $(-1, i)$, reversing the complex κ orientation; the Jacobian directions retain theirs. In genus two, use (κ, χ) at

negative rank drop. The same reversal occurs in the κ plane, while the chosen orientation of P is opposite to complex ξ there and $d\chi = d\xi$. These two reversals cancel. The determinant of B_ρ is nonzero on the connected regular branch, so its sign is constant. Moving the two P directions past c, ρ , or multiplying those two coordinate vectors by positive ρ , changes no sign. This proves (366). $\square$

The same observation estimate shows that two analytic kernel maps on the same actual solution have orientation-preserving transition if both have the stated limiting cap observations. This includes changes of the right inverse that preserve the estimates. It does not identify separately normalized solutions without the previously proved chart-overlap comparison.

We next specify the remaining excision sign. Concatenate the paths of Proposition 10.1, including the rotation identification, in the direction

$$(368) \quad \mathcal{D}_0 = D_0 \oplus C_E \longrightarrow \mathcal{D}_1 = D_{\text{gl}} \oplus C_\alpha.$$

Trivialize its ordinary source and target Hilbert bundles over the interval, retaining those identifications. Every operator has index $m = 4 + \dim_{\mathbb{R}} T_p M_g$, and both endpoints are surjective. Put $V = K_E \oplus T_p M_g$ with its complex orientation and choose a complex splitting j of attaching evaluation. On the endpoint kernels prescribe rows

$$(369) \quad \begin{aligned} r_0(0, h) &= (h - j \operatorname{dev}_\infty h, \operatorname{dev}_\infty h), \\ r_1(v, a) &= (\mathcal{T}_\rho^{-1} v, a). \end{aligned}$$

Extend them to bounded finite-rank maps on the ordinary sources by first projecting onto the finite-dimensional endpoint kernels, and join the extensions by a continuous family r_t . No point or raw holonomy evaluation on arbitrary H^1 fields is required. The augmented operators

$$(370) \quad \mathcal{A}_t = (\mathcal{D}_t, r_t) : X \longrightarrow Y \oplus V$$

have index zero and invertible endpoints. Write Exc for the original determinant comparison from $\det D_{\text{gl}}$ to $\det D_0 \otimes \det K_E$, after cancelling the attaching factor with its complex orientation.

For an index-zero Fredholm path with invertible endpoints, its *orientation parity* is the sign of its determinant transport relative to the canonical orientations at those endpoints.

Lemma 10.4 (criterion for the excision orientation). *Let ϵ_{exc} be the orientation parity of (370). Then the excision transport, after cancelling the $T_p M_g$ factor, satisfies*

$$(371) \quad \text{Exc}^{-1}(o_{D_0}^{\text{can}} \otimes o_{K_E}) = \epsilon_{\text{exc}} \mathcal{T}_\rho(o_{K_E}).$$

For a finite target stabilization $(\mathcal{A}_t, j_t) : X \oplus F \rightarrow Y \oplus V$ that is onto throughout, choose a continuous frame of its kernel bundle W_t and a fixed basis of F . If B_i is the matrix of the isomorphism $\pi_F : W_i \rightarrow F$ at $i = 0, 1$, then

$$(372) \quad \epsilon_{\text{exc}} = \operatorname{sgn}(\det B_0 \det B_1).$$

The sign is independent of the row extension inside the interval. In particular, invertibility of $\mathcal{A}_t$ throughout the interval would imply $\epsilon_{\text{exc}} = +1$.

Proof. Finite cokernel representatives and a finite cover provide the stated stabilization. Since $\mathcal{A}_t$ has index zero, W_t has rank $\dim F$ and has a continuous frame. The stabilization sequence identifies $\det \mathcal{A}_t = \det W_t \otimes (\det F)^*$. At an invertible endpoint π_F is an isomorphism, and its determinant sign compares this transported frame with the canonical determinant orientation.

This proves (372). A continuous change of frame has the same sign at both ends and cancels. Enlarging the stabilization adds cancelling factors.

Adding the row gives the exact sequence

$$0 \rightarrow \ker \mathcal{A}_t \rightarrow \ker \mathcal{D}_t \rightarrow V \rightarrow \operatorname{coker} \mathcal{A}_t \rightarrow \operatorname{coker} \mathcal{D}_t \rightarrow 0.$$

With the kernel-first determinant convention it identifies $\det \mathcal{D}_t$ with $\det \mathcal{A}_t \otimes \det V$ continuously through kernel jumps. At the endpoints, these are exactly the orientations prescribed by (369). The first is the complex compact-kernel orientation; the second is the analytic kernel orientation times the constant complex factor. Since $\dim V$ is even, moving its factor introduces no graded sign. This proves (371). Changing the row extension gives a homotopy with fixed invertible endpoints. If the entire augmented path is invertible, its canonical determinant orientations already give continuous transport, proving the final assertion. $\square$

Define the remaining coherent sign by

$$\operatorname{Exc}(o_{\text{DFL},\text{gl}}) = n(x_0)\epsilon_{\text{coh}} o_{D_0}^{\text{can}} \otimes o_{K_E}.$$

Equations (366) and (371) give

$$(373) \quad \eta_g = \sigma_g \epsilon_{\text{exc}} \epsilon_{\text{coh}}.$$

Formula (372) specifies the excision comparison whose sign is evaluated below. The subsequent comparison with DFL's coherent cylinder prescription determines ϵ_{coh} .

The existing geometric right inverse does not apply directly to every ordinary operator in (368). In particular, the weak-adjoint surjectivity argument used gauge covariance at an exact solution. Along a non-solution configuration path the gauge variation instead includes the equation error. The proof below controls the commutator in the complete geometric norms and constructs the augmented kernel frames along the path.

10.3.1. The Neumann gauge block along excision. The gauge part of this problem admits a uniform estimate. Let ∇_t be the derivative on negative adjoint zero-forms along the excision path. On its rotation segment this is a metric connection on the direct sum, with an off-diagonal term on the identified negative annuli. Impose covariant Neumann boundary conditions and put

$$\Delta_t = \nabla_t^* \nabla_t, \quad \|\phi\|_{\mathcal{H}_t}^2 = \|\phi\|_2^2 + \|\nabla_t \phi\|_2^2.$$

These gauge spaces have no finite critical variables.

Proposition 10.5 (uniform Neumann estimate along excision). *For the deformations in Proposition 10.1, the cap metrics of Lemma 6.11, and the fixed compact regular parameter patch, the paths may be chosen so that*

$$(374) \quad \|\nabla_t \phi\|_2^2 \geq \lambda \|\phi\|_2^2$$

with $\lambda > 0$ independent of t and sufficiently small ρ . Consequently the weak Neumann inverse $\mathcal{H}_t^ \rightarrow \mathcal{H}_t$ is uniformly bounded. One sufficiently small common positive end weight, chosen at the outset and equal to one on the compact component, has the same property. The negative gauge projection is uniformly bounded on L^2 , also with that weight. At each fixed positive scale an L^2 source has an H^2 Neumann solution. No uniform H^2 bound is asserted.*

Proof. First consider the fixed residual connection A_0 . Its limiting zero-form operators on the ends have positive gap by irreducibility. A normalized sequence whose covariant derivatives tend to zero cannot lose mass on those ends. Local Rellich compactness would therefore

produce a nonzero parallel adjoint section if the global gap vanished. Irreducibility on a seam fiber rules this out. The fixed Sobolev inequality and this gap give

$$(375) \quad \|[a, \phi]\|_2 \leq C(\|a_{\text{dec}}\|_{L^4} + \|a_{\text{fin}}\|_{L^\infty})\|d_{A_0}\phi\|_2, \quad a = a_{\text{dec}} + a_{\text{fin}}.$$

Here a_{fin} will be a smooth extension of the changes in limiting critical representatives.

In the retained base-flat negative frame, the primitive surface connection is bounded and the reconstructed compact base coefficient is $O(\delta)$, where $\delta \leq C_J \rho^2 R_1^{J+8}$ for a fixed sufficiently large J . The global insertion is confined to radius $O(\rho R_2)$. Its L^4 connection difference from A_0 is consequently $O(\rho^{1/2} T^N)$ for a fixed N ; the conversion of a compact base coefficient contributes its factor ρ^{-1} . The residual flattening has the same smallness. For the correction path, (230) and (231) give, with $M_1 \leq C R_1^2$,

$$\|a/f\|_{L^4_{\text{phys}}} \leq C \rho^{1/2} M_1 \|a/M_1\|_{H_B^1 H_\Sigma^k}, \quad \|b\|_{L^4_{\text{phys}}} \leq C \rho^{1/2} M_1^{1/2} \|b\|_{H_B^1 H_\Sigma^k}.$$

The outer weighted H^1 remainder has global L^4 norm $O(T^{-37})$. The finite exact end profiles have L^∞ norm $O(T^{-37})$ in the fixed smooth end charts. The nonlinear chart remainder satisfies the same bounds. Thus throughout the chosen graph-configuration interpolation,

$$(376) \quad \|a_{\text{dec}}\|_{L^4} + \|a_{\text{fin}}\|_{L^\infty} \leq C T^{-37} + C \rho^{1/2} T^N = o(1).$$

Only the finite profiles, rather than an arbitrary decaying H^1 remainder, are estimated in L^∞ . Choose the smooth configuration replacement sufficiently close in these coefficient norms in local unitary gauges, relative to its endpoint representatives. The form inequality is open in these norms and invariant under the gauge identifications. Equations (375) and (376) prove the uniform gap for the global configuration segments.

For the compact component put $R = R_1$ and $a = \rho R$. The negative metric $\rho^2 f_R^2 g_B$ is $a^2 h$, where h is one fixed smooth hemisphere metric after the dilation $w = Rv$. Indeed, near $\zeta = 1/w = 0$, write $f_\infty = c(\zeta)|\zeta|^{-2}$, so that $c^2 g_B = |d\zeta|^2$. With $u = R\zeta$, the explicit cap formula gives

$$(377) \quad R^{-2} f_R^2 g_B = [\chi(|u|)|u|^{-2} + 1 - \chi(|u|)]^2 |du|^2.$$

Away from this chart the metric in v is Euclidean. The expressions agree on their overlap. The fixed product Sobolev inequality, followed by the surface gap of the attaching connection α , gives

$$(378) \quad \|\phi\|_4 \leq C a^{-1/2} (\|\nabla^\alpha \phi\|_2 + \|\phi\|_2) \leq C a^{-1/2} \|\nabla^\alpha \phi\|_2.$$

The L^4 and L^2 scaling factors are $a^{1/2}$ and a ; the base derivative has its additional inverse factor a^{-1} . The displayed coarser estimate is valid for $a \leq 1$.

In the retained negative bundle frame, the limiting surface coefficient satisfies $\beta_0 - \alpha = O(|\zeta|)$ at infinity and is bounded elsewhere. Its base coefficient is zero. The reconstruction changes both compact coefficients by $O(\delta)$. Since

$$\int_{|\zeta| < r_*} f_R^2 |\zeta|^4 d\text{vol}_B \leq C, \quad \int_B f_R^2 d\text{vol}_B \leq C R^2,$$

with a bounded contribution from the remaining fixed chart, the physical connection satisfies

$$(379) \quad \|(A_E - \alpha)_\Sigma\|_4 \leq C \rho^{1/2} + C a^{1/2} \delta, \quad \|(A_E)_B\|_4 \leq C \rho^{-1/2} \delta.$$

For the base term its fourth power is $\rho^{-2} \int_B f_R^{-2} |(A_E)_B|^4$ in compact coordinates. Flattening toward α preserves these coefficient bounds: the small surface chart coordinates and base

coefficients are multiplied by bounded cutoffs. No derivative of a cutoff enters this estimate. Hölder and (378) now give

$$(380) \quad \|(\nabla^{A_E} - \nabla^\alpha)\phi\|_2 \leq Cq_\rho \|\nabla^\alpha \phi\|_2, \quad q_\rho = R^{-1/2} + \delta + \frac{\delta}{\rho\sqrt{R}} = o(1).$$

This proves the compact gap, including negative rank drop. No irreducibility of each interior surface connection is needed.

It remains to consider the rotation commutator. Let ∇_{sp} be the split flattened derivative. On the common annulus both its summands equal ∇^α , and its commutator path has derivative

$$\nabla_t = \nabla_{\text{sp}} - tJ d\vartheta, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

For negative zero-forms, the rotation extends across the paired inner negative discs: use the retained identifications with the attaching surface bundle, set $V_t = R(t\vartheta)$ on the annulus, the identity outside, and $R(t\pi/2)$ inside. The inner physical metrics agree because $R_2 \ll R_1$. Thus V_t is unitary on the direct sum of negative L^2 spaces. Its normal derivative is zero on the seam. This extension makes no assertion about an identification of the full mixed domains on the inner discs.

Conjugation cancels the derivative of ϑ exactly:

$$V_t^{-1} \nabla_t V_t = \nabla_{\text{ref}} + B_t.$$

The reference is the flattened residual on the global component and ∇^α on the compact component. Both equal ∇^α on the paired inner discs. There B_t is the orthogonal conjugate of the compact connection difference; it vanishes on the common annulus. Its exterior part is the original diagonal difference. The identical inner metrics preserve (379) for the off-diagonal coefficients. Use (378) for compact inputs and the fixed Sobolev estimate for global inputs. Since $a \leq 1$, the former constant bounds both. It follows that

$$\|B_t \phi\|_2 \leq Cq_\rho \|\nabla_{\text{ref}} \phi\|_2$$

uniformly in t . The reference gap and unitarity prove (374). In particular, no small coefficient estimate for $d\vartheta$ has been used.

Finally choose an end function h , linear on the global tails and zero on the surgery region and compact component, with zero normal derivative on the boundary. The weak form of $e^{\omega h} \Delta_t e^{-\omega h}$ is

$$b_\omega(u, v) = \langle \nabla_t u - \omega dh u, \nabla_t v + \omega dh v \rangle.$$

For real adjoint fields its diagonal is $\|\nabla_t u\|_2^2 - \omega^2 \|dh u\|_2^2$. The uniform gap gives coercivity for one sufficiently small $\omega > 0$. Lax–Milgram proves the weighted weak inverse. At each fixed positive scale, smooth-coefficient Neumann regularity gives the stated H^2 solution. For a negative ordinary one-form a with zero normal trace, the weak equation for $\phi = \Delta_t^{-1} \nabla_t^* a$ is $\langle \nabla_t \phi, \nabla_t \psi \rangle = \langle a, \nabla_t \psi \rangle$. It gives $\|\nabla_t \phi\|_2 \leq \|a\|_2$ and $\|\phi\|_2 \leq \lambda^{-1/2} \|a\|_2$. The map $a \mapsto a - \nabla_t \phi$ extends to the orthogonal projection off the closed gradient range on negative L^2 . For a general L^2 input its boundary condition is understood in this weak sense. The weighted form gives the same bound with a uniform constant in place of one. $\square$

10.3.2. Coulomb reduction of the orientation parity. Write the ordinary excision family as $\mathcal{D}_t = (L_t, C_t)$, where L_t comprises the geometric equations and $C_t = -\nabla_t^*$ is its gauge-fixing row. Let Z_t be the weighted H^2 zero-form space with covariant Neumann condition and no finite critical variables. The gradient $G_t \phi = (\nabla_t \phi, 0)$ maps Z_t into the ordinary matching domain X_t : its normal trace vanishes, and its surface trace is exact, with zero harmonic part.

The rotation term has no surface component and has zero normal component at the seam, so this remains true throughout. At fixed positive scale, Proposition 10.5 therefore supplies

$$(381) \quad H_t = (C_t G_t)^{-1} = -\Delta_t^{-1}, \quad P_t = I - G_t H_t C_t, \quad X_t^C = \ker C_t.$$

Here P_t is a bounded projection in the ordinary H^1 domain. Fixed-scale elliptic regularity suffices for this assertion; no uniform bound for this projection in the geometric graph norms is implied.

Proposition 10.6 (removal of the gauge determinant). *With the finite rows of (370), the path*

$$(382) \quad \mathcal{A}_t^C = (L_t|_{X_t^C}, r_t|_{X_t^C}) : X_t^C \longrightarrow Y_t^{\text{geo}} \oplus V$$

is Fredholm of index zero, has invertible endpoints, and satisfies

$$(383) \quad \epsilon_{\text{exc}} = \text{par}(\mathcal{A}_t^C).$$

The eliminated Neumann block contributes +1 to orientation transport.

Proof. The isomorphism $(\xi, \phi) \mapsto \xi + G_t \phi$ from $X_t^C \oplus Z_t$ to X_t has inverse $x \mapsto (P_t x, H_t C_t x)$. These are continuous at fixed positive scale; the images of P_t form a continuous Hilbert subbundle. Replace r_t by $r_t P_t$, using the homotopy $r_t[(1-s)I + sP_t]$. The projection is the identity on $\ker \mathcal{D}_t$. Thus the endpoint kernel rows stay fixed, the augmented endpoints remain invertible throughout this homotopy, and the orientation parity is unchanged.

Set $B_t = L_t G_t$, without assuming it vanishes. In the displayed source splitting, the augmented operator is

$$\begin{pmatrix} L_t|_{X_t^C} & B_t \\ 0 & C_t G_t \\ r_t|_{X_t^C} & 0 \end{pmatrix}.$$

The target shear $(f, h, v) \mapsto (f - B_t H_t h, h, v)$ is joined to the identity through invertible shears. It gives the direct sum of $\mathcal{A}_t^C$ and the invertible Neumann block $C_t G_t$. The induced kernel and cokernel identifications yield a continuous determinant-line isomorphism. The invertible block has its canonical determinant orientation at every parameter, so it cancels with parity +1. At invertible endpoints all these kernel and cokernel identifications preserve canonical orientations. This proves (383), as well as the Fredholm and endpoint assertions. The cut-and-paste maps preserve the corresponding gauge rows and Neumann gradients, so the reductions concatenate with the specified identifications in the original excision path. $\square$

The equation-error term has not disappeared from the remaining geometric problem. If Q_t is a geometric right inverse in spaces on which the following compositions are bounded, then

$$(384) \quad L_t P_t Q_t = I - E_t, \quad E_t = (L_t G_t) H_t C_t Q_t.$$

Only if $\|E_t\| < 1$ on that geometric target does the formula $P_t Q_t (I - E_t)^{-1}$ give a right inverse on the Coulomb slice. At an exact solution gauge covariance gives $L_t G_t = 0$. At a non-solution it gives the infinitesimal gauge action on the equation error. The rotation connection itself is flat on the constant annulus, but the flattened and interpolated pieces can still have such errors. We therefore estimate $H_t C_t Q_t$ in the surface-sensitive graph norms along each part of the path. The annular estimates below, followed by Propositions 10.11 and 10.14, bound the Schur error and construct continuous frames of the full kernels. These frames evaluate (383).

10.3.3. *Gauge action on the annular equation error.* The smooth flattenings and interpolation to the common constant annulus have errors supported on a bounded number of annuli of radius comparable to $r = \rho R_2$. The nonlinear correction path has an additional distributed error and is excluded from the following annular assertions. Put $a = \rho R_1$ and use the fixed compact metric h from (377). For a compact zero-form write $\phi_c = \eta_c + \psi_c$, where $\eta_c = m_h \phi_c$ is its base average in the retained negative bundle frame.

Lemma 10.7 (annular error multiplication). *Let $D \times \Sigma$ be a fixed global negative collar containing the shrinking annuli, and put*

$$h_\rho = R_2^{-1} + \rho R_2 + \frac{\delta}{\rho R_2^2}.$$

For the global annular geometric error e_g ,

$$(385) \quad \| [e_g, \phi_g] \|_{Y_\rho} \leq C r^{-\varepsilon} h_\rho \sqrt{1 + |\log r|} \|\phi_g\|_{H_B^1 H_\Sigma^k(D \times \Sigma)}.$$

For the compact error in its normalized target Y_{cap} ,

$$(386) \quad \| [e_c, \phi_c] \|_{Y_{\text{cap}}} \leq C R_2^\varepsilon h_\rho \left(\|\eta_c\|_{H_\Sigma^k} + \sqrt{1 + \log(R_1/R_2)} \|\psi_c\|_{H_h^1 H_\Sigma^k} \right).$$

The constants are uniform for $0 \leq \varepsilon \leq 1/4$. The compact average is not omitted from this estimate.

Proof. For a Hilbert-valued $u \in H^1(D)$ and a fixed rescaled annulus A_1 , extension to the plane and Fourier splitting at r^{-1} give

$$\|u(r \cdot)\|_{H^1(A_1)} \leq C \sqrt{1 + |\log r|} \|u\|_{H^1(D)}.$$

Indeed the low-frequency supremum is bounded by $C(\int_{|\xi| \leq r^{-1}} (1 + |\xi|^2)^{-1} d\xi)^{1/2} \|u\|_{H^1}$. The high-frequency L^2 norm is at most $C r \|u\|_{H^1}$, and the base gradient norm is invariant under this rescaling. This proof uses no point value of an H^1 field. For a coefficient e_r with $\|e_r(r \cdot)\|_{C_B^1 H_\Sigma^k} \leq M_r$, the surface Sobolev algebra and fixed-domain multiplication imply

$$\| [e_r, u] \|_{H_B^\varepsilon H_\Sigma^k} \leq C M_r r^{1-\varepsilon} \sqrt{1 + |\log r|} \|u\|_{H_B^1 H_\Sigma^k}.$$

A fixed rescaled cutoff supplies the extension outside the annulus and its fractional cross terms. Reflection handles a half-annulus.

For the actual error put $v = R_2^{-1}$. The interpolated surface and positive coordinates have size $O(v)$, and the compact base coefficient has size $O(\delta + \rho^2/v^3)$, with their fixed rescaled derivatives. The normalized mixed error is $O(1)$; the normalized scalar error is $O(\delta v/\rho + \rho/v^2)$. These are the same curvature cancellations used in the pregluing estimate, also when the constant configuration is one of the interpolated pairs. Conversion to physical curvature divides by ρf , where $f \asymp v^{-2}$. Thus $r \|e_g(r \cdot)\|_{C_B^1 H_\Sigma^k} \leq C h_\rho$. The global supported source norm is the local mixed Sobolev norm, proving (385).

On the compact component, the physical annulus is at radius R_2/R_1 in the h coordinate. Apply the concentration estimate to ψ_c there and retain the constant η_c separately. The normalized compact target has coordinate radius v and H^ε scale $v^{1-\varepsilon}$. The two rescaled coordinates differ by inversion on a fixed annulus. Its bounded derivatives give (386). In particular $r^{-\varepsilon}$ and R_2^ε are bounded at $\varepsilon = T^{-1}$, and $h_\rho = O(T^{-40}) + O(\rho T^N)$. $\square$

10.3.4. *The compact gauge potential and its base average.* For a compact negative variation $U^- = \rho u + b$ in the original compact coordinate, define

$$X_k(U) = \|u/M_1\|_{H_B^{1+\varepsilon} H_\Sigma^k} + \|b\|_{H_B^{1+\varepsilon} H_\Sigma^k}.$$

These are terms of the existing raw cap norm. The next estimate controls the potential rather than the full differentiated gauge variation.

Proposition 10.8 (compact Neumann potential). *Let A be the exact compact connection or one of its smooth flattenings toward the attaching constant. If U^- has zero normal trace, the solution of $\Delta_A \phi = d_A^* U^-$ satisfies*

$$(387) \quad \|m_h \phi\|_{H_\Sigma^{k+1}} \leq C_k T^{1/2} X_k(U), \quad \|\phi - m_h \phi\|_{H_h^1 H_\Sigma^k} \leq C_k \rho T^{N_k} X_k(U).$$

Here $k \geq 2$ is fixed, N_k is finite, and the average uses the retained bundle frame. The estimate includes negative rank drop.

Proof. First impose ordinary base Coulomb gauge for this estimate. The small base Neumann construction, normalized to be the identity at infinity, applies to the flattened base coefficient. Every fixed compact-coordinate bound for it and its gauge is $\hat{\delta} = \rho^2 T^N$ after increasing a fixed N . In the h coordinate,

$$d_h^* A_B = 0, \quad A_B(n) = 0, \quad e := \|A_B\|_{L_h^\infty C_\Sigma^{k+2}} \leq C R_1 \hat{\delta}, \quad e/a = o(1).$$

The two-dimensional conformal covariance of base divergence justifies this gauge condition after dilation. Writing $B_\Sigma = \beta - \alpha$, the retained decay $B_\Sigma = O(|\zeta|) + O(\hat{\delta})$ at infinity gives

$$(388) \quad \int_B \|B_\Sigma\|_{C_\Sigma^{k+2}} d \text{vol}_h \leq C(R_1^{-1} + \hat{\delta}).$$

Indeed $d \text{vol}_h = R_1^{-2} f_{R_1}^2 d \text{vol}_B$, and the integral of $f_{R_1}^2 |\zeta|$ is $O(R_1)$.

Normalize the fixed h volume in the notation for averaging. Let m be averaging, J constant extension, and $Q_b = I - Jm$. Put

$$N = \Delta_A, \quad N_0 = a^{-2} \Delta_h + N_\alpha, \quad E = N - N_0, \quad N_\alpha = d_\alpha^* d_\alpha.$$

On zero-average fields use

$$\|z\|_{\mathcal{W}_{k,a}} = \|z\|_{L_h^2 H_\Sigma^{k+1}} + a^{-1} \|z\|_{H_h^1 H_\Sigma^k}.$$

The base Poincaré inequality makes its second term equivalent to the base-gradient term. Product spectral decomposition, with nonconstant base eigenvalue μ and surface eigenvalue λ , gives

$$(389) \quad \|T_0 q\|_{\mathcal{W}_{k,a}} \leq C a \|q\|_{L_h^2 H_\Sigma^k}, \quad T_0 = (N_0|_{\ker m})^{-1}.$$

The inverse multiplier is $(a^{-2} \mu + \lambda)^{-1}$ and μ has a positive lower bound. The same calculation gives

$$\|T_0 Q_b(a^{-2} d_h^* v + d_\alpha^* b)\|_{\mathcal{W}_{k,a}} \leq C(a^{-1} \|v\|_{L_h^2 H_\Sigma^k} + \|b\|_{L_h^2 H_\Sigma^k}).$$

Flat surface Hodge intertwining proves this estimate without an additional derivative of b .

The surface part $E_\Sigma = N_\beta - N_\alpha$ is first order with bounded surface coefficients. The base part is

$$E_B z = -2a^{-2} [A_B, d_h z] - a^{-2} [A_B, [A_B, z]].$$

There is no base divergence term or mixed derivative of β . The preceding inverse estimate and $\|z\|_{H_\Sigma^k} + \|d_h z\|_{H_\Sigma^k} \leq Ca\|z\|_{\mathcal{W}_{k,a}}$ show

$$\|T_0 Q_b E z\|_{\mathcal{W}_{k,a}} \leq C(a + e + e^2)\|z\|_{\mathcal{W}_{k,a}}.$$

Thus $T = (Q_b N|_{\ker m})^{-1}$ exists by a Neumann series. In particular

$$\|T Q_b N J \eta\|_{\mathcal{W}_{k,a}} \leq C(a + e^2/a)\|\eta\|_{H_\Sigma^{k+1}} \leq Ca\|\eta\|_{H_\Sigma^{k+1}}.$$

The potentially dangerous average of the base derivative vanishes:

$$(390) \quad \int_B [A_B, d_h z] d\text{vol}_h = 0.$$

This follows by integration by parts using both base gauge conditions. For zero-average z it implies

$$\|m N z\|_{H_\Sigma^{k-1}} \leq C(1 + e^2/a^2)\|z\|_{L_h^2 H_\Sigma^k}.$$

Consequently the exact surface Schur complement $S = mN(J - T Q_b N J)$ satisfies

$$(391) \quad \|S - N_\alpha\|_{H_\Sigma^{k+1} \rightarrow H_\Sigma^{k-1}} \leq C(R_1^{-1} + \widehat{\delta} + e^2/a^2 + a^2) = o(1).$$

The irreducible surface inverse therefore makes S^{-1} uniformly bounded.

In the h coordinate write the base input as v . For $f = d_A^* U^- = a^{-2} d_{A_B, h}^* v + d_{\beta, \Sigma}^* b$, the zero normal trace of v gives

$$\|m f\|_{H_\Sigma^{k-1}} \leq C(e a^{-2} \|v\|_{L_h^2 H_\Sigma^k} + \|b\|_{L_h^2 H_\Sigma^k}).$$

The zero-average inverse gives, for $z_0 = T Q_b f$, $\|z_0\|_{\mathcal{W}_{k,a}} \leq C(a^{-1} \|v\|_{L_h^2 H_\Sigma^k} + \|b\|_{L_h^2 H_\Sigma^k})$. The solution is $\eta = S^{-1}(m f - m N z_0)$ and $\psi = z_0 - T Q_b N J \eta$. Hence

$$(392) \quad \begin{aligned} \|\eta\|_{H_\Sigma^{k+1}} &\leq C((1 + e/a^2)\|v\|_{L_h^2 H_\Sigma^k} + \|b\|_{L_h^2 H_\Sigma^k}), \\ \|\psi\|_{H_h^1 H_\Sigma^k} &\leq C(\|v\|_{L_h^2 H_\Sigma^k} + a\|b\|_{L_h^2 H_\Sigma^k} + a^2\|\eta\|_{H_\Sigma^{k+1}}). \end{aligned}$$

Conformal invariance of the base one-form L^2 norm gives $\|v\|_{L_h^2 H_\Sigma^k} \leq C\rho M_1 X_k(U)$. The fixed-volume norm of b is bounded by its base supremum, hence by $C\varepsilon^{-1/2} X_k(U)$. Moreover $(e/a^2)\rho M_1 \leq C\widehat{\delta} R_1/\rho = \rho T^N$. This proves (387) in the auxiliary gauge. Changing back has fixed surface norm $O(\widehat{\delta})$ and base H_h^1 multiplier error $O(R_1 \widehat{\delta})$. The additional surface derivative of the mean-zero solution has at worst the polynomial bound in its $\mathcal{W}_{k,a}$ estimate. All resulting frame errors are absorbed by $\rho T^{N_k} X_k(U)$, proving the assertion in the retained frame. $\square$

10.3.5. The global potential in mixed Sobolev norms.

Proposition 10.9 (global Neumann potential near attachment). *For the preglued connection and its smooth unrotated annular deformations, choose fixed nested negative half-discs $D_0 \Subset D_1 \Subset D_2 \Subset D_3$ near attachment. For an input U in the complete gluing domain, the Neumann solution $\Delta_A \phi = d_A^* U^-$ satisfies*

$$(393) \quad \|\phi\|_{W_B^{1,4} H_\Sigma^k(D_0 \times \Sigma)} + \|\phi\|_{L_B^4 H_\Sigma^{k+1}(D_0 \times \Sigma)} \leq C_k \|U\|_{X_\rho}.$$

In particular it is a uniformly bounded multiplier on the local $H_B^\varepsilon H_\Sigma^k$ source for $0 \leq \varepsilon \leq 1/4$.

Proof. Let p be the finite end parameters and U_{dec} their subtracted remainder. First use

$$\|U\|_{\mathcal{U}_k} = \|U_{\text{dec}}^-\|_{L_\omega^2} + |p| + \|U^-\|_{L_B^4 H_\Sigma^k(D_3 \times \Sigma)}.$$

The global weak inverse bounds ϕ in weighted H^1 by the first two terms. The finite profiles produce fixed smooth decaying gauge sources of size $C|p|$. Surface differentiation of the weak equation, with nested fixed base cutoffs, then gives

$$\|\phi\|_{H_B^1 H_\Sigma^k(D_2 \times \Sigma)} + \|\phi\|_{L_B^2 H_\Sigma^{k+1}(D_2 \times \Sigma)} \leq C_k \|U\|_{\mathcal{U}_k}.$$

The commutators contain surface derivatives of the connection and lower surface derivatives of ϕ . Their uniform L^4 coefficient bounds and the four-dimensional H^1 embedding bound the weak terms by the preceding surface energy order. The fixed reference commutators and differentiated cutoffs have the same lower-order control. No base derivative of the rapidly varying surface connection is taken.

For the constant product operator $N_\alpha^{\text{prod}} = \Delta_B + N_\alpha$ on the negative half-plane, the Neumann inverse has the estimate

$$(394) \quad \|z\|_{W_B^{1,4} H_\Sigma^k} + \|z\|_{L_B^4 H_\Sigma^{k+1}} \leq C_k (\|v\|_{L_B^4 H_\Sigma^k} + \|f\|_{L_B^2 H_\Sigma^k}), \quad N_\alpha^{\text{prod}} z = d_\alpha^* v + f.$$

Here d_α includes the base derivative. Reflect the normal input oddly and the other components evenly. The product heat formula gives operator-valued kernel bounds $C|z - z'|^{-2}$ and $C|z - z'|^{-3}$ for the twice-differentiated inverse and its first base derivative. Its L^2 bound is the product spectral bound. The Calderón–Zygmund decomposition applied to the Hilbert norm of the input gives weak type $(1, 1)$: use the L^2 bound for the good part and kernel cancellation for the bad parts outside their doubled supports. Interpolation and the adjoint estimate give the required L_B^4 bound. Flat Hodge intertwining handles surface-form inputs. The once-differentiated kernels have integrable heat bounds at zero and exponential decay at infinity by the surface gap. For the scalar input f , the product spectral estimate gives $H_B^2 H_\Sigma^k$ and $H_B^1 H_\Sigma^{k+1}$ bounds; two-dimensional Sobolev embedding completes (394).

Denote its potential norm by V_k . Morrey embedding gives $V_k \hookrightarrow C_B^0 H_\Sigma^k$. On a sufficiently small fixed D_3 the connection difference from α has small $L_B^4 C_\Sigma^{k+2}$ norm b : the residual contributes $O(r_*^{3/2})$, and the primitive insertion contributes $\rho^{1/2} T^N$. Extend this difference by a fixed cutoff and write $d_A = d_\alpha + B$. The weak perturbation is $d_\alpha^* B + B^* d_\alpha + B^* B$. Its divergence input Bz has $L_B^4 H_\Sigma^k$ bound $Cb\|z\|_{V_k}$, while the other two terms have scalar $L_B^2 H_\Sigma^k$ bounds $C(b + b^2)\|z\|_{V_k}$. Equation (394) therefore gives an inverse by a Neumann series. The same smallness gives uniqueness in the weak energy class.

Apply that inverse to a cutoff of ϕ which equals ϕ on D_0 . Its Laplacian commutator is a divergence of a fixed cutoff derivative times ϕ plus a zeroth-order cutoff term. The preliminary surface energy estimate bounds the former in $L_B^4 H_\Sigma^k$ and the latter in $L_B^2 H_\Sigma^k$. The localized input is controlled by $\|U\|_{\mathcal{U}_k}$. Thus (393) holds with that norm on the right. The physical inner $L_B^4 H_\Sigma^k$ bounds have factors $\rho^{1/2} M_1$ and $\rho^{1/2} M_1^{1/2}$ for base and surface fields. Both tend to zero. The outer part has its fixed base H^1 bound and retained finite parameters. Consequently $\|U\|_{\mathcal{U}_k} \leq C\|U\|_{X_\rho}$.

Finally $W_B^{1,4}$ embeds in $C_B^{3/8}$. The fractional difference formula shows that multiplication by this field is bounded on $H_B^\varepsilon H_\Sigma^k$: the additional radial integral is bounded uniformly because $3/8 - \varepsilon \geq 1/8$. $\square$

10.3.6. Orientation transport on the unrotated annular paths.

Proposition 10.10 (the annular configuration paths). *The compact primitive flattening and the global interpolation from the prescribed pregluing to its flattened form have surjective ordinary full-Coulomb operators throughout. Their kernels have continuous frames whose observations converge to the corresponding compact deformation coordinates. On the compact path the resulting orientation is transported from the complex orientation of $H^1(\text{End}_0 E)$. On the global path it is the orientation specified by K_E and the fixed-attaching observations. Finite rows defined from these kernel frames make the augmented paths invertible. The residual flattening remains invertible. These assertions exclude the coupled rotation and the path from the pregluing to the actual corrected solution.*

Proof. In the existing geometric norms each annular configuration displacement is $O(T^{-40}) + O(\rho T^N)$. For the surface and positive fields this is the $H^{1+\varepsilon}$ norm of an $O(v)$ field on the compact coordinate disc of radius $v = R_2^{-1}$. The base change divided by ρM_1 is exponentially small. The weak exterior surface row has its additional small ρM_1 factor. Reimposing base Coulomb gauge in the compact realization changes these bounds only by an exponentially small polynomial. The matching-chart transports preserve that base condition and the compact exact attaching normalization.

The nonlinear constant CT^4 therefore bounds the transported operator change by $CT^{-36} + C\rho T^N$. When comparing with the actual covariant linearization, retain the derivative of positive target transport times the equation error. The strong-times-weak product estimate bounds this additional term by $C\varepsilon^{-1/2}$ times the error norm, which is included in the stated bound. The existing compact and preglued geometric inverses thus extend by

$$Q_t = Q_0(I + (L_t - L_0)Q_0)^{-1}, \quad \|Q_t\| \leq CT^2.$$

Propositions 10.8 and 10.9, together with Lemma 10.7, give

$$(395) \quad \|(L_t G_t) H_t C_t U\|_Y \leq \mu_\rho \|U\|_X, \quad \mu_\rho = CT^{-79/2} + C\rho T^N.$$

For the global term its multiplier estimate even gives CT^{-40} . For the compact term its average costs $T^{1/2}$, while the zero-average contribution is exponentially small. The compact estimate uses the normalized compact target, as in (386). Consequently

$$E_t = (L_t G_t) H_t C_t Q_t, \quad \|E_t\| \leq CT^{-75/2} + C\rho T^N = o(1), \quad Q_t^C = P_t Q_t (I - E_t)^{-1}$$

satisfies $C_t Q_t^C = 0$ and $L_t Q_t^C = I$. No bound for $P_t : X \rightarrow X$ is needed. At fixed positive scale X embeds into the ordinary H^1 domain, and P_t is bounded there by Neumann regularity. The ordinary Coulomb-slice operator is Fredholm and its closed image contains the dense geometric target. It is therefore onto. Its kernel dimension is four on the global path and $4 + \dim T_p M_g$ on the compact path.

The negative affine-averaged holonomy differentials extend to $L_B^2 H_\Sigma^k$ on the fixed rescaled observation region. Indeed, for a loop $(w(\tau), y(\tau))$, the map $(a, b) \mapsto aw(\tau) + b$ has Jacobian $|\text{Im } w(\tau)|$ bounded below. Changing variables in its averaging integral, Duhamel's formula and $H^k(\Sigma) \hookrightarrow C^0(\Sigma)$ prove the claim. It does not assert boundedness on arbitrary ordinary four-dimensional H^1 fields. The Neumann potentials used above have their base and surface derivatives in this local space. In the compact case use also the finite $\mathcal{W}_{k,a}$ bound for the extra surface derivative. Approximation by smooth potentials therefore extends gauge invariance to

$$\ell_t G_t \phi = 0.$$

The positive attaching evaluation added in the compact case has the same property. Its graph-norm bound is at most $CT^{1/2}$.

For the compact problem choose a complex-oriented basis of the based compact kernel. For the global problem use the fixed-attaching pregluing derivatives, with norm $O(1)$ and geometric error $O(T^{-40})$. Transport these finite fields by the configuration chart and call them W_t . Put

$$(396) \quad w_t = (I - Q_t L_t) W_t, \quad J_t = P_t w_t + Q_t^C (L_t G_t) H_t C_t w_t.$$

Then $L_t w_t = 0$, and both $L_t J_t$ and $C_t J_t$ vanish. The second term of J_t cancels the non-solution error left by the Coulomb projection. The geometric operator estimate gives $w_t - W_t = o(1)$. Their observations converge to the same invertible limiting matrix M . Gauge invariance gives

$$(397) \quad \ell_t J_t = \ell_t w_t + \ell_t Q_t (I - E_t)^{-1} (L_t G_t) H_t C_t w_t = M + o(1).$$

The last correction is bounded by $CT^{1/2}T^2\mu_\rho = O(T^{-37}) + O(\rho T^N)$. Thus J_t is an isomorphism onto the ordinary kernel; its vectors are smooth by fixed-scale elliptic regularity. At the exact compact endpoint its value is the ordinary Coulomb representative of the holomorphic deformation. It has the complex compact orientation there and transports it continuously. On the global path the same formula transports the fixed-attaching observation orientation. Projecting the ordinary source onto these smooth finite kernels and composing with J_t^{-1} supplies bounded rows with invertible augmented operators.

Finally the residual flattening is an ordinary small perturbation of the fixed invertible residual operator. Its connection change is L^4 -small. Its positive principal coefficient change is $O(\rho R_2)$, and its zeroth-order coefficients and derivatives of the seam-domain transports are bounded on a disc of radius $O(\rho R_2)$. The $H_B^1 \rightarrow L_B^4$ estimate makes their multiplication norm to L_B^2 of order $(\rho R_2)^{1/2}$. The surface transports have the same fixed surface operator bounds. Hence the transported ordinary operator differs from the residual one by $o(1)$ and remains invertible. $\square$

The preceding kernel frames evaluate the annular configuration segments. Propositions 10.11 and 10.14 treat the correction from the preglued frame to $\mathcal{T}_\rho$ and the coupled rotation. The correction path has a distributed nonlinear equation error, which requires the estimates below. If ε_{rot} denotes the comparison between the flattened endpoint frames through the rotation segment, and $\varepsilon_{\text{corr}}$ the comparison from the preglued frame to $\mathcal{T}_\rho$ through the correction segment, concatenation of determinant transport gives

$$(398) \quad \varepsilon_{\text{exc}} = \varepsilon_{\text{rot}} \varepsilon_{\text{corr}}.$$

The correction factor can be evaluated by choosing a path whose equation error remains on the pregluing annulus.

10.3.7. Orientation along a path of forced equations.

Proposition 10.11. *Under the regular primitive gluing hypotheses, the ordinary correction transport takes the preglued kernel orientation of Proposition 10.10 to the analytic kernel orientation $\mathcal{T}_\rho$. In the notation of (398),*

$$(399) \quad \varepsilon_{\text{corr}} = +1, \quad \varepsilon_{\text{exc}} = \varepsilon_{\text{rot}}.$$

This holds on the same fixed compact regular patches, including the negative rank-drop locus.

Proof. Use the augmented equation and observation map $\mathbb{F}_\rho$ on the fixed complete subspace $\mathcal{B}_\rho$ from Proposition 7.23. Write

$$\mathbb{F}_\rho(U) = \mathbf{e}_\rho + \mathbb{A}_\rho U + \mathbb{N}_\rho(U).$$

Its linear inverse $\mathbb{Q}_\rho$ has norm at most CT^3 , the initial error is at most CT^{-40} , and the quadratic difference constant is at most CT^5 . For $0 \leq s \leq 1$ solve, on the same contraction ball,

$$(400) \quad U_s = -\mathbb{Q}_\rho(s\mathbf{e}_\rho + \mathbb{N}_\rho(U_s)).$$

The contraction constant is uniformly smaller than $1/2$. Thus

$$\mathbb{F}_\rho(U_s) = (1-s)\mathbf{e}_\rho, \quad U_0 = 0, \quad U_1 = U_\rho^{\mathcal{O}}, \quad \|U_s\|_{X_\rho} + \|\dot{U}_s\|_{X_\rho} \leq CT^{-37}.$$

The derivative follows by differentiating the contraction; the inverse of $I + \mathbb{Q}_\rho d\mathbb{N}_\rho(U_s)$ has norm at most two. The same construction at any prescribed fixed surface order has identical iterates, by Lemma 7.26. The bounds are uniform in s at each such order, with a threshold for ρ allowed to depend on that order. In particular the endpoint is exactly the observation-normalized solution used to define $\mathcal{T}_\rho$.

Put $x_s = (A_s, u_s) = \Phi_\rho(U_s)$. The negative target trivialization is the fixed bundle frame; only the positive target is parallel transported. Consequently

$$(401) \quad \mathcal{F}^-(x_s) = (1-s)e_{\text{pre}}^-.$$

The positive equation is $(1-s)$ times its initial error, transported to u_s . Both errors are supported on the original pregluing annulus. For $G_s\phi = (d_{A_s}\phi, 0)$, gauge covariance therefore gives

$$L_s G_s \phi = ([(1-s)e_{\text{pre}}^-, \phi], 0),$$

with the fixed infinitesimal gauge convention. Although the correction is distributed, its equation error along this path is annular.

The higher surface bounds just proved and Proposition 10.9 control the Neumann potential in the retained frame. Its local $C_B^{3/8} H_\Sigma^k$ multiplier bound and Lemma 10.7 imply

$$(402) \quad \|L_s G_s (C_s G_s)^{-1} C_s V\|_{Y_\rho} \leq C(T^{-40} + \rho T^N) \|V\|_{X_\rho}, \quad C_s = -d_{A_s}^*.$$

The weak estimate extends to the retained coefficients by approximation in the coefficient norms of that proposition. No smoothing gauge is used in this uniform bound.

We give the fixed-scale ordinary realization before applying Fredholm determinant transport. Set

$$p = 2 + \varepsilon/4, \quad r = p + 2, \quad q = r/2.$$

Then $r > 4$, $2 < q < p$, and $r < 4p/(4-p)$. The graph path is continuous in local $W^{1,p}$. Local relative Coulomb gauges, with normal condition at the seam and constant gauge freedom fixed, can be chosen continuously in $W^{2,p}$ on a small parameter interval. In such a gauge g_s the negative forcing becomes

$$(403) \quad f_s^- = (1-s) \text{Ad}(g_s^{-1}) e_{\text{pre}}^-.$$

It is continuous in $W^{2,p}$ and hence in $W^{1,r}$. The positive forcing is a smooth function $f^+(s, z, u_s(z))$. Neither g_s nor (403) is assumed smooth.

The component regularity proof of [DFL24, §3.1] applies with these sources as follows. Add f_s^- to its equation (3.7), and retain its components in (3.13), (3.20), and (3.23). The boundary-flatness identity (3.10) is unchanged. The second entry of (3.23) is now $f^+(s, z, u_s)$. At an exponent $2 < t < 4$, that proof estimates the quadratic source in $W^{1,4t/(8-t)}$ and improves the pair to $W^{1,2t/(4-t)}$, using its Proposition B.10. The negative forcing is available in $W^{1,r}$ with $r > 4$. The positive forcing is in $W^{1,t}$ by composition, and $4t/(8-t) < t$. Thus the same finite iteration reaches r , truncating its last exponent there. The $k = 1$ case of Proposition B.4 and the other component estimates then give $W^{2,r/2} = W^{2,q}$. At this order

the coefficients form an algebra, since $q > 2$. The negative forcing belongs to $W^{2,q}$ because $q < p$, and the positive forcing has that regularity by composition. The $k = 2$ component estimates and Proposition B.4 yield

$$(404) \quad (g_s^* A_s, u_s) \in W_{\text{loc}}^{3,q}, \quad q > 2.$$

This finite improvement suffices. Arbitrary smoothness does not follow by treating (403) as a smooth source.

The difference estimates in [DFL24, §3.2] and the sequential statements of Propositions B.4 and B.10 give continuity in (404). At each step the sources converge at the required preceding order. The perturbed interior argument is unchanged, with the two derivatives of the compact forcing retained. On the ends the forcing vanishes, so the ordinary equation, exact finite critical profiles, and common weight give the usual weighted regularity and continuity. Transitions between the local $W^{3,q}$ representatives satisfy the first-order gauge equation. Starting from $W^{2,p}$, this equation improves them to $W^{4,q}$. They therefore define a path in the ordinary Sobolev configuration quotient, in the original bundle class and with the prescribed end conventions. Successive local configuration charts along the interval suffice; no uniform graph estimate for a global smoothing gauge is required.

We also specify the Fredholm extension at this finite regularity. The negative coefficients in (404) are $C^{1,\alpha}$ for some $\alpha > 0$, and the positive map is $C^{2,\alpha}$ after decreasing α . The actual surface Hodge projections and harmonic matching maps admit local domain identifications bounded on H^1 and L^2 , with their first base derivatives bounded on L^2 . To check the coefficient dependence, differentiate the surface inverse using $\dot{N}^{-1} = -N^{-1}\dot{N}N^{-1}$. Derivatives of the connection variation in $\dot{N}$ are read in divergence form $H^1 \rightarrow H^{-1}$; the adjacent inverses supply the derivative. The same identity gives the required continuity of the projections. Irreducibility keeps their surface inverse norms bounded.

Freeze the boundary data and positive principal coefficient on a sufficiently short mixed chart. The frozen flat connection can be put into a smooth flat representative by a C^2 surface gauge. The constant-data mixed estimate then applies. The remaining continuous principal changes are small and are absorbed; derivatives of the domain transports are bounded zeroth-order terms. Localization gives the H^1 estimate with compact L^2 remainder on the compact middle. The same argument applies to the adjoint domain. Together with the fixed end estimates, including the finite tangent parameters, these estimates prove Fredholmness. They also prove operator continuity under $W^{3,q}$ changes and under smooth approximation in the ordinary configuration charts. Thus this family has the usual determinant line. A sufficiently close smooth path in those charts, relative to the smooth endpoint classes, induces the same transport. DFL orientability identifies it with the correction transport in Proposition 10.1.

The $W^{2,p}$ gauges act boundedly on ordinary H^1 variations, because $dg_s \in L^{4p/(4-p)}$ with exponent greater than four, and act unitarily on L^2 targets. They intertwine both the actual geometric and Coulomb rows, also at non-solutions. Hence the retained family has the same ordinary Fredholm index, equal to four.

It remains to check onto operators and their finite frames. Include the configuration derivative and target transport when writing L_s in the fixed geometric spaces. If S_s is the target transport, the actual operator is

$$S_s L_s D\Phi_\rho(U_s) = dF_\rho(U_s) - (d_U S_s) \mathcal{F}(x_s).$$

The last term is retained. The nonlinear constant CT^4 and $\|U_s\| \leq CT^{-37}$ give an operator change $O(T^{-33}) + \rho T^N$ from pregluing. The pregluing inverse of norm CT^2 therefore gives

actual right inverses Q_s of norm at most CT^2 . Set

$$B_s = L_s G_s, \quad H_s = (C_s G_s)^{-1}, \quad P_s = I - G_s H_s C_s, \quad E_s = B_s H_s C_s Q_s.$$

By (402), $\|E_s\| \leq CT^{-38} + \rho T^N = o(1)$. The formula

$$Q_s^C = P_s Q_s (I - E_s)^{-1}$$

satisfies $L_s Q_s^C = I$ and $C_s Q_s^C = 0$ on the dense geometric target.

At fixed scale the retained connection is $H^{1+\varepsilon}$. In the scalar Neumann expansion, its first-order coefficients are in $L^{4+\delta}$ and their derivatives in $L^{2+\delta'}$, for positive δ, δ' . Their actions $H^2 \rightarrow L^2$ are relatively compact by subcritical Sobolev embedding. The normal coefficient is fixed. Elliptic regularity and the weak gap give an H^2 potential, with scale-dependent constants allowed. Consequently P_s maps ordinary H^1 to itself. The ordinary Coulomb-slice operator is Fredholm by Proposition 10.6; its closed range contains the dense geometric target and hence is the whole ordinary target. The full operator is onto and its kernel has dimension four.

Transport the finite pregluing fields to W_s and put

$$(405) \quad w_s = (I - Q_s L_s) W_s, \quad J_s = P_s w_s + Q_s^C B_s H_s C_s w_s.$$

Both ordinary rows annihilate J_s . The averaged observations annihilate the gauge gradients in the local regularity class supplied by Proposition 10.9. The same argument as for (397) gives $\ell_s J_s = M_E + o(1)$ uniformly. The geometric frame change costs at most $\sqrt{T} T^2 T^{-33}$ in observations; the last term in (405) costs at most $\sqrt{T} T^2 T^{-40}$. Both tend to zero. Thus J_s is a continuous basis of the entire ordinary kernel. Projection onto that kernel followed by J_s^{-1} supplies bounded rows with invertible augmented operators throughout the path.

At $s = 0$ this is the prescribed preglued frame. At $s = 1$ the equation error vanishes and the frame is the analytic map $\mathcal{T}_\rho$, with the compatible inverse just constructed. Other inverses with the same limiting observations give the same orientation. The finite-row transport therefore has sign +1, proving (399). $\square$

We next compare the coupled rotation with these flattened endpoint frames. The coherent DFL comparison follows the rotation argument.

10.3.8. The scalar estimate and attaching map during rotation. For the split rotation domain $D_0^f \oplus C_E^f$, give an input U the norm X_{split} obtained from the global complete cap/residual norm with constant inner cap, together with the compact primitive raw-cap norm. Use the corresponding sum of the global and native compact geometric target norms, denoted by Y_{split} . The actual normal and matching conditions are retained.

Proposition 10.12. *The compact flattening in Proposition 10.1 can be chosen so that, along its rotation path $D_t = (L_t, C_t)$,*

$$(406) \quad \|L_t G_t (C_t G_t)^{-1} C_t U\|_{Y_{\text{split}}} \leq C(T^{-79/2} + \rho T^N) \|U\|_{X_{\text{split}}}, \quad 0 \leq t \leq 1.$$

Here G_t is the covariant gradient of the negative zero-form connection of the rotation path. The constant and the finite exponent N are uniform on the fixed compact regular patch.

Proof. The exact compact base coefficient is in ordinary base Coulomb gauge, with zero normal trace. On the negative hemisphere it has a stream function $A_{E,B} = *_B d_B \Psi$, $\Psi|_{\partial B} = 0$. For the primitive flattening cutoff τ , replace its base coefficient by

$$A_{E,B}^f = *_B d_B (\tau \Psi).$$

This preserves zero divergence and normal trace and makes the coefficient exactly zero on the constant annulus and outer cap. Its change from affine flattening is bounded, at every required finite order, by ρ^2 times a polynomial in T . Near the attaching point $\Psi = O(\delta|\zeta|)$, so differentiated cutoffs satisfy the same bounds. Proposition 10.10 applies to this choice and its interpolation with the former one. It does not change the excision transport.

Write $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. On the identified negative annulus,

$$\nabla^t = \nabla^{\text{split}} - tJ d\vartheta, \quad C_t G_t = -(\nabla^t)^* \nabla^t.$$

The connection on the annulus is exactly $\text{diag}(\alpha, \alpha)$, which commutes with J . Since $d(d\vartheta) = 0$ and $d\vartheta \wedge d\vartheta = 0$,

$$(407) \quad F_{\nabla^t} = F_{\nabla^{\text{split}}}.$$

This is a curvature identity for an orthogonal connection on the paired vector bundles; no Lie-algebra automorphism property of the rotation is assumed. On perturbation supports G_t is the original gauge gradient. Gauge covariance there, together with (407), gives

$$L_t G_t(\phi_0, \phi_E) = ([e_0^-, \phi_0], [e_E^-, \phi_E], 0).$$

The errors are supported on their original flattening annuli. No new equation-error term occurs where $d\vartheta \neq 0$.

Conjugate zero-forms by $V_t = R(t\vartheta)$, extended over the paired negative inner discs as in Proposition 10.5. Put $a = \rho R_1$, $E = N_E^f - N_\alpha^c$, and $N_\alpha^c = a^{-2}\Delta_h + N_\alpha$. The coefficients of E are supported where V_t is the constant rotation $R(t\pi/2)$ and the two metrics agree. Thus the conjugated scalar operator is exactly

$$(408) \quad \tilde{N}_t = \begin{pmatrix} N_0^f + s^2 E & scE \\ scE & N_\alpha^c + c^2 E \end{pmatrix}, \quad s = \sin(t\pi/2), \quad c = \cos(t\pi/2).$$

Its source is $\tilde{\nabla}_t^* U'$, where $U' = V_t^{-1} U$; its off-diagonal zero-order source terms are included below. Set $X = \|U\|_{X_{\text{split}}}$. Value rotation, conformal invariance of the base one-form L^2 norm and the raw-cap evaluation estimate give

$$\|U'_g\|_{\mathcal{U}_k} \leq CX, \quad \|v'_c\|_{L_h^2 H_\Sigma^k} \leq C\rho M_1 X, \quad \|b'_c\|_{L_h^2 H_\Sigma^k} \leq C\sqrt{T} X.$$

For example, the compact input entering the global component has physical $L_B^4 H_\Sigma^k$ norm at most $\rho^{1/2}$ times a fixed polynomial in T times X . The global input entering the compact component is measured in its existing raw-cap coordinates on the same inner disc. No derivative of V_t is used in these input estimates, and the end parameters are unchanged.

Write $z = V_t^{-1}\phi = (z_g, \eta + \psi)$ with $\eta = m_h z_c$ and $m_h \psi = 0$. Let G be the sum of the global weak energy norm of z_g and its local norms in (393), let $H = \|\eta\|_{H_\Sigma^{k+1}}$, and let $W = \|\psi\|_{\mathcal{W}_{k,a}}$, where $\mathcal{W}_{k,a}$ is the mean-zero norm in the proof of Proposition 10.8.

The compact diagonal in (408) has the same base spectral inverse and surface Schur complement as that proposition. Multiplication of its perturbation by c^2 changes none of the bounds. Its first-order base term still has zero average. The global diagonal has the estimate of Proposition 10.9. Indeed on the inner region it is the convex combination of the scalar Laplacians at α and A_E^f . It equals the Laplacian of the interpolated connection plus the nonnegative variance term $s^2(1-s^2)B^*B$, $B = A_E^f - \alpha$. The latter is small in the global L^2 coefficient norm. Weak commutation estimates use $L^2 \times L^4 \times L^4$, and the local base estimate bounds it by $\|B\|_{L_B^4 C_\Sigma^{k+2}}^2 \|z_g\|_{C_B^0 H_\Sigma^k}$. Thus the same absorption argument applies.

We record the off-diagonal estimates to retain their scale factors. The surface coefficient B_Σ is uniformly bounded, has physical L^4 norm $O(\rho^{1/2}T^N)$, physical L^2 norm $O(\rho T^N)$, and normalized compact average $O(R_1^{-1} + \delta)$, also at the fixed surface derivative orders used here. The physical base coefficient has L^∞ norm at most $C\delta\rho^{-1}T^N$ and L^4 norm at most $C\delta\rho^{-1/2}T^N$, where $\delta = \rho^2T^N$. Also

$$\|\psi\|_{H_h^1 H_\Sigma^k} \leq aW, \quad \|\psi\|_{L_{\text{phys}}^4 H_\Sigma^k} \leq Ca^{3/2}W, \quad \|\psi\|_{L_{\text{phys}}^2 H_\Sigma^{k+1}} \leq aW.$$

Writing Ez_c as a divergence source and a scalar source in the global equation now gives

$$(409) \quad G \leq CX + C\rho^{1/2}T^N H + C\rho T^N W.$$

For instance $B_\Sigma \eta$ has L^4 norm at most $C\rho^{1/2}T^N H$, while $B_\Sigma \nabla_\Sigma \psi$ has L^2 norm at most CaW . The base derivative term costs at most $C\delta R_1 T^N W$. The off-diagonal coefficient terms in $\tilde{\nabla}_t^* U'$ have the same bounds on the input and are absorbed into CX .

For the compact mean-zero equation its inverse costs a on an $L_h^2 H_\Sigma^k$ source. Hence

$$a\|Ez_g\|_{L_h^2 H_\Sigma^k} \leq C\rho^{1/2}T^N G.$$

In the surface derivative term the conversion cancels a , leaving $\|B_\Sigma \nabla_\Sigma z_g\|_{L_{\text{phys}}^2 H_\Sigma^k}$, bounded by the two physical L^4 factors. The base derivative term is bounded by $\|A_{E,B}^f \nabla_B z_g\|_{L_{\text{phys}}^2 H_\Sigma^k} \leq C\delta\rho^{-1/2}T^N G$. Thus

$$(410) \quad W \leq CT^N X + CaH + C\rho^{1/2}T^N G.$$

For the compact average the divergence-free choice is essential:

$$\int_B [A_{E,B}^f, d_h z_g] d\text{vol}_h = 0.$$

There is no normal boundary term, and the coefficient vanishes near the artificial boundary of the paired inner region. In the first-order surface term the H_Σ^{k-1} target needs only the $C_B^0 H_\Sigma^k$ bound for z_g . Its coefficient average is $O(R_1^{-1} + \delta)$. The base square term has coefficient at most $C\delta^2\rho^{-2}T^N$. The compact Schur complement therefore gives

$$(411) \quad H \leq C\sqrt{T} X + C(R_1^{-1} + \rho T^N)G.$$

The ordinary divergence of the compact input has zero average. Its off-diagonal base coefficient term costs at most $(e/a^2)\rho M_1 X = \rho T^N X$, with $e \leq R_1 \delta T^N$; the surface input costs $C\sqrt{T} X$. The mean-zero inverse contribution has the extra factor a , as in the compact proof.

Substitution of (410) and (411) into (409) leaves only $\rho^{1/2}$ times fixed polynomials multiplying G . Absorption gives

$$G \leq CX, \quad H \leq C\sqrt{T} X, \quad W \leq CT^N X, \quad \|\psi\|_{H_h^1 H_\Sigma^k} \leq C\rho T^N X.$$

The weak inverse already exists by Proposition 10.5; density extends these estimates from smooth inputs.

On the residual error annulus $V_t = I$. On the primitive error annulus it is a constant rotation, so the original compact potential is a combination of z_c and z_g . The former has the compact mean/mean-zero bounds just proved; the latter has the global $C_B^{3/8} H_\Sigma^k$ multiplier bound. The two native annular estimates of Lemma 10.7, applied to the exact errors in (407), prove (406). $\square$

The constant harmonic sector of the same rotation has an explicit matching map. Put $V = T_p M_g$. If h_0 is the attaching value supplied by the residual inverse and $a \in V$ is the free value of the outer constant cap, its inner values obey

$$(b_0, b_E) = R(t\pi/2)(h_0, a).$$

This follows by applying the actual annular operator to $R(t\vartheta)h$: both derivatives of the rotation cancel and the normal and matching conditions are preserved. For $\theta = t\pi/2$, the two matching rows and the row prescribing a have matrix

$$(412) \quad \begin{pmatrix} I & 0 & \sin \theta I \\ 0 & I & -\cos \theta I \\ 0 & 0 & I \end{pmatrix} \quad \text{on } (b_0, b_E, a) \in V^3.$$

Its real determinant is one, and its inverse is uniformly bounded. The zero-attaching parameter $k \in K_E$ and its row add an identity block. At the second endpoint $(b_0, b_E) = (-a, 0)$ for homogeneous sources; undoing the signed interchange sends the constant inner value to $+a$ and retains the zero-attaching parameter. This finite map has positive orientation. Lemma 10.13 supplies the outer inverse used in the geometric construction. Proposition 10.14 then constructs the full inverse in $X_{\text{split}}, Y_{\text{split}}$, identifies its kernel with (412), and evaluates ε_{rot} .

10.3.9. Localization of the outer constant cap.

Lemma 10.13. *The constant compact operator supplies an outer geometric inverse normalized at its inner puncture, followed by a based local gauge and stream cutoff, with norm at most CT^3 in the native compact norms. If its source vanishes on $|w| < cR_0$, its stream commutator at radius comparable to R_- has target norm at most $CT^2(R_-/R_0)$ times the source norm. Its exact and harmonic attaching profiles have size at most CT^2 times that norm. The profile commutator is retained separately from the stream error.*

Proof. First note that for the constant cap the field

$$\mathcal{P}_\alpha(e, h) = (0, e + h, \kappa_\alpha^{-1}h)$$

is a geometric kernel field whenever e is exact and h harmonic on Σ . Its raw-cap norm is bounded by $C(\|e\|_{H^k} + |h|)$: the weak derivative is $d_\alpha(e + h) = 0$, not a full derivative of e . The constant compact kernel after exact normalization is the space of constant harmonic fields. The constant-cap specialization of the cap inverse, with zero total attaching value at infinity, has norm at most CT . Subtracting $\mathcal{P}_\alpha(E_\alpha b(z_*), H_\alpha b(z_*))$ from its output moves the total attaching normalization to any other fixed seam point z_* . Evaluation costs $C\sqrt{T}$, so the new inverse has norm at most $CT^{3/2}$. It is unique: a homogeneous field in base Coulomb gauge is a constant profile, and a profile with zero value at z_* vanishes.

Use $x = w/R_1$ and $a = \rho R_1$. The metric is $a^2h + g_\Sigma$ on the fixed hemisphere, Euclidean near $x = 0$. Write $U^- = av + b$ and use

$$\|U\|_{X_h} = \|(v, b, \nu)\|_{H_h^{1+\varepsilon} H_\Sigma^k} + a\|d_\alpha v\|_{H_h^\varepsilon H_\Sigma^k} + a\|d_\alpha b\|_{H_h^\varepsilon H_\Sigma^k}.$$

Its normalized target is $Y_h = H_h^\varepsilon H_\Sigma^k$, with the positive finite factor. Coordinate conversion gives

$$(413) \quad \|F_h\|_{Y_h} \leq CR_1^{C\varepsilon} \|F\|_{Y_{\text{cap}}}, \quad \|U\|_{X_{\text{cap}}^{\text{raw}}} \leq C(1 + \log R_1)^{1/2} R_1^{C\varepsilon} \|U_h\|_{X_h}.$$

For the target, one-form pullback and the scalar conformal weight preserve L^2 up to fixed metric constants. Their H^1 bounds are fixed powers of R_1 ; interpolation proves the fractional

estimate. For scalar domain fields the Dirichlet integral is conformally invariant. The pulled volume has density at most $CR_1^2/(1+R_1^2r^2)^2$ near its concentrating point. Integrating the annular H^1 restriction bound gives the displayed square-root logarithm. An H^2 chain-rule bound and interpolation give the strong fractional estimate. The original normalized base one-form is R_1/M_1 times pullback of v . One-form L^2 is conformally invariant; its exterior and co-exterior derivative norms, after this normalization, are bounded because the conformal dilation is at most CR_1 and $R_1/M_1 = O(R_1^{-1})$. The normal-trace Gaffney estimate and interpolation give its strong bound. The weak rows have the same conformal target weights. In particular (413) contains no negative power of ρ .

Apply the constant-cap inverse at $x = 0$ in this fixed geometry, with parameter a . Its output has X_h norm at most $CT^{3/2}\|F_h\|$ and base divergence zero. To impose full Coulomb gauge solve

$$(a^{-2}\Delta_h + N_\alpha)\phi = d_\alpha^*b, \quad \partial_n\phi = 0,$$

and subtract $(a^{-1}d_h\phi, d_\alpha\phi, 0)$. This projection is uniformly bounded in X_h . For a Neumann base eigenvalue μ and surface frequency λ , its two gradient multipliers are

$$(414) \quad \frac{a^2\lambda^2}{\mu + a^2\lambda^2} \leq 1, \quad \frac{a\sqrt{\mu}\lambda}{\mu + a^2\lambda^2} \leq \frac{1}{2}.$$

Multiplying the second by $a\lambda$ gives at most $\sqrt{\mu}$, controlled by the strong base norm of b . The added surface field is exact, so its exterior derivative vanishes. These estimates hold in the stated fractional norms: on the fixed doubled base, scalar Neumann fields extend evenly and their gradients have odd normal and even tangential components. At $1 + \varepsilon \leq 5/4$ no normal derivative trace is required of the scalar input. The doubled Hodge Laplacian intertwines the scalar gradient. This identifies the spectral and Sobolev norms uniformly at the orders used here. This bounded graph projection uses the constant connection and is not asserted for the primitive.

The source-free radius in h coordinates is $L = cR_0/R_1$, with physical radius $aL = c\rho R_0$. Evaluation and the preceding inverse give the rescaled size bound $B = CT^2\|F_h\|$. The homogeneous full-Coulomb estimate and based local gauge construction used in the residual inverse apply here with surface mass $aL\lambda$. Their constants are uniform in that mass. The local gauge variation in h coordinates is $(d_h\zeta, a d_\alpha\zeta, 0)$. Its strong and weak cap rows cost at most $CL^{-\varepsilon}B$; for the weak base row use $ad_\alpha d_h\zeta = d_h(ad_\alpha\zeta)$. It restores base divergence zero locally and makes the base value zero at $x = 0$, while preserving the surface and positive attaching values. These values have size at most CB by the rescaled estimate, without another evaluation loss.

Subtract their constant profile P locally and write $U^0 = U^G - P$. Every component vanishes at zero. Its base stream satisfies $\psi(LZ) = L\Psi(Z)$ with $\Psi(0) = d\Psi(0) = 0$. At $r = c'R_-/R_1$, the Taylor and stream estimates give

$$\|U^0(r\cdot)\|_{H^{1+\varepsilon}H^k} + aL\|U^0(r\cdot)\|_{H^{1+\varepsilon}H^{k+1}} \leq C(r/L)B, \quad \|\Psi((r/L)\cdot)\|_{H^{2+\varepsilon}H^k} \leq C(r/L)^2B.$$

The extra surface derivative is weighted by the physical radius aL . For the outer stream cutoff put $V = \mathcal{C}_-U^0 + \chi P$, where χ vanishes inside r and equals one outside $2r$. Since P is a geometric kernel field and $\chi F_h = F_h$, direct expansion gives

$$C_\alpha^{\text{geo}}V - F_h = E_- + [C_\alpha^{\text{geo}}, \chi]P, \quad E_- = C_\alpha^{\text{geo}}\mathcal{C}_-U^0 - \chi C_\alpha^{\text{geo}}U^0.$$

The displayed local bounds give $\|E_-\|_{Y_h} \leq Cr^{-\varepsilon}(r/L)B$. All fractional radius factors are bounded, so this is $CT^2(R_-/R_0)\|F_h\|$. The cutoff field has X_h norm at most CB . Equation (413) then gives the coarse native bound CT^3 . The profile commutator remains for cancellation with the inserted inner profile.

A two-component phase on the disjoint rotation annulus obeys the same bounds. Its target multiplier has norm $CR_2^{C\varepsilon}$. The term with its domain derivative is estimated by the rescaled value bound B on that annulus, before the coordinate conversion. The phase is constant on the stream cutoff annulus. Thus it adds no larger power of T . $\square$

10.3.10. *The rotation inverse and its orientation.*

Proposition 10.14. *On the fixed compact regular primitive patches the rotation path admits geometric right inverses satisfying*

$$(415) \quad L_t Q_t = I, \quad \|Q_t\|_{Y_{\text{split}} \rightarrow X_{\text{split}}} \leq CT^3.$$

Its ordinary operators are onto, and their determinant transport agrees with the compact complex and flattened analytic kernel orientations. Consequently

$$(416) \quad \varepsilon_{\text{rot}} = +1, \quad \varepsilon_{\text{exc}} = +1.$$

The coherent DFL comparison is a further assertion.

Proof. Use four reference pieces: the inner constant and exact primitive caps with zero total attaching value, the fixed residual with its complete source and finite end variables, and the outer constant cap of Lemma 10.13. The inner blocks use the cap inverses and stream localization in the proof of Proposition 7.13. The fourth block includes the full-Coulomb and based local gauge steps just proved.

On $|w| \geq R_-$ use the paired matching-domain and target identifications with the common α model. Denote these by P^X, P^Y for each background. In physical coordinates the outer transport is

$$\mathcal{A}_t^X = P_{\text{actual}}^X R(t\vartheta)(P_{\text{outer}}^X)^{-1}, \quad \mathcal{A}_t^Y = P_{\text{actual}}^Y R(t\vartheta)(P_{\text{outer}}^Y)^{-1}.$$

It is used only outside the inner cutoff. For the inner cap pair use the phase $R(t(\vartheta - \pi/2))$, which is the identity on the deep inner region and extends there on the actual inner bundles. No identification with a constant mixed domain is made over a nonconstant core. On the support of $d\vartheta$ the actual background is exactly constant and $P_{\text{actual}} = I$. The derivative of either phase cancels precisely the tK term of L_t . All other derivatives are those of the paired background identifications. At both stream cutoff annuli the phase is constant.

Partition every source as $F_{\text{in}} = \theta_0 F$ and $F_{\text{out}} = (1 - \theta_0)F$ at radius ρR_0 . The native cutoff/source conversions are bounded on the complete target. Convert F_{in} to the two inner cap targets, apply their zero-attaching inverses and the stream cutoff at affine radius $R_1/4$, then apply the inner transport. Call this map P_{in} . For the outer pair use the source

$$G_{\text{out}} = (\mathcal{A}_t^Y)^{-1} F_{\text{out}},$$

extended by zero on the smaller source-free discs, with all native row conversions included. Its norm is at most $C\|F\|_{Y_{\text{split}}}$. Apply the fixed residual inverse with zero Coulomb source, retaining its exact end parameters, and the outer constant-cap construction. Put the two outputs in their specified based local gauges. Their attaching data satisfy

$$\|e_0\|_{H^k} + |h_0| + \|e_c\|_{H^k} + |h_c| \leq CT^2\|F\|_{Y_{\text{split}}} =: B.$$

The residual alone has the smaller bound $C\sqrt{T}\|F\|$. No second evaluation loss is used.

Let $P = (P_0, P_c)$ be the reference attaching profiles. The harmonic part of P_0 can have nonzero geometric source; P_c is a constant geometric kernel field. Rotate their attaching data by $R(t\pi/2)$ and lift them to the inner constant and primitive cap profiles, denoting the result by B_t . These lifts use only bounded sines and cosines. They are exact geometric kernel fields on the region where inserted. For the cutoff χ at R_- define

$$V_j = \mathcal{C}_-(U_j^G - P_j) + \chi P_j, \quad P_{\text{out}}F = \mathcal{A}_t^X V + (1 - \chi)B_t.$$

Both terms satisfy the actual matching and normal conditions. The outer finite end parameters are unchanged.

Set $E_{-,j} = L_j \mathcal{C}_-(U_j^G - P_j) - \chi L_j (U_j^G - P_j)$. Since $\chi G_j = G_j$, the exact identity is $L_j V_j = G_j + E_{-,j} + [L_j, \chi]P_j$. The terms $\chi L_j P_j$ cancel, including the nonzero residual harmonic-profile source. With $K_t = L_t \mathcal{A}_t^X - \mathcal{A}_t^Y L_{\text{out}}$, principal base-symbol intertwining gives

$$(417) \quad L_t P_{\text{out}}F - F_{\text{out}} = \mathcal{A}_t^Y E_- + K_t V + [L_t, \chi](\mathcal{A}_t^X P - B_t).$$

The last inner source vanishes because B_t is inserted entirely in the exact inner region. This identity applies to every source in the same completed target.

We retain the quantitative error counts. The two inner errors are bounded by

$$C\varepsilon^{-3}R_0^8(R_1^{-1+2\varepsilon} + R_2^{-2+C\varepsilon})\|F\| + C\rho T^N\|F\|.$$

These are the stream and paired inner background estimates in the proof of Proposition 7.13. The phase derivative has canceled before these estimates are taken; target rotation has bounded fractional norm. For (417), the stream term is at most $C(R_-/R_0)B$, by the residual estimate and Lemma 10.13. Every physical outer coefficient triple has rescaled $H_B^1 H_\Sigma^k$ norm at most CB on the relevant annuli. For the constant cap its physical base coefficient is controlled by the fixed- h construction; the old a/M_1 bound is not substituted for this fact. The paired dyadic estimate therefore gives $\|K_t V\| \leq CR_-^{-1}B + C\rho T^N B$. The profile difference at R_- has zero attaching value. Its surface and positive terms are $O(R_-^{-1} + \rho R_-)$, and its physical base term is $O(\rho R_- + (\delta/\rho)R_-^{-2})$, times B . The commutator calculation gives

$$\|[L_t, \chi](\mathcal{A}_t^X P - B_t)\| \leq C(R_-^{-1} + \rho R_- + (\delta/\rho)R_-^{-2})B.$$

Only the H^k norm of each exact attaching datum is used: its gauge potential gains one surface derivative, and the base dependence of its profile has a $C^0 H^k$ bound without another evaluation factor.

For the chosen radii the inner errors are $O(T^{-13})\|F\|$ and the three outer errors are $O(T^{-2})\|F\|$, apart from exponentially small polynomial terms. Thus, for $P_t = P_{\text{in}} + P_{\text{out}}$,

$$\|L_t P_t - I\| \leq CT^{-2} + C\rho T^N = o(1), \quad \|P_t\| \leq CT^3.$$

The domain bound includes the local gauges, profiles and the phase's rescaled value bound, before coordinate conversion. All error sources belong to Y_{split} . The Neumann series $Q_t = P_t(L_t P_t)^{-1}$ proves (415). These maps are continuous in t at fixed scale.

By Proposition 10.12,

$$\|L_t G_t (C_t G_t)^{-1} C_t Q_t\| \leq CT^{-73/2} + C\rho T^N = o(1).$$

Write $P_t^C = I - G_t (C_t G_t)^{-1} C_t$, to distinguish it from the geometric parametrix. Then

$$Q_t^C = P_t^C Q_t (I - L_t G_t (C_t G_t)^{-1} C_t Q_t)^{-1}$$

solves both ordinary rows on the dense geometric source class. At fixed scale P_t^C maps ordinary H^1 to itself. The ordinary path is Fredholm by the compact-commutator construction,

and removing the invertible Neumann block preserves Fredholm closed range. Hence D_t is onto. Its kernel has dimension $m = 4 + \dim_{\mathbb{R}} V$.

To orient that kernel use parameters $(k, a) \in K_E \oplus V$. For a , take outer profiles $(0, \mathcal{P}_\alpha(0, a))$ and inner harmonic values $(-\sin \theta a, \cos \theta a)$, where $\theta = t\pi/2$. Join them by the preceding construction, using for these finite fields the radius $R_b = T^{20}$ in place of R_- . It lies between R_0 and R_2 . For k use the finite zero-attaching cap derivatives and their inner localization. Their sum $W_t(k, a)$ satisfies

$$(418) \quad \|W_t\| \leq C, \quad \|L_t W_t\| \leq CT^{-20} + C\rho T^N.$$

The k part has the stronger T^{-40} error. These are fixed finite smooth profiles, so no growing-source inverse factor is used in their norm.

Let ℓ_E be the four averaged negative holonomy differentials on the fixed inner E region, and ℓ_∞ positive evaluation at the outer compact attaching point. Write the limiting compact observation matrix as $\ell_E(k + j_E h) = M_E k + N_E h$ and set

$$\ell_t = (\ell_E - \cos \theta N_E \ell_\infty, \ell_\infty).$$

It has graph norm at most $C\sqrt{T}$, and $\ell_t W_t = \text{diag}(M_E, I_V) + o(1)$. It annihilates $G_t \phi$ in the potential regularity class just proved: the negative observation lies where the gradient is the ordinary split gauge gradient, and the positive observation has zero gauge variation. The averaged-observation approximation argument applies as before.

Put

$$w_t = (I - Q_t L_t) W_t, \quad J_t = P_t^C w_t + Q_t^C L_t G_t (C_t G_t)^{-1} C_t w_t.$$

Both ordinary rows annihilate J_t . The geometric correction in its observations costs at most $\sqrt{T} T^3 T^{-20} = T^{-33/2}$, and the last term costs at most $\sqrt{T} T^3 T^{-79/2} = T^{-36}$. Thus $\ell_t J_t = \text{diag}(M_E, I_V) + o(1)$. Kernel dimension makes J_t a continuous basis of the whole ordinary kernel. Projection onto that kernel followed by J_t^{-1} supplies bounded finite rows with invertible augmented operators.

At $t = 0$ this is the compact complex splitting $K_E \oplus V$, with the invertible residual factor. At $t = 1$ the signed interchange sends the constant inner value $-a$ to $+a$ on the original constant cap and sends k to the glued primitive. The latter has the same limiting observations as the flattened analytic frame. Its transition is orientation preserving. The attaching map is precisely (412); the even real indices add no factor permutation sign. Hence $\varepsilon_{\text{rot}} = +1$. Proposition 10.11 then gives $\varepsilon_{\text{exc}} = +1$. $\square$

The actual excision transport now agrees with the analytic kernel map and the gluing derivative in the stated regular sector. The remaining local sign comparison is with DFL's coherent prescription. In particular the physical signs still satisfy $\eta_g = \sigma_g \varepsilon_{\text{coh}}$; we next evaluate the genus-two factor in the selected sphere normalization.

10.4. The genus-two coherent sign. The comparison in this subsection takes place at a fixed strictly positive metric scale. Its sphere lies a fixed positive distance from the seam.

Lemma 10.15. *Fix a very stable genus-two attaching value and one of its four sublimes. The complex orientation of the compact primitive determinant agrees, by a path of actual matching configurations, with the canonical orientation of a positive generator sphere inserted into the constant mixed cap. The same assertion holds after removing the complex attaching factor $T_p M_2$.*

Proof. Write the extension line as $F : \mathbb{P}^1 \rightarrow M_2$, with coordinate ξ and $F(0) = p$. Its tangent bundle splits as $\mathcal{O}(2) \oplus \mathcal{O} \oplus \mathcal{O}$ by Proposition 8.21. Fix a strictly inside the positive disc and put

$$f_b(z) = F(b/(z - a)), \quad b \neq 0.$$

All surface fibers of its extension bundle E_b are stable. For $\pi : \mathbb{P}^1 \times \Sigma \rightarrow \mathbb{P}^1$, stability gives $R^0\pi_* \text{End}_0 E_b = 0$ and $R^1\pi_* \text{End}_0 E_b = f_b^* TM_2$. There are no higher fiber cohomology groups. Hence Leray gives the complex-linear identification

$$(419) \quad H^1(\mathbb{P}^1 \times \Sigma, \text{End}_0 E_b) = H^0(\mathbb{P}^1, f_b^* TM_2).$$

The kernel map of Proposition 6.4 followed by this identification is the actual map variation on the positive disc.

Let $C_\alpha^\perp$ be the nonharmonic part of the constant compact mixed operator. It is an isomorphism in the ordinary spaces by the constant calculation in Lemma 6.3. Choose global holomorphic sections $\sigma_1, \sigma_2, \sigma_3$ of $F^* TM_2$: take $\sigma_1 = \partial_\xi$ on $\mathcal{O}(2)$ and frames of the two trivial summands. They form a frame near p ; σ_1 has a double zero at $\xi = \infty$. Their pullbacks give a holomorphic map $S_b : \mathbb{C}^3 \rightarrow f_b^* TM_2$. It is a uniformly bounded invertible frame outside any fixed disc about a when b is small.

Use the reference operator

$$D_b^* = C_\alpha^\perp \oplus \bar{\partial}_{f_b^* TM_2}.$$

On the negative disc express its harmonic part in the holomorphic frame S_b and match its trace to the positive part. This gives a well-defined ordinary mixed reference domain. The operator is onto and has real kernel dimension ten.

Here only a localized inverse estimate is needed. Fix a positive disc about a disjoint from the seam, and let Ω contain its exterior and the whole negative domain. Allow sources in a fixed closed subset of Ω , denoting their target by Y_Ω . There is an inverse on these sources with

$$(420) \quad D_b^* Q_\Omega F = F, \quad \|Q_\Omega F\|_{H^1(\Omega)} \leq C \|F\|_{L^2}, \quad F \in Y_\Omega,$$

uniformly as $b \rightarrow 0$. On nonharmonic sources use $(C_\alpha^\perp)^{-1}$. On harmonic sources use

$$Q_\Omega F = S_b Q_0 (S_b^{-1} F),$$

where the coefficient source is extended by zero across the disc and Q_0 is a fixed right inverse for $\bar{\partial}$ on the trivial rank-three sphere bundle. Such an inverse exists since $H^{0,1}(\mathbb{P}^1) = 0$. Holomorphicity of S_b proves the equation exactly. Its coefficients are uniformly controlled on Ω . Inside a smaller source-free disc the scalar solution has a bounded C^1 norm by the interior holomorphic estimate. The sections S_b have uniformly bounded values and Dirichlet energy, being pullbacks by a degree-one conformal map. Thus the product is an ordinary H^1 section globally. This also bounds its values at any fixed finite set of points away from the source support and from a .

Choose $\chi = 0$ on the sphere core and $\chi = 1$ near the seam and on the negative hemisphere. On the positive side take the actual configuration path

$$u_{s,b}(z) = F((1 - s\chi(z))b/(z - a)), \quad 0 \leq s \leq 1.$$

On the negative side use Dirichlet reconstruction for the holomorphic family with residue $(1 - s)b$. Its boundary agrees with $u_{s,b}$, including at $s = 1$, when the negative connection is α . At the fixed metric scale, Dirichlet invertibility at the constant pair and smooth dependence give $A_{s,b} - \alpha = O(|b|)$ in every fixed C^k norm. The positive map equals f_b on the whole core; outside it all its derivatives differ from the constant map by $O(|b|)$.

Use the smooth exact/harmonic seam transports and complex-linear positive bundle identifications, equal to the identity on the core, to identify the actual domains with the reference domain. On the exterior these maps send $\sigma_j(f_b)$ to $\sigma_j(u_{s,b})$ and differ from their constant values by $O(|b|)$. The normal condition is retained, and Coulomb gauge remains an output row. The full operators become

$$(421) \quad D_{s,b} = D_b^* + K_{s,b}, \quad \text{supp } K_{s,b} \subset \Omega, \quad \|K_{s,b}U\|_{L^2} \leq C|b|\|U\|_{H^1(\Omega)}.$$

This includes all negative surface derivatives and the first-order terms from the transports. It is a fixed-scale estimate; its constant need not be uniform in the physical gluing scale.

Equations (420) and (421) give $\|K_{s,b}Q_\Omega\| \leq C|b|$ on Y_Ω . For any fixed- b right inverse Q_b^* of D_b^* ,

$$Q_{s,b} = Q_b^* - Q_\Omega(I + K_{s,b}Q_\Omega)^{-1}K_{s,b}Q_b^*$$

is a right inverse of $D_{s,b}$. No uniform bound for Q_b^* on arbitrary sources is being assumed.

A complex basis of the reference kernel is

$$S_b e_1, \quad \frac{S_b e_1}{z-a}, \quad \frac{S_b e_1}{(z-a)^2}, \quad S_b e_2, \quad S_b e_3.$$

The double zero of σ_1 cancels these apparent poles. This basis is uniformly bounded on Ω , though its global norms may grow. Observe its tangent component at three distinct fixed positive points in the unchanged core neighborhood, away from a , and its two normal components at one of them. In the frame S_b the matrix is a fixed invertible matrix in the powers of $(z-a)^{-1}$, together with I_2 . The localized inverse has bounded values at these points by the preceding interior estimate.

For the real frame W_b underlying this complex basis, the corrected frame

$$J_{s,b} = W_b - Q_\Omega(I + K_{s,b}Q_\Omega)^{-1}K_{s,b}W_b$$

is annihilated by $D_{s,b}$. Its observed matrix differs by $O(|b|)$ from the reference matrix. Surjectivity and index therefore make it a frame of the full ordinary kernel throughout the path. At $s = 0$, (419) and the positive observations show that this is the compact complex orientation.

At $s = 1$ the negative connection and positive seam collar are constant. The operator splits exactly into $C_\alpha^\perp$ and a complex Cauchy–Riemann operator on the closed harmonic sphere. For the smooth map obtained by extending $u_{1,b}$ constantly over the negative disc, a Kähler connection gives such a complex-linear operator; other linearization conventions give its canonical orientation by lower-order homotopy. In the reference identifications, $K_{1,b}$ has no nonharmonic coupling and is complex-linear on the harmonic part. The corrected frame is therefore complex-linear at this endpoint. It has the canonical sphere orientation.

Removing the same complex attaching factor $T_p M_2$ at both ends preserves the comparison. The full-rank parameter set is connected; the complex compact orientation varies continuously on it. Negative rank drop is also regular and is obtained by continuity. This extends the comparison from the small- b family to the stated branch. No extension through positive rank drop or uniform nodal collar is used. $\square$

Proposition 10.16. *Choose the marked genus-two component for the sphere normalization of [DFL26, Remark 4.76]. For every regular residual and each of the four very-stable branches in Proposition 9.8,*

$$(422) \quad \varepsilon_{\text{coh},2,j} = +1, \quad \eta_{2,j} = +1.$$

The corresponding actual finite-center artificial boundary count is $4n(x_0)$. This is a local count in the selected normalization.

Proof. We first record the needed naturality of determinant transport. The construction in Proposition 10.1 works in compact families of smooth configurations at positive scales, using a common finite target stabilization. For two surgeries on disjoint constant collars, retain all pieces in one labelled direct sum and use the signed interior interchanges. After relabelling, the two commutator defects K_1, K_2 have disjoint supports on a common ordinary domain. The family

$$D_{r,t} = D_{0,0} + rK_1 + tK_2, \quad (r,t) \in [0,1]^2,$$

is Fredholm, since both defects are compact. Its edges give the two surgery orders. A common finite stabilization over the square shows that the two orientation transports agree. Inserting an intermediate constant collar gives the same statement for nested cuts. The kernel-first determinant exact sequence cancels intermediate factors associatively. The sphere and attaching factors have even real dimensions.

Consequently, when a mixed cap is constant on its negative side and near its seam, excising it agrees with first removing its constant part and then inserting its positive sphere. The constant nonharmonic block is invertible; its harmonic block and the sphere block carry their complex orientations. The scalar rotations and harmonic matching sequence are complex-linear, so they add no sign.

DFL's selected sphere compatibility applies to an arbitrary mixed residual as follows. Flatten a small matching neighborhood and insert the sphere strictly inside its positive half. Use [DFL26, Lemma 4.64] to deform the residual to a symplectically constant pair. Carry the insertion during this deformation: over the compact path choose a common small matching disc, flatten it in the actual charts, and transport the sphere by a target isotopy along its attaching value. This uses smooth configurations and the sphere's homotopy class; it requires no family of holomorphic spheres. The flattening is trivial at a symplectically constant endpoint.

At that endpoint the residual is a gauge cylinder followed by a constant mixed pair. Move the sphere through the constant positive region into the outgoing strip. Cylinder gluing, mixed-to-strip gluing and sphere insertion have disjoint collars, so the preceding determinant square identifies their compositions. Section 4.7 of [DFL26] defines the strip orientation by gluing to this constant mixed pair, whose operator is invertible. It defines the mixed orientation by the gauge cylinder, compatibly with cylinder composition. Remarks 2.39 and 4.76 of that source make the selected positive generator sphere insertion orientation preserving for these strip orientations. It is therefore orientation preserving for the mixed pair. Transporting back along the simultaneous configuration homotopy gives the residual DFL orientation times the canonical sphere factor. This uses the coherent prescription, not orientability alone.

Now move a full-rank primitive in the chosen branch to the family in Lemma 10.15. That lemma identifies its compact complex orientation with the pure positive-sphere cap orientation. Family naturality of excision commutes with this cap path relative to the flattened residual. Thus its coherent factor is $+1$. This calculation can be made at any fixed positive scale, then transported in the smooth configuration family. It makes no simultaneous limiting assertion in the sphere residue and physical gluing scale. At the original glued solution, Proposition 10.14 and the correction comparison identify this configuration-level transport with the actual gluing orientation.

Negative rank drop follows by continuity in the regular compact line. Each of the four extension lines is a positive generator in the same selected component, so the argument applies to all four using one coherent normalization, independently of the residual. Finally $\sigma_2 = +1$

and $\varepsilon_{\text{exc}} = +1$ give $\eta_{2,j} = +1$ in (373). Each branch has seam degree one, so Proposition 9.8 gives $4n(x_0)$. $\square$

Reversing the single index-four normalization reverses all these relative signs, as in Proposition 10.2. The torus comparison is proved in Proposition 12.7; simultaneous genus-two positivity is proved in Proposition 12.12, including rationally acyclic cases. The global boundary identity is proved below in Theorem 11.5.

10.5. Interior excision and the closed relative orientation. The following two lemmas isolate an ordinary closed orientation calculation relevant to the remaining torus comparison. They do not evaluate the mixed coherent sign. All scales in the excision construction are positive.

Lemma 10.17. *Let $X = \mathbb{P}^1 \times \Sigma_g$, let Λ have degree one, and let E_0 be the pullback of a stable rank-two bundle with determinant Λ . Let E be a primitive extension*

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^1}(1) \boxtimes L \longrightarrow E \longrightarrow \mathcal{O}_{\mathbb{P}^1} \boxtimes (\Lambda L^{-1}) \longrightarrow 0, \quad \deg L = 0.$$

On the relative determinant line of the closed ASD operators Q_E, Q_0 , Donaldson’s orientations for these integral lifts and the Kähler homology orientation satisfy

$$(423) \quad o_E^D \otimes (o_0^D)^* = (-1)^g o_E^C \otimes (o_0^C)^*.$$

The right-hand orientations are induced by the closed Dolbeault operators and transported through the space of connections. The relative sign is unchanged by line twists of either rank-two lift.

Proof. Write h, f for the positive generators pulled back from $\mathbb{P}^1, \Sigma_g$. Then $h^2 = f^2 = 0$, $\langle hf, [X] \rangle = 1$, and $K_X = -2h + (2g - 2)f$. The constant and primitive bundles have first Chern classes f and $h + f$, and second Chern classes 0 and hf , respectively.

The closed Kähler orientation comparison is

$$o_c^D = (-1)^{\langle c^2 + K_X c, [X] \rangle / 2} o_c^C;$$

see [Don87, Proposition 3.25 and its proof, pp. 418–419]. Its exponent is the parity $\chi(\mathcal{L}_c^{-1}) - \chi(\mathcal{O}_X)$ in the reduction calculation, evaluated by Riemann–Roch. The Kähler operator identification and its compatibility with ordinary instanton addition are given in Section 3(c) and Proposition 3.20 of that source. This orientation is defined for the determinant operator without requiring an ASD solution at the chosen connection.

The two exponents are

$$\frac{1}{2} \langle f^2 + K_X f, [X] \rangle = -1, \quad \frac{1}{2} \langle (h + f)^2 + K_X (h + f), [X] \rangle = g - 1.$$

Taking their difference proves (423). In particular, the constant reference contributes a minus sign. For a replacement $c \mapsto c + 2a$, the exponent changes by $\langle 2ca + 2a^2 + K_X a, [X] \rangle$, which is even since K_X is even. This proves the lift assertion. $\square$

Lemma 10.18. *Let C_E, C_0 be the compact primitive and constant mixed operators, with their retained clutching identification. The contraction and interior cut in Lemma 6.3 determine a continuous isomorphism of orientation double covers*

$$(424) \quad \text{Or}(\det C_E \otimes \det Q_0) \cong \text{Or}(\det C_0 \otimes \det Q_E).$$

It is compatible with homotopies of those choices on a fixed small parameter chart. Combined with Proposition 10.1, it gives

$$(425) \quad \text{Or}(\det D_{\text{gl}} \otimes \det Q_0) \cong \text{Or}(\det D_{\text{res}} \otimes \det Q_E).$$

The construction retains the original framing; independence from another clutching component is not asserted.

Proof. Transport the compact determinant along the configuration path which makes the positive map constant and the negative collar equal to the attaching flat connection. The boundary identification is transported, and is not extended over the negative hemisphere. The closed primitive Pontryagin number therefore remains -2 .

Cut on a circle strictly inside this negative collar, and cap both negative interiors with the same ordinary cap. The signed two-by-two rotation in the proof of Lemma 6.3 interchanges these interiors in the sum of one mixed and one closed operator. It is the identity near the complete mixed seam and on the positive component. Consequently it preserves the actual mixed Sobolev domain. After the source and target identifications, the two operators differ by a smooth zero-order commutator K , compact from H^1 to L^2 .

Use the Fredholm path $D_t = D_{\text{start}} + tK$. A finite-dimensional target supplement F makes (D_t, ι_F) onto for every t : take a finite sum of local cokernel supplements over the compact interval. Its kernels form a vector bundle, and the exact sequences give

$$\det D_t \cong \det \ker(D_t, \iota_F) \otimes (\det F)^*.$$

Orientation transport in this bundle gives (424), including any cokernel crossings. Enlarging F cancels the additional factor in the exact sequence. The same construction over a square proves invariance under the stated homotopies. It requires no regularity along the flattening or the compact-perturbation path.

The actual mixed excision gives $\det D_{\text{gl}} \otimes \det C_0 \cong \det D_{\text{res}} \otimes \det C_E$ on orientation double covers. Tensor with $\det Q_0$, apply (424), and cancel $\det C_0$. This proves (425). The compact and closed indices are $6g - 6$ and $6g - 2$, all even, so these cancellations introduce no odd interchange. For two compatible surgery orders, retain all pieces in one labelled sum and use the Fredholm square $D_{0,0} + rK_1 + tK_2$ on disjoint collars. Its stabilized determinant has trivial boundary monodromy. Nested cuts are separated by an intermediate constant collar, whose determinant then cancels. This proves the required composition compatibility. $\square$

To specify the remaining comparison, put

$$L_{\text{mix}} = \det C_E \otimes (\det C_0)^*, \quad L_{\text{cl}} = \det Q_E \otimes (\det Q_0)^*, \quad L_{\text{gl}} = \det D_{\text{gl}} \otimes (\det D_{\text{res}})^*.$$

Let $J : L_{\text{mix}} \rightarrow L_{\text{cl}}$ and $G : L_{\text{gl}} \rightarrow L_{\text{mix}}$ be the two orientation transports just used. Give L_{mix} its regular-kernel complex orientation and L_{cl} the closed complex orientation. Define the signs $\delta_{\text{cap}}, \delta_{\text{ref}}$ by

$$J(o_{\text{mix}}^{\mathbb{C}}) = \delta_{\text{cap}} o_{\text{cl}}^{\mathbb{C}}, \quad (JG)(o_{\text{gl}}^{\text{DFL}}) = \delta_{\text{ref}} o_{\text{cl}}^{\text{D}},$$

where the DFL orientation system is held fixed. Lemma 10.17 gives

$$(426) \quad \varepsilon_{\text{coh},g} = (-1)^g \delta_{\text{cap}} \delta_{\text{ref}}.$$

Since $\sigma_1 = -1$ and $\varepsilon_{\text{exc}} = +1$, the desired torus boundary sign is $\eta_1 = \delta_{\text{cap}} \delta_{\text{ref}}$. Neither new sign is evaluated by these lemmas. They are locally constant on the specified regular framed families; changing J by a sign changes both and leaves their product unchanged.

In particular, the smooth kernel identification in Proposition 6.4 does not compute δ_{cap} along a flattening path which need not consist of solutions. The mixed boundary domain is not complex-linear, even though the interior rotation is scalar. A determinant comparison compatible with the compact Dolbeault orientation is needed; the next subsection constructs one in genus one. Likewise, δ_{ref} compares (425) with DFL's reference-cylinder prescription

in the chosen ordinary gauge normalization. The closed parity calculation does not perform that comparison.

10.6. The compact complex orientation in genus one. The first sign in (426) can be evaluated in genus one. We use an integrable flattening and rotation, with a coupled boundary problem for the gauge potential. The connections in the flattening need not be ASD. For an integrable connection family on the excision sum, write F_r for the holomorphic coefficient bundle on $\mathbb{P}^1 \times T$ obtained by adjoining the omitted constant positive adjoint summand, where $T = \Sigma_1$. It has complex rank six.

Lemma 10.19. *Let $T = \Sigma_1$ and let B be the stable attaching bundle. Replace the compact re-constructed connection by an integrable unitary connection on the same holomorphic $\text{End}_0 E$, with the same flat surface restriction at the seam. Its compact mixed operator satisfies*

$$\ker C_A \cong H^1(\mathbb{P}^1 \times T, \text{End}_0 E), \quad \text{coker } C_A = 0.$$

The identification is continuous along smooth families of such connections at positive scale. The same assertion holds along the integrable interior-rotation family of $C_E \oplus Q_0$, with its holomorphically capped rank-six bundle F_r in place of $\text{End}_0 E$.

Proof. Put $F_\alpha = \text{End}_0 B$. Stability and Serre duality give $H^0(T, F_\alpha) = H^1(T, F_\alpha) = 0$. The flat real adjoint cohomology in degrees zero and one also vanishes. Thus the mixed conditions are

$$a_{\mathbf{n}} = 0, \quad a_\Sigma = d_\alpha \xi$$

for a real adjoint zero-form ξ , with no positive map variation.

For the direct sum, retain both adjoint summands on the negative hemisphere and the second on the positive hemisphere. Omit the first positive summand and impose the preceding mixed condition on its boundary. The second summand has ordinary transmission across the seam. The connection preserves this splitting near the seam but may mix the summands strictly inside the negative hemisphere. Adjoining the omitted holomorphic cap gives F_r . Its surface fibers in a fixed collar are $F_\alpha \oplus F_\alpha$.

For either capped bundle F , a global holomorphic section vanishes on that collar. The same holds for $F^* \otimes K_X$, since K_T is trivial. Therefore $H^0(X, F) = H^2(X, F) = 0$. The real index is four for the primitive and its direct sum with Q_0 , and zero for the constant operator. For the sum it is unchanged by interior zero-order perturbations. In each case it equals $-2\chi(F) = 2h^1(X, F)$.

Write a complex zero-form as $\phi = \chi + i s$ in the unitary real form. On the omitted summand impose

$$(427) \quad s|_\Gamma = 0, \quad \nabla_{\mathbf{n}} \chi|_\Gamma = 0.$$

Use ordinary transmission of the value and covariant normal derivative on the retained summand. The real operator $L = \bar{\partial}^* \bar{\partial}$ has scalar Laplace principal symbol. These are elliptic Dirichlet and covariant Neumann conditions on its imaginary and real parts. All curvature terms are retained; the potentials are not decoupled.

The omitted-summand Green boundary term, up to a fixed positive normalization, is

$$\text{Re} \int_{\Gamma \times T} \langle \phi, \nabla_{\mathbf{n}} \psi + i \nabla_{J\mathbf{n}} \psi \rangle.$$

The trace of ϕ is real. The normal derivative of ψ is purely imaginary, and its tangential derivative has real trace. Thus both displayed terms have zero real part. Transmission terms

cancel. Consequently L is self-adjoint as a real elliptic operator and

$$(428) \quad \operatorname{Re}(L\phi, \phi) = \|\bar{\partial}\phi\|^2.$$

A kernel element is holomorphic. Its omitted seam trace is a holomorphic section of F_α , hence zero. Extending it by zero on the omitted cap gives a global holomorphic section of F , which vanishes. Thus L is invertible. Mixed-boundary regularity gives smooth solutions and a continuous inverse in the fixed-scale smooth families.

The Kähler identities identify the real operator on one-forms with $\bar{\partial}^* \oplus \bar{\partial}$ on $a^{0,1}$, in the fixed target convention; see [Don87, Section 3(c), pp. 412–413]. Integrability makes this a Dolbeault complex; $\Lambda F_A = 0$ is unnecessary. The identity also holds for the real orthogonal connection on the sum.

Double a smooth kernel element by subtracting $\bar{\partial}$ of an extension of ξ on the omitted side and setting the corrected form to zero on the positive cap. The vertical traces agree, so there is no distributional Dolbeault seam term, as in Proposition 6.4. The retained summand continues across the seam. This defines a class in $H^1(X, F)$.

If the class is zero, its potential vanishes on the omitted positive cap. Undoing the ξ correction gives a potential $\phi = \chi + is$ with real seam trace. The real one-form associated to $\bar{\partial}\phi$ is $d_A\chi + d_A^c s$, where $d_A^c = i(\bar{\partial}_A - \partial_A)$. Its second term has normal component equal, up to sign, to a tangential derivative of the zero trace of s . The zero normal condition therefore gives (427). Co-closedness gives $L\phi = 0$, and the kernel element vanishes. The same mixed elliptic problem regularizes weak potentials.

Conversely, a closed Dolbeault representative v is exact on the omitted cap and a collar: solve $\bar{\partial}_\Sigma \zeta = v_\Sigma$ using $H^0 = H^1 = 0$ on each surface fiber. The cocycle equation makes $v_B - \bar{\partial}_B \zeta$ a vertical holomorphic zero-form, hence zero. Subtract a cutoff extension of ζ so that the omitted component of v vanishes on the positive cap and near the seam. Solve $L\phi = \bar{\partial}^* v$ with (427). Then $v - \bar{\partial}\phi$ is closed and co-closed, and its real one-form satisfies the mixed and transmission conditions. Doubling returns the original cohomology class. This proves the kernel isomorphism, and the index calculation proves surjectivity.

The vertical inverses, the inverse of L , and the closed Hodge projections vary continuously in these families. The capped cohomology dimensions are constant by the preceding vanishing. This proves the family assertion. $\square$

Proposition 10.20. *In genus one, the framed interior excision of Lemma 10.18 can be chosen with*

$$(429) \quad \delta_{\text{cap}} = +1.$$

For this comparison, $\varepsilon_{\text{coh},1} = -\delta_{\text{ref}}$ and $\eta_1 = \delta_{\text{ref}}$.

Proof. The stable complex slice has dimension zero in genus one, and its trace-free base coefficient vanishes. The holomorphic adjoint bundle on a stable seam collar is therefore the pullback of F_α . The local projective identifications form a covering with discrete automorphism group. The given seam identification is a section over the circle, so extends over its collar; local uniqueness makes this extension holomorphic. This retains the original framing without extending it over the unstable negative fiber.

In this identification, the reconstructed and product adjoint metrics agree at the seam by uniqueness of the NS metric. Interpolate Hermitian metrics relative to that boundary value to make the metric product on a smaller collar. Their Chern connections are integrable and retain the flat surface boundary value. The positive map is already constant. Lemma 10.19

transports the compact complex orientation through this flattening. For the capped cohomology one may extend each negative metric smoothly; its positive extension need not be product during the interpolation.

On the interior cut annulus both connections are the product α . Write the rotation as $R = e^{\theta S}$, where

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \otimes \text{id}_{F_\alpha},$$

and θ depends only on the base. After the bundle identification, the connection difference is $S d\theta$, up to its fixed conjugation sign. Interpolating this term realizes the first-order commutator path of the excision. It remains integrable because

$$d(S d\theta) = 0, \quad (S d\theta) \wedge (S d\theta) = 0, \quad [\alpha, S] = 0.$$

It is a real orthogonal connection on the sum. A Lie algebra structure on the mixed interior summands is not required, and their seam splitting is unchanged.

Apply the partially capped case of Lemma 10.19. Every kernel is identified continuously with the complex $H^1(X, F_r)$ and every cokernel vanishes. At the bundle identification, R intertwines the conjugated Dolbeault operators complex-linearly. It is the identity near the omitted seam, so it also intertwines the doubling maps. The determinant transport sends

$$o_{C_E}^{\mathbb{C}} \otimes o_{Q_0}^{\mathbb{C}} \quad \text{to} \quad o_{C_0}^{\mathbb{C}} \otimes o_{Q_E}^{\mathbb{C}}.$$

All real factor indices are even, so interchange and cancellation introduce no odd permutation. This proves (429). The remaining assertions follow from (426). $\square$

The compact complex-orientation comparison is therefore proved on the specified positive-scale framed torus patches. Agreement with DFL’s fixed ordinary gauge reference, namely $\delta_{\text{ref}} = +1$, is proved in Section 12.4.

11. COMPACTIFICATION OF THE POINT CUT AND ITS BOUNDARY

We exclude energy loss down the mixed ends and ordinary positive bubbles using the chosen incidence conditions. Every remaining bubbling end then lies in a regular primitive gluing neighborhood. Finite regularization of the compact complement gives Theorem 11.5, with the coherent orientation signs retained.

11.1. Exclusion of energy loss on the mixed ends. Fix regular residual moduli spaces through index three, with no negative-index solutions, and regular ordinary Floer data. Use the compact secondary perturbations and dependency convention of Section 9.1. The union $\mathcal{C}$ of the index-zero residual moduli spaces for the finite sets of ordinary generators is finite by the index-zero compactness argument of [DFL26, Section 5.4]. Choose the point sections as in the proof of Proposition 8.25: they are independent on every completed residual insertion path and at its ordinary critical limits. The negative sections use finite compact holonomy dependencies, and the seam sections use the actual universal identification and normal parallel transport. These independence conditions can be imposed in every genus: their parameter spaces have dimension at most one and the rank-one stratum has real codimension four. In genus two retain also the line-avoidance conditions of that proposition.

Proposition 11.1. *For this choice of sections, every sequence in the index-four point cut has a subsequence convergent on both mixed ends in the weighted topology of [DFL26, equation (5.36)], including its exact critical representatives. In particular, a concentration center cannot escape to infinity along the matching line in the point cut.*

If such a sequence has a finite matching atom, the atom has charge one on exactly one surface component, exhausts the global defect, and has an index-zero residual with the original ordinary generators. There are no other atoms, ordinary bubbles or nonconstant ordinary Floer levels. The physical outer and end hypotheses of Theorem 8.19 then hold.

Proof. Apply local mixed compactness and ordinary Floer chain compactness, before imposing an index-based exclusion. The global defect is (11). On either mixed end, Proposition 5.29 and equations (5.36)–(5.37) of [DFL26] give a subsequential alternative: weighted convergence, including the critical representative, or loss of a strictly positive amount ϵ of topological energy. Only positivity of ϵ is needed.

Suppose the latter alternative occurs on at least one mixed end. Then $\Delta > 0$, and Proposition 3.1 gives $\Delta = \frac{1}{2}$, an index-zero mixed residual $x_0 \in \mathcal{C}$, and no nonconstant ordinary levels. Thus x_0 has the original ordinary generators. By Theorem 3.2, any finite matching atom costs at least $\frac{1}{2}$. An ordinary sphere or disc costs at least $\frac{1}{2}$, and a gauge bubble costs at least one; the same bounds hold on the translated ordinary ends. These nonnegative losses add on disjoint regions in the energy comparison. Any one of them together with the positive mixed-end loss would exceed $\Delta = \frac{1}{2}$. There are consequently no finite atoms and no ordinary-end bubbles.

Local compactness now gives smooth convergence on every compact gauge dependency region and positive domain region, including the seam crossing. The ordinary ends have neither bubbles nor nonconstant levels. Ordinary Floer chain compactness gives energy tightness there: otherwise a translated nonconstant level or a bubble would be extracted. The small-energy entry and exit argument and the critical spectral gap, as in Proposition 8.18, give uniform ordinary-end convergence. This use of the ordinary ends does not presume convergence on the mixed ends.

Pass to a subsequence in the insertion parameter. If t_ν has a finite limit, the evaluated sections converge to the independent residual pair. For a negative insertion, convergence holds on the full compact holonomy dependency; at the crossing it respects the actual seam identification and normal transport. If $t_\nu \rightarrow -\infty$, the prescribed gauge-end pullback tends to the independent pair at the incoming critical point. If $t_\nu \rightarrow +\infty$, the positive evaluation tends to the independent pair at the outgoing critical point. Each case contradicts the point incidence. Hence both mixed ends satisfy the convergence alternative. When $\Delta = 0$, that alternative follows immediately from the strict-loss inequality.

Finally, if a finite matching atom occurs, its local half-unit lower bound and index exhaustion force it to be the unique charge-one atom on one component and to consume the whole defect. The other losses vanish. The small-tail argument of Proposition 8.18 supplies convergence on all ends, and Theorem 8.19 supplies the outer data under its stated algebraic hypotheses. $\square$

The compactness inequalities used here occur before the final index-at-most-one contradiction in [DFL26, Section 5.4, pp. 71–74]. The proof replaces that contradiction by Proposition 3.1; it does not extend the stated regularity range of DFL Proposition 3.23. It proves no classification of positive bubbles when there is no mixed-end energy loss. For a finite matching atom, Theorem 8.26 now applies with its remaining low-genus and section hypotheses, without separately assuming the global defect or ordinary-end convergence of the incoming sequence.

11.2. Interior evaluation and ordinary sphere attachments.

Lemma 11.2. *At a smooth regular index-zero residual $x = (A, u)$, evaluation at finitely many distinct positive interior points is jointly submersive under the compact negative holonomy variations of Section 9.1. These evaluations may be combined with finitely many seam evaluations. The almost complex structure may remain fixed.*

Proof. For one interior point z , write $K_z \xi = \xi_u(z)$. It is bounded on the H^2 mixed domain and onto on smooth compact positive configuration variations. Suppose (λ, ν) annihilates $(D_2 \xi + P_x \dot{\eta}, K_z \xi)$. For a compact domain-preserving cutoff χ zero near z , the compatible lower inverse gives

$$(\chi \lambda)(f) = -\lambda([D, \chi] D_2^{-1} f), \quad |(\chi \lambda)(f)| \leq C_\chi \|\lambda\|_{(\mathcal{Y}^1)^*} \|f\|_{\mathcal{Y}^0}.$$

The differential and nonlocal commutators have the bounds proved in Proposition 9.1. Thus λ is locally L^2 off z , including every seam point. The same Coulomb splitting, holonomy separation and interior unique continuation remove its scalar and self-dual components. The weak-adjoint matching theorem gives zero positive adjoint trace on a seam arc. Zero extension and unique continuation remove that adjoint on the connected positive domain punctured at z . Local L^2 representation excludes residual boundary functionals.

The remaining functional is supported at z . In a local positive coordinate choose χ_ϵ equal to one for $r \leq \epsilon^2$, zero for $r \geq \epsilon$, and logarithmic between them, with a smooth approximation. Then

$$\|\chi_\epsilon\|_2 = O(\epsilon), \quad \|d\chi_\epsilon\|_2^2 = O(|\log \epsilon|^{-1}).$$

For smooth compact f , support at z gives $\lambda(f) = \lambda(\chi_\epsilon f) \rightarrow 0$. Thus $\lambda = 0$, and onto configuration evaluation gives $\nu = 0$. Subtracting the graph of the invertible D_2 reduces the range question to a finite-dimensional target, so this proves surjectivity.

For several evaluations, use cutoffs near their finite union, the same interior point argument, and the seam capacity argument of Proposition 9.1. The positive domain minus finitely many points remains connected. Disjoint compact configuration variations make the joint evaluation onto. $\square$

The compactness and finite-direction argument of Proposition 9.1 now makes residual evaluations on a compact positive insertion interval avoid the genus-two exceptional set B and the higher-genus images $i_g(N_g)$, simultaneously in all factors. The seam endpoint uses the seam version of the lemma. Small variations preserve previously achieved avoidance on the completed seam. This imposes no avoidance condition on the outgoing ordinary critical points.

Let $z_t = (t, 0)$ lie where compact variations of the positive almost complex structure are allowed. For a simple $J(z_t)$ -holomorphic sphere F with nodal point a and $F(a) = u(z_t)$, the universal coupled derivative is

$$(430) \quad (\xi, \zeta, \dot{\eta}, Y) \longmapsto (D_x \xi + P_x \dot{\eta} + R_x Y, D_F \zeta + \tfrac{1}{2} Y(z_t, F) dF j, \zeta(a) - K_{z_t} \xi).$$

This operator is onto. Indeed, simple-sphere universal transversality solves the second row using (ζ, Y_0) ; one may also prescribe $\zeta(a)$ [OZ09, Theorem 6.1]. Extend Y_0 by a compact domain cutoff equal to one at z_t . Lemma 11.2 solves the first and third rows, including the full residual source $-R_x Y$, using $(\xi, \dot{\eta})$. The holonomy variation has no sphere component. Quotienting by sphere reparametrization preserves surjectivity. For $c_1(F) = 2$, the resulting one-nodal-mark family, including the real parameter t , has dimension

$$0 + (2n - 6 + 4 + 2) - 2n + 1 = 1, \quad 2n = \dim_{\mathbb{R}} M.$$

One further point moving on the sphere gives parameter dimension three. We next verify the compactness and end conditions needed to use this family for incidence exclusion.

Proposition 11.3. *For a fixed insertion point q , the low-index regular equation data and the sections in Proposition 11.1 can be chosen so that no index-four point-cut sequence has an ordinary positive sphere or outer-boundary disc bubble. This includes bubbles escaping down the outgoing ordinary end. The fixed standard complex matching collar and completed-seam residual avoidance are retained.*

Proof. Write $2n = \dim_{\mathbb{R}} M$ for the full product target. Every nonconstant compatible holomorphic sphere has normalized area at least $\frac{1}{2}$. A sphere of area $\frac{1}{2}$ has $c_1 = 2$ and is simple: a nontrivial multiple cover would have a nonconstant underlying sphere of smaller positive area. Simple spheres are somewhere injective [Zin17, Proposition 4.11]. Gromov compactness for varying target structures [IS99, Theorem 1.1] shows that a sequence of one-marked minimal spheres has a stable limit with exactly one nonconstant component. There is no stable constant tree with only one external mark and one nonconstant component. Thus the one-marked moduli space is proper over compact parameter families, including a prescribed continuous nodal value. For two marks, their collision is recorded by allowing the second point to equal the node on the nonconstant sphere. Its evaluation remains continuous.

There are only finitely many possible homology classes under this energy bound in a compact family of compatible structures. Indeed, integration of any fixed closed two-form is bounded by a constant times the energy. A finite integral cohomology basis bounds the nontorsion class in a lattice, and the torsion subgroup is finite. No product condition on a varying J is used.

First choose the ordinary family J_θ generically also for simple-sphere nodal evaluation at each of the finitely many outgoing critical points β , for J_0 . Theorem 6.1 of [OZ09] supplies this generic condition. A variation of J_0 extends to J_θ by a cutoff equal to one at $\theta = 0$, so it can be imposed together with ordinary Floer regularity as in [DFL26, Lemma 2.35]. The constrained one-marked sphere spaces have dimension $(2n - 6 + 4 + 2) - 2n = 0$. They are compact and hence finite. Their constrained operators modulo reparametrization are isomorphisms. The implicit function theorem continues their elements uniquely to nearby nodal values and structures, and properness excludes additional nearby spheres.

Make this choice small, interpolate it into the positive equation away from the fixed matching collar, and keep the far outgoing end translation invariant. The ordinary generators remain fixed and their spectral gaps remain bounded below. The regular index-zero residual set $\mathcal{C}$ continues under these small data changes. To see exhaustion, a sequence of additional index-zero solutions with data tending to the original data would have a residual for the original data. Positive loss would force index at most -4 , and a nonconstant ordinary level would force negative residual index. Both are excluded. The sequence must therefore converge to an original regular index-zero solution, where the implicit function theorem applies.

On the compact part $0 \leq t \leq a$ of the insertion path in the fixed J_* collar, use Lemma 11.2 and its seam version to avoid the exceptional sets in every target factor. Genus-one factors have no spheres. In higher genus, minimal standard spheres attach through $i_g(N_g)$ and so cannot occur at these values. In genus two, the four regular extension lines have, including the constant target factors,

$$F^*TM = \mathcal{O}(2) \oplus \mathcal{O}^{n-1}, \quad H^1(F^*TM(-a_{\text{node}})) = 0.$$

Thus the sphere equation with nodal evaluation is onto at fixed J_* . Modulo automorphisms preserving the node it is an isomorphism. Compactness gives a buffer beyond the fixed collar

on which the matched sphere problem remains regular. Small changes retain the completed-seam avoidance already established.

Consider the matched one-marked family

$$(431) \quad \mathcal{S} = \left\{ (x, t, [F, a_{\text{node}}]) : \begin{array}{l} x \in \mathcal{C}, \ t \geq 0, \ \bar{\partial}_{J(t,0)} F = 0, \\ \mathcal{E}(F) = \frac{1}{2}, \ F(a_{\text{node}}) = u_x(t, 0) \end{array} \right\}.$$

Simplicity makes the reparametrization action free; local slices can be chosen at three distinct injective points. This family is regular in the fixed collar. It is regular for all sufficiently large t by the finite continuation branches at β , uniform end convergence of $\mathcal{C}$, and $J(t, 0) = J_0$ on the far end. On the remaining compact interval, the universal operator (430) is onto. Properness and finiteness of $\mathcal{C}$ give a compact set of background matched spheres. A finite cover supplies one finite set of compact positive J variations and compact negative holonomy variations making the universal problem regular. Enlarge the permitted compact positive support if necessary. Properness excludes extra solutions outside these neighborhoods for small parameters. Sard’s theorem now makes (431) regular on the middle interval as well, while preserving the regular buffers.

The resulting $\mathcal{S}$ is a smooth one-dimensional manifold with boundary at $t = 0$, proper over $[0, \infty)$. Its projection to t may have critical points. On a sufficiently long tail it consists of finitely many graphs continued from the matched spheres at the ordinary critical points. Adding those limiting spheres gives a compact completion. The universal sphere $\mathcal{W} \rightarrow \mathcal{S}$, with an additional point b allowed to equal a_{node} , has dimension three and evaluation

$$(x, t, [F, a_{\text{node}}], b) \longmapsto (F(b), q).$$

Its completion adds two-dimensional end spheres, and the evaluation image is compact. Evaluation continuity includes colliding marks [Zin17, Section 1.3]. No smooth coordinate at $t = \infty$ is required.

Choose finite global section variations spanning the complex rank-three evaluated bundle. Two independent coefficient rows span the pair fiber everywhere. Parameterized Sard on $\mathcal{W}$, its boundary, and its two-dimensional end strata gives a pair σ^S independent on all their images, since the rank-one stratum has real codimension four. Use a countable coordinate cover of the noncompact smooth $\mathcal{W}$. Impose simultaneously independence on the completed residual insertion paths and critical limits. Their union with the sphere images is compact, so the wedge has a positive minimum. At $t = 0$ the sphere family includes the genus-two line families required by Proposition 8.25. The gauge and compact middle sections are then chosen by that proposition with the same holonomy dependencies and seam identification.

Suppose now that an actual point-cut sequence has an ordinary positive bubble. Index exhaustion gives a residual in $\mathcal{C}$, the original ordinary generators, no nonconstant ordinary levels, and total defect $\frac{1}{2}$. The positive bubble consumes this defect, leaving no gauge or matching atom and no second nonconstant bubble. The negative compact dependency and incoming ordinary end converge to the residual, so every negative, seam, or incoming-end insertion limit sees an independent pair.

A bounded positive insertion away from the bubble sees the independent residual pair. If it enters an interior sphere scale, the node lies on the insertion path and marked Gromov convergence places its limiting value on that minimal sphere, including its nodal value. These are precisely the evaluations of $\mathcal{W}$. The actual positive universal bundle and sections are pullbacks by target evaluation, so the limiting pair is again independent. The insertion path is locally separated from the outer Lagrangian boundary; it cannot enter a boundary-disc scale or a sphere attached at that boundary.

If $t_\nu \rightarrow +\infty$, translate by the insertion parameter and use ordinary chain compactness. There are no nonconstant strip levels, so the residual strip is the constant β . The insertion value tends either to β or to a point on a minimal J_0 sphere attached at β . Both are in the avoided sets. A boundary disc remains separated from $\theta = 0$. Every case contradicts the point incidence. $\square$

This is a choice for fixed q , or for a prescribed finite set of insertion points. Parameter families varying q require their own argument. The proof establishes neither index-four regularity of the full mixed moduli space nor an ambient nodal gluing theorem.

11.3. Exhaustion of the bubbling ends. The point bundle for $q \in \Sigma_q$ is the pullback of the universal adjoint point bundle from the marked target factor $M_{g(q)}$. Choose the positive sections in Proposition 11.3 from this factor. Their coefficient variations still span every pair fiber on the sphere evaluation family, so its avoidance proof is unchanged. Make the same choice for every seam-dependent section and model refinement in Section 9.2. Its equivariant extension uses the marked surface connection and the actual normal transport. The compact interior holonomy dependencies remain at positive distance from the matching line. In genus one the positive factor is a point and its pair is a fixed independent pair.

Proposition 11.4. *For these fixed equation and point-section choices, the only positive-defect limits of the index-four cut are the covered charge-one primitive ends on the marked component, at the insertion crossing, in genus one or two. There are none when the marked component has genus at least three. After removing the finite collection of small-scale neighborhoods of Proposition 9.8, every sequence in the remainder has zero energy defect after ordinary chain extraction.*

Proof. First suppose a finite matching atom lies on a component different from Σ_q . By index exhaustion and local charge it is the unique charge-one atom and consumes the entire half-unit defect. The residual has index zero, the original ordinary generators and full end convergence. The local energy measure on Σ_q has no defect. Restricting the mixed equation to that factor in the fixed product collar, local compactness and regularity give smooth convergence of the marked surface connection and map through the apparent full-target concentration point. All compact interior holonomy dependencies and normal segments at q are disjoint from the atom. The positive sections depend only on the marked map component, and the seam-dependent negative sections depend only on the marked connection and normal transport. Every bounded insertion therefore sees the independent residual pair. Full end convergence treats both insertion infinities. This contradicts incidence.

We also verify higher-genus attachment support without a regular normalized primitive limit. Let p be the residual value at a finite matching atom on a component of genus h . The finite doubled families and coherent charge of Section 4.4, with the small caps used in the physical converse, have alternatives $(d_\nu, \ell_\nu) = (0, 1)$ or $(1, 0)$. In the first, the stable replacement has vanishing area and boundary values tending to p on a shrinking retained disc. Holomorphic monotonicity makes its entire image tend to p . At the modification point its stable fiber contains the degree-zero quotient line from the Langton step. Its class belongs to $i_h(N_h)$. This image is closed, since N_h is compact, so $p \in i_h(N_h)$.

In the second alternative the stable classifying maps are holomorphic on the fixed doubled disc and converge to the residual map on punctured compact annuli. Their capped degree and cap convergence give concentrating area exactly $\frac{1}{2}$. Ordinary holomorphic-map compactness extracts a sphere tree through p . Minimal spherical area leaves exactly one nonconstant sphere. Muñoz's extension-line classification puts its image in $i_h(N_h)$, so again $p \in i_h(N_h)$.

This uses the cap only for the degree and area comparison; it does not assert that the cap is holomorphic. For $h \geq 3$ the completed residual seam avoids $i_h(N_h)$ by Proposition 9.2. Such an atom is impossible.

Now take any positive-defect cut sequence. Index exhaustion gives defect $\frac{1}{2}$, an index-zero residual and no nonconstant ordinary levels. Propositions 11.1 and 11.3 exclude mixed-end loss and ordinary positive bubbles; the energy bound excludes gauge bubbles. Comparing energies on compact truncations and then using the tight ends leaves a finite matching atom. Local charge makes it unique. The preceding arguments put it on the marked component in genus one or two.

Proposition 11.1 supplies the physical defect and end hypotheses. Global residual avoidance and the line-independent positive pair supply the remaining genus-two conditions of Theorem 8.26. In genus one that theorem applies before incidence. The section limits of Section 9.2 then put the center at the crossing, bound the normalized center ratio, and place the insertion in a compact negative middle interval. The normalized limit is a zero of the compact relative model. Its transverse refinement is still available with the factor-dependent sections, since their coefficient variations span every pair fiber.

There are finitely many residuals and model zeros. The neighborhoods in Proposition 9.8 therefore exhaust every such sequence, with its scale tending to zero. A sequence remaining after all regions $0 < \rho < \epsilon$ have been removed cannot have positive defect. $\square$

The artificial boundary counts remain $8n(x_0)\eta_1(x_0)$ and $n(x_0) \sum_j \eta_{2,j}(x_0)$. For the selected genus-two normalization the latter is $4n(x_0)$. A zero-defect remainder may still have ordinary breaking or an escaping insertion; its global boundary formula does not follow from the proposition alone.

11.4. The regular cut and its global boundary. For an ordinary Floer trajectory of index r , let Q_r^G and Q_r^S denote the point cuts at the zero slice. These use parameterized trajectories and have dimension $r - 4$ when transverse. Write H_r for the mixed point cut with its insertion coordinate, of expected dimension $r - 3$. Thus $H_3 = \mathcal{L}_0^q$ and $H_4 = \mathcal{L}_1^q$.

Theorem 11.5. *For fixed q , the equation and section choices of Proposition 11.4 can be made so that the index-four point cut is regular. Its truncation by the finite primitive neighborhoods compactifies to an oriented one-manifold whose ordinary boundary consists of*

$$(432) \quad \begin{aligned} H_3(\alpha, \gamma) \times \check{\mathcal{M}}_1^S(\gamma, \beta), \quad \check{\mathcal{M}}_1^G(\alpha, \gamma) \times H_3(\gamma, \beta), \\ \mathcal{M}_0(\alpha, \gamma) \times Q_4^S(\gamma, \beta), \quad Q_4^G(\alpha, \gamma) \times \mathcal{M}_0(\gamma, \beta). \end{aligned}$$

Their product orientation signs are respectively $+, +, +, -$. The other boundary points are the artificial primitive scale slices. Consequently the actual chain counts satisfy

$$(433) \quad Nb_q^G - b_q^S N = d_S K_q^{p_1} + K_q^{p_1} d_G + B_q,$$

where

$$(434) \quad B_q(\alpha) = \sum_{\beta} \sum_{x_0 \in \mathcal{M}_0(\alpha, \beta)} n(x_0) \begin{cases} 8\eta_1(x_0), & g(q) = 1, \\ \sum_{j=1}^4 \eta_{2,j}(x_0), & g(q) = 2, \beta. \\ 0, & g(q) \geq 3 \end{cases}$$

Regularity is asserted at the index-four cut points, not at every uncut index-four solution.

Proof. First make the lower mixed cuts, their $t = 0$ restrictions, and the ordinary point cuts through index four transverse. The positive and seam coefficient variations on the marked factor span every pair fiber. On the incoming end use finite holonomy frames on the

compact set of relevant slice evaluations. Ordinary gauge bubbling costs eight index units and cannot occur here; chain compactness makes this set compact, with irreducible limits. Irreducibility of trajectory slices and of the sufficiently negative mixed slices is proved in [DFL26, Section 7.2, pp. 89–90]. The compact middle restrictions through mixed index three are likewise compact after ordinary breaking, since no bubbling occurs. They admit finitely many interior holonomy frames. Together with the seam and end coefficients these span the pair fiber along the full interpolation.

Parameterized Sard on countably many smooth coordinate charts makes all the indicated cuts transverse, including the regular positive-scale primitive charts. Smallness retains the independent sphere and residual images and all model side boundaries. Rank zero is absent by its real codimension twelve. The dimension count gives

$$H_r = \emptyset \quad (r \leq 2), \quad Q_r^G = Q_r^S = \emptyset \quad (r \leq 3).$$

The zero-manifolds H_3, Q_4^G, Q_4^S are compact. For H_3 , a broken limit with finite insertion would lie on an H_r with $r \leq 2$; an escaping insertion would lie on an ordinary Q_r with $r \leq 3$ or at an independent critical pair. Bubbling is excluded in index three. For Q_4^G , a nontrivial broken limit puts the insertion on a lower-index ordinary level or a critical point. There is no gauge bubble. For Q_4^S the only further possibility is a minimal positive bubble with constant index-zero residual strip. Its visible sphere is attached at a critical point for J_0 , and is excluded by the end-sphere choice in Proposition 11.3. A boundary disc is separated from the insertion.

Remove the small primitive neighborhoods. Proposition 11.4 gives zero defect for every remaining sequence. If r_0 is its mixed residual index and $r_i \geq 1$ are its nonconstant ordinary-level indices, then

$$r_0 + \sum_i r_i = 4.$$

A bounded insertion in a broken limit forces $r_0 \geq 3$, hence $r_0 = 3$ and one ordinary level of index one. An escaping insertion sees either a critical pair or an ordinary level after translation. Critical pairs are avoided. A level carrying the point incidence has index at least four, so it is the sole ordinary level and the mixed residual has index zero. These are exactly (432), and all four sets are finite.

Their ordinary gluing collars are regular. The factors are onto and the intermediate critical point has a fixed gap. The ordinary Floer inverse on kernel complements and the long-neck correction give onto glued operators. This construction uses regular factors, not a numerical index bound [DFL26, Section 5.4 and Proposition 7.12]. For the first two boundary sets, the full compact incidence dependency stays on the mixed factor. For the last two, translate the insertion with the ordinary level and use the prescribed end pullback. Gluing gives convergence of these sections and their first kernel-parameter derivatives on the whole dependency. Transversality of the zero-dimensional factor cut and the implicit function theorem therefore give one half-interval at each boundary point. Ordinary converse gluing covers every sequence with that limit.

After removing smaller ordinary collars as well, the remaining set is compact, without assuming index-four regularity there. Indeed, every sequence has zero defect. A broken or insertion-escaping limit is one of the finite boundary points just listed and forces the sequence into its removed collar. Otherwise weighted compactness gives an unbroken cut solution with finite insertion in the remaining closed set.

At every point of this compact set, the local cokernel separation in [DFL26, Section 6.2, pp. 82–83] supplies finitely many compact holonomy and positive J directions making the universal equation onto. That local argument is independent of the index. Independent section coefficients then solve its normal incidence row. A finite cover gives one finite-dimensional universal family for the compact set, also treating $t = 0$ separately.

We give the uniformity needed to exclude new solutions outside this cover. Keep the ordinary equation data fixed and let a range over a small closed ball in this finite family. Index-zero compactness and the implicit function theorem give all residuals as finitely many smooth branches $x_j(a)$. Their completed seam avoidance and end gaps persist. The sphere independence margin is open: a violating sequence for $a_\nu \rightarrow 0$ would have a marked minimal-sphere limit in the original compact avoided image, including its outgoing-end strata. The same statement holds for the residual and model side-boundary independence conditions.

On the selected finite regular primitive patches all gluing and converse estimates are uniform in a . The local equation and cutoff profiles are unchanged. The compact surface and primitive inverses, their finite observations, and the ordinary residual inverse vary continuously on compact regular parameter sets. Their gaps are positive and norms bounded. Outer coefficients, holonomy bounds and decay estimates are uniform in each fixed order. The existing quantitative Neumann and contraction estimates therefore hold with common constants and the same logarithmic scale powers. At positive scale the parameter maps are smooth by the ordinary implicit function theorem.

For converse sequences with $a_\nu \rightarrow 0$, use the corresponding $x_j(a_\nu)$, including its exact end profiles, as reference. The diagonal outer radii are chosen for convergence to these references; no fixed difference from $x_j(0)$ is divided by a shrinking scale. The finite-family extraction and the uniform observation-correction estimates give the same converse coverage. Thus every positive-defect sequence for nearby data enters the continued primitive neighborhoods. The artificial scale slices are at fixed positive regular scale and stay transverse by smooth dependence. Ordinary gluing gives the analogous uniform statement at the finite ordinary boundary. Any new cut outside these neighborhoods for $a_\nu \rightarrow 0$ would therefore converge to the old compact interior set. The finite universal cover is consequently exhaustive for small parameters.

Sard’s theorem now makes the fixed-parameter equation and incidence derivative onto at every index-four cut point. Its first row is the mixed operator, which is therefore onto at these points. Rank zero is absent. At $t = 0$ the fixed-insertion equation and cut have index zero and are onto. On each side the cut is locally a graph $x = x(t)$. Their values and their section derivatives in the x directions agree at zero by the actual seam identification. They join to an interval with coordinate t , with agreeing induced orientations. No smoothness of the sections across the seam beyond these properties is required.

This produces the compact oriented one-manifold in the statement. The ordinary gluing determinant maps are DFL’s. In Proposition 7.12 of [DFL26], replacing the surface insertion by a point and increasing the trajectory index by two preserves all relevant parities. The incidence normal has dimension four, so commuting it with the boundary normal introduces no sign. Finite-insertion index-one breaking gives the two positive differential terms. For insertion escape, write $t = +T + \tau$ on the outgoing end and $t = -T + \tau$ on the incoming end, with T increasing neck length and τ the ordinary trajectory coordinate. Their Jacobian signs are $+1$ and -1 , giving the last two signs in the statement.

The artificial primitive boundary counts are those of Proposition 9.8, with outward normal $-\partial_\rho$. The signed boundary count of the compact one-manifold is zero, which is (433) with (434). $\square$

For the genus-two component selected in DFL's sphere normalization, Proposition 10.16 gives $B_q = 4N$. For $g(q) \geq 3$, $B_q = 0$. Thus the rational fixed-point comparison is proved with correction 1 in the selected genus-two case and 0 in higher genus. Proposition 12.7 below identifies the torus factors in (434), giving $B_q = 8N$. Proposition 12.12 gives simultaneous genus-two normalization as well. No all-uncut index-four regularity or ambient positive nodal collar is used.

12. COMPARISON OF THE REFERENCE ORIENTATIONS

The signed boundary identity reduces the point comparison to the torus and genus-two orientation factors. The ordinary torus tube and closed reference determine the torus factor. Homologous matching surfaces have the same closed reference; the local sphere comparison and a stabilized torus example then determine both genus-two factors in one coherent system. Theorem 12.13 follows from these local signs and the geometric boundary identity.

12.1. Component signs and simultaneous point choices.

Proposition 12.1. *Fix one coherent DFL orientation system $\mathcal{O}$. For every genus-two component Σ_j , there is a sign $\chi_j(\mathcal{O}) \in \{+1, -1\}$ such that*

$$\eta_{2,k}(x_0) = \chi_j(\mathcal{O}) \quad (1 \leq k \leq 4), \quad B_{q_j} = 4\chi_j(\mathcal{O})N$$

for every regular residual and every point $q_j \in \Sigma_j$. The sign is also the compatibility sign for insertion of a positive generator sphere in this component. Reversing the reference index-four gauge-path orientation reverses all χ_j simultaneously; their pairwise products are unchanged.

Proof. Keep the reference generator, reference paths, their orientations, and the standard S^4 orientation fixed. Let $\mathcal{O}_j$ be the coherent system obtained by selecting Σ_j in [DFL26, Remark 4.76]. The corresponding gauge path represents the index-four class from the reference connection to its image under the distinguished involution. If the original reference has index $8k + 4$, compare it using the prescribed S^4 stabilization. The orientation recipe on pp. 21–22 of [DFL26], using cylinder composition and S^4 gluing, shows that $\mathcal{O}_j$ and $\mathcal{O}$ differ only in the orientation of this one torsor.

In $\mathcal{O}_j$, Proposition 10.16 gives $\eta_{2,k}(x_0) = +1$ for every branch and residual. The normalization law in Proposition 10.2 reverses every residual-relative primitive sign when that torsor is reversed. It accounts for the change of the residual orientation as well. This proves the asserted uniform sign and the formula for B_{q_j} by Theorem 11.5. The same law applied to the index-four sphere insertion identifies its compatibility sign. Reference-generator orientation changes cancel in these ratios. $\square$

Lemma 12.2. *For any fixed finite list of points on Σ , the identities in Theorem 11.5 can be realized with one equation, one pair of ordinary Floer complexes, and one comparison map N . Each point has its own section pair and cut.*

Proof. Impose residual attaching-value avoidance on all components in the list. The matched minimal-sphere families depend on the equation; each point has a separate evaluated adjoint bundle and section pair. The independent section choices of Proposition 11.3 can therefore be made for all these finitely many bundles. Include every prescribed ordinary insertion coordinate in the finite end-regularity requirements. Choose the negative and model pairs separately for each point, retaining its marked-factor dependence.

Take the union of the finite holonomy and coefficient directions for the lower cuts. Sard's theorem makes each cut transverse while preserving all independence margins. After the finite

primitive and ordinary collars are removed, the remaining sets in the proof of Theorem 11.5 have compact union. A finite cover supplies common equation directions and separate section parameters. At a configuration belonging to several cuts, the enlarged family is universally transverse to each cut separately. No transversality of their intersection is needed.

For this specified finite family, the nearby-data exhaustion in that theorem applies to each cut. Intersect its finitely many parameter neighborhoods and use common scale and neck thresholds. Sard for the finitely many projections now gives one final parameter for all cuts. DFL’s surface and loop cuts for a finite generating set can be made transverse for the same data by their index-at-most-three construction in Section 7.2 of [DFL26]. This argument does not concern a continuum of moving point insertions. $\square$

12.2. The torus-only splitting.

Proposition 12.3. *Suppose both matching components have genus one. There is one coherent gauge normalization in which the integral point comparison has correction $+8N$ on both components. The rational correction is $+2N$. If the complexes are nonzero, they have one generator, $N = \text{id}$, the point operations are $b_{q_j}^G = 8\text{id}$ and $b_{q_j}^S = 0$, and every surface and loop operation vanishes.*

Proof. The target $M_1 \times M_1$ is a point. An embedded three-manifold Lagrangian is empty or a single point. Here embeddedness includes injectivity of the restriction map, as required by DFL; it does not discard repeated preimages. If either Lagrangian is empty the assertion is vacuous. Otherwise $C_G = C_S = \mathbb{Z}\alpha$, with zero differential, concentrated in one relative $\mathbb{Z}/4$ degree. DFL Proposition 3.23(iii) gives one index-zero residual. It is the continuation of the constant pair, whose operator is invertible by their Proposition 3.19, and its orientation is $+1$. Thus $N = \text{id}$, as also follows from the diagonal calculation in their Theorem 3.29. The final small perturbations preserve this unique residual and its sign.

Choose points on both components using Lemma 12.2. The symplectic point operations vanish because the target is a point. The point homotopies vanish by their odd relative degree. Theorem 11.5 gives

$$b_{q_j}^G = 8\eta_j \text{id}, \quad \eta_j \in \{+1, -1\}.$$

The two gauge point operations agree. Indeed, join the points by a smooth path in the connected three-manifold $Y_\#$. Interpolate the universal section pairs along this path, independently at the critical connection. Such an independent interpolation exists because a bundle over an interval is trivial and the space of independent complex two-frames in $\mathbb{C}^3$ is connected. Extend it to a neighborhood of the critical connection and make the remaining ordinary cuts transverse.

The parameterized index-four gauge trajectories, the path parameter, and the point incidence give a one-dimensional cut. Gauge bubbling would cost eight index units and cannot occur. Broken insertion limits are excluded by the lower cuts or the independent critical pair. Ordinary gauge regularity through this index is provided by DFL Lemma 2.56. The oriented boundary gives

$$b_{q_1}^G - b_{q_0}^G = d_G L + L d_G = 0.$$

Thus $\eta_0 = \eta_1$, with equality deduced over $\mathbb{Z}$ before any base change. If the common sign is negative, reverse the single reference index-four gauge-path orientation. Proposition 10.2 reverses both signs and leaves the unique index-zero residual orientation unchanged. This proves the result in one coherent system. The surface and loop operations have relative degrees two and three, so vanish on these one-generator complexes. $\square$

The magnitude eight in this argument comes from the clutching calculation and the global boundary theorem. The torus cylinder has the same positive normalization, but its ring formula is not used to transfer a sign between different three-manifolds. When a torus is paired with a higher-genus component, the one-generator argument does not apply.

12.3. The ordinary torus tube and its orientation.

Proposition 12.4. *Suppose an oriented torus $T \subset Y_\#$ has odd pairing with $w_2(E_\#)$. Let $\tilde{I}(Y_\#; \mathbb{Q})$ be the ordinary relatively $\mathbb{Z}/8$ -graded instanton homology, before the distinguished involution is divided out. There is one coherent lift τ_T of that involution for which the positive integral point operation satisfies*

$$b^G = 8\tau_T \quad \text{on } \tilde{I}(Y_\#; \mathbb{Q}), \quad b^G = 8 \text{ id} \quad \text{on } I(Y_\#; \mathbb{Q}).$$

The second group uses the quotient $\tau_T = 1$. For a splitting with T as a matching component, this is a choice of DFL's one index-four gauge-path orientation.

Proof. Remove an open tube $D^2 \times T$ about $\{1/2\} \times T$ from $[0, 1] \times Y_\#$. The ordinary instanton cobordism map is

$$\Psi : \tilde{I}(S^1 \times T) \otimes \tilde{I}(Y_\#) \longrightarrow \tilde{I}(Y_\#).$$

If e is the relative invariant of the filling $D^2 \times T$, coherent gluing with the product cobordism gives $\Psi(e \otimes v) = v$. This is the tube and filling construction in [KM10, proof of Corollary 7.2, arXiv version, pp. 53–54]. We use the unreduced theory at this step. The flat filling with trivial meridional holonomy gives the nonzero vector e ; its adjoint deformation problem is acyclic. Moving the point insertion through the connected cobordism gives

$$b^G \Psi(a \otimes v) = \Psi(b_T a \otimes v).$$

The unreduced torus group has two generators of degrees differing by four. In the positive torus normalization its involution τ_0 satisfies $b_T e = 8\tau_0 e$. The unreduced matrix is recorded in [CS18, Appendix A, arXiv version, p. 26]; their u is $\beta = -4\mu(\text{point})$, the positive p_1 generator. Its quotient is the relation $\beta = 8$ in [Muñ99a, Lemma 11].

To compare the involutions, continue T from one radial ray of the removed tube to the outgoing boundary. This gives a properly embedded three-manifold P in the tube complement, with boundary T on outgoing $Y_\#$ and $\{\theta_0\} \times T$ on $S^1 \times T$, and with no incoming $Y_\#$ boundary. Its mod-two Poincaré dual defines a real line bundle ξ . On the three ends its first Stiefel–Whitney classes are respectively 0, $\text{PD}(\{\theta_0\} \times T)$, and $\text{PD}(T)$. Tensoring a $U(2)$ lift by ξ therefore induces the torus involution and the distinguished outgoing involution, and is the identity on incoming $Y_\#$. The determinant of the lift is unchanged, since $\xi^{\otimes 2}$ has its canonical trivialization, and the adjoint differential expressions are naturally identified.

Ordinary determinant gluing commutes with this tensoring: restriction, cutoff multiplication, and the finite kernel/cokernel matching sequence commute under the adjoint identification. Fix the tube-end reference orientation as τ_0 , and choose the outgoing index-four reference-path orientation to make this determinant square orientation preserving. This is one choice for the whole map. Gluing reference cylinders propagates it to every pair of generators. Changing the relative instanton number adds the standard S^4 factor, on whose ball ξ is trivial. Thus the same square commutes with that factor. These are exactly the reference-cylinder and S^4 extensions of DFL's gauge orientation recipe on pp. 21–22 of [DFL26]. They give a coherent lift τ_T with

$$\Psi(\tau_0 a \otimes v) = \tau_T \Psi(a \otimes v).$$

Both involutions have even degree; no odd factor has been interchanged.

Combining these identities yields

$$b^G v = \Psi(b_T e \otimes v) = 8\Psi(\tau_0 e \otimes v) = 8\tau_T v.$$

The quotient by $\tau_T = 1$ proves the second assertion. Its nice homology class is the one represented by T , as in [KM10, Section 7.1] and DFL’s construction. The cited complex-coefficient tube and torus identities are rational identities by faithful extension of scalars from $\mathbb{Q}$ to $\mathbb{C}$. Integral descent through the finite gauge quotient is not asserted here. $\square$

This proposition concerns the operation on homology. It does not identify every mixed primitive sign or prescribe an integral mixed point-cut homotopy.

12.4. The torus reference orientation and the geometric homotopy. We compare the ordinary reference in the preceding proposition with the relative closed determinant in (425). The comparison is made on determinant lines before passing to Floer homology.

Let T be a matching torus and let B represent a nondegenerate ordinary generator. On a normal collar of T , B is pulled back from the odd flat projective connection α . A smooth degree-one rank-two lift on T has a smooth splitting $L \oplus (\Lambda L^{-1})$ with $\deg L = 0$. Rotate the first summand once in the clutching around an oriented disc in the time/normal plane. Fix the direction of the clutching by requiring that its closure be the bundle of Lemma 10.17:

$$(435) \quad c_1(E) = h + f, \quad c_2(E) = hf, \quad p_1(\text{ad } E) = -2hf \quad \text{on } \mathbb{P}^1 \times T.$$

The splitting used here is smooth; the stable attaching bundle is not asserted to split holomorphically.

The clutching is projectively a loop. Its determinant-one lift begins at I and ends at $-I$. After the disc is inserted in $\mathbb{R} \times Y_\#$, the outgoing framing therefore differs by a projective gauge transformation supported on the normal collar of T . Its determinant-one lifts change sign across T , so its lifting obstruction is $\text{PD}(T)$. This is the distinguished involution ι_T . The absolute cylinder bundle is unchanged; the compactly supported relative class becomes zero when the end framings are forgotten. The framed replacement defines an ordinary path $p_T(B)$ from B to $\iota_T B$ of index four, with the sign of its relative charge fixed by (435).

This model represents every retained primitive framing in the torus comparison. On $(D^2 \times T, S^1 \times T)$ the relative determinant class lies in $H^2 = \mathbb{Z}$. With that class fixed, the relative second Chern class in $H^4 = \mathbb{Z}$ classifies the rank-two lift: the determinant fiber $BSU(2)$ is three-connected with $\pi_4 = \mathbb{Z}$. Here the displayed cohomology groups are relative groups, and the relative H^1 is zero, so no loop of determinant identifications changes the classification. Capping reads the two classes as (435). There is therefore a bundle isomorphism relative to the retained boundary framing. Affine connection transport on this bundle commutes with the same excision square. The closed Donaldson orientation is retained; changing the isomorphism has positive action by [Don87, Corollary 3.27].

For an ordinary cylinder connection A and its replacement A^E , the interior rotation of Lemma 10.18 gives

$$(436) \quad \text{Or}(\det D_{A^E} \otimes \det Q_0) \longrightarrow \text{Or}(\det D_A \otimes \det Q_E).$$

Use the ordinary Donaldson orientations of Q_E, Q_0 , with the product homology orientation and the lifts already specified. At a constant cylinder B , D_B is invertible. Define $\lambda_T^{\text{cl}}(B)$ to be the orientation of $p_T(B)$ for which (436) becomes positive after cancellation of $\det D_B$ and the constant closed factor. Thus the relative closed line L_{cl} determines the reference orientation. Place all operators in the same target convention as in Lemma 10.17. Changing to DFL’s

(d^*, d^+) convention uses a fixed target isomorphism on every family and every gluing map; no untransported identification with $-D$ is used.

Lemma 12.5. *The orientations $\lambda_T^{\text{cl}}(B)$ arise from one choice of DFL's index-four reference orientation, with the reference-generator paths and the standard S^4 orientations held fixed. Denote this coherent system by $\mathcal{O}_T^{\text{cl}}$. In this system the comparison in (426) satisfies $\delta_{\text{ref}} = +1$ at every torus residual.*

Proof. We first record the determinant transport for moving the insertion. In a product band of tori choose a normal rectangle, an oriented smaller disc, and the based clutching of (435). Translate the disc through the rectangle. Its isotopy identifies the relatively framed modified bundles. Connections on the rectangle can be interpolated relative to its outer boundary, since their space is affine; metrics can be interpolated through positive metrics. These are configuration families, not families of ASD solutions.

Combine the disc motion with the signed rotation in (436). After the stated identifications this is a Fredholm family on a square, with fixed invertible end operators. A common finite-dimensional target supplement makes the whole family surjective. Its kernel determinant, with the supplement determinant cancelled, shows that the two boundary transports agree. This includes cokernel crossings. Cylinder gluing on a disjoint collar commutes with this square. For nested collars insert and cancel a constant product collar, as in the proof of Lemma 10.18.

Apply this construction along a reference cylinder q from B to B' . The two end decompositions are $p_T(B)$ followed by the ι_T transform of q , and q followed by $p_T(B')$. Use the natural adjoint identification on the transformed cylinder. The determinant square equates these two gluing maps with their relative closed factors. The orientation of q cancels, so $\lambda_T^{\text{cl}}(B')$ is determined by $\lambda_T^{\text{cl}}(B)$.

Endpoint product framings can be extended over this band. Indeed, the odd flat torus connection has holonomies given by half-turns about two orthogonal axes. Its four parallel projective automorphisms realize all four lifting obstructions in $H^1(T; \mathbb{Z}/2)$: this follows by taking commutators of their $SU(2)$ lifts with the two holonomy lifts. These are all components of the surface gauge group, because $\pi_1(SO(3)) = \mathbb{Z}/2$ and $\pi_2(SO(3)) = 0$. An endpoint parallel automorphism therefore removes the obstruction to extending a framing. Such changes preserve the closed orientation by [Don87, Corollary 3.27]. Line twists of the lifts preserve the relative orientation by Lemma 10.17. The meridional clutching is retained; no framing is extended across the modified disc.

The doubled reference agrees with the standard index-eight orientation. Two copies of the clutching close to a bundle E_2 with

$$c_1(E_2) = 2h + f, \quad c_2(E_2) = 2hf, \quad p_1(\text{ad } E_2) = -4hf.$$

Close the insertions by the common product caps. The disjoint rotation square compares their composite with

$$\det Q_{E_1} \otimes \det Q_{E_1} \longrightarrow \det Q_{E_2} \otimes \det Q_0.$$

This closed excision preserves complex orientations. On the Kähler product, the ordinary first-order operator in the fixed target identification is the complex-linear operator $\bar{\partial}^* \oplus \bar{\partial}$ on $(0, 1)$ -forms. This first-order identity does not require $\bar{\partial}^2 = 0$. The scalar rotation intertwines the complex structures, and its commutator path remains complex-linear. Its determinant transport is therefore positive through any cokernel crossings. The Donaldson/complex exponents for E_0, E_1, E_2 are $-1, 0, 1$, since for $c = kh + f$ one has $\langle c^2 + Kc, [\mathbb{P}^1 \times T] \rangle / 2 = k - 1$. The

product signs on both sides are $+1$. The map consequently preserves the specified Donaldson orientations as well.

Tensoring E_2 by $\mathcal{O}(-1)$ gives $c_1 = f$, $c_2 = hf$, with no orientation sign by Lemma 10.17. It differs from E_0 by one instanton with the first Chern class fixed. Its relative orientation is therefore the standard instanton-addition orientation used to define the Donaldson orientations [Don87, Proposition 3.20]. This verifies the squared-reference compatibility with the standard S^4 factor. Further S^4 factors can be placed in disjoint balls and commute with the same rotation square. All inserted factors have even index. DFL’s path classification and orientation recipe in [DFL26, Section 2.4, pp. 20–22] therefore give the asserted coherent system.

For the mixed comparison, first use the smooth configuration transports in the construction of (425). They replace the glued solution and residual by representatives differing by the ordinary framed insertion in a strictly negative disc times T . The positive torus map is constant. Proposition 10.20 supplies the chosen integrable flattening with its product collar. The other matching component and all global ends remain in the common exterior. Naturality for nested excision identifies this relative comparison with the specified composite JG .

Prepare the two configurations simultaneously for DFL Lemma 4.62, with preparation away from the insertion. They have the same positive map and may use the same positive extension A_u in DFL’s formula (4.63). In their Φ_- coordinates the translation $A_\tau(s, y) = A(s + \tau, y)$ moves the insertion to the left. For a finite sufficiently large shift both pairs are symplectically constant and the entire insertion lies in the ordinary incoming gauge cylinder. The original disc framing and the orientation of its normal plane move by this same translation.

The shifted pairs are ordinary cylinders, with or without p_T , glued to constant mixed pairs. If the joining representatives differ by ι_T , use the correspondingly transformed constant pair. Its operator and the original constant operator are invertible by DFL Proposition 3.19; their determinant units agree under the adjoint identification. The framing change remains in the p_T factor.

DFL Section 4.7 defines the mixed orientation by this cylinder gluing, with determinant isomorphism canonical up to positive scalar. The family shifting lemma, DFL Lemma 4.64, and the finite-supplement square commute this gluing with the interior excision. After the constant mixed factors cancel, JG is (436) for the shifted residual cylinder. Its reference factor is positive by the coherence just proved. This gives $\delta_{\text{ref}} = +1$ in $\mathcal{O}_T^{\text{cl}}$ for every residual. $\square$

Lemma 12.6. *The system $\mathcal{O}_T^{\text{cl}}$ agrees with the tube normalization in Proposition 12.4.*

Proof. Let W° be the tube complement and $P \cong [0, 1] \times T$ the three-manifold used in that proposition. Its trivial normal line, together with the interval direction, gives a band of discs crossed with T . Orient the disc plane by the orientations of W° and T . At the incoming tube boundary take r increasing from the removed disc into W° and θ its positive meridian. The plane has orientation $dr \wedge d\theta$, with the incoming cylinder direction first. It also agrees with the outgoing time/normal orientation. Consequently sliding a disc through the band retains the clutching in (435).

The odd bundle near P is pulled back from T . Extend compatible endpoint framings as in the preceding lemma, and flatten a cobordism connection A to the product α on a smaller band, relative to its ends and exterior. Slide the framed insertion from the outgoing collar to the tube collar. The two descriptions of the resulting connection are

$$A \text{ followed by } p_T(B_{\text{out}}), \quad p_{T^3}(B_{\text{tube}}) \text{ followed by } A^\xi,$$

where ξ is the real line with $w_1(\xi) = \text{PD}(P)$. To see this last identification, cut along P . The determinant-one lifts of the rotating-summand clutching differ by $-I$ across this cut and agree elsewhere. These are exactly the central transition functions of ξ . They are trivial on incoming $Y_\#$ and induce the distinguished involutions on the tube and outgoing $Y_\#$. The same based clutching has been moved by isotopy, so no additional relative instanton class is introduced.

The disc-motion and rotation square identifies the determinant maps for these two factorizations. In both, the reference determinant is oriented by the same closed relative line. The natural adjoint identification of A with A^ξ also identifies the end matching factors. After cancellation the square is positive. A reference factor has index four, so passing it through a cobordism factor of arbitrary index produces no odd sign. The overall cobordism orientation cancels. Reference cylinders and disjoint S^4 factors give the same extension to all configuration components as in the preceding lemma. Thus, at the level of determinant orientations, the tube satisfies

$$(437) \quad \Psi(\tau_0^{\text{cl}} a \otimes v) = \tau_T^{\text{cl}} \Psi(a \otimes v).$$

Positivity here is proved by the determinant square, independently of whether Ψ or the ambient Floer groups are nonzero.

It remains to identify the tube-end reference. Use the actual torus-only DFL splitting of T^3 into two product cylinders. The preceding lemma gives $\delta_{\text{ref}} = +1$ in $\mathcal{O}_0^{\text{cl}}$, and Proposition 10.20 gives $\delta_{\text{cap}} = +1$. Since $\sigma_1 = -1$ and the closed relative sign is -1 , we obtain $\eta_1 = +1$. The single-generator calculation in the proof of Proposition 12.3 and Theorem 11.5 then give $b_T = 8 \text{ id}$ on the quotient in this already fixed orientation.

The unreduced torus complex has two generators of degrees differing by four, zero differential, and homology $\mathbb{Z} \oplus \mathbb{Z}$ [KM10, Section 7.1, pp. 52–53]. The filling vector e is one generator up to sign. Its image in the quotient by $\tau_0^{\text{cl}} = 1$ generates a free copy of $\mathbb{Z}$. The degree and the quotient formula therefore give

$$b_T e = 8 \tau_0^{\text{cl}} e.$$

This identifies τ_0^{cl} with the positive τ_0 used in Proposition 12.4. Equation (437) then identifies the outgoing reference with that proposition's reference orientation. The nonzero torus group calibrates the tube end only; no nonvanishing assumption on $I(Y_\#)$ has entered the determinant transport. $\square$

Proposition 12.7. *For a point q on a matching torus, use the coherent tube normalization of Proposition 12.4 and the regular point-cut data of Theorem 11.5. Every primitive torus contribution has $\eta_1 = +1$ relative to its residual. The original integral geometric homotopy satisfies*

$$(438) \quad N b_q^G - b_q^S N = d_S K_q^{p_1} + K_q^{p_1} d_G + 8N.$$

The assertion holds for simultaneous finite point cuts and after any extension of coefficients from $\mathbb{Z}$.

Proof. Lemmas 12.5 and 12.6 give $\delta_{\text{ref}} = +1$ in the prescribed system. Proposition 10.20 and (426) then give $\eta_1 = +1$. The actual gluing and finite parameter orientations have already been compared, so the multiplicity of every torus contribution is $+8$. Theorem 11.5 therefore has $B_q = 8N$, proving (438). Lemma 12.2 gives the finite simultaneous version. If both components are tori, their normalizations agree by the integral path-of-points argument in Proposition 12.3. $\square$

This identifies the prescribed geometric homotopy on a torus component without a rational chain correction. Simultaneous agreement with a genus-two sphere normalization is proved in Section 12.6.

12.5. The sphere character and the genus-two relation. For any component of genus at least two, comparison with the system which selects that component in DFL Remark 4.76 gives a sphere compatibility sign χ_j . The proof of Proposition 12.1 applies to the sphere insertion itself in every such genus: its relative index is four, and the two systems differ only in the orientation of p_0 . Thus χ_j is independent of the strip class and its generators. Disjoint sphere gluing extends these signs to the character

$$\chi : \pi_2(M) = \bigoplus_{g_j \geq 2} \mathbb{Z}A_j \longrightarrow \{+1, -1\}, \quad \chi \left(\sum_j d_j A_j \right) = \prod_j \chi_j^{d_j}.$$

Associativity of determinant gluing proves multiplicativity; the sphere factors have even index and inverse classes have dual orientation. DFL Remark 2.39 makes the comparison depend only on homotopy class.

In the quantum-module convention for this coherent strip orientation, a closed sphere term of class A carries the weight $\chi(A)$. Indeed, the ordinary strip-breaking terms in the oriented module boundary argument use the coherent strip gluing, while the closed-sphere terms use its comparison with the complex sphere orientation. Their discrepancy is precisely $\chi(A)$. Retaining formal sphere variables and then specializing the variable of A_j to χ_j gives this product. The character law preserves its associativity, unit, and graded commutativity. We use this convention for the rational module structure recalled on p. 5 of [DFL26]; an all-positive specialization on unselected components is not assumed.

More explicitly, choose auxiliary spin structures on L, L' and let λ_p^{sp} be their strip orientations. Existence of the spin structures is stated in DFL Remark 2.39. Write $\lambda_p^{\mathcal{O}} = \epsilon(p)\lambda_p^{\text{sp}}$. Comparison of the strip and sphere gluing maps gives

$$\epsilon(p \# p') = \epsilon(p)\epsilon(p'), \quad \epsilon(p \# A) = \epsilon(p)\chi(A).$$

Marked strips inherit their determinant orientations from the unmarked problem. A broken strip in total class $p = p_1 \# p_2$ therefore changes orientation by $\epsilon(p_1)\epsilon(p_2) = \epsilon(p)$. A boundary with a closed sphere A and remaining strip p' changes by $\epsilon(p')\chi(A)$, again equal to $\epsilon(p)$. Applying this comparison to the reference oriented module identity gives exactly the sphere weight above. Any disc terms in its compactification are retained with their total relative-class signs. The pair is monotone with minimal Maslov number four by DFL Proposition 2.27, so changing signs introduces no Maslov-two obstruction. This is a classwise comparison of oriented identities; it does not assume that $\epsilon(p)$ depends only on the endpoints, or that the chosen spin structures induce the DFL orientations.

Lemma 12.8. *For a genus-two component put $a_j^S = -\frac{1}{2}b_{\Sigma_j}^S$, and retain the geometric point operation b_j^S . On rational symplectic instanton homology,*

$$(a_j^S)^2 + b_j^S = 4\chi_j \text{ id}.$$

Proof. In [Muñ99b, Example 24], the geometric generators satisfy $\widehat{\beta} = \beta + 4$ and $\alpha^2 + \widehat{\beta} - 8 = 0$ on both summands of the displayed quantum presentation. Here $\alpha = -\frac{1}{2}p_1/[\Sigma_j]$ and $\beta = p_1/[q_j]$. Thus the relation is $\alpha * \alpha + \beta = 4$ in the complex sphere orientation. Before specialization the right side is $4q_j$. The minimal Chern number is two, so q_j has degree four; a degree-four relation has only its classical term and this one-sphere constant term.

The coefficient can also be checked geometrically. In the intersection of two quadrics, $\alpha = h_2$, $\beta = -4h_4$, and $\alpha \smile \alpha = 4h_4$. A minimal sphere is a line of hyperplane degree one. The degree-four attaching map of Proposition 5.2 counts four lines through a generic point. The divisor equation then gives $\text{GW}_{0,3,A}(\alpha, \alpha, [\text{point}]) = 4$, which is the stated constant term.

The pulled-back relation in the product target is the factor relation. With product data the other factor contributes its degree-zero unit, so no other sphere variable occurs. Specializing q_j to χ_j in the module convention just described gives the assertion. This uses the geometric β , not the corrected $\widehat{\beta}$, and no quantum-to-Floer phase change. $\square$

12.6. Homologous surfaces and the genus-two reference. For an odd connected surface $S \subset Y_\#$, define $\lambda_S^{\text{cl}}(B)$ at a nondegenerate generator B by the ordinary excision

$$(439) \quad \text{Or}(\det D_B^S \otimes \det Q_{S,0}) \longrightarrow \text{Or}(\det D_B \otimes \det Q_{S,1}).$$

Here D_B^S is the degree-one framed local replacement, and $Q_{S,0}, Q_{S,1}$ are the closed product pair of Lemma 10.17, with determinant degree normalized to one. Require positivity for their relative Donaldson orientation, cancelling the invertible constant cylinder. This extends the definition in Section 12.4; the constant closed determinant is retained in every genus.

Lemma 12.9. *If two odd connected surfaces in $Y_\#$ bound an intervening three-manifold, their closed references agree in the same ordinary index-four path torsor. The comparison is independent of a reversal of either surface orientation and commutes with reference cylinders and the standard S^4 orientation.*

Proof. At the closed product pair, deform the connections to preserve the smooth splittings

$$E_0 = \mathbb{C} \oplus \Lambda, \quad E_1 = \mathcal{O}(1) \oplus \Lambda.$$

The adjoint coefficient then splits as $\mathbb{R} \oplus \text{Hom}(\Lambda, \mathcal{O}(k))$, $k = 0, 1$. The second term is a complex line regarded as a real plane; its ASD operator is complex-linear with respect to the coefficient complex structure, independently of a Kähler structure on the base.

These reductions give the specified Donaldson orientations. Indeed, $E_k = (\mathbb{C} \oplus \mathcal{O}(k)\Lambda^{-1}) \otimes \Lambda$. The reduction comparison with the same determinant lift and trivial second line has sign $(-1)^{\langle c_1(\Lambda)^2, [\mathbb{P}^1 \times S] \rangle} = +1$ by [Don87, Proposition 3.25]. Thus the relative closed orientation is the coefficient-complex orientation of

$$(440) \quad \det_{\mathbb{R}} D_{\mathcal{O}(1)\Lambda^{-1}} \otimes (\det_{\mathbb{R}} D_{\Lambda^{-1}})^*,$$

after cancellation of the identical real coefficient operators. Kernels and cokernels are both retained. The complex relative index is two. The real coefficient operator on $\mathbb{P}^1 \times S$ has even index $2g(S) - 2$, so these cancellations have no odd sign.

First orient S_0, S_1 to be homologous. An odd connected surface is nonseparating and primitive in integral homology. Tensoring a rank-two lift by an appropriate line therefore normalizes its determinant degree to one on both surfaces. This leaves the adjoint unchanged; the closed lift independence was checked in Lemma 10.17. The normalized lift on $Y_\#$ has a smooth nowhere-zero section, since its real rank is four and the base has dimension three. Write it as $\mathbb{C} \oplus \Lambda$.

In a bounded time interval deform a cylinder connection to preserve this splitting, using a larger interval to interpolate to the original connection. Turn off the lower-order perturbations in the smaller interval with a common cutoff for the two operators being compared. The asymptotic operators stay fixed and invertible. This gives Fredholm deformations even when the intermediate slice connections are reducible.

Close a copy of the smaller interval to $Z = S^1 \times Y_\#$. A line bundle K_i trivialized outside a disc times S_i , with relative first Chern class the Thom class of $\{t_0\} \times S_i$, describes the local replacement as

$$\mathbb{C} \oplus \Lambda \longmapsto K_i \oplus \Lambda.$$

Writing z for the positive class of $H^1(S^1)$ gives $c_1(K_i) = z \smile \text{PD}(S_i)$. Its square is zero and $\langle c_1(K_i)c_1(\Lambda), [Z] \rangle = 1$, so the relative charge is the prescribed one.

Choose a common bounded neighborhood of the intervening three-manifold and both collars. The two Thom classes agree in its relative H^2 , since their difference is dual to the boundary of that three-manifold. Classification of line bundles with exterior trivialization gives $K_0 \cong K_1$ relative to the exterior. Thus the framed modifications are identified, in addition to having equal index.

The interior excision and the cylinder-closing excision commute by the stabilized square in Lemma 10.18. On the closed reducible operators they preserve the real and complex coefficient splitting. The unchanged real block contributes the unit: its rotation extends to a homotopy on the globally unchanged real coefficient, and conjugation followed by the commutator path is a contractible loop of compact perturbations. The complex block preserves its coefficient-complex orientation. The real index on $S^1 \times Y_\#$ is zero, and on $\mathbb{P}^1 \times S_i$ it is $2g(S_i) - 2$. Both are even.

Consequently (439) is computed on Z by the relative complex determinants for $K_i \oplus \Lambda$ and $\mathbb{C} \oplus \Lambda$. The unitary line isomorphism $K_0 \cong K_1$ and an affine path of line connections identify these orientations positively. The same relative bundle identification gives the comparison path for the global cylinders. Naturality proves equality of their reference orientations. For a reference cylinder between different generators the identical core reduction retains that cylinder as a common exterior factor, whose orientation cancels. Disjoint S^4 insertions commute with the same square; their orientations are the fixed instanton-addition orientations. The doubled clutching is normalized to a fixed lift as in Lemma 12.5, using the reduction comparison above.

Finally reverse a surface orientation, keeping the ambient orientation fixed. The normal plane reverses too, and exchanging the two line summands supplies the degree-one normalization. The two descriptions are $K \oplus \Lambda$ and $\mathbb{C} \oplus \Lambda K^{-1}$, identified projectively by tensoring the second with K . Exchanging the summands reverses the real adjoint axis and conjugates the complex relative determinant. The real change cancels between the modified and constant operators. The complex relative index is two, so conjugation contributes $(-1)^2 = +1$. The line twist has $c_1(K)^2 = 0$ and introduces no Donaldson orientation sign. The reference is therefore unchanged. $\square$

Let λ_S^{sph} be the ordinary reference orientation defined by a positive generator sphere in DFL Remark 4.76, with reference-generator paths and S^4 orientations fixed.

Lemma 12.10. *There is a sign $\delta_{\text{sph},2} \in \{+1, -1\}$, independent of the ambient splitting and reference generator, such that*

$$\lambda_S^{\text{sph}} = \delta_{\text{sph},2} \lambda_S^{\text{cl}} \quad \text{when } g(S) = 2.$$

Proof. Fix an attaching flat connection α , put $V = H^1(S; \text{ad } \alpha) = T_{[\alpha]}M(S)$, and choose a based smooth positive generator sphere s , constant near the boundary of its representing disc. Insert it strictly in the positive interior of the compact constant mixed cap C_α , giving C_s . The positive-interior excision in the proof of Proposition 10.16 orients the relative line $\det C_s \otimes (\det C_\alpha)^*$ by

$$\det D_s \otimes (\det V)^*.$$

Here D_s is the closed parametrized sphere Cauchy–Riemann operator. The node value is removed once; no sphere automorphism quotient is introduced. The harmonic space V has complex dimension three, and the relative real index is four. The nonharmonic constant block is invertible. The harmonic matching sequence and sphere determinant have their canonical complex orientations, including when s is not holomorphic or D_s has cokernel.

Insert this cap into a global constant mixed pair with attaching value α . The full uncut constant residual operator is invertible by DFL Proposition 3.19, so its determinant has the canonical unit orientation. In the relative excision square we retain both the cut exterior determinant and the constant-cap harmonic factor $\det V$ until the matching factors cancel. The disjoint cylinder, mixed-to-strip and sphere gluing squares in Proposition 10.16 identify the displayed orientation with exactly the sphere orientation used in DFL Remark 4.76.

Now perform DFL’s shifting simultaneously on the sphere insertion and the constant pair. The only varying positive target factor is $M(S)$, and the sphere is constant near the upper and lower strip edges. Choose its surface lift on a supporting rectangle, fixed on three edges. The fourth edge retains the nonextendible surface gauge loop described in [DFL26, Section 2.2, p. 8]. That loop has half-unit charge. In the ordinary normal collar it produces the distinguished outgoing involution. No lift fixed on all four edges is assumed.

For two ambient extensions of the same local data, use the same surface lifts and fourth-edge transition in the positive extension of DFL (4.63). The ambient piece connections are constant outside the supporting collar. The outgoing constant mixed pair is involution-transformed if required by the retained transition; its determinant unit is unchanged. A finite shift puts the entire support in the ordinary negative cylinder. Any coordinate preparation and exterior deformation is performed on both members of each relative pair.

This cancellation is an identity of determinant maps. Choose a common outer constant collar over the compact shifting interval. The compact mixed domains keep the harmonic modes, gauge block and normal condition; their smooth family remains elliptic Fredholm. Combine shifting with the signed excision rotation. A common finite target supplement over the parameter square gives equality of the two boundary transports through all cokernel crossings. Nested cuts are separated by a constant collar before the factors are cancelled.

At the shifted endpoint the ordinary insertion has the retained nontrivial meridional class and relative $p_1 = -2$. After the base line normalization its closed lift has $c_1 = h + f$, $c_2 = hf$. The relative bundle classification used in Section 12.4 identifies it with the model defining λ_S^{cl} . Thus the ratio of the sphere reference to the closed reference is the sign of one local compact determinant transport, independent of the ambient exterior.

This sign is continuous in the local data. The surface moduli space is connected and simply connected, with $\pi_2 = \mathbb{Z}$ [DFL26, Section 2.2, p. 8]. Move α along a path and carry the based positive sphere with it. Representatives of the same based class are homotopic; compatible almost complex structures and the surface metrics vary in connected families. The stabilized determinant comparison includes nonregular configurations along these paths.

Gauge changes preserve the canonical sphere and harmonic orientations and the closed Donaldson orientation [Don87, Corollary 3.27]. Lift changes preserve the latter relative orientation by Lemma 10.17. Orientation-preserving identifications of genus-two surfaces carry the odd bundle and positive generator to the corresponding data. Hence the sign is the single $\delta_{\text{sph},2}$ in the statement. No torus acyclicity is used for the genus-two harmonic block. $\square$

Lemma 12.11. *The sign in Lemma 12.10 is $\delta_{\text{sph},2} = +1$.*

Proof. Use $T^3 = S^1 \times T$ with the odd bundle pulled back from T . Cut at a product torus T_0 and stabilize the product relative Heegaard splitting of $T \times I$ once in a three-ball. The

new surface S has genus two. Both pieces are compression bodies from T to S , obtained by adding one one-handle to $T \times I$. Rejoining their negative torus boundaries gives the original T^3 split along $T_0 \sqcup S$. These surfaces are homologous up to boundary orientation and remain odd.

We check embeddedness. A compression body C retracts to T with one additional one-cell. Every flat adjoint local system restricts to the acyclic odd torus system. The relative cellular complex gives

$$H^0(C; \text{ad } B) = H^2(C; \text{ad } B) = 0, \quad \dim H^1(C; \text{ad } B) = 3.$$

Poincaré–Lefschetz duality gives $H^1(C, \partial C; \text{ad } B) = 0$. The flat moduli space is therefore regular, compact and three-dimensional, with a Lagrangian immersion into the six-dimensional $M(S)$; see also DFL Propositions 2.10 and 2.21. Restriction is injective because $\pi_1(S) \rightarrow \pi_1(C)$ is surjective. For the liftable gauge quotient, the obstruction to lifting an equivalence which lifts on S vanishes, since $H^1(C; \mathbb{Z}/2) \rightarrow H^1(S; \mathbb{Z}/2)$ is injective. A parallel boundary automorphism extends by the same π_1 surjectivity, so an equivalence can be chosen with the prescribed liftable boundary restriction. The compact injective immersion is an embedding, on both sides.

The adjoint cohomology of T^3 vanishes by its torus factor. The intersection is consequently transverse by DFL Lemma 2.47. Small regularizing data preserve these compact embeddings and the isolated intersection. The unreduced torus group is $\mathbb{Z} \oplus \mathbb{Z}$; the quotient by the nice involution is $\mathbb{Z}$ [KM10, Section 7.1]. In particular this example has nonzero rational Floer homology.

Use the fixed torus normalization. Let χ be the genus-two sphere compatibility sign in this system. The torus operation and the signed global genus-two boundary give

$$b^G = 8 \text{ id}, \quad N_H b^G = (b_S^S + 4\chi \text{ id}) N_H.$$

The torus surface operation is zero. Homology of the two matching surfaces and DFL’s surface comparison imply $a_S^S = 0$. The signed genus-two quantum relation in Section 12.5 therefore gives $b_S^S = 4\chi \text{ id}$. Since N_H is an isomorphism and this homology is nonzero, $8 \text{ id} = 8\chi \text{ id}$ implies $\chi = +1$.

Lemma 12.6 identifies this normalization with the closed torus reference. Lemma 12.9 identifies the closed references of T_0 and S . Thus the positive genus-two sphere reference is its closed reference, and $\delta_{\text{sph},2} = +1$ by locality. Only the explicit T^3 example is required to have nonzero homology. No invariance of a mixed point count under Heegaard stabilization is assumed. $\square$

Proposition 12.12. *Choose the torus normalization when a torus is present; otherwise select a genus-two component when there is one. Keep that coherent system fixed for both components. Every torus primitive and each of the four genus-two branches has positive orientation relative to its residual. For the simultaneous regular point cuts,*

$$(441) \quad B_{q_j} = c_{g_j} N, \quad c_1 = 8, \quad c_2 = 4, \quad c_{g \geq 3} = 0.$$

No Floer homology nonvanishing assumption is required.

Proof. The preceding two lemmas identify each genus-two positive-sphere reference with its closed reference. Lemma 12.9 identifies the two matching closed references. They agree with the torus normalization when a torus is present. Thus each genus-two component has $\chi_j = +1$ in the chosen single system. Proposition 12.1 gives positivity of every branch at every residual. The torus assertion is Proposition 12.7, and the higher-genus contribution already vanishes. Substitute these signs into Theorem 11.5, using Lemma 12.2. $\square$

12.7. Evaluation of the local coefficient. For genus at least three, attachment support and residual avoidance in Proposition 11.4 exclude the primitive contribution, so

$$(442) \quad c_g^{\text{loc}} = 0, \quad g \geq 3.$$

Propositions 9.6 and 9.7 compute the actual relative seam-incidence magnitudes eight and four. The latter is one on each of the four attaching branches. Proposition 12.12 compares these local orientations with DFL's coherent orientation at every residual in one system. Consequently

$$(443) \quad c_g^{\text{loc}} = \begin{cases} 8, & g = 1, \\ 4, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

The comparison is on determinant lines and applies also when rational Floer homology vanishes. Substitution in the global boundary formula gives the point theorem.

12.8. The oriented boundary count.

Theorem 12.13 (integral geometric point comparison). *In the embedded DFL setting, use the single coherent normalization of Proposition 12.12. For simultaneous regular point-cut data, before division by four, the operations represented by the positive integral degeneracy cycle and the original geometric counts satisfy*

$$(444) \quad Nb_q^G - b_q^S N = d_S K_q^{p_1} + K_q^{p_1} d_G + c_{g_j} N, \quad c_g = \begin{cases} 8, & g = 1, \\ 4, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

The same equation data and comparison map work for both matching components. The identity extends from $\mathbb{Z}$ to any commutative coefficient ring.

Proof. Theorem 11.5 proves the integral identity with B_q in place of $c_{g_j} N$. Proposition 12.12 identifies these maps in one coherent system, without a homology nonvanishing assumption. Lemma 12.2 gives the simultaneous data. All coefficients in the oriented boundary count are integral, so extension of coefficients preserves the identity. $\square$

13. COEFFICIENT CHANGES AND OBSERVABLE OPERATIONS

Theorem 13.1 (rational point-class comparison). *Let $(Y_\#, E_\#)$ have an admissible splitting $Y_\# = Y \cup_\Sigma (-Y')$ with two matching components in the embedded-Lagrangian DFL setting. Assume the flat connections are irreducible and use compatible standard perturbations that vanish near the matching line. Let $q \in \Sigma_j$ and write g_j for the genus of that component. Over $\mathbb{Q}$, there is one coherent normalization and a rational chain homotopy for which the comparison map and the positive quarter-Pontryagin point operations satisfy*

$$(445) \quad Nm_q^G - m_q^S N = d_S K_q + K_q d_G + \kappa_{g_j} N, \quad \kappa_g = \begin{cases} 2, & g = 1, \\ 1, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

The choices can be made for both components with the same equation and comparison map. The homotopy can be chosen as the original integral point-cut count divided by four.

Proof. Base-change Theorem 12.13 to $\mathbb{Q}$ and divide its identity by four. Put

$$(446) \quad K_q = \frac{1}{4} K_q^{p_1}.$$

Since $c_g = 4\kappa_g$, this gives (445) with the stated geometric homotopy. $\square$

Corollary 13.2. *Define*

$$\widehat{m}_q^S = m_q^S + \kappa_{g_j} \text{id}.$$

Then N intertwines m_q^G and $\widehat{m}_q^S$ up to chain homotopy:

$$(447) \quad Nm_q^G - \widehat{m}_q^S N = d_S K_q + K_q d_G.$$

In particular, the induced Atiyah–Floer isomorphism intertwines these corrected point operations on homology.

Proof. Move $\kappa_{g_j} N$ to the left-hand side of (445) and use the definition of $\widehat{m}_q^S$. $\square$

13.1. The full corrected operation algebra. We now prove the operation relations over arbitrary coefficients. The only two-insertion moduli spaces required are ordinary instanton trajectory spaces of index at most seven.

We use the cohomological indexing introduced in Section 1. An insertion of codimension p has degree p , and its comparison homotopy has degree $p - 1$. Give $H_k(\Sigma; \mathbb{Z})$ degree $4 - k$. For a commutative ring R , put $V_R = H_*(\Sigma; \mathbb{Z}) \otimes_{\mathbb{Z}} R$ and define

$$(448) \quad \mathbb{A}_{p_1}(\Sigma; R) = T_R(V_R) / \langle x \otimes y - (-1)^{|x||y|} y \otimes x \rangle =: \text{Sym}_R^{\text{gr}}(V_R).$$

The relations are imposed on homogeneous generators. In particular, they impose $2x^2 = 0$, but not $x^2 = 0$, for odd x . If two is invertible in R , this is the familiar presentation

$$(449) \quad \mathbb{A}_{p_1}(\Sigma; R) \cong \text{Sym}_R(H_0(\Sigma; R) \oplus H_2(\Sigma; R)) \otimes_R \Lambda_R H_1(\Sigma; R).$$

For $\sigma \in H_k(\Sigma; \mathbb{Z})$ let b_σ^G, b_σ^S be the operations associated with the positive integral cycle $p_1/[\sigma]$. Choose oriented representative of an integral basis of $H_*(\Sigma; \mathbb{Z})$ and extend the constructions linearly. For $q \in \Sigma_j$, set

$$(450) \quad \widehat{b}_q^S = b_q^S + c_{g_j} \text{id}, \quad c_g = \begin{cases} 8, & g = 1, \\ 4, & g = 2, \\ 0, & g \geq 3. \end{cases}$$

Write $A_x = b_x^G$ and let B_x be the corrected symplectic operator; thus $B_x = b_x^S$ for a surface or loop. The degree of each operation is $|x| = 4 - \dim x$. The point correction has the same degree in the relative $\mathbb{Z}/4$ grading. We use the graded chain convention

$$d_G A_x = (-1)^{|x|} A_x d_G, \quad d_S B_x = (-1)^{|x|} B_x d_S.$$

For the point operation, the latter identity also follows from Theorem 12.13 and the fact that N is a chain isomorphism.

Lemma 13.3 (integral generator comparisons). *The data of Theorem 12.13 can be chosen so that, for every generator x , there is an integer map K_x of degree $|x| - 1$ satisfying*

$$(451) \quad N A_x - B_x N = d_S K_x + (-1)^{|x|} K_x d_G.$$

All identities use the same integral chain isomorphism N .

Proof. Theorem 12.13 proves the point assertion with the geometric homotopy. The surface construction in [DFL26, Section 7.2, pp. 88–92, Proposition 7.12] uses the integral two-section dependence cycle with its p_1 coorientation, and its counts are integers. There is no division by four in those cutdowns. The final paragraph on p. 92 constructs the loop comparison by increasing the indices by one. The homotopies have degrees one and two for surfaces and loops, respectively. With the graded chain convention, moving the insertion past the odd differential gives the sign $(-1)^{|x|}$ in the displayed boundary identity. In particular it is negative for a loop.

Lemma 12.2 permits these finitely many section choices for the same equation and N . Finally [DFL26, Theorem 3.29] is an integral chain isomorphism theorem: the matrix of N is triangular in the energy ordering, with diagonal entries ± 1 . Its inverse is an integer chain map. $\square$

Lemma 13.4 (integral operation relations). *On gauge and symplectic Floer homology with coefficients in any commutative ring R , the integral operations satisfy*

$$b_x b_y = (-1)^{|x||y|} b_y b_x$$

for homogeneous $x, y \in V_R$. The corrected symplectic operations satisfy the same relations. All these commutators are induced by integer chain homotopies. For an odd loop class the assertion is $2b_x^2 = 0$; no odd-square vanishing in characteristic two is used.

Proof. Work first over $\mathbb{Z}$ and on the ordinary gauge side. Let X, Y represent generators of dimensions k, l , and write $p = 4 - k$, $q = 4 - l$. Suppose first that $p + q \leq 7$. For an ordinary parametrized trajectory A and $T \in \mathbb{R}$, place the first insertion at time zero along X and the second at time T along Y . At actual integer Fredholm index r the uncut space is

$$\mathcal{M}_G(\alpha, \beta)_r \times \mathbb{R}_T \times X \times Y.$$

The two labeled section-dependence conditions each have codimension four. Hence the cut dimension is

$$(452) \quad r + 1 - p - q.$$

Use $r = p + q - 1$ for the count H_{xy} and $r = p + q$ for its one-dimensional boundary.

We describe the relative section choices. At a broken end where the marks lie on different nonconstant levels, prescribe the original single-insertion pairs on those levels. Where they lie on the same level, prescribe the lower-index two-insertion data there. Induction on index and iterated products give compatible data at multiple-breaking corners. Extend from these ordinary gluing neighborhoods and make the two pulled-back section pairs independently transverse on the remaining parts. On a bounded T interval the pairs may depend on T . Even if $x = y$ and $T = 0$, their variations remain independent. Thus coincident labeled marks form an interior parameter value and create no diagonal boundary.

At every critical connection keep each pair independent on its entire cycle, uniformly in T . The complexified adjoint bundle has $c_1 = 0$ and is trivial over the three-manifold, so the necessary frames exist. The original critical frames can be kept fixed throughout this parameter family, with independence on their critical neighborhoods. No interpolation of unrelated global framings is required. These choices exclude a cut in a long constant critical segment. The rank-zero stratum of either pair has codimension twelve, whereas the maximal uncut dimension above is nine. Relative generic choices therefore avoid it on all the relevant strata.

All ordinary trajectories of index at most seven are regular by [DFL26, Lemma 2.56]. The ordinary index-energy formula bounds their energy. On a small ball the bundle lifts to $SU(2)$,

so a local Uhlenbeck bubble removes eight index units; the distinguished global index-four involution is not a local bubble. Bubbling would therefore leave a negative-index solution, which regularity excludes. Admissibility excludes reducible ends. Consequently the only limits are ordinary broken trajectories with the two marks retained on nonconstant levels.

For a broken configuration with n nonconstant levels and marks on different levels, the cut dimension is

$$r - (p + q) - (n - 2).$$

If the marks lie on the same level with finite separation, the dimension is

$$r + 1 - (p + q) - (n - 1).$$

These formulas subtract the translation on every unmarked level. A level carrying a single insertion has index at least p or q ; a level carrying both insertions and their separation parameter has index at least $p + q - 1$. At $r = p + q$ the only boundary possibilities are two marked levels of indices p, q with $T \rightarrow +\infty$ or $T \rightarrow -\infty$, or a two-marked level of index $p + q - 1$ and an unmarked index-one level. They give the two composites and the differential terms, respectively. Additional breakings have negative cut dimension. At $r = p + q - 1$ there are no broken ends.

Ordinary instanton gluing supplies a collar at each listed boundary point; the compactness, gluing and orientation conventions are those of [Don02, Theorem 4.17, Proposition 5.11 and Sections 5.1, 5.4–5.6] and the admissible setup in [DFL26, Section 2.4]. Iterated gluing gives the compatible product neighborhoods used for the relative section choices above. The section pairs are prescribed to converge with their derivatives to the product pairs there. Transversality and the implicit function theorem identify each collar with the indicated product cut. Compactness after removing these ends and relative section transversality give a compact oriented one-manifold with exactly this boundary. No change to the equation or to N is needed.

For the sign, let ν_x be the oriented rank-four normal bundle to the first dependence stratum. The determinant of its effective normal condition after integration along X is $\det \nu_x \otimes (\det TX)^*$, of parity p . The second effective condition has parity q . Interchanging their labels has sign $(-1)^{pq}$. Equivalently the sign is $(-1)^{kl}$, since the rank-four factors are even. The two separation ends have opposite outward normals. With the ordinary coherent gluing orientations and the corresponding orientation of H_{xy} , the boundary identity is

$$(453) \quad A_x A_y - (-1)^{pq} A_y A_x = d_G H_{xy} + (-1)^{p+q} H_{xy} d_G.$$

The differential sign is that of the graded Hom differential on a map of degree $p + q - 1$. These are integer counts, including when $x = y$.

Only a pair of points has $p + q > 7$. Join q_0, q_1 by a path in the connected $Y_\#$, and evaluate the universal bundle at slice zero along that path. An interpolating pair can be kept independent at each critical connection, because the space of independent complex two-frames in $\mathbb{C}^3$ is connected. Keep the given endpoint pairs, extend over critical neighborhoods, and choose the remaining interpolation generically. Parameterized index-three cuts define a degree-three map L . The boundary of the parameterized index-four cut consists of the two endpoint point cuts and ordinary index-one breakings. The same bubbling and constant-level exclusions apply. Thus

$$(454) \quad A_{q_1} - A_{q_0} = d_G L + L d_G.$$

Writing $P = A_{q_0}$, $Q = A_{q_1}$, direct substitution gives

$$(455) \quad PQ - QP = d_G J + J d_G, \quad J = PL - LP.$$

Indeed P is an even chain map. A repeated point generator commutes with itself identically. No index-eight compactification has been used.

For completeness the symplectic commutator homotopy can be written explicitly. Put

$$F_{xy} = K_x A_y + (-1)^p B_x K_y, \quad s = (-1)^{pq}.$$

The generator identities and graded Leibniz rule give

$$d_S F_{xy} + (-1)^{p+q} F_{xy} d_G = N A_x A_y - B_x B_y N.$$

Let H be the gauge homotopy just constructed, including J in the two-point case. Then

$$(456) \quad H_{xy}^S = (NH - F_{xy} + sF_{yx})N^{-1}$$

satisfies

$$B_x B_y - s B_y B_x = d_S H_{xy}^S + (-1)^{p+q} H_{xy}^S d_S.$$

The point corrections are even central scalars, so the same equation gives the raw symplectic commutator. All maps, including N^{-1} , are integral. Tensoring these actual chain identities with R proves the assertion for every R , without a flatness assumption or an identification of $H(C_R)$ with $H(C_{\mathbb{Z}}) \otimes R$. $\square$

Denote the resulting homology actions by ρ_{G,p_1}^R and $\hat{\rho}_{S,p_1}^R$, and write N_R for the base change of N .

Corollary 13.5 (all-coefficient operation algebra). *Under the hypotheses of Theorem 12.13, for every commutative ring R the comparison isomorphism is an isomorphism of $\mathbb{A}_{p_1}(\Sigma; R)$ -modules:*

$$(457) \quad N_R \rho_{G,p_1}^R(a) = \hat{\rho}_{S,p_1}^R(a) N_R, \quad a \in \mathbb{A}_{p_1}(\Sigma; R).$$

For each chosen word representing a , the equality is induced by an explicit chain homotopy. No strict chain-level graded-commutativity or homotopy-coherence assertion is included.

Proof. Lemma 13.3 gives the integer homotopy for every generator. If it holds for homogeneous words u, v , set

$$(458) \quad K_{uv} = K_u A_v + (-1)^{|u|} B_u K_v.$$

Then the graded Leibniz rule gives

$$\begin{aligned} N A_u A_v - B_u B_v N &= (N A_u - B_u N) A_v + B_u (N A_v - B_v N) \\ &= d_S K_{uv} + (-1)^{|u|+|v|} K_{uv} d_G. \end{aligned}$$

Induction and linearity prove the identity for each ordered polynomial expression. Lemma 13.4 makes the homology actions factor through (448). The homotopy may depend on the representing word. Tensor the maps, homotopies, N and its integer inverse with R to obtain (457). $\square$

Corollary 13.6 (change of normalization). *Under the hypotheses of Theorem 12.13, suppose four is invertible in R and put $m_\sigma = \frac{1}{4} b_\sigma$. The comparison intertwines the normalized observable actions, with corrected point operator*

$$(459) \quad \hat{m}_q^S = m_q^S + \kappa_{g_j} \text{id}, \quad (\kappa_1, \kappa_2, \kappa_{g \geq 3}) = (2, 1, 0).$$

The observable algebra has the presentation (449).

Proof. Divide each integral generator identity and its homotopy by four. The coefficients 8, 4, 0 become 2, 1, 0. Since two is invertible as well, (449) applies. $\square$

Corollary 13.7. *With the normalization of Theorem 13.1, the DFL isomorphism intertwines the full rational observable algebra*

$$\mathbb{A}(\Sigma) = \text{Sym}(H_0(\Sigma; \mathbb{Q}) \oplus H_2(\Sigma; \mathbb{Q})) \otimes \Lambda H_1(\Sigma; \mathbb{Q})$$

after correcting each symplectic point generator by $\kappa_{g_j} \text{id}$. The action kernels agree, their image algebras are conjugate by N_H , and each chosen word has a chain comparison homotopy.

Proof. Apply Corollary 13.6 with $R = \mathbb{Q}$. Its generator homotopies use the same geometric counts as (446). Invertibility of N_H identifies the kernels and conjugates the image operation algebras. $\square$

Corollary 13.8 (operation images and kernels). *For every coefficient ring covered by Corollary 13.5, conjugation by N_R identifies the two image operation algebras. Moreover,*

$$(460) \quad \ker \rho_{G,p_1}^R = \ker \widehat{\rho}_{S,p_1}^R,$$

so the identity on $\mathbb{A}_{p_1}(\Sigma; R)$ induces an isomorphism of the two operation quotients.

Proof. Equation (457) and invertibility of N_R give (460); the same conjugacy identifies the images. $\square$

Remark 13.9. Corollary 13.5 concerns the action of the full observable algebra. It does not say that the mixed comparison map preserves a gauge-theoretic pair-of-pants product, and the wordwise homotopies (458) do not constitute an A_∞ module morphism. An integral operation representing $\frac{1}{4}p_1/[\sigma]$ still requires the divisibility data discussed in Remark 13.11; the all-coefficient statement instead uses the integral p_1 cycle.

Remark 13.10. The correction in genera one and two cannot be removed by changing the compactification. Proposition 5.5 detects it algebraically on $\Sigma_g \times S^1$. It is therefore part of the point operation itself under the DFL comparison, rather than an artifact of the proof.

Remark 13.11. The quarter-Pontryagin statement is proved over $\mathbb{Q}$. An integral operation representing $\frac{1}{4}p_1/[q]$ requires a divisibility choice for the universal bundle that is not used here. The integral geometric comparison is Theorem 12.13, formulated directly with the degeneracy cycle $p_1/[q]$. Its original geometric homotopy follows from the simultaneous local sign comparison.

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